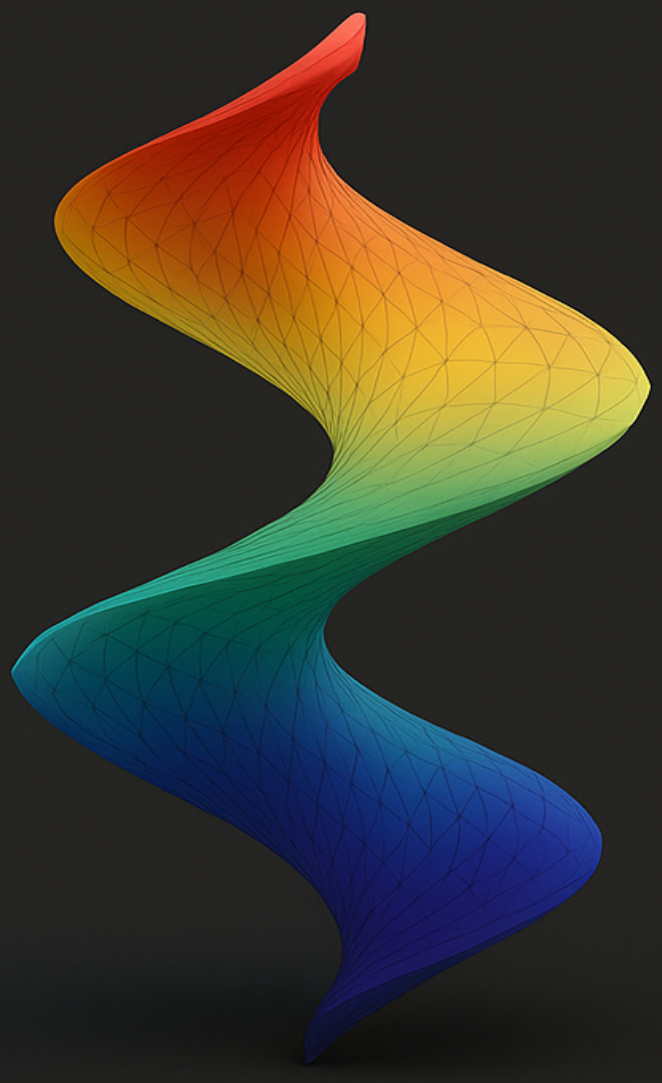
CALCULUS IN THREE DIMENSIONS

A Modern Approach



SAM HOUSTON STATE UNIVERSITY – FALL 2025

Preface

This workbook collects the core materials for Calculus III at Sam Houston State University: lecture notes, in-class activities, homework assignments, weekly quizzes, and exams. It is meant to serve both as a day-to-day course notebook and as a standalone reference after the semester.

The course is organized into *cycles*. Each cycle corresponds to a single homework set and a coherent block of ideas. In the live class, the pattern for a typical cycle is:

- **Friday:** Introduce the main idea of the cycle (for example, parametric curves, the cross product, or partial derivatives) and motivate it with geometry or applications.
- **Monday:** Develop the theory further, push into more substantial examples, and begin connecting the new idea to previously learned material.
- **Wednesday:** Consolidate the cycle with structured classwork and problem-solving time. By the end of Wednesday, students have seen the tools needed for the corresponding homework set.
- **Following Monday:** A short quiz over the previous cycle, usually at the beginning of class, to check understanding before moving fully into the next block of material.

In this book, each cycle is presented in the following order:

1. A brief *overview* summarizing the main themes and learning objectives.
2. The *lecture notes* for the relevant class meetings, including definitions, examples, and worked derivations.
3. The associated *classwork*, intended to be completed collaboratively during the Wednesday meeting or shortly thereafter.
4. The *homework assignment*, which consolidates and extends the ideas from the cycle.
5. The corresponding *quiz*, given the following Monday.

Homework solutions for all cycles are collected at the end of the book in the part titled "Homework Solutions". This preserves the feel of a traditional textbook with answers in the back. Students are encouraged to attempt each homework problem fully before consulting the worked solutions.

Exams and their learning objectives are collected in the part titled "Exams and Review Material". These documents give a global view of what the course is building toward: the exam learning objectives are, in effect, the long-term goals obtained by composing the goals of the individual cycles.

You can treat each cycle as a self-contained chapter: it introduces a set of concepts, develops them through lectures and in-class work, and then tests your understanding through homework and a quiz. Later cycles return to and build on earlier ones (for example, gradients and directional derivatives reappear inside constrained optimization; double integrals reappear in mass and center-of-mass computations; vector fields, divergence, and curl unify much of the earlier geometry). The course is deliberately cumulative and spiral in structure.

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Part I

Course Cycles

CHAPTER 1

Cycle 1: Parametric Motion and Polar Coordinates (Homework 1)

Overview

This cycle introduces parametric curves and polar coordinates as a bridge from single-variable calculus to curves traced by moving particles in the plane. The Friday and Monday lectures develop the language of parametrization, arc length, and polar graphs; Wednesday's classwork focuses on concrete computations and conceptual questions; the homework consolidates the techniques; and the following Monday's quiz checks core skills before moving on.

Lecture Notes and Classwork

August 25: Introduction to Parametric Equations

Introduction to Parametric Equations

Learning Objectives

1. Understand the definition of parametric equations, $x = f(t), y = g(t)$, and interpret them both geometrically as a curve and physically as the motion of a particle.

2. Eliminate the parameter t to derive the corresponding Cartesian equation (in the form of $y = F(x)$ or $G(x, y) = 0$).

3. Recognize that the parametrization of a curve is not unique and that different parametrizations can describe the same path, potentially with different speeds or orientations.

4. Utilize Desmos to graph parametric curves, observe the orientation (direction of motion), and manipulate sliders to understand the effect of parameters.

5. Interpret parametric curves in the context of both abstract geometric settings and simple motion problems.

Lecture

- **(0-5 min) Introduction & Administrative**
  - Brief syllabus reminders (upcoming quizzes, homework due dates).
  - Motivation: Why do we need parametric equations? (e.g., describing motion, curves that are not functions of y in terms of x).
- **(5-25 min) Core Concepts & Definitions**

  [Parametric Curve] Suppose that x and y are both given as continuous functions of a third variable t on an interval I. This variable t is called the **parameter**. The equations

  $$x = f(t), \quad y = g(t)$$

  are called **parametric equations**. The set of all points $(x, y) = (f(t), g(t))$ in the plane, traced as t varies over the interval I, is called a **parametric curve**.

Example 1: The Unit Circle

We are given the parametric equations:

$$x = \cos t, \quad y = \sin t, \quad \text{for } 0 \leq t \leq 2\pi$$

Your Task:

1. Eliminate the parameter t to find the Cartesian (x, y) equation.
2. Determine the orientation (direction of motion) of the curve as t increases.
3. Sketch the curve and add arrows.

1. **Identify the relationship between the functions.** We are given $x = \cos t$ and $y = \sin t$. When you see cosine and sine paired like this, your first thought should be the fundamental Pythagorean Identity:

$$\cos^2 t + \sin^2 t = 1$$

2. **Substitute the parametric equations into the identity.** Since we know $x = \cos t$ and $y = \sin t$, we can replace $\cos t$ with x and $\sin t$ with y in the identity.

$$(\cos t)^2 + (\sin t)^2 = 1$$

$$\downarrow \quad \downarrow$$

$$(x)^2 + (y)^2 = 1$$

3. **State the final Cartesian equation and identify the curve.** The resulting equation is:

$$x^2 + y^2 = 1$$

This is the standard equation for a circle centered at the origin $(0, 0)$ with a radius of $r = 1$. We call this the **unit circle**.

Important Idea: What is"Eliminating the Parameter?"
Eliminating the parameter is the process of converting a set of parametric equations into a single Cartesian equation (in terms of only x and y). This is useful because it allows us to identify the geometric *shape* of the path being traced. However, be aware that the Cartesian equation loses information about the *motion* (direction and speed) along the path.

4. **Understand Orientation.** The parameter t often represents time. "Orientation" is the direction the point (x, y) moves as t increases.
5. **Plot points for key values of t.** We check points at the beginning, middle, and end of the given domain for t, which is $[0, 2\pi]$.
   - **At $t = 0$ (Starting Point):**

$$x = \cos(0) = 1$$

$$y = \sin(0) = 0$$

   The starting point is $(1, 0)$.

- **At $t = \pi/2$:**

$$x = \cos(\pi/2) = 0$$

$$y = \sin(\pi/2) = 1$$

  The curve has moved to $(0, 1)$.

- **At $t = \pi$:**

$$x = \cos(\pi) = -1$$

$$y = \sin(\pi) = 0$$

  The curve has moved to $(-1, 0)$.

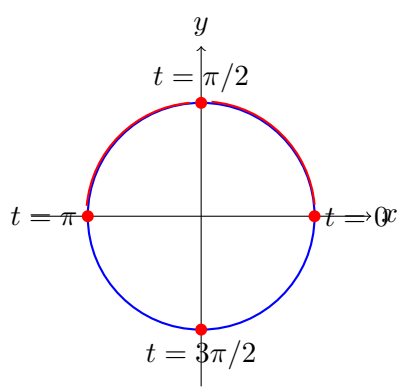
- **At $t = 2\pi$ (Ending Point):**

$$x = \cos(2\pi) = 1$$

$$y = \sin(2\pi) = 0$$

  The curve returns to the starting point $(1, 0)$.

6. By plotting these points in order, we can see the direction of travel.



The point moves **counter-clockwise** around the circle as t increases.

Example 2: An Ellipse

We are given the parametric equations:

$$x = 2\cos t, \quad y = 3\sin t, \quad \text{for } 0 \le t \le 2\pi$$

Your Task:

1. Eliminate the parameter t to find the Cartesian (x, y) equation.
2. Determine the orientation (direction of motion) of the curve as t increases.
3. Sketch the curve and add arrows.

1. **Identify the strategy.** Just like in the previous example, the presence of $\cos t$ and $\sin t$ points us toward the Pythagorean Identity: $\cos^2 t + \sin^2 t = 1$. However, we cannot directly substitute x and y. We must first isolate the trigonometric terms.
2. **Isolate the trigonometric functions.** We solve each parametric equation for its trigonometric part:
$$x = 2\cos t \quad \Longrightarrow \quad \cos t = \frac{x}{2}$$
$$y = 3\sin t \quad \Longrightarrow \quad \sin t = \frac{y}{3}$$
3. **Substitute these expressions into the identity.** Now we can replace $\cos t$ and $\sin t$ in the identity with our new expressions.
$$(\cos t)^2 + (\sin t)^2 = 1$$
$$\downarrow \qquad \downarrow$$
$$\left(\frac{x}{2}\right)^2 + \left(\frac{y}{3}\right)^2 = 1$$
4. **State the final Cartesian equation and identify the curve.** The resulting equation is:
$$\frac{x^2}{4} + \frac{y^2}{9} = 1$$
This is the standard equation for an **ellipse** centered at the origin $(0, 0)$. The denominator under x^2 is $4 = 2^2$, so the ellipse extends 2 units horizontally from the center. The denominator under y^2 is $9 = 3^2$, so it extends 3 units vertically.

General Form for Ellipses
The equations $x = a\cos t, y = b\sin t$ generally trace an ellipse centered at the origin.

- a is the semi-axis length in the x-direction.
- b is the semi-axis length in the y-direction.
- If $a = b$, the ellipse becomes a circle of radius a.

5. **Understand Orientation.** We will determine the direction of motion by tracking the point's position as t increases.

6. **Plot points for key values of t.** We use the same key values from the domain $[0, 2\pi]$.

   - **At $t = 0$ (Starting Point):**

   $$x = 2\cos(0) = 2(1) = 2$$

   $$y = 3\sin(0) = 3(0) = 0$$

   The starting point is $(2, 0)$.

   - **At $t = \pi/2$:**

   $$x = 2\cos(\pi/2) = 2(0) = 0$$

   $$y = 3\sin(\pi/2) = 3(1) = 3$$

   The curve has moved to $(0, 3)$.

   - **At $t = \pi$:**

   $$x = 2\cos(\pi) = 2(-1) = -2$$

   $$y = 3\sin(\pi) = 3(0) = 0$$

   The curve has moved to $(-2, 0)$.

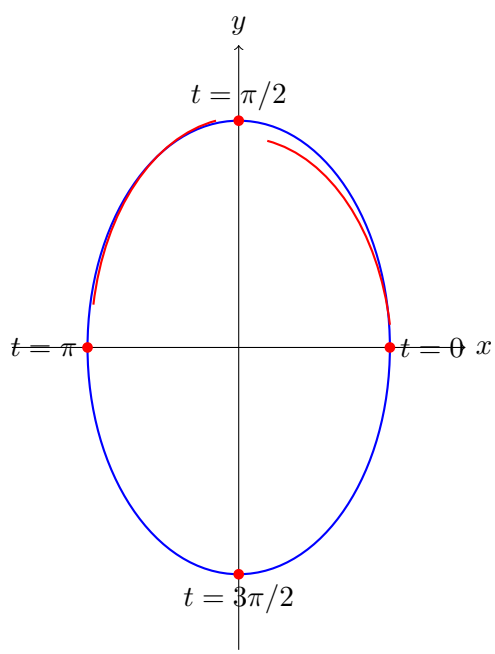
   - **At $t = 2\pi$ (Ending Point):**

   $$x = 2\cos(2\pi) = 2(1) = 2$$

   $$y = 3\sin(2\pi) = 3(0) = 0$$

   The curve returns to the starting point $(2, 0)$.

7. By plotting these points sequentially, we can sketch the curve and determine its orientation.



The point moves **counter-clockwise** around the ellipse as t increases.

Example 3: A Parabola and Orientation

We are given the parametric equations:

$$x = t^2 - 1, \quad y = 2t + 3$$

(No domain for t is specified, so we assume t can be any real number, $t \in (-\infty, \infty)$.)

Your Task:

1. Eliminate the parameter t to find the Cartesian (x, y) equation.
2. Determine the orientation (direction of motion) of the curve as t increases.
3. Sketch the curve and add arrows.

1. **Identify the strategy.** These equations do not involve trigonometric functions. Therefore, our strategy will be algebraic substitution. We will solve one of the equations for the parameter t and then substitute that expression into the other equation.
2. **Choose the easier equation and solve for t.** The equation $y = 2t + 3$ is linear with respect to t, making it much simpler to solve than the quadratic equation $x = t^2 - 1$.

$$y = 2t + 3$$

$$y - 3 = 2t$$

$$t = \frac{y - 3}{2}$$

3. **Substitute this expression for t into the other equation.** Now we take our expression for t and substitute it into the equation for x.

$$x = t^2 - 1$$

$$\downarrow$$

$$x = \left(\frac{y-3}{2}\right)^2 - 1$$

4. **State the final Cartesian equation and identify the curve.** Simplifying the expression gives us the Cartesian equation:

$$x = \frac{(y-3)^2}{4} - 1$$

This is the equation of a **parabola**. Since the y term is squared, it is a "sideways" parabola. The vertex occurs where the squared term is zero, so at $y = 3$, which gives $x = -1$. The vertex is at $(-1, 3)$. Because the coefficient on the squared term $(1/4)$ is positive, the parabola opens to the right.

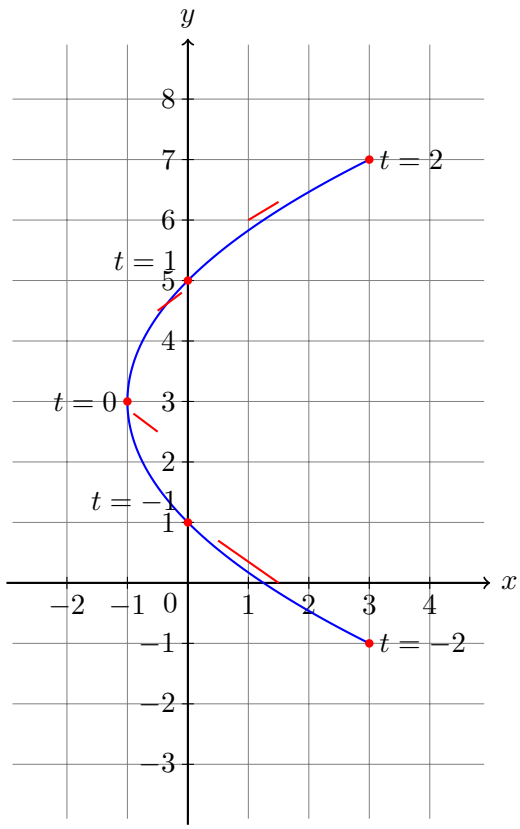
> **Strategy: Substitution vs. Identities**
> Your method for eliminating the parameter depends on the functions involved.
> - If you see $\cos t$ and $\sin t$, think Pythagorean Identity: $\cos^2 t + \sin^2 t = 1$.
> - If you see $\sec t$ and $\tan t$, think Pythagorean Identity: $\sec^2 t - \tan^2 t = 1$.
> - If the equations are polynomials or rational functions, use algebraic substitution: solve for t in the simpler equation and substitute into the other.

5. **Understand Orientation.** Since t can be any real number, we will plot points for several values of t (negative, zero, and positive) to trace the path of the curve.

6. **Create a table of values.**

t	$x = t^2 - 1$	$y = 2t + 3$	**Point** (x, y)
-2	$(-2)^2 - 1 = 3$	$2(-2) + 3 = -1$	(3, -1)
-1	$(-1)^2 - 1 = 0$	$2(-1) + 3 = 1$	(0, 1)
0	$(0)^2 - 1 = -1$	$2(0) + 3 = 3$	(-1, 3) **(Vertex)**
1	$(1)^2 - 1 = 0$	$2(1) + 3 = 5$	(0, 5)
2	$(2)^2 - 1 = 3$	$2(2) + 3 = 7$	(3, 7)

7. By plotting these points in order of increasing t, we can visualize the motion and sketch the curve.



As t increases from $-\infty$ to ∞, the point moves up the parabola. The orientation is from the bottom arm, through the vertex, and up the top arm in the direction of increasing y.

(25-40 min) Desmos Demonstration: Visualizing Parametric Curves

Demonstration Goal: From Static Equations to Dynamic Motion

Our algebraic work helps us find the static *path* of a curve. Now, we will use the free, browser-based tool Desmos (desmos.com/calculator) to visualize the *motion*—the dynamic tracing of that path over time.

Demonstration Steps:

1. Plot a basic parametric curve and adjust its domain.
2. Use sliders to explore how constants affect the shape of a curve.
3. Animate the drawing of a curve to reveal its orientation.
4. Preview how these concepts extend to 3D space curves.

Step 1: Plotting a Curve and Adjusting its Domain

To graph a parametric curve in Desmos, you simply type an ordered pair using the parameter t.

1. In a new graph, type $(\cos t, \sin t)$ into the expression list.

2. **Initial Observation:** Desmos automatically creates a domain for t, typically $0 \leq t \leq 1$. This partial curve is not what we want.

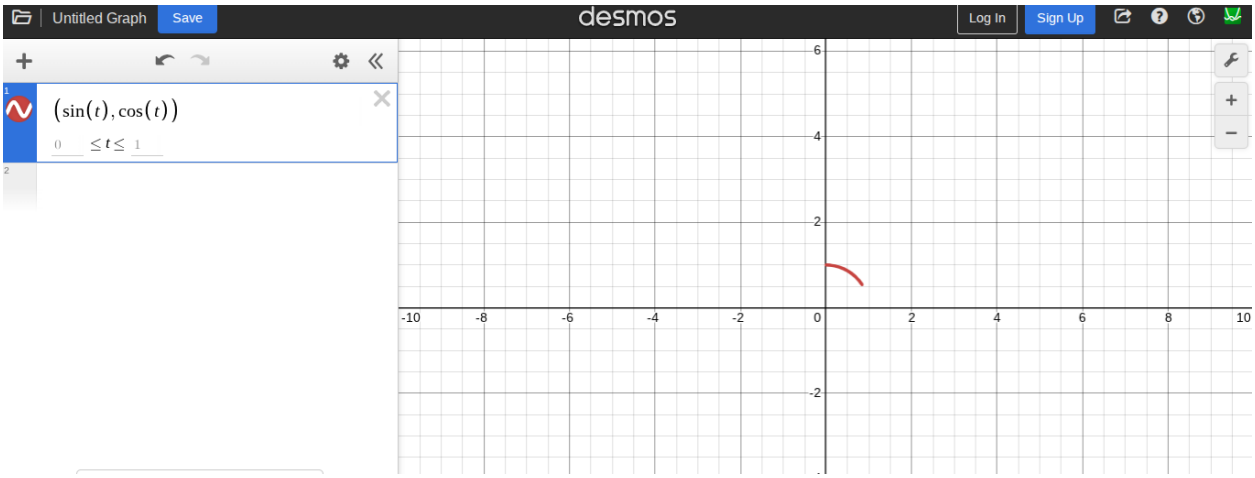


Figure 1.1: Initial plot of $(\cos t, \sin t)$. The curve is only a small arc because Desmos defaults to the domain $0 \leq t \leq 1$.

3. **Correcting the Domain:** Click on the numbers in the domain entry (e.g., $0 \leq t \leq 1$) to edit them.

4. Change the upper bound from 1 to 2π. You can type 'pi' and Desmos will convert it to the symbol π. The complete unit circle now appears.

5. **Experimentation:** Show how changing the domain to $0 \leq t \leq \pi$ produces just the top half, reinforcing the link between the parameter's value and the point's position.

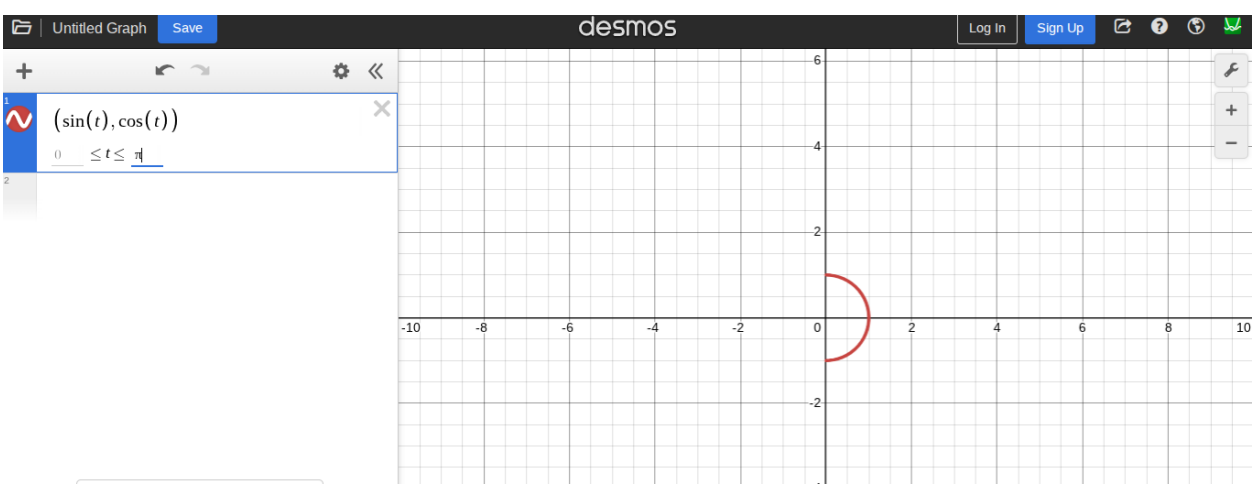


Figure 1.2: Editing the domain of t to $0 \leq t \leq \pi$ produces the half of the curve.

Step 2: Using Sliders to Explore a Family of Curves

Sliders allow us to see how constants affect a graph in real-time, which is excellent for building intuition.

1. Enter the general parametric equation for an ellipse: $(a\cos t, b\sin t)$.
2. Desmos will recognize that a and b are undefined variables and will offer to "add slider" for them. Click the "all" button.
3. Sliders for a and b appear. Play them to show how they control the horizontal and vertical semi-axes of the ellipse.
4. Set $a = b$ to demonstrate how the ellipse becomes a perfect circle.

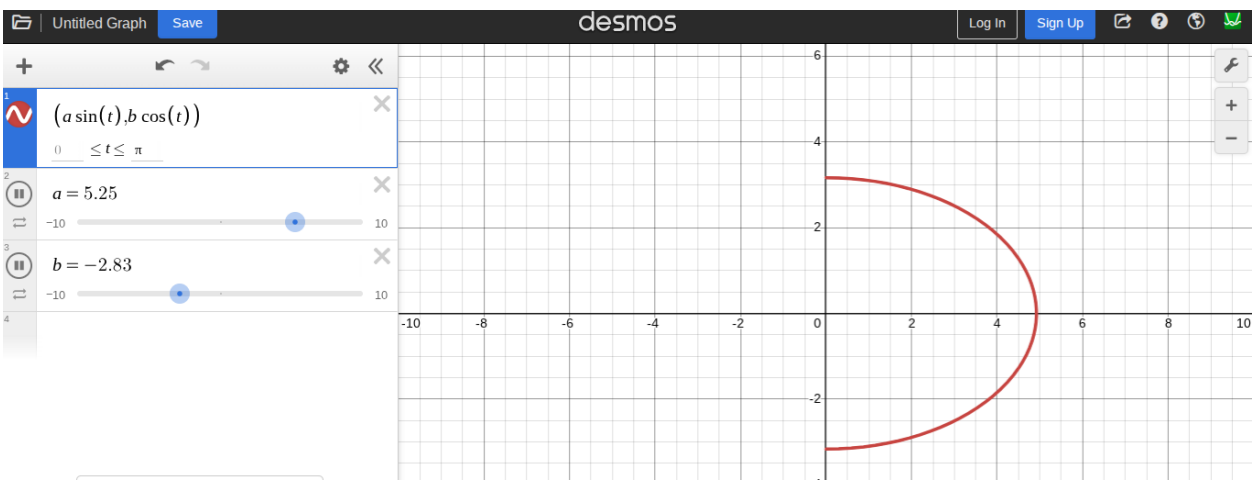


Figure 1.3: Using sliders for a and b to dynamically change the shape of the ellipse.

Step 3: The Best Way to See Orientation

A static graph doesn't show the direction of motion. We can use a slider to animate the drawing process, making the orientation crystal clear.

1. Graph our unit circle again: $(\cos t, \sin t)$.
2. Set the domain for t, but this time, use a variable for the upper bound: $0 \leq t \leq c$.
3. Desmos will prompt you to "add slider" for c. Do this.
4. Set the range of the slider c to be from 0 to 2π. Press the "play" button next to the c slider. You will see the circle being drawn dynamically. This animation clearly shows the **counter-clockwise orientation**.

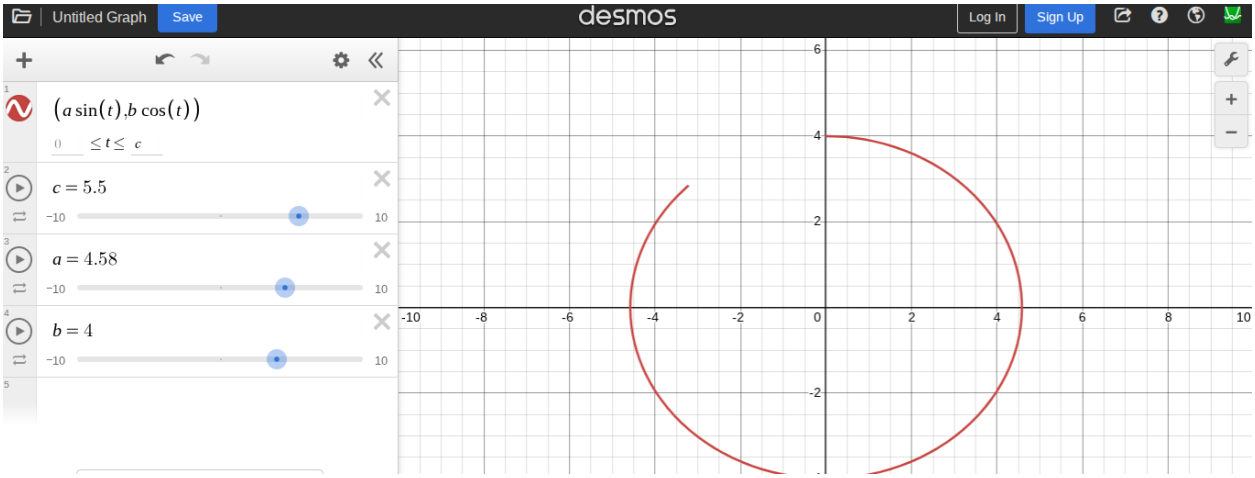


Figure 1.4: Setting the upper bound of the parameter t to a slider c and pressing play animates the drawing of the curve, revealing its orientation.

Step 4: A Glimpse into 3D Space Curves

This entire concept extends beautifully into three dimensions. To describe a curve in space, we simply add a third equation for the z-coordinate.

$$x = f(t), \quad y = g(t), \quad z = h(t)$$

Desmos has a separate 3D calculator for this, available at desmos.com/3d. A classic example is a **helix** (like a spring). We can graph it in the 3D calculator by entering the point $(\cos t, \sin t, t)$. As t increases, the point circles in the xy-plane while also rising in the z-direction. This is a topic we will explore in detail later in the course.

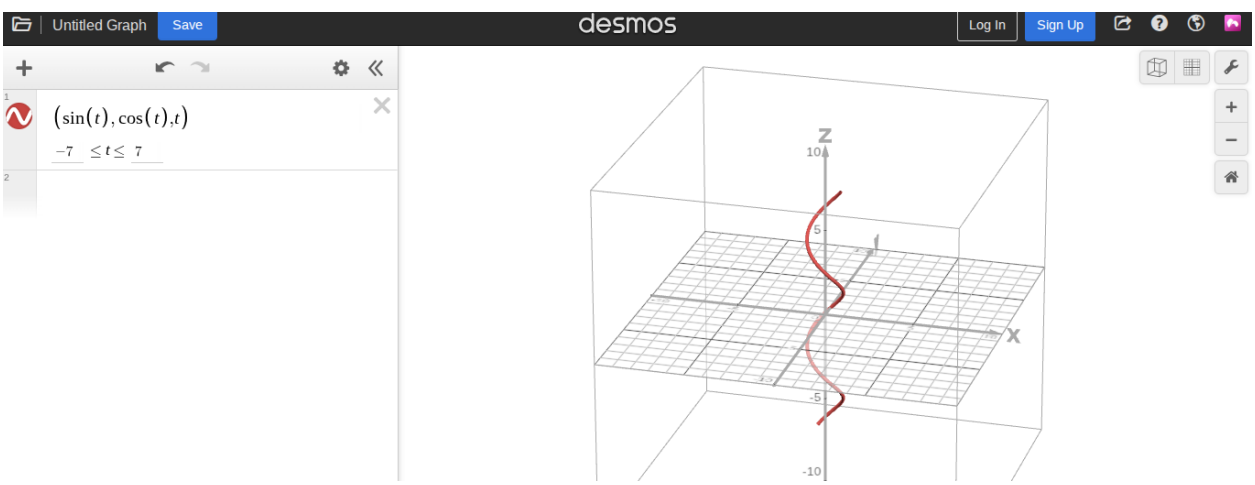


Figure 1.5: A preview of a 3D parametric curve (a helix) in the Desmos 3D Calculator.

Classwork - Parametric Equations

Name:_____

Problem 1

We are given the parametric equations:

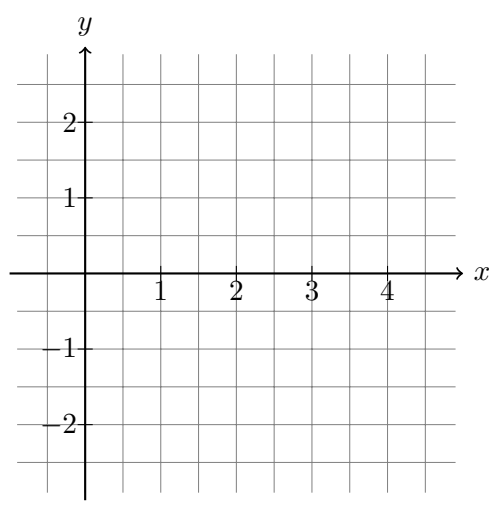
$$x = t^2, \quad y = t^3 - t$$

1. Eliminate the parameter t to find the Cartesian equation.

2. Complete the table to find points on the curve.

t	$x = t^2$	$y = t^3 - t$	**Point** (x, y)
-1.5			
-1			
0			
1			
1.5			

3. Sketch the curve. Use arrows to show the orientation (direction of increasing t).



Problem 2

Consider two parametrizations:

- **Curve A:** $x = \cos t, \quad y = \sin t$, for $0 \leq t \leq 2\pi$
- **Curve B:** $x = \cos(2t), \quad y = \sin(2t)$, for $0 \leq t \leq \pi$

1. Show that both curves trace the same path by finding their Cartesian equations.

   Curve A:

   Curve B:

2. Complete the tables to compare the motion of each particle.

Curve A: $t \in [0, 2\pi]$

t	**Point** (x, y)
0	
$\pi/2$	
π	
2π	

Curve B: $t \in [0, \pi]$

t	**Point** (x, y)
0	
$\pi/4$	
$\pi/2$	
π	

3. Which particle moves faster? Explain your reasoning.

August 27: Calculus of Parametric Curves

Learning Objectives

By the end of this lecture, students will be able to:

1. Compute the first derivative dy/dx of a parametrically defined curve.
2. Find the equation of a tangent line to a parametric curve at a given point.
3. Identify points where a parametric curve has horizontal or vertical tangents.
4. Compute the second derivative d^2y/dx^2 and use it to determine the concavity of a curve.
5. Set up and evaluate integrals to find the arc length of a parametric curve.

Lecture: Calculus of Parametric Curves

The First Derivative: Slope of the Tangent Line

Given a parametric curve $x = x(t)$ and $y = y(t)$, how do we find the slope of the tangent line, dy/dx? We use the Chain Rule:

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$$

Assuming $\frac{dx}{dt} \neq 0$, we can solve for $\frac{dy}{dx}$:

[First Derivative of a Parametric Curve] If $x = x(t)$ and $y = y(t)$ are differentiable functions, then

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}, \quad \text{provided } \frac{dx}{dt} \neq 0$$

Horizontal and Vertical Tangents

- A **horizontal tangent** occurs when the slope is zero. This happens when $\frac{dy}{dt} = 0$ (and $\frac{dx}{dt} \neq 0$).
- A **vertical tangent** occurs when the slope is undefined. This happens when $\frac{dx}{dt} = 0$ (and $\frac{dy}{dt} \neq 0$).

Example 1: Finding a Tangent Line

Find the equation of the tangent line to the ellipse $x = 2\cos t$, $y = 3\sin t$ at $t = \pi/6$.

1. **Find the derivatives with respect to t:**

$$\frac{dx}{dt} = -2\sin t \qquad \frac{dy}{dt} = 3\cos t$$

2. **Find the formula for dy/dx:**

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{3\cos t}{-2\sin t} = -\frac{3}{2}\cot t$$

3. **Evaluate the slope at $t = \pi/6$:**

$$m = \left.\frac{dy}{dx}\right|_{t=\pi/6} = -\frac{3}{2}\cot(\pi/6) = -\frac{3}{2}(\sqrt{3}) = -\frac{3\sqrt{3}}{2}$$

4. **Find the coordinates of the point at $t = \pi/6$:**

$$x(\pi/6) = 2\cos(\pi/6) = 2\left(\frac{\sqrt{3}}{2}\right) = \sqrt{3}$$

$$y(\pi/6) = 3\sin(\pi/6) = 3\left(\frac{1}{2}\right) = \frac{3}{2}$$

The point is $(\sqrt{3}, 3/2)$.

5. **Write the equation of the tangent line** using point-slope form, $y - y_1 = m(x - x_1)$:

$$y - \frac{3}{2} = -\frac{3\sqrt{3}}{2}(x - \sqrt{3})$$

The Second Derivative: Concavity

To find the second derivative, we must differentiate the *first* derivative (dy/dx) with respect to x, again using the chain rule.

[Second Derivative of a Parametric Curve]

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{\frac{dx}{dt}}$$

Common Mistake Warning!
The second derivative is NOT $\frac{d^2y/dt^2}{d^2x/dt^2}$. You must differentiate the first derivative (dy/dx) with respect to t and then divide by the original dx/dt.

Example 2: Second Derivative and Concavity

Find $\frac{d^2y}{dx^2}$ for the curve $x = t^2$, $y = t^3 - 3t$.

1. **Find the first derivative, dy/dx:**

$$\frac{dx}{dt} = 2t \qquad \frac{dy}{dt} = 3t^2 - 3 \implies \frac{dy}{dx} = \frac{3t^2 - 3}{2t} = \frac{3}{2}t - \frac{3}{2t}$$

2. **Differentiate dy/dx with respect to t:**

$$\frac{d}{dt}\left(\frac{dy}{dx}\right) = \frac{d}{dt}\left(\frac{3}{2}t - \frac{3}{2}t^{-1}\right) = \frac{3}{2} - \frac{3}{2}(-1)t^{-2} = \frac{3}{2} + \frac{3}{2t^2}$$

3. **Divide by dx/dt:**

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{dx/dt} = \frac{\frac{3}{2} + \frac{3}{2t^2}}{2t} = \frac{\frac{3t^2+3}{2t^2}}{2t} = \frac{3(t^2+1)}{4t^3}$$

The curve is concave up when $d^2y/dx^2 > 0$, which occurs when $t > 0$.

Arc Length

The formula for the length of a parametric curve comes from the Pythagorean theorem, just like in single-variable calculus.

[Arc Length of a Parametric Curve] The length L of a curve defined by $x = x(t), y = y(t)$ for $\alpha \leq t \leq \beta$ is given by

$$L = \int_\alpha^\beta \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$$

Example 3: Arc Length of a Circle

Find the circumference of a circle of radius r parametrized by $x = r\cos t, y = r\sin t$ for $0 \leq t \leq 2\pi$.

1. **Find derivatives:** $\frac{dx}{dt} = -r\sin t$ and $\frac{dy}{dt} = r\cos t$.

2. **Set up the integrand:**

$$\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} = \sqrt{(-r\sin t)^2 + (r\cos t)^2} = \sqrt{r^2\sin^2 t + r^2\cos^2 t} = \sqrt{r^2(1)} = r$$

3. **Integrate:**

$$L = \int_0^{2\pi} r\, dt = [rt]_0^{2\pi} = r(2\pi) - r(0) = 2\pi r$$

Classwork - Calculus of Parametric Curves

Name: _____

Problem 1: Tangent Line Equation

Consider the parametric curve given by $x(t) = t^2 + 1$ and $y(t) = t^3 + t$.

Find the equation of the tangent line to the curve at the point corresponding to $t = 1$.

1. Find the derivatives $\dfrac{dx}{dt}$ and $\dfrac{dy}{dt}$.

2. Find the slope $\dfrac{dy}{dx}$ at the specific value $t = 1$.

3. Find the coordinates (x, y) of the point on the curve at $t = 1$.

4. Use the point-slope formula to write the equation of the tangent line.

Problem 2: Horizontal and Vertical Tangents

Consider the curve given by $x(t) = t^3 - 12t$ and $y(t) = t^2 - 2t$.

Find all points (x, y) where the curve has a horizontal tangent and all points where it has a vertical tangent.

1. **For Horizontal Tangents:** Find the values of t that make $\frac{dy}{dt} = 0$. Then, plug these t-values into the original equations for $x(t)$ and $y(t)$ to find the (x, y) coordinates.

2. **For Vertical Tangents:** Find the values of t that make $\frac{dx}{dt} = 0$. Then, plug these t-values into the original equations for $x(t)$ and $y(t)$ to find the (x, y) coordinates.

Problem 3: Setting Up an Arc Length Integral

Set up, but **do not evaluate**, the integral that gives the length of the curve

$$x(t) = t^2, \quad y(t) = \sin(t)$$

from $t = 0$ to $t = \pi$.

1. Write down the general formula for the arc length L of a parametric curve.

2. Find the necessary derivatives, $\dfrac{dx}{dt}$ and $\dfrac{dy}{dt}$.

3. Substitute your derivatives and the given bounds into the arc length formula to get the final integral.

August 29: Polar Coordinates and Area

Learning Objectives

By the end of this lecture, students will be able to:

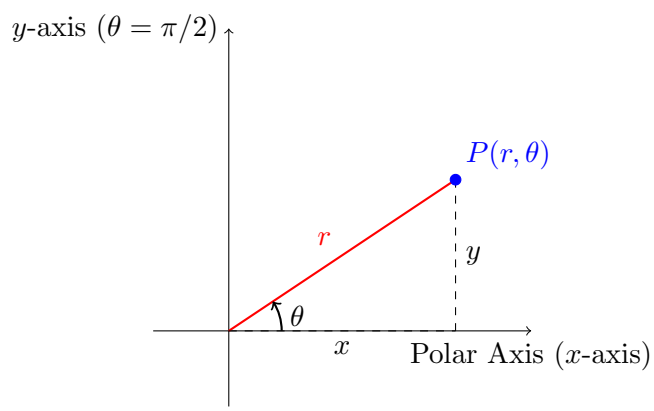
1. Find the area of a region bounded by a polar curve.
2. Find the area of the region between two polar curves.
3. Plot points in 3D rectangular and cylindrical coordinates.
4. Recognize that coordinate representations are not always unique.

A Quick Review of Polar Coordinates

Before we can find the area of regions described by polar functions, we must be comfortable with the polar coordinate system itself. While the rectangular (Cartesian) system uses (x, y) to locate a point on a grid, the polar system uses a distance and an angle.

[Polar Coordinates] A point P in the polar coordinate system is represented by an ordered pair (r, θ), where:

- **r** is the **directed distance** from the origin (called the pole) to the point P.
- θ is the **angle** measured counterclockwise from the positive x-axis (called the polar axis) to the ray connecting the pole to the point P.



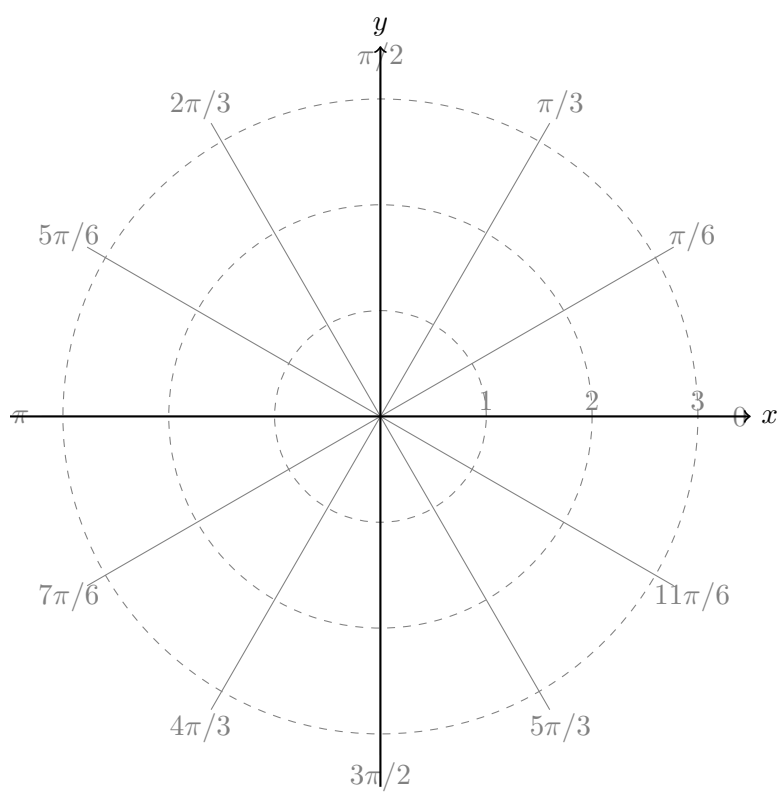
Polar-Rectangular Conversion Formulas These formulas allow us to move between the two coordinate systems.

$$x = r\cos\theta \qquad r^2 = x^2 + y^2$$
$$y = r\sin\theta \qquad \tan\theta = \frac{y}{x}$$

Example: Plotting Points and Non-Uniqueness

(a) Plot the point $(2, \pi/3)$**.**
First, rotate counterclockwise by an angle of $\pi/3$. Then, move 2 units out from the pole along that ray.
(b) Plot the point $(-3, 5\pi/6)$**.**
First, find the ray for the angle $5\pi/6$. Because r is negative, we move 3 units in the *opposite* direction from the pole (along the ray for $11\pi/6$).
Important Note on Non-Uniqueness: Unlike rectangular coordinates, a single point can have infinitely many polar coordinate representations. For example, the point from part (a) can also be described as:

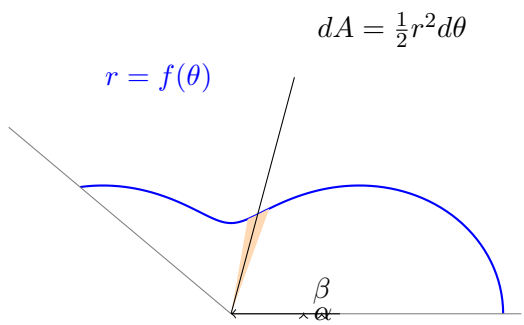
- $(2, \pi/3 + 2\pi) = (2, 7\pi/3)$ (adding a full rotation)
- $(-2, \pi/3 + \pi) = (-2, 4\pi/3)$ (using a negative radius and adding a half rotation)



In single-variable calculus, we found the area under a curve by summing up infinitely many small rectangles. In polar coordinates, our fundamental shape is not a rectangle, but a **sector** of a circle.

Recall that the area of a sector of a circle with radius r and angle θ (in radians) is $A = \frac{1}{2}r^2\theta$.

To find the area of a region defined by a polar function $r = f(\theta)$, we can imagine slicing the region into infinitely many tiny sectors. Each small sector has an angle $d\theta$, and its area is approximately $dA = \frac{1}{2}r^2 d\theta = \frac{1}{2}[f(\theta)]^2 d\theta$.



‘‘‘

Summing these infinitesimal areas with a definite integral gives us our main formula.

[Area of a Polar Region] The area of the region bounded by the graph of $r = f(\theta)$ between $\theta = \alpha$ and $\theta = \beta$ is given by:

$$A = \int_\alpha^\beta \frac{1}{2}[f(\theta)]^2\, d\theta = \int_\alpha^\beta \frac{1}{2}r^2\, d\theta$$

Example 1: Area of a Cardioid

Find the area of the region enclosed by the cardioid $r = 1 + \cos\theta$.

Solution Strategy: First, we determine the bounds of integration. The cardioid completes a full loop as θ goes from 0 to 2π. Then we apply the area formula.

1. **Set up the integral:** The function is $r = 1 + \cos\theta$. The bounds are $\alpha = 0$ and $\beta = 2\pi$.

$$A = \int_0^{2\pi} \frac{1}{2}(1 + \cos\theta)^2\, d\theta$$

2. **Expand the integrand:**

$$A = \frac{1}{2}\int_0^{2\pi} (1 + 2\cos\theta + \cos^2\theta)\, d\theta$$

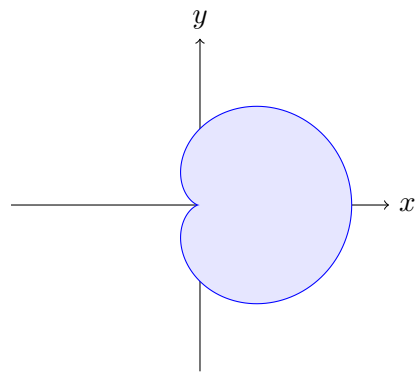
To integrate $\cos^2\theta$, we must use the half-angle identity: $\cos^2\theta = \frac{1}{2}(1 + \cos(2\theta))$.

$$A = \frac{1}{2}\int_0^{2\pi} \left(1 + 2\cos\theta + \frac{1}{2}(1 + \cos(2\theta))\right) d\theta$$

$$A = \frac{1}{2}\int_0^{2\pi} \left(\frac{3}{2} + 2\cos\theta + \frac{1}{2}\cos(2\theta)\right) d\theta$$

3. **Evaluate the integral:**

$$\begin{aligned} A &= \frac{1}{2}\left[\frac{3}{2}\theta + 2\sin\theta + \frac{1}{4}\sin(2\theta)\right]_0^{2\pi} \\ &= \frac{1}{2}\left[\left(\frac{3}{2}(2\pi) + 2\sin(2\pi) + \frac{1}{4}\sin(4\pi)\right) - (0+0+0)\right] \\ &= \frac{1}{2}[3\pi] = \frac{3\pi}{2} \end{aligned}$$



Area Between Two Polar Curves

To find the area of the region between two curves, $r_{\text{outer}} = f(\theta)$ and $r_{\text{inner}} = g(\theta)$, we subtract the area of the inner region from the area of the outer region.

[Area Between Two Polar Curves]

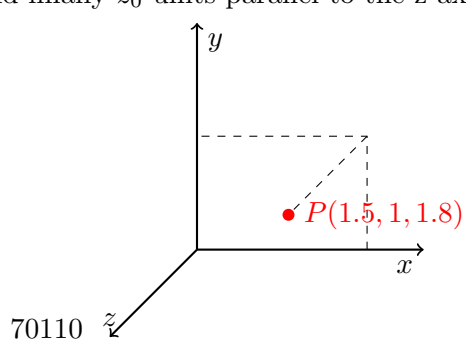
$$A = \int_\alpha^\beta \frac{1}{2}\left((r_{\text{outer}})^2 - (r_{\text{inner}})^2\right)\, d\theta$$

3D Coordinate Systems

Now we move from the 2D plane to 3D space. While we will eventually do calculus in 3D, today's goal is to become familiar with the coordinate systems themselves.

Rectangular Coordinates (x, y, z)

This is the most direct extension of the 2D Cartesian plane. We add a third axis, z, perpendicular to both the x and y axes. The three axes are mutually orthogonal and typically form a right-handed system. To plot a point $P(x_0, y_0, z_0)$, you start at the origin, move x_0 units along the x-axis, y_0 units parallel to the y-axis, and finally z_0 units parallel to the z-axis.



In this system, every point in space has a **unique** address (x, y, z).

Cylindrical Coordinates (r, θ, z)

This system simply combines 2D polar coordinates with a vertical z-axis. It's excellent for describing objects with rotational symmetry, like cylinders or cones.

- r: distance from the z-axis ($r \geq 0$).
- θ: angle in the xy-plane from the positive x-axis.
- z: height above the xy-plane.

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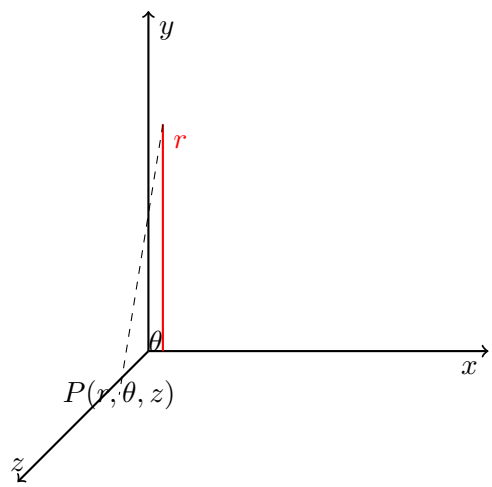


Classwork - Polar Area and 3D Coordinates

Name: _____

Problem 1: Area of a Circle in Polar Coordinates

Set up and evaluate a polar integral to find the area of the region enclosed by the circle $r = 4$.

(a) What are the bounds for θ to trace the entire circle?

(b) Set up the integral using the polar area formula $A = \int_\alpha^\beta \frac{1}{2} r^2 \, d\theta$.

(c) Evaluate the integral to find the area. Does your answer match the geometric formula $A = \pi r^2$?

Problem 2: Area of a Single Rose Petal

Find the area of a single petal of the rose curve given by $r = \cos(2\theta)$.

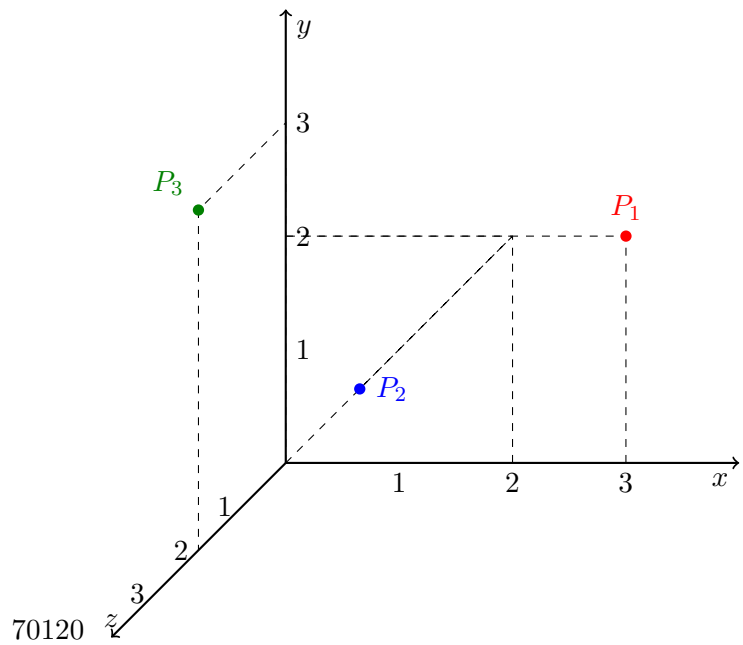
(a) Find the bounds for θ that trace out the petal in the first quadrant. (Hint: When is $r = 0$?)

(b) Set up the integral for the area.

(c) Evaluate the integral. You will need the identity: $\cos^2(x) = \frac{1}{2}(1 + \cos(2x))$.

Problem 3: Identifying 3D Coordinates

Three points, P_1, P_2, and P_3, are plotted below. Determine the rectangular coordinates (x, y, z) for each point. Then convert to cylindrical coordinates.



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$P_1 = ($_____, _____, _____$)$

$P_2 = ($_____, _____, _____$)$

$P_3 = ($_____, _____, _____$)$

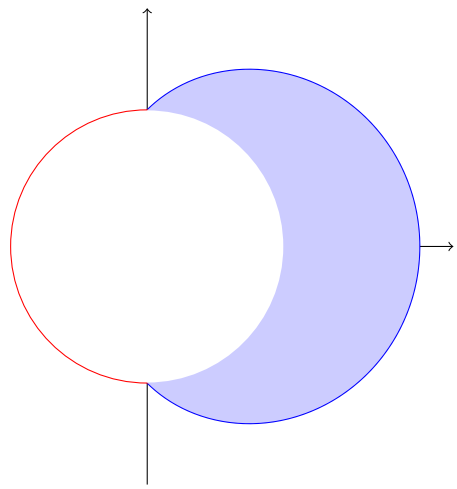
Problem 4: Area Between Two Polar Curves

Consider the region that lies inside the cardioid $r = 2 + 2\cos\theta$ and outside the circle $r = 2$, as shaded in the figure. Set up the definite integral that gives the area of this region. **Do not evaluate the integral.**

(a) Find the intersection points of the two curves to determine the bounds of integration.

(b) Identify which curve is r_{outer} and which is r_{inner}.

(c) Write the definite integral for the area.



Homework 1

Homework - 1

Work solutions on blank paper, showing all steps. Scan and submit by Sunday at 11:59 pm on Blackboard. Solutions will be posted 1–2 days before the deadline.

Problem 1

We are given the parametric equations for an ellipse:

$$x = 2\cos t, \quad y = 3\sin t, \quad 0 \le t \le 2\pi$$

1. Eliminate the parameter to find the Cartesian equation.
2. Sketch the curve and indicate orientation.

Problem 2

We are given the parametric equations for a line segment:

$$x = 1 + 2t, \quad y = t, \quad 0 \leq t \leq 3$$

1. Eliminate t to find the Cartesian equation.
2. Find the start and end points of the segment.

Problem 3

We are given:

$$x = t^3 - t, \quad y = t^2$$

1. Sketch the curve for $t \in [-2, 2]$, showing direction.
2. Find the coordinates of the self-intersection point.

Problem 4

We are given:

$$x = 4 + \cos t, \quad y = -2 + \sin t$$

1. Eliminate t to find the Cartesian equation.
2. Find the center and radius of the circle.

Problem 5

The unit circle is parametrized by $x = \cos t, y = \sin t$. Provide two **different** parametrizations. For each, describe orientation and speed.

Problem 6

We are given:

$$x = \sin t, \quad y = \cos t, \quad 0 \leq t \leq 2\pi$$

1. Show this is the unit circle.
2. Determine its orientation by checking points at $t = 0, \pi/2$.

Problem 7 (Desmos Required)

Use Desmos to graph:

$$\text{Curve 1: } (\cos t, \sin t), \quad 0 \leq t \leq 2\pi$$

$$\text{Curve 2: } (\cos(3t), \sin(3t)), \quad 0 \leq t \leq 2\pi$$

Describe the motion and how many times each particle travels around the circle.

Problem 8

We are given:

$$x = \sqrt{t}, \quad y = 1 - t, \quad t \geq 0$$

1. Eliminate t to relate x, y.
2. Sketch the curve. Why is it only part of the parabola?

Problem 9

Projectile motion:

$$x = 10t, \quad y = 40t - 5t^2$$

1. Eliminate t to express y in terms of x.
2. What is the shape of the path?

Problem 10

We are given:

$$x = \sec t, \quad y = \tan t, \quad -\tfrac{\pi}{2} < t < \tfrac{\pi}{2}$$

Eliminate t using $1 + \tan^2 t = \sec^2 t$.

Problem 11

Find the equation of the tangent line to the parametric curve $x(t) = t^2 - 3$, $y(t) = 3t - t^3$ at the point $(1, 2)$.

Problem 12

Find the equation of the tangent line to the parametric curve $x(t) = \cos t$, $y(t) = \sin(2t)$ at $t = \pi/6$.

Problem 13

Find all points on the parametric curve $x(t) = t^2 - 2t$, $y(t) = t^3 - 12t$ where the tangent line is horizontal.

Problem 14

Find all points on the parametric curve $x(t) = 2\cos t$, $y(t) = \sin(2t)$ where the tangent line is vertical.

Problem 15

For the parametric curve $x(t) = t^2$ and $y(t) = t^3 - 3t$, find the second derivative d^2y/dx^2 and determine for which values of t the curve is concave upward.

Problem 16

For the parametric curve $x(t) = te^{-t}$ and $y(t) = t^2e^{-t}$, find the second derivative d^2y/dx^2.

Problem 17

Find the length of the parametric curve $x(t) = t^2$ and $y(t) = t^3$ for $0 \leq t \leq 1$.

Problem 18

Find the length of the parametric curve $x(t) = \cos t + t \sin t$ and $y(t) = \sin t - t \cos t$ for $0 \leq t \leq 2\pi$.

Problem 19

Find the area of the surface generated by revolving the curve $x(t) = t^3$ and $y(t) = t^2$ for $0 \leq t \leq 1$ about the x-axis.

Problem 20

Find the area of the surface generated by revolving the curve $x(t) = \cos^2 t$ and $y(t) = \sin^2 t$ for $0 \leq t \leq \pi/2$ about the y-axis.

Quiz 1

Quiz 1

Name: _____

(Show all your work for full credit.)

Problem 1: Polar Coordinates

(a) Convert the rectangular coordinates $(x, y) = (2, 2)$ to polar coordinates (r, θ), where $r > 0$ and $0 \leq \theta < 2\pi$.

(b) Find the area of a single petal of the rose curve $r = \cos(2\theta)$. (Hint: One petal is centered on the x-axis, traced from $\theta = -\pi/4$ to $\theta = \pi/4$). You may need the identity $\cos^2(u) = \frac{1+\cos(2u)}{2}$.

Problem 2: Vector Operations

Let $\vec{u} = \langle 2, 1, -2 \rangle$ and $\vec{v} = 3\vec{i} - 4\vec{k}$. Compute the following.

(a) The vector $2\vec{u} - \vec{v}$ in component form.

(b) The magnitude of $\vec{v}$, i.e., $||\vec{v}||$.

(c) The unit vector in the same direction as $\vec{u}$.

CHAPTER 2

Cycle 2: Vectors, Dot Products, and Cross Products (Homework 2)

Overview

This cycle develops vectors in $\mathbb{R}^3$, the dot product, and the cross product. The focus is on geometric quantities such as lengths, angles, areas, and volumes, and on interpreting vector operations both algebraically and geometrically.

Lecture Notes and Classwork

September 3: Introduction to Vectors

Learning Objectives

By the end of this lecture, students will be able to:

1. Represent vectors in component form $\langle x, y, z \rangle$ and as a linear combination of unit vectors $\vec{i}, \vec{j}, \vec{k}$.
2. Calculate the result of vector operations (addition, scalar multiplication) and interpret them graphically.
3. Determine the magnitude of a vector and find the corresponding unit vector.

Lecture: Introduction to Vectors

Representing Vectors

What is a vector? A vector is a mathematical object that has both **magnitude** (length or size) and **direction**. We represent a vector graphically as a directed line segment or arrow.

[Component Form] A vector is in **standard position** if its initial point is at the origin. A vector in standard position is uniquely determined by the coordinates of its terminal point (v_1, v_2, v_3). We write this vector in **component form** as:

$$\vec{v} = \langle v_1, v_2, v_3 \rangle$$

The numbers v_1, v_2, and v_3 are the **components** of the vector.

To find the vector between two arbitrary points, we subtract the coordinates of the initial point from the coordinates of the terminal point.

Vector Between Two Points
The vector $\vec{AB}$ from an initial point $A(x_1, y_1, z_1)$ to a terminal point $B(x_2, y_2, z_2)$ is given by:

$$\vec{AB} = \langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle$$

Example 1: Finding a Vector from Points

Find the vector that starts at point $A(1, 4, -2)$ and ends at point $B(3, 1, 1)$.

Solution: We subtract the coordinates of A from the coordinates of B.

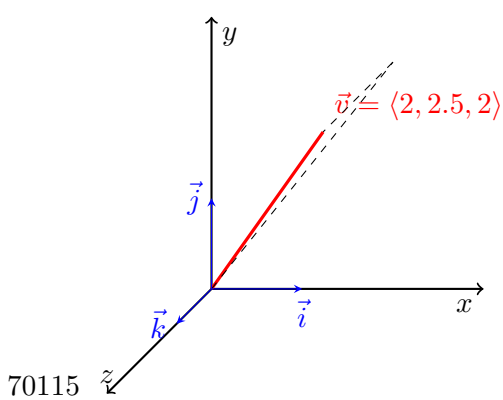
$$\vec{AB} = \langle 3 - 1, 1 - 4, 1 - (-2) \rangle = \langle 2, -3, 3 \rangle$$

Another way to represent vectors is using the **standard unit vectors**. These are vectors of length 1 that point along the positive coordinate axes.

$$\vec{i} = \langle 1, 0, 0 \rangle \quad \vec{j} = \langle 0, 1, 0 \rangle \quad \vec{k} = \langle 0, 0, 1 \rangle$$

Any vector can be written as a sum of scalar multiples of these standard unit vectors. This is called a **linear combination**.

$$\vec{v} = \langle v_1, v_2, v_3 \rangle = v_1\vec{i} + v_2\vec{j} + v_3\vec{k}$$



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Vector Operations

Vector operations are performed component by component. Let $\vec{u} = \langle u_1, u_2, u_3 \rangle$, $\vec{v} = \langle v_1, v_2, v_3 \rangle$, and let c be a scalar (a real number).

Vector Operations

- **Addition:** $\vec{u} + \vec{v} = \langle u_1 + v_1, u_2 + v_2, u_3 + v_3 \rangle$
- **Subtraction:** $\vec{u} - \vec{v} = \langle u_1 - v_1, u_2 - v_2, u_3 - v_3 \rangle$
- **Scalar Multiplication:** $c\vec{u} = \langle cu_1, cu_2, cu_3 \rangle$

Scalar multiplication changes the magnitude (length) of the vector. If $c > 0$, the direction is the same. If $c < 0$, the direction is reversed.

Example 2: Combined Vector Operations

Let $\vec{a} = 2\vec{i} + 3\vec{j} - \vec{k}$ and $\vec{b} = \vec{i} - 4\vec{j} + 5\vec{k}$. Find $2\vec{a} - 3\vec{b}$ in component form.

Solution: First, convert to component form and perform scalar multiplication.

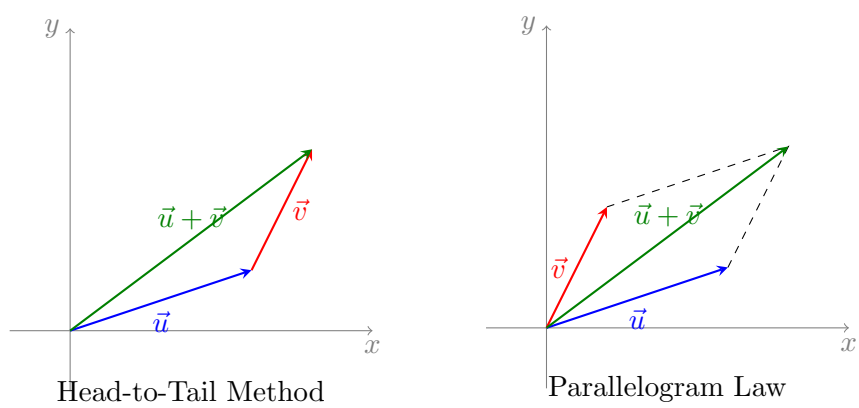
$$\vec{a} = \langle 2, 3, -1 \rangle \implies 2\vec{a} = \langle 4, 6, -2 \rangle$$
$$\vec{b} = \langle 1, -4, 5 \rangle \implies 3\vec{b} = \langle 3, -12, 15 \rangle$$

Now, subtract the resulting vectors.

$$2\vec{a} - 3\vec{b} = \langle 4 - 3, 6 - (-12), -2 - 15 \rangle = \langle 1, 18, -17 \rangle$$

Graphical Interpretation of Addition:

- **Head-to-Tail Method:** Place the tail of $\vec{v}$ at the head of $\vec{u}$. The resultant vector $\vec{u} + \vec{v}$ goes from the tail of $\vec{u}$ to the head of $\vec{v}$.
- **Parallelogram Law:** Place both vectors at the same initial point. The resultant is the diagonal of the parallelogram formed by the two vectors.



Magnitude and Unit Vectors

[Magnitude] The **magnitude** (or length, or norm) of a vector $\vec{v} = \langle v_1, v_2, v_3 \rangle$ is denoted by $\|\vec{v}\|$ and is calculated using the distance formula:

$$\|\vec{v}\| = \sqrt{v_1^2 + v_2^2 + v_3^2}$$

A **unit vector** is a vector with a magnitude of 1. To find the unit vector $\vec{u}$ in the direction of a non-zero vector $\vec{v}$, we "normalize" $\vec{v}$ by dividing it by its own magnitude.

Finding a Unit Vector
The unit vector $\vec{u}$ in the direction of $\vec{v}$ is:

$$\vec{u} = \frac{\vec{v}}{\|\vec{v}\|} = \frac{1}{\|\vec{v}\|}\langle v_1, v_2, v_3\rangle$$

Example 3: Finding a Vector of Specific Length

Find a vector that has the same direction as $\vec{a} = \langle -2, 4, 2\rangle$ but has a magnitude of 6.

Solution Strategy: This is a two-step process. First, find the unit vector in the direction of $\vec{a}$. Second, scale that unit vector by the desired magnitude.

1. **Find the magnitude of $\vec{a}$:**

$$\|\vec{a}\| = \sqrt{(-2)^2 + 4^2 + 2^2} = \sqrt{4 + 16 + 4} = \sqrt{24} = 2\sqrt{6}$$

2. **Find the unit vector $\vec{u}$ in the direction of $\vec{a}$:**

$$\vec{u} = \frac{\vec{a}}{\|\vec{a}\|} = \frac{1}{2\sqrt{6}}\langle -2, 4, 2\rangle = \left\langle \frac{-1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}} \right\rangle$$

3. **Scale the unit vector by the desired magnitude (6):** The new vector, let's call it $\vec{b}$, is $6\vec{u}$.

$$\vec{b} = 6\left\langle \frac{-1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}} \right\rangle = \left\langle \frac{-6}{\sqrt{6}}, \frac{12}{\sqrt{6}}, \frac{6}{\sqrt{6}} \right\rangle$$

After rationalizing the denominator, we get:

$$\vec{b} = \langle -\sqrt{6}, 2\sqrt{6}, \sqrt{6}\rangle$$

Classwork - Vectors in 2D and 3D

Name: _____

Problem 1: Basic Vector Arithmetic

Let $\vec{u} = \langle 2, 0, -3\rangle$ and $\vec{v} = \langle -4, 1, 5\rangle$. Compute the following vectors.

(a) $\vec{u} + \vec{v}$

(b) $\vec{v} - \vec{u}$

(c) $5\vec{u}$

Problem 2: Magnitude and Unit Vectors

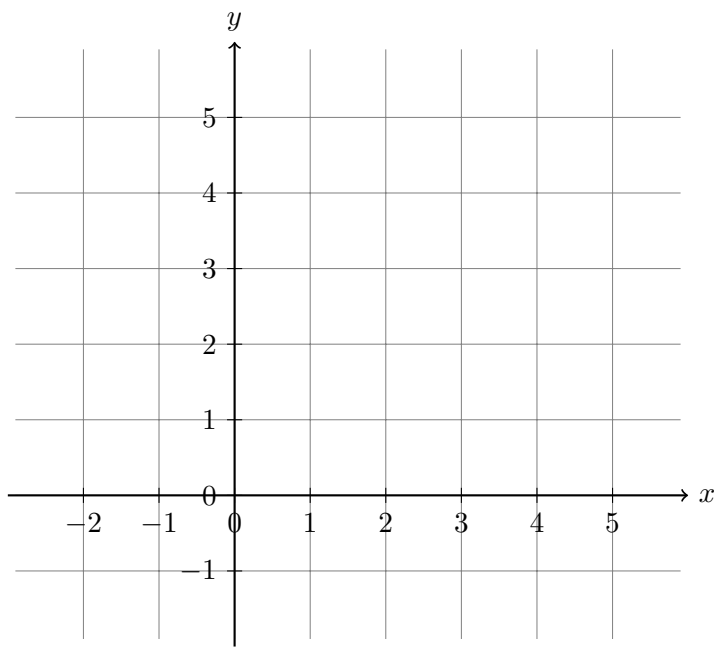
Let $\vec{w} = 2\vec{i} - \vec{j} + 2\vec{k}$.

(a) Find the magnitude (length) of $\vec{w}$.

(b) Find the unit vector that has the same direction as $\vec{w}$.

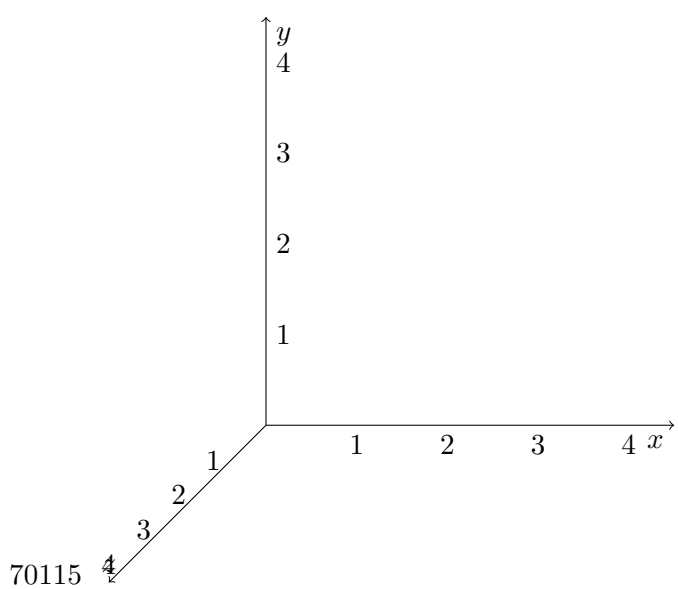
Problem 3: Graphical Vector Addition in 2D

Consider the vectors $\vec{a} = \langle 4, 1 \rangle$ and $\vec{b} = \langle -1, 3 \rangle$. On the grid provided, draw the vectors $\vec{a}$, $\vec{b}$, and the resultant vector $\vec{a} + \vec{b}$ using the Parallelogram Law.



Problem 4: Sketching a Vector in 3D

Draw the vector $\vec{v} = \langle 3, 4, 2 \rangle$ in standard position (starting at the origin) on the coordinate system below. Include dashed lines to show its position. Then identify the magnitude of $\vec{v}$.



September 5: The Dot Product

Learning Objectives

By the end of this lecture, students will be able to:

1. Compute the dot product of two vectors using both the component and angle formulas.
2. Use the dot product to find the angle between two vectors and to test for orthogonality.
3. Find the scalar and vector projection of one vector onto another.

Lecture: The Dot Product

Defining and Computing the Dot Product

So far, we have added vectors and multiplied them by scalars. The **dot product** is a form of vector multiplication that results in a **scalar** (a single number), not another vector.

[The Dot Product - Component Form] If $\vec{a} = \langle a_1, a_2, a_3 \rangle$ and $\vec{b} = \langle b_1, b_2, b_3 \rangle$, their dot product is:

$$\vec{a} \cdot \vec{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

Example 1: Basic Dot Product Calculation

Given $\vec{u} = \langle 2, 1, -4 \rangle$ and $\vec{v} = \langle 3, -1, 5 \rangle$, find $\vec{u} \cdot \vec{v}$.

Solution: We multiply the corresponding components and add the results.

$$\vec{u} \cdot \vec{v} = (2)(3) + (1)(-1) + (-4)(5) = 6 - 1 - 20 = -15$$

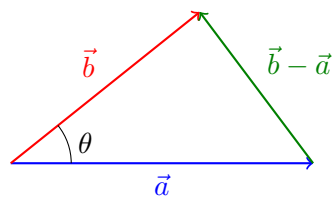
Properties of the Dot Product Let $\vec{a}, \vec{b}$, and $\vec{c}$ be vectors and k be a scalar.

1. $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$ (Commutative Property)
2. $\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$ (Distributive Property)
3. $(k\vec{a}) \cdot \vec{b} = k(\vec{a} \cdot \vec{b}) = \vec{a} \cdot (k\vec{b})$
4. $\vec{a} \cdot \vec{a} = a_1^2 + a_2^2 + a_3^2 = \|\vec{a}\|^2$

The fourth property provides a crucial link between the dot product and magnitude.

Geometric Derivation and the Law of Cosines

The component-wise definition of the dot product seems purely algebraic. So where does its geometric meaning come from? We can derive it using the Law of Cosines. Consider two vectors, $\vec{a}$ and $\vec{b}$, placed at the origin. They form a triangle with the vector $\vec{a} - \vec{b}$.



From trigonometry, the **Law of Cosines** for this triangle states:

$$\|\vec{b}-\vec{a}\|^2 = \|\vec{a}\|^2 + \|\vec{b}\|^2 - 2\|\vec{a}\|\|\vec{b}\|\cos\theta$$

We can also compute the left side using the properties of the dot product:

$$\begin{aligned}\|\vec{b}-\vec{a}\|^2 &= (\vec{b}-\vec{a})\cdot(\vec{b}-\vec{a}) \\ &= (\vec{b}\cdot\vec{b}) - (\vec{b}\cdot\vec{a}) - (\vec{a}\cdot\vec{b}) + (\vec{a}\cdot\vec{a}) \\ &= \|\vec{b}\|^2 - 2(\vec{a}\cdot\vec{b}) + \|\vec{a}\|^2\end{aligned}$$

Now, we set our two expressions for $\|\vec{b}-\vec{a}\|^2$ equal to each other:

$$\underbrace{\|\vec{a}\|^2 + \|\vec{b}\|^2 - 2\|\vec{a}\|\|\vec{b}\|\cos\theta}_{\text{Law of Cosines}} = \underbrace{\|\vec{b}\|^2 - 2(\vec{a}\cdot\vec{b}) + \|\vec{a}\|^2}_{\text{Dot Product Property}}$$

Canceling the $\|\vec{a}\|^2$ and $\|\vec{b}\|^2$ terms from both sides leaves us with:

$$-2\|\vec{a}\|\|\vec{b}\|\cos\theta = -2(\vec{a}\cdot\vec{b})$$

Dividing by -2 gives us the profound geometric interpretation of the dot product.

The Angle Between Vectors

The derivation above leads directly to our next theorem.

[The Dot Product - Angle Form] If θ is the angle between two nonzero vectors $\vec{a}$ and $\vec{b}$ (where $0 \le \theta \le \pi$), then:

$$\vec{a}\cdot\vec{b} = \|\vec{a}\|\|\vec{b}\|\cos\theta$$

By rearranging this formula, we get a way to find the angle between two vectors.

$$\cos\theta = \frac{\vec{a}\cdot\vec{b}}{\|\vec{a}\|\|\vec{b}\|}$$

Example 2: Finding the Angle Between Two Vectors
Find the angle θ between $\vec{a} = \langle 1, 2, 2\rangle$ and $\vec{b} = \langle 3, 0, -4\rangle$.

Solution: We need three pieces of information: the dot product and the two magnitudes.

1. **Dot Product:** $\vec{a}\cdot\vec{b} = (1)(3) + (2)(0) + (2)(-4) = 3 + 0 - 8 = -5$.
2. **Magnitudes:**

$$\|\vec{a}\| = \sqrt{1^2 + 2^2 + 2^2} = \sqrt{1+4+4} = \sqrt{9} = 3$$
$$\|\vec{b}\| = \sqrt{3^2 + 0^2 + (-4)^2} = \sqrt{9+0+16} = \sqrt{25} = 5$$

3. **Calculate** $\cos\theta$**:**

$$\cos\theta = \frac{-5}{(3)(5)} = \frac{-5}{15} = -\frac{1}{3}$$

4. **Find** θ**:**

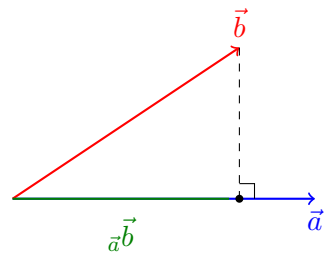
$$\theta = \arccos\left(-\frac{1}{3}\right) \approx 109.5^\circ$$

[Orthogonal Vectors] Two vectors $\vec{a}$ and $\vec{b}$ are **orthogonal** (perpendicular) if the angle between them is $\theta = \pi/2$ (or 90°). Since $\cos(\pi/2) = 0$, this means that two nonzero vectors are orthogonal if and only if their dot product is zero.

$$\vec{a} \perp \vec{b} \iff \vec{a} \cdot \vec{b} = 0$$

Vector Projections

A projection is like a vector's "shadow" on another vector. It tells us how much of one vector points in the direction of another.



The vector projection of $\vec{b}$ onto $\vec{a}$

[Scalar and Vector Projections] The **scalar projection** of $\vec{b}$ onto $\vec{a}$ is a signed length representing the size of the shadow. It is also called the component of $\vec{b}$ along $\vec{a}$.

$${}_{\vec{a}}\vec{b} = \frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\|}$$

The **vector projection** of $\vec{b}$ onto $\vec{a}$ is the vector representing the shadow itself. It is the scalar projection multiplied by the unit vector in the direction of $\vec{a}$.

$${}_{\vec{a}}\vec{b} = \left(\frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\|}\right) \frac{\vec{a}}{\|\vec{a}\|} = \left(\frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\|^2}\right) \vec{a}$$

Example 3: Finding Projections

Find the scalar and vector projections of $\vec{b} = \langle 1, 1, 2 \rangle$ onto $\vec{a} = \langle -2, 3, 1 \rangle$.

Solution: We need the dot product and the magnitude of the vector we are projecting onto, $\vec{a}$.

1. **Dot Product:** $\vec{a} \cdot \vec{b} = (-2)(1) + (3)(1) + (1)(2) = -2 + 3 + 2 = 3$.

2. **Magnitude of** $\vec{a}$**:** $\|\vec{a}\| = \sqrt{(-2)^2 + 3^2 + 1^2} = \sqrt{4 + 9 + 1} = \sqrt{14}$.

3. **Scalar Projection:**

$$_{\vec{a}}\vec{b} = \frac{\vec{a}\cdot\vec{b}}{\|\vec{a}\|} = \frac{3}{\sqrt{14}}$$

4. **Vector Projection:**

$$_{\vec{a}}\vec{b} = \left(\frac{\vec{a}\cdot\vec{b}}{\|\vec{a}\|^2}\right)\vec{a} = \frac{3}{(\sqrt{14})^2}\langle -2,3,1\rangle = \frac{3}{14}\langle -2,3,1\rangle = \left\langle -\frac{6}{14}, \frac{9}{14}, \frac{3}{14}\right\rangle = \left\langle -\frac{3}{7}, \frac{9}{14}, \frac{3}{14}\right\rangle$$

Classwork - The Dot Product

Name: _____

Problem 1: Dot Product Calculations

Let $\vec{a} = \langle -1,4,2\rangle$, $\vec{b} = \langle 3,1,-1\rangle$, and $\vec{c} = 2\vec{i} - 5\vec{k}$. Compute the following:

(a) $\vec{a}\cdot\vec{b}$

(b) $\vec{b}\cdot\vec{c}$ (Hint: First write $\vec{c}$ in component form.)

Problem 2: Angle Between Vectors

Find the angle θ between the vectors $\vec{u} = \langle 1,\sqrt{2},1\rangle$ and $\vec{v} = \langle -1,\sqrt{2},-1\rangle$.

Problem 3: Checking for Orthogonality

Determine whether the following pairs of vectors are orthogonal, parallel, or neither. Justify your answer.

(a) $\vec{a} = \langle 2, 6, -4 \rangle$, $\vec{b} = \langle -3, -9, 6 \rangle$

(b) $\vec{u} = 2\vec{i} - 3\vec{j} + \vec{k}$, $\vec{v} = \vec{i} + \vec{j} + \vec{k}$

Problem 4: Projections

Let $\vec{u} = \langle 4, 2 \rangle$ and $\vec{v} = \langle 1, 3 \rangle$.

(a) Find the scalar projection of $\vec{v}$ onto $\vec{u}$ (${}_{\vec{u}}\vec{v}$).

(b) Find the vector projection of $\vec{v}$ onto $\vec{u}$ (${}_{\vec{u}}\vec{v}$).

September 8: The Cross Product (Part 1)

Calculus III: The Cross Product (Part 1)

Lecture Notes

So far, we have studied the dot product, which takes two vectors and produces a **scalar**. The cross product is our second way of multiplying vectors. It takes two vectors and produces a new **vector**. This new vector has a unique and powerful geometric property: it is orthogonal (perpendicular) to both of the original vectors.

Learning Objectives

By the end of this lecture, you will be able to:

1. Compute the determinant of 2x2 and 3x3 matrices.
2. Calculate the cross product of two 3D vectors using the determinant formula.
3. Relate the magnitude of the cross product to the area of a parallelogram.

1. The Determinant: The Tool for the Cross Product

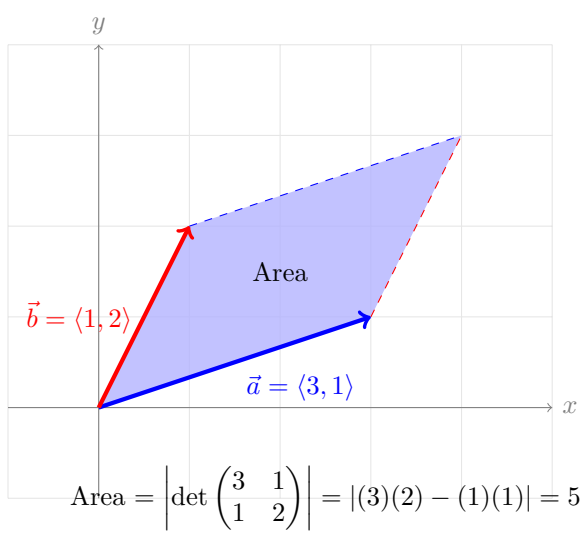
Before we define the cross product, we need to learn a computational tool called the **determinant**.

The 2x2 Determinant

The determinant of a 2x2 matrix is found by cross-multiplying and subtracting.

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

Geometric Meaning: The absolute value of the determinant gives the **area of the parallelogram** formed by its row vectors.

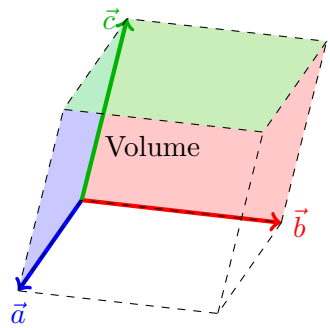


The 3x3 Determinant

We compute a 3x3 determinant using **cofactor expansion**. We expand along the first row, creating a sum of terms involving 2x2 determinants. Remember the pattern of signs: $(+, -, +)$.

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix} - a_2 \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix} + a_3 \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}$$

Geometric Meaning: The absolute value of the 3x3 determinant gives the **volume of the parallelepiped** formed by its row vectors.



Example: Calculating a 3x3 Determinant

Compute the determinant of the matrix $A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 1 & 0 & 6 \end{pmatrix}$.

$$\begin{aligned}
\det(A) &= \begin{vmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 1 & 0 & 6 \end{vmatrix} \\
&= +1 \cdot \begin{vmatrix} 4 & 5 \\ 0 & 6 \end{vmatrix} - 2 \cdot \begin{vmatrix} 0 & 5 \\ 1 & 6 \end{vmatrix} + 3 \cdot \begin{vmatrix} 0 & 4 \\ 1 & 0 \end{vmatrix} \\
&= 1 \cdot ((4)(6) - (5)(0)) - 2 \cdot ((0)(6) - (5)(1)) + 3 \cdot ((0)(0) - (4)(1)) \\
&= 1 \cdot (24 - 0) - 2 \cdot (0 - 5) + 3 \cdot (0 - 4) \\
&= 1(24) - 2(-5) + 3(-4) \\
&= 24 + 10 - 12 = 22
\end{aligned}$$

The volume of the parallelepiped formed by the vectors $\langle 1, 2, 3 \rangle$, $\langle 0, 4, 5 \rangle$, and $\langle 1, 0, 6 \rangle$ is 22.

2. The Cross Product: Definition and Computation

Definition: The Cross Product

Given two vectors $\vec{a} = \langle a_1, a_2, a_3 \rangle$ and $\vec{b} = \langle b_1, b_2, b_3 \rangle$, their **cross product** $\vec{a} \times \vec{b}$ is a new vector defined by the following "formal" determinant:

$$\vec{a} \times \vec{b} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix}$$

Expanding this out gives the component form:

$$\vec{a} \times \vec{b} = \langle (a_2 b_3 - a_3 b_2), -(a_1 b_3 - a_3 b_1), (a_1 b_2 - a_2 b_1) \rangle$$

Example: Calculating a Cross Product

Let $\vec{u} = \langle 1, 2, 3 \rangle$ and $\vec{v} = \langle 4, 5, 6 \rangle$. Find $\vec{u} \times \vec{v}$.

$$\begin{aligned}
\vec{u} \times \vec{v} &= \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{vmatrix} \\
&= \begin{vmatrix} 2 & 3 \\ 5 & 6 \end{vmatrix} - \begin{vmatrix} 1 & 3 \\ 4 & 6 \end{vmatrix} + \begin{vmatrix} 1 & 2 \\ 4 & 5 \end{vmatrix} \\
&= ((2)(6) - (3)(5)) - ((1)(6) - (3)(4)) + ((1)(5) - (2)(4)) \\
&= (12 - 15) - (6 - 12) + (5 - 8) \\
&= -3 - (-6) - 3 \\
&= -3 + 6 - 3 = \langle -3, 6, -3 \rangle
\end{aligned}$$

3. A Deeper Look at the Geometric Meaning

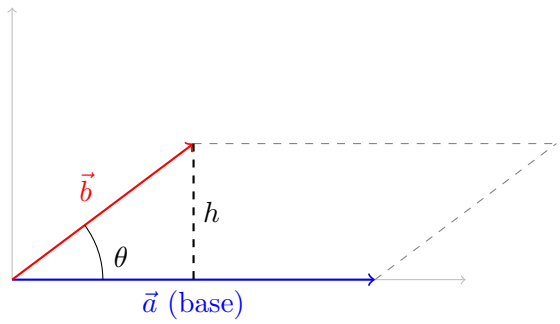
The cross product has two key geometric properties. We will see later that the vector $\vec{a} \times \vec{b}$ is **orthogonal** to both $\vec{a}$ and $\vec{b}$. Today, we focus on the meaning of its **magnitude**.

Informal Reasoning: Connecting to Area = base $\times$ height

Why should we believe that $||\vec{a} \times \vec{b}|| = ||\vec{a}|| \cdot ||\vec{b}|| \sin\theta$? Let's start with a formula everyone knows for the area of a parallelogram:

$$\text{Area} = \text{base} \times \text{height}$$

Consider the parallelogram spanned by vectors $\vec{a}$ and $\vec{b}$. Let θ be the angle between them.



- We can choose the **base** to be the length of vector $\vec{a}$, which is $||\vec{a}||$.
- The **height**, h, can be found using trigonometry. In the right triangle formed, the hypotenuse is $||\vec{b}||$. The height is the side opposite the angle θ. Therefore:

$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{h}{||\vec{b}||} \implies h = ||\vec{b}||\sin\theta$$

- Combining these, we get the familiar geometric formula for the area:

$$\text{Area} = (\text{base}) \times (\text{height}) = ||\vec{a}|| \cdot ||\vec{b}||\sin\theta$$

This shows that the geometric formula for the area of a parallelogram is *exactly the same* as the sine formula for the magnitude of the cross product. The only remaining step is to believe that the algebraic definition of the cross product's magnitude is equivalent to this sine formula.

Proof Sketch: Connecting the Algebra to the Geometry

Here, we will show that the component-form definition of $||\vec{a} \times \vec{b}||$ is equivalent to $||\vec{a}|| \cdot ||\vec{b}||\sin\theta$. The key is to work with the squares of the magnitudes to avoid square roots, and to use a powerful link between the cross product and the dot product.

Key Identity: Lagrange's Identity

For any two vectors $\vec{a}$ and $\vec{b}$ in $\mathbb{R}^3$, the following identity holds:

$$||\vec{a} \times \vec{b}||^2 = ||\vec{a}||^2||\vec{b}||^2 - (\vec{a} \cdot \vec{b})^2$$

(Proving this identity is a matter of expanding both sides into their component forms and showing they are equal—a tedious but straightforward algebraic exercise.)

Now, let's use this identity. We know the geometric definition of the dot product: $\vec{a} \cdot \vec{b} = ||\vec{a}||||\vec{b}||\cos\theta$. Let's substitute this into Lagrange's Identity.

$$\begin{aligned}
||\vec{a} \times \vec{b}||^2 &= ||\vec{a}||^2||\vec{b}||^2 - (\vec{a} \cdot \vec{b})^2 \\
&= ||\vec{a}||^2||\vec{b}||^2 - (||\vec{a}||||\vec{b}||\cos\theta)^2 \\
&= ||\vec{a}||^2||\vec{b}||^2 - ||\vec{a}||^2||\vec{b}||^2\cos^2\theta \\
&= ||\vec{a}||^2||\vec{b}||^2(1 - \cos^2\theta) \quad \text{(Factoring out common terms)}
\end{aligned}$$

Using the fundamental Pythagorean identity, $1 - \cos^2\theta = \sin^2\theta$, we get:

$$||\vec{a} \times \vec{b}||^2 = ||\vec{a}||^2||\vec{b}||^2 \sin^2\theta$$

Taking the square root of both sides gives us the result we wanted to prove:

$$||\vec{a} \times \vec{b}|| = ||\vec{a}||||\vec{b}|| \sin\theta$$

This confirms that the magnitude of the cross product, as calculated from its components, is precisely the area of the parallelogram spanned by the two vectors.

Theorem: Magnitude of the Cross Product

The magnitude (length) of the cross product vector $\vec{a} \times \vec{b}$ is equal to the **area of the parallelogram** spanned by the vectors $\vec{a}$ and $\vec{b}$.

$$\text{Area} = ||\vec{a} \times \vec{b}|| = ||\vec{a}|| \cdot ||\vec{b}|| \sin\theta$$

Example: Finding the Area of a Parallelogram

Find the area of the parallelogram formed by $\vec{u} = \langle 1, 2, 3\rangle$ and $\vec{v} = \langle 4, 5, 6\rangle$. From our previous calculation, we found $\vec{u} \times \vec{v} = \langle -3, 6, -3\rangle$. The area is the magnitude of this resulting vector:

$$\text{Area} = ||\langle -3, 6, -3\rangle|| = \sqrt{(-3)^2 + 6^2 + (-3)^2} = \sqrt{9 + 36 + 9} = \sqrt{54} = 3\sqrt{6}$$

The area of the parallelogram is $3\sqrt{6}$ square units.

Homework 2

Homework - 2

Work solutions on blank paper, showing all steps. Scan and submit on or before 9/7 at 11:59 pm on Blackboard.

Problem 1: Coordinate Conversion (Polar to Rectangular)

Find the rectangular coordinates (x, y) for the point given in polar coordinates by $(r, \theta) = (6, 5\pi/6)$.

Problem 2: Coordinate Conversion (Rectangular to Polar)

Find the polar coordinates (r, θ) for the point given in rectangular coordinates by $(x, y) = (-\sqrt{3}, 1)$. Provide an answer where $r > 0$ and $0 \leq \theta < 2\pi$.

Problem 3: Sketching a Polar Curve

Sketch the graph of the polar equation $r = 4\sin\theta$. What shape is the curve?

Problem 4: Area of a Simple Polar Region

Set up and evaluate an integral to find the area of the region enclosed by the polar curve $r = 2\sin\theta$.

Problem 5: Area of a Cardioid

Find the total area of the region enclosed by the cardioid $r = 3 - 3\cos\theta$.

Problem 6: Area of a Rose Petal

Find the area of a single petal of the rose curve $r = 2\sin(3\theta)$.

Problem 7: Area Between Curves

Find the area of the region that lies inside the circle $r = 3\cos\theta$ and outside the cardioid $r = 1 + \cos\theta$.

Problem 8: Plotting in 3D Rectangular Coordinates

Draw a 3D rectangular coordinate system and plot the point $P(2, 3, 4)$. Include dashed lines to show its position relative to the axes, similar to the examples in the lecture notes.

Problem 9: Cylindrical to Rectangular Conversion

Convert the cylindrical coordinates $(r, \theta, z) = (4, 4\pi/3, -2)$ to rectangular coordinates (x, y, z).

Problem 10: Rectangular to Cylindrical Conversion

Convert the rectangular coordinates $(4, -4, 3)$ to cylindrical coordinates (r, θ, z). Give an answer with $r > 0$ and $0 \leq \theta < 2\pi$.

Problem 11: Vectors Between Points

Find the vector $\vec{AB}$ that starts at point $A(1, 4, -2)$ and ends at point $B(3, 1, 1)$.

(a) Write your answer in component form $\langle x, y, z\rangle$.

(b) Write your answer as a linear combination of the standard unit vectors $\vec{i}, \vec{j}, \vec{k}$.

Problem 12: Vector Operations

Given the vectors $\vec{u} = \langle 2, -5, 1\rangle$ and $\vec{v} = \langle -3, 0, 4\rangle$, compute the following:

(a) $\vec{u} + \vec{v}$

(b) $\vec{u} - \vec{v}$

(c) $4\vec{u}$

Problem 13: Linear Combinations of Vectors

Let $\vec{a} = 2\vec{i} + 3\vec{j} - \vec{k}$ and $\vec{b} = \vec{i} - 4\vec{j} + 5\vec{k}$. Find the vector $2\vec{a} - 3\vec{b}$ and write it in component form.

Problem 14: Graphical Vector Addition

Given the 2D vectors $\vec{v} = \langle 3, 1 \rangle$ and $\vec{w} = \langle 1, 4 \rangle$:

(a) Draw both vectors starting from the origin on a 2D Cartesian plane.

(b) Draw the resultant vector $\vec{v} + \vec{w}$ using the head-to-tail method.

(c) Draw the resultant vector $\vec{v} + \vec{w}$ using the parallelogram law.

Problem 15: Magnitude of a Vector

Find the magnitude (or length) of the vectors $\vec{w} = \langle 4, -2, 3\rangle$, $\vec{u} = \langle 4, -2, 5\rangle$, $\vec{v} = \langle -1, -2, 5\rangle$.

Problem 16: Finding a Unit Vector

Find the unit vector that has the same direction as the vectors $\vec{v} = \langle 1, -2, 2\rangle$, $\vec{w} = \langle 4, -2, 5\rangle$, $\vec{w} = \langle -1, -2, 2\rangle$.

Problem 17: Normalizing a Vector

Find the unit vector in the direction of $\vec{u} = 5\vec{j} - 12\vec{k}$.

Problem 18: Creating a Vector of Specific Length

Find a vector that has the same direction as $\vec{a} = \langle -2, 4, 2\rangle$ but has a magnitude of 6.

Problem 19: Application of Vector Operations

Two forces are acting on an object at the origin. Force $\vec{F_1} = \langle 10, 5, -2\rangle$ and Force $\vec{F_2} = \langle -3, -4, 9\rangle$. Find the resultant force vector $\vec{F_R} = \vec{F_1} + \vec{F_2}$ and its magnitude.

Problem 20: Finding an Endpoint

The vector $\vec{v} = 7\vec{i} - \vec{j} + 3\vec{k}$ starts at the point $P(2, 5, -1)$. Find the coordinates of the terminal point Q.

Quiz 2

Quiz 2: The Dot and Cross Products

Name: _____

(Show all your work for full credit.)

Problem 1: Dot Product Applications

Let $\vec{a} = \langle 2, -2, 1\rangle$ and $\vec{b} = \langle 4, 0, 3\rangle$.

(a) Find the angle θ between vectors $\vec{a}$ and $\vec{b}$.

(b) Find the vector projection of $\vec{b}$ onto $\vec{a}$, which is ${}_{\vec{a}}\vec{b}$.

Problem 2: Cross Product Applications

Consider the points $P(1,0,0)$, $Q(0,1,0)$, and $R(0,0,1)$.

(a) Find the area of the triangle with vertices P, Q, and R.

(b) Find the volume of the parallelepiped with adjacent edges given by the vectors $\vec{PQ}$, $\vec{PR}$, and $\langle 1,0,0\rangle$.

CHAPTER 3

Cycle 3: Cross Product Geometry and Applications (Homework 3)

Overview

This cycle deepens the study of the cross product and scalar triple product, using them to compute areas and volumes and to model physical quantities such as torque. The goal is to solidify geometric intuition for vector operations in three dimensions.

Lecture Notes and Classwork

September 10: The Cross Product (Part 2)

Calculus III: The Cross Product (Part 2)
Properties and Applications

On Monday, we introduced the cross product $\vec{a} \times \vec{b}$, defined it using determinants, and saw that its magnitude $||\vec{a} \times \vec{b}||$ represents the area of a parallelogram. Today, we explore the properties of the resulting vector itself, particularly its direction.

Learning Objectives

By the end of this lecture, you will be able to:

1. Apply the algebraic properties of the cross product (e.g., anticommutativity, distributivity).
2. Use the scalar triple product to find the volume of a parallelepiped.
3. Apply the cross product to solve physics problems, such as finding torque.

1. The Direction of the Cross Product

The most important geometric property of the cross product is its direction.

Theorem: Orthogonality of the Cross Product

The vector $\vec{a} \times \vec{b}$ is orthogonal (perpendicular) to both $\vec{a}$ and $\vec{b}$.

We can prove this by showing the dot product with each original vector is zero. Let $\vec{a} \times \vec{b} = \langle c_1, c_2, c_3 \rangle = \langle a_2b_3 - a_3b_2, a_3b_1 - a_1b_3, a_1b_2 - a_2b_1 \rangle$.

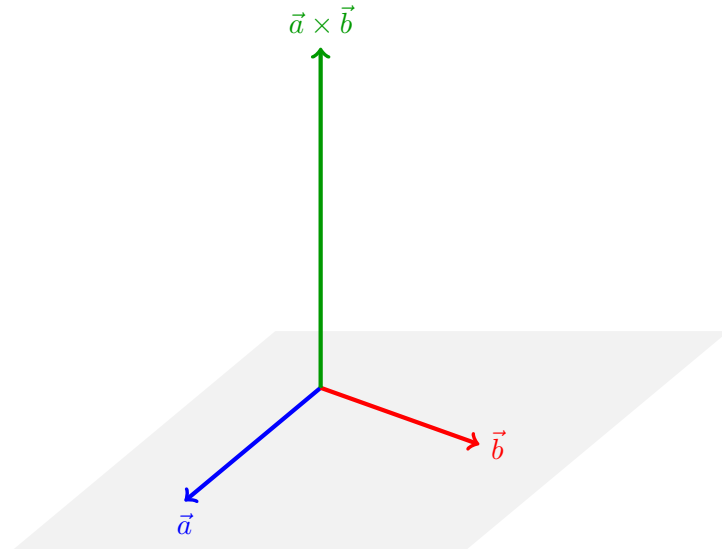
$$(\vec{a} \times \vec{b}) \cdot \vec{a} = a_1(a_2b_3 - a_3b_2) + a_2(a_3b_1 - a_1b_3) + a_3(a_1b_2 - a_2b_1)$$

$$= a_1a_2b_3 - a_1a_3b_2 + a_2a_3b_1 - a_2a_1b_3 + a_3a_1b_2 - a_3a_2b_1 = 0$$

All terms cancel out. A similar calculation shows $(\vec{a} \times \vec{b}) \cdot \vec{b} = 0$.

The Right-Hand Rule: Since $\vec{a} \times \vec{b}$ is orthogonal to the plane containing $\vec{a}$ and $\vec{b}$, it must point "up" or "down". We determine which direction using the **right-hand rule**:

1. Point the fingers of your right hand in the direction of the first vector, $\vec{a}$.
2. Curl your fingers toward the direction of the second vector, $\vec{b}$.
3. Your thumb will point in the direction of $\vec{a} \times \vec{b}$.



2. Algebraic Properties of the Cross Product

Unlike multiplication of real numbers, the cross product has some surprising properties.

Properties of the Cross Product

If $\vec{a}$, $\vec{b}$, and $\vec{c}$ are vectors and k is a scalar:

1. $\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$ (**Anticommutative**)
2. $(k\vec{a}) \times \vec{b} = k(\vec{a} \times \vec{b}) = \vec{a} \times (k\vec{b})$
3. $\vec{a} \times (\vec{b} + \vec{c}) = \vec{a} \times \vec{b} + \vec{a} \times \vec{c}$ (**Distributive**)

4. $\vec{a} \times \vec{a} = \vec{0}$ (The cross product of a vector with itself is the zero vector)

Warning: The cross product is **not** associative! In general, $(\vec{a} \times \vec{b}) \times \vec{c} \neq \vec{a} \times (\vec{b} \times \vec{c})$.

Example: Anticommutativity

From Monday, we found for $\vec{u} = \langle 1, 2, 3 \rangle$ and $\vec{v} = \langle 4, 5, 6 \rangle$ that $\vec{u} \times \vec{v} = \langle -3, 6, -3 \rangle$. Let's compute $\vec{v} \times \vec{u}$:

$$\vec{v} \times \vec{u} = \begin{vmatrix} 4 & 5 & 6 \\ 1 & 2 & 3 \end{vmatrix} = (15 - 12) - (12 - 6) + (8 - 5) = 3 - 6 + 3 = \langle 3, -6, 3 \rangle$$

As expected, $\vec{v} \times \vec{u} = -(\vec{u} \times \vec{v})$.

3. The Scalar Triple Product and Volume

Combining the dot product and the cross product gives a useful geometric tool.

The Scalar Triple Product

The **scalar triple product** of vectors $\vec{a}$, $\vec{b}$, and $\vec{c}$ is $\vec{a} \cdot (\vec{b} \times \vec{c})$. Its value can be computed as the 3x3 determinant of the matrix formed by the vectors' components.

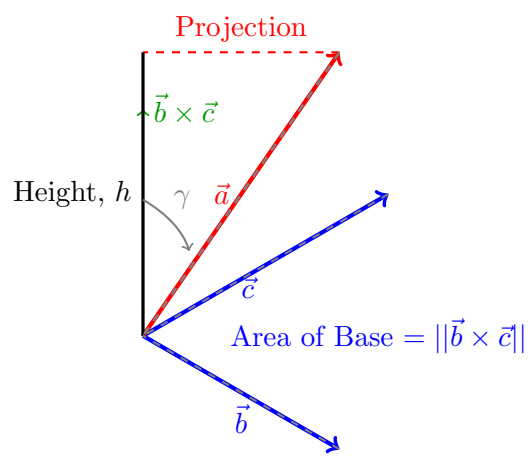
$$\vec{a} \cdot (\vec{b} \times \vec{c}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

The absolute value of the scalar triple product gives the **volume of the parallelepiped** spanned by the three vectors.

$$V = |\vec{a} \cdot (\vec{b} \times \vec{c})|$$

Geometric Proof: Volume = (Area of Base) × (Height)

Why does this formula work? We can derive it by thinking about the basic formula for the volume of a prism. Let's consider the parallelepiped spanned by vectors $\vec{a}$, $\vec{b}$, and $\vec{c}$. We can choose the parallelogram formed by $\vec{b}$ and $\vec{c}$ as the base.



1. **Area of the Base:** The base is the parallelogram spanned by $\vec{b}$ and $\vec{c}$. From our study of the cross product, we know its area is given by the magnitude of the cross product:

$$\text{Area}_{\text{base}} = ||\vec{b} \times \vec{c}||$$

2. **Height:** The height h of the parallelepiped is the perpendicular distance from the base plane to the tip of vector $\vec{a}$. As shown in the diagram, this is the length of the projection of $\vec{a}$ onto the normal vector of the base, which is $\vec{b} \times \vec{c}$. Let γ be the angle between $\vec{a}$ and the normal vector $\vec{b} \times \vec{c}$. From trigonometry, the height is:

$$h = ||\vec{a}|| \cdot |\cos(\gamma)|$$

(We use the absolute value of cosine because height must be positive).

3. **Volume:** Now, we multiply the base area by the height:

$$V = (\text{Area}_{\text{base}}) \cdot (h) = ||\vec{b} \times \vec{c}|| \cdot ||\vec{a}|| |\cos(\gamma)|$$

This final expression is precisely the definition of the absolute value of the dot product between the vector $\vec{a}$ and the vector $(\vec{b} \times \vec{c})$.

$$V = |\vec{a} \cdot (\vec{b} \times \vec{c})|$$

This confirms that the scalar triple product gives the volume of the parallelepiped.

Example: Finding Volume

Find the volume of the parallelepiped with adjacent edges given by the vectors $\vec{a} = \langle 1, 4, -7 \rangle$, $\vec{b} = \langle 2, -1, 4 \rangle$, and $\vec{c} = \langle 0, -9, 18 \rangle$. We just need to compute the determinant of the matrix formed by these vectors.

$$\begin{aligned} V = \left|\begin{vmatrix} 1 & 4 & -7 \\ 2 & -1 & 4 \\ 0 & -9 & 18 \end{vmatrix}\right| &= \left|1 \begin{vmatrix} -1 & 4 \\ -9 & 18 \end{vmatrix} - 4 \begin{vmatrix} 2 & 4 \\ 0 & 18 \end{vmatrix} + (-7) \begin{vmatrix} 2 & -1 \\ 0 & -9 \end{vmatrix}\right| \\ &= |1(-18 - (-36)) - 4(36 - 0) - 7(-18 - 0)| \\ &= |1(18) - 4(36) - 7(-18)| \\ &= |18 - 144 + 126| = |0| = 0 \end{aligned}$$

Since the volume is 0, this means the three vectors are **coplanar** (they all lie on the same plane).

Condition for Coplanarity

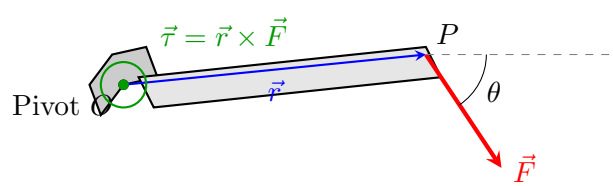
Three vectors $\vec{a}$, $\vec{b}$, and $\vec{c}$ are **coplanar** (meaning they lie in the same plane) if and only if their scalar triple product is zero.

$$\vec{a} \cdot (\vec{b} \times \vec{c}) = 0$$

This is because a zero volume implies the parallelepiped they span is "flat."

4. Application: Torque

In physics, the cross product is used to describe torque, which is the rotational force on an object. The classic example is using a wrench to turn a bolt.



Definition: Torque

If a force $\vec{F}$ is applied at a point P, the torque $\vec{\tau}$ about a pivot point O is given by:

$$\vec{\tau} = \vec{r} \times \vec{F}$$

where $\vec{r} = \vec{OP}$ is the position vector from the pivot to the point where the force is applied (the "lever arm").

- The **direction** of the torque vector $\vec{\tau}$ is the axis of rotation (given by the right-hand rule, pointing in to the page in the diagram above).
- The **magnitude** of the torque is $||\vec{\tau}|| = ||\vec{r}|| \cdot ||\vec{F}|| \sin\theta$. Maximum torque occurs when the force is applied perpendicularly to the lever arm ($\theta = 90°$, so $\sin\theta = 1$).

Example: Calculating Torque

A wrench is tightening a bolt at the origin. A force of $\vec{F} = \langle 0, -10, 3\rangle$ Newtons is applied to the wrench handle at the point $P(0, 0.2, 0)$ meters. Find the torque vector and its magnitude.

1. **Find the position vector $\vec{r}$:** The vector goes from the pivot (origin O) to the point of application P.
$$\vec{r} = \vec{OP} = \langle 0-0, 0.2-0, 0-0\rangle = \langle 0, 0.2, 0\rangle$$
2. **Calculate the cross product $\vec{\tau} = \vec{r} \times \vec{F}$:**
$$\vec{\tau} = \begin{vmatrix} 0 & 0.2 & 0 \\ 0 & -10 & 3 \end{vmatrix} = (0.2 \cdot 3 - 0 \cdot (-10)) - (0) + (0) = 0.6 = \langle 0.6, 0, 0\rangle$$
3. **Find the magnitude:**
$$||\vec{\tau}|| = ||\langle 0.6, 0, 0\rangle|| = \sqrt{0.6^2} = 0.6$$

The torque vector is $\langle 0.6, 0, 0\rangle$, and its magnitude is 0.6 Newton-meters. The rotation occurs around the x-axis.

Classwork: Properties of the Cross Product

Name: _____

(Show all your work for full credit.)

Problem 1: Orthogonality and Anticommutativity

Given the vectors $\vec{a} = \langle 2, 1, 0\rangle$ and $\vec{b} = \langle 0, 1, 3\rangle$.

(a) Calculate the cross product $\vec{c} = \vec{a} \times \vec{b}$.

(b) Calculate $\vec{b} \times \vec{a}$. How does it relate to your answer in part (a)?

(c) Show that your result for $\vec{c}$ from part (a) is orthogonal to both $\vec{a}$ and $\vec{b}$ by using the dot product.

Problem 2: Scalar Triple Product

Find the volume of the parallelepiped with adjacent edges given by the vectors $\vec{u} = \langle 1, 2, 3 \rangle$, $\vec{v} = \langle -1, 1, 2 \rangle$, and $\vec{w} = \langle 2, 1, 4 \rangle$.

Problem 3: Application of Torque

A force $\vec{F} = \langle 4, 5, 0 \rangle$ is applied to the end of a lever at point $P(3, 0, 0)$. The lever is attached to a pivot at the origin $O(0, 0, 0)$.

(a) Find the position vector (lever arm) $\vec{r} = \vec{OP}$.

(b) Calculate the torque vector $\vec{\tau} = \vec{r} \times \vec{F}$.

(c) Find the magnitude of the torque.

September 12: Equations of Planes in Space

Calculus III: Equations of Planes in Space

Lecture Notes

How do we describe a flat surface, or a plane, in 3D space? We need two key ingredients: a single point $P_0(x_0, y_0, z_0)$ that we know is on the plane, and a vector that defines the plane's tilt or orientation. The most effective way to define this orientation is with a **normal vector**, $\vec{n}$, which is a vector orthogonal (perpendicular) to the plane.

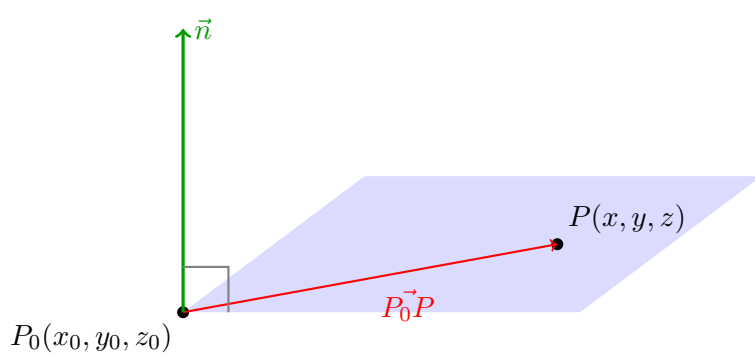
Learning Objectives

By the end of this lecture, you will be able to:

1. Derive the equation of a plane using a point on the plane and a normal vector.
2. Find the equation of a plane given three non-collinear points.
3. Identify parallel and orthogonal planes by analyzing their normal vectors.

1. The Equation of a Plane

Let a plane contain a known point $P_0(x_0, y_0, z_0)$. Let $P(x, y, z)$ be *any other point* in the plane. The vector from P_0 to P, which is $\vec{P_0P} = \langle x - x_0, y - y_0, z - z_0 \rangle$, must lie *in* the plane. The plane's orientation is defined by its normal vector, $\vec{n}$.



Since the normal vector $\vec{n}$ is orthogonal to every vector lying in the plane, it must be orthogonal to our vector $\vec{P_0P}$. This gives us a simple but powerful geometric condition:

$$\vec{n} \cdot \vec{P_0P} = 0$$

Let's express this using components. We define our normal vector as $\vec{n} = \langle a, b, c \rangle$ and our vector in the plane as $\vec{P_0P} = \langle x - x_0, y - y_0, z - z_0 \rangle$. Taking the dot product gives the equation of the plane.

Equations of a Plane

- **Vector Equation:** $\vec{n} \cdot (\vec{r} - \vec{r}_0) = 0$
- **Standard Form (or Point-Normal Form):** The coefficients a, b, c are the components of the normal vector.

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

- **General Form:** After distributing and collecting constants.

$$ax + by + cz = d$$

(where $d = ax_0 + by_0 + cz_0$)

2. Finding the Equation of a Plane in Practice

Example 1: Given a Point and a Normal

Find the equation of the plane passing through the point $(5, 1, -2)$ with normal vector $\vec{n} = \langle 2, -3, 4 \rangle$.

Solution: We have the point $(x_0, y_0, z_0) = (5, 1, -2)$ and the normal vector components $\langle a, b, c \rangle = \langle 2, -3, 4 \rangle$. We plug directly into the standard form:

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

$$2(x - 5) - 3(y - 1) + 4(z - (-2)) = 0$$

$$2x - 10 - 3y + 3 + 4z + 8 = 0$$

$$2x - 3y + 4z + 1 = 0$$

This is the equation of the plane in general form.

Example 2: Given Three Points

Find the equation of the plane passing through the points $P(1, 3, 2)$, $Q(3, -1, 6)$, and $R(5, 2, 0)$.

Solution Strategy:

1. We need a point (we have three!) and a normal vector.
2. To get a normal vector, we first create two vectors that lie in the plane, for example, $\vec{PQ}$ and $\vec{PR}$.
3. The cross product $\vec{PQ} \times \vec{PR}$ will be orthogonal to both, giving us the normal vector $\vec{n}$.

Step 1: Create two vectors in the plane.

$$\vec{PQ} = Q - P = \langle 3 - 1, -1 - 3, 6 - 2 \rangle = \langle 2, -4, 4 \rangle$$

$$\vec{PR} = R - P = \langle 5 - 1, 2 - 3, 0 - 2 \rangle = \langle 4, -1, -2 \rangle$$

Step 2: Find the normal vector $\vec{n}$ using the cross product.

$$\vec{n} = \vec{PQ} \times \vec{PR} = \begin{vmatrix} 2 & -4 & 4 \\ 4 & -1 & -2 \end{vmatrix} = (8 - (-4)) - (-4 - 16) + (-2 - (-16))$$

$$\vec{n} = 12 + 20 + 14 = \langle 12, 20, 14 \rangle$$

Any non-zero scalar multiple of this vector will also work. Let's simplify it by dividing by 2: $\vec{n} = \langle 6, 10, 7 \rangle$.

Step 3: Use a point (e.g., P) and the normal vector to write the equation.

$$6(x - 1) + 10(y - 3) + 7(z - 2) = 0$$

$$6x - 6 + 10y - 30 + 7z - 14 = 0$$

$$6x + 10y + 7z - 50 = 0$$

3. Relationships Between Planes

The orientation of a plane is entirely determined by its normal vector. We can use this fact to determine if two planes are parallel, orthogonal, or neither.

Let two planes be given by:

- Plane 1: $a_1x + b_1y + c_1z = d_1 \quad \Longrightarrow \quad \vec{n_1} = \langle a_1, b_1, c_1 \rangle$
- Plane 2: $a_2x + b_2y + c_2z = d_2 \quad \Longrightarrow \quad \vec{n_2} = \langle a_2, b_2, c_2 \rangle$

Conditions for Parallel and Orthogonal Planes

- Two planes are **parallel** if their normal vectors are parallel.

$$\vec{n_1} = k\vec{n_2} \quad \text{for some scalar } k$$

- Two planes are **orthogonal** if their normal vectors are orthogonal.

$$\vec{n_1} \cdot \vec{n_2} = 0$$

If neither of these conditions is met, the planes intersect at an angle.

Example 3: Identifying Parallel and Orthogonal Planes

Consider the following planes:

- Plane A: $2x - 3y + z = 5$
- Plane B: $-4x + 6y - 2z = 1$
- Plane C: $x + 2y + 4z = 0$

(a) Determine if Plane A and Plane B are parallel, orthogonal, or neither.

Solution: First, we extract their normal vectors from the coefficients of x, y, z.

$$\vec{n_A} = \langle 2, -3, 1 \rangle \quad \text{and} \quad \vec{n_B} = \langle -4, 6, -2 \rangle$$

We check if they are parallel by seeing if one is a scalar multiple of the other.

$$\vec{n_B} = -2 \cdot \langle 2, -3, 1 \rangle = -2\vec{n_A}$$

Since $\vec{n_B}$ is a scalar multiple of $\vec{n_A}$, the normal vectors are parallel. Therefore, **Plane A and Plane B are parallel**.

(b) Determine if Plane A and Plane C are parallel, orthogonal, or neither.

Solution: We extract the normal vector for Plane C.

$$\vec{n_A} = \langle 2, -3, 1 \rangle \quad \text{and} \quad \vec{n_C} = \langle 1, 2, 4 \rangle$$

First, we check for parallelism. Is $\langle 2, -3, 1 \rangle = k\langle 1, 2, 4 \rangle$? Looking at the x-components, k would have to be 2. But $2 \cdot \langle 1, 2, 4 \rangle = \langle 2, 4, 8 \rangle \neq \vec{n_A}$. So, the planes are not parallel.

Next, we check for orthogonality using the dot product.

$$\vec{n_A} \cdot \vec{n_C} = (2)(1) + (-3)(2) + (1)(4) = 2 - 6 + 4 = 0$$

Since the dot product of their normal vectors is 0, **Plane A and Plane C are orthogonal**.

Classwork: Equations of Planes

Name: _____

Problem 1: Finding the Equation from a Point and Normal

Find the equation of the plane that passes through the point $P(2, -1, 5)$ and has a normal vector $\vec{n} = \langle 3, 4, -1 \rangle$. Write your final answer in the general form $ax + by + cz = d$.

Problem 2: Angle Between Planes

Find the angle between the planes $x + y + z = 1$ and $x - 2y + 3z = 1$.

Problem 3: Parallel and Orthogonal Planes

Consider the following three planes:

- Plane 1: $x - 2y + 3z = 5$
- Plane 2: $-2x + 4y - 6z = 1$
- Plane 3: $4x + 5y + 2z = 0$

(a) Write down the normal vector for each plane: $\vec{n_1}$, $\vec{n_2}$, and $\vec{n_3}$.

(b) Which pair of planes is parallel? Justify your answer.

(c) Which pair of planes is orthogonal? Justify your answer.

Problem 4: Finding the Equation from Three Points

Find the equation of the plane that contains the points $A(1, 1, 0)$, $B(0, 2, 1)$, and $C(3, 2, -1)$.

(a) Find two vectors in the plane, such as $\vec{AB}$ and $\vec{AC}$.

(b) Compute the normal vector $\vec{n} = \vec{AB} \times \vec{AC}$.

(c) Write the equation of the plane.

September 15: Lines in Space and an Introduction to Vector-Valued Functions

Calculus III: Lines in Space and an Introduction to Vector-Valued Functions

Lecture Notes

In 2D, we define a line with a point and a slope. In 3D, the concept of "slope" is ambiguous. Instead, we need a point on the line and a **direction vector**, $\vec{v}$, which is a vector parallel to the line that dictates its orientation.

This new way of thinking about lines as being "traced out" by a moving point is our first step into the world of **vector-valued functions**, where the position of a point is a function of a single parameter, like time.

Learning Objectives

By the end of this lecture, you will be able to:

1. Derive the vector equation of a line from a point and a direction vector.
2. Write the parametric and symmetric equations for a line in 3D space.
3. Find the equation of a line that passes through two given points.

1. The Vector Equation of a Line

Let's build the equation of a line L that passes through a point $P_0(x_0, y_0, z_0)$ and is parallel to a direction vector $\vec{v} = \langle a, b, c\rangle$. Let $P(x, y, z)$ be any other point on the line.

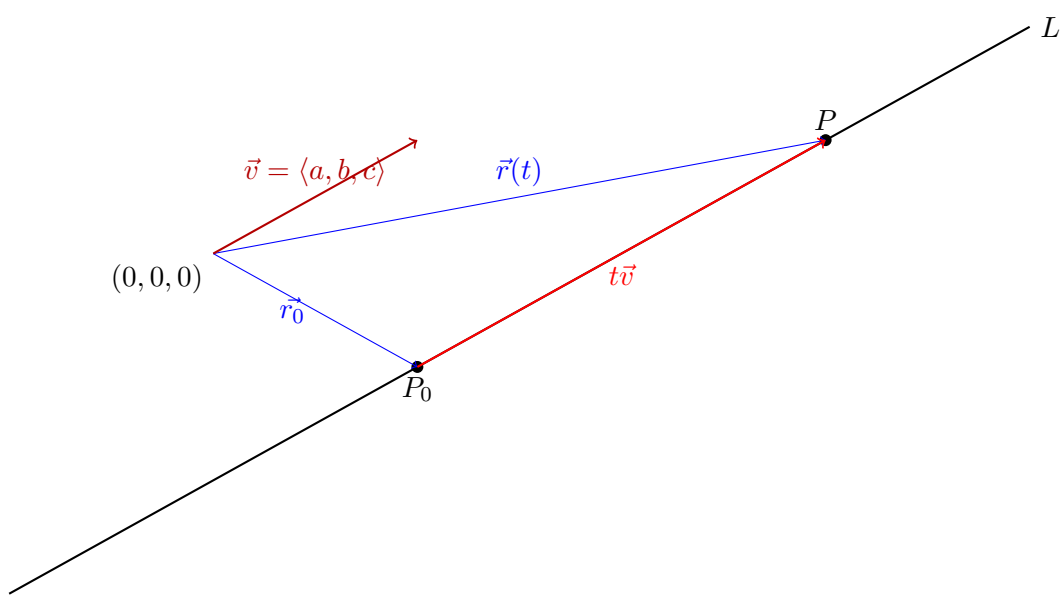
The key insight is that the vector from our known point P_0 to our arbitrary point P, $\vec{P_0P}$, must be parallel to the direction vector $\vec{v}$. This means $\vec{P_0P}$ must be a scalar multiple of $\vec{v}$.

$$\vec{P_0P} = t\vec{v}, \quad \text{for some scalar } t$$

Now, let's use position vectors. Let $\vec{r_0} = \vec{OP_0}$ be the position vector of P_0, and $\vec{r} = \vec{OP}$ be the position vector of P. We know that $\vec{P_0P} = \vec{r} - \vec{r_0}$. Substituting this in gives:

$$\vec{r} - \vec{r_0} = t\vec{v}$$

Solving for $\vec{r}$ gives us the vector equation of the line.



Vector Equation of a Line

The equation of a line passing through the point with position vector $\vec{r_0} = \langle x_0, y_0, z_0 \rangle$ and parallel to the direction vector $\vec{v} = \langle a, b, c \rangle$ is:

$$\vec{r}(t) = \vec{r_0} + t\vec{v}$$

This is a **vector-valued function** of the parameter t. As t varies, the tip of the vector $\vec{r}(t)$ traces out the entire line L.

2. Parametric and Symmetric Equations

While the vector equation is conceptually elegant, we often use its component forms for computation.

Parametric Equations

We can expand the vector equation $\vec{r}(t) = \vec{r_0} + t\vec{v}$ into its components:

$$\langle x, y, z \rangle = \langle x_0, y_0, z_0 \rangle + t\langle a, b, c \rangle = \langle x_0 + at, y_0 + bt, z_0 + ct \rangle$$

Equating the components gives us the **parametric equations** of the line.

Parametric Equations of a Line

$$x = x_0 + at, \quad y = y_0 + bt, \quad z = z_0 + ct$$

Symmetric Equations

If we solve each parametric equation for the parameter t (assuming $a, b, c \neq 0$), we find:

$$t = \frac{x - x_0}{a}, \quad t = \frac{y - y_0}{b}, \quad t = \frac{z - z_0}{c}$$

Since t is the same in all three, we can set them equal to get the **symmetric equations**.

Symmetric Equations of a Line

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$$

Note: If a direction number is zero (e.g., if $a = 0$), the corresponding parametric equation is simply written on its own. For example, if $\vec{v} = \langle 0, b, c \rangle$, the equations would be $x = x_0$, $\frac{y-y_0}{b} = \frac{z-z_0}{c}$.

Example 1: Find the Equation of a Line Through Two Points

Find the parametric and symmetric equations of the line that passes through the points $A(2, 4, -3)$ and $B(3, -1, 1)$.

Solution Strategy:

1. We have points, but we need a *direction vector*. We can find one by creating the vector between the two given points.
2. Choose one of the points to be our base point P_0.
3. Plug the point and direction vector into the formulas.

1. Find the direction vector $\vec{v}$.

$$\vec{v} = \vec{AB} = B - A = \langle 3-2, -1-4, 1-(-3)\rangle = \langle 1, -5, 4\rangle$$

2. Choose a point. Let's use $A(2, 4, -3)$ as our (x_0, y_0, z_0). **3. Write the equations.** The **parametric equations** are:

$$x = 2 + 1t, \quad y = 4 - 5t, \quad z = -3 + 4t$$

The **symmetric equations** are:

$$\frac{x-2}{1} = \frac{y-4}{-5} = \frac{z+3}{4}$$

Classwork: Equations of Lines in Space

Name: _____

(Show all your work for full credit.)

Problem 1: Finding Equations from a Point and Vector

A line passes through the point $P(5, 1, 3)$ and is parallel to the vector $\vec{v} = \langle 2, -3, 4\rangle$.

(a) Write the parametric equations for the line.

(b) Write the symmetric equations for the line.

Problem 2: Finding Equations from Two Points

Find the parametric equations for the line that passes through the points $A(-2, 1, 0)$ and $B(1, 3, 5)$.

Problem 3: Checking for a Point on a Line

Does the point $Q(3, -1, 4)$ lie on the line given by the parametric equations:

$$x = 1 + 2t, \quad y = 2 - 3t, \quad z = -2 + 3t$$

Justify your answer with a calculation. (Hint: Does a single value of t work for all three components?)

Homework 3

Homework 3 - The Dot and Cross Products

Work solutions on blank paper, showing all steps. Scan and submit by the due date on Blackboard.

Problem 1: Dot Product Calculation

Given the vectors $\vec{u} = \langle 2, -3, 5 \rangle$ and $\vec{v} = \langle 1, 4, -2 \rangle$, compute the dot product $\vec{u} \cdot \vec{v}$.

Problem 2: Dot Product with Standard Unit Vectors

Let $\vec{a} = 3\vec{i} + 2\vec{j} - \vec{k}$ and $\vec{b} = 4\vec{i} + 5\vec{k}$. Find $\vec{a} \cdot \vec{b}$.

Problem 3: Finding the Angle Between Vectors

Find the angle θ between the vectors $\vec{p} = \langle 1, 1, 0 \rangle$ and $\vec{q} = \langle 2, 0, 2 \rangle$. State your answer in both radians and degrees.

Problem 4: Angles and Orthogonality

Consider the vectors $\vec{a} = \langle 2, 2, -1 \rangle$, $\vec{b} = \langle 5, -4, 2 \rangle$, $\vec{u} = \langle 2, 1, 4 \rangle$ and $\vec{v} = \langle 3, -2, -1 \rangle$.

(a) Find the cosine of the angle between $\vec{a}$ and $\vec{b}$.

(b) Determine if the vectors $\vec{u}$ and $\vec{v}$ are orthogonal. Justify your answer.

Problem 5: Finding an Unknown for Orthogonality

Consider the vectors $\vec{u} = \langle -6, k, 2 \rangle$ and $\vec{v} = \langle 2, 5, -1 \rangle$. Find the value of the constant k such that $\vec{u}$ and $\vec{v}$ are orthogonal.

Problem 6: Scalar and Vector Projections

Let $\vec{b} = \langle 2, 3, 1 \rangle$ and $\vec{a} = \langle -1, 4, 8 \rangle$.

(a) Find the scalar projection of $\vec{b}$ onto $\vec{a}$, which is ${}_{\vec{a}}\vec{b}$.

(b) Find the vector projection of $\vec{b}$ onto $\vec{a}$, which is ${}_{\vec{a}}\vec{b}$.

Problem 7: Vector Decomposition

Decompose the vector $\vec{u} = \langle 3, 1, -2 \rangle$ into two orthogonal vectors, one parallel to $\vec{v} = \langle 1, 1, 1 \rangle$ and one orthogonal to $\vec{v}$.

(a) First, find $\vec{u}_{\parallel} =_{\vec{v}} \vec{u}$.

(b) Then, find the orthogonal component $\vec{u}_{\perp} = \vec{u} - \vec{u}_{\parallel}$. Verify that $\vec{u}_{\perp}$ is indeed orthogonal to $\vec{v}$ by computing their dot product.

Problem 8: Application - Work

A constant force is given by the vector $\vec{F} = \langle 4, 3, 5 \rangle$ (in Newtons). If this force is applied to an object that moves along a straight line from point $P(1, 0, 2)$ to point $Q(3, 1, 5)$ (in meters), find the work done by the force. (Recall that Work $= \vec{F} \cdot \vec{D}$, where $\vec{D}$ is the displacement vector.)

Problem 9: Cross Product Calculation

Given the vectors $\vec{a} = \langle 1, 2, -1 \rangle$ and $\vec{b} = \langle 3, 0, 4 \rangle$, compute the cross product $\vec{a} \times \vec{b}$.

Problem 10: Cross Product with Standard Unit Vectors

Let $\vec{u} = \vec{i} + 3\vec{j} - 2\vec{k}$ and $\vec{v} = 4\vec{i} - \vec{j}$. Find $\vec{u} \times \vec{v}$ and write your answer in terms of $\vec{i}, \vec{j}, \vec{k}$.

Problem 11: Orthogonality Property

Let $\vec{p} = \langle 2, -1, 3 \rangle$ and $\vec{q} = \langle 1, 1, 1 \rangle$. Let $\vec{w} = \vec{p} \times \vec{q}$. Show by calculation that $\vec{w}$ is orthogonal to both $\vec{p}$ and $\vec{q}$.

Problem 12: Anticommutativity Property

For the vectors $\vec{a}$ and $\vec{b}$ from Problem 9:

(a) Compute $\vec{b} \times \vec{a}$.

(b) How does your result from part (a) compare to the answer for $\vec{a} \times \vec{b}$ in Problem 9?

Problem 13: Area of a Parallelogram

Find the area of the parallelogram with adjacent sides given by the vectors $\vec{u} = \langle 3, 2, -1 \rangle$ and $\vec{v} = \langle 1, 2, 3 \rangle$.

Problem 14: Area of a Triangle

Find the area of the triangle with vertices at the points $P(1,0,1)$, $Q(2,1,3)$, and $R(3,4,2)$. (Hint: The area of the triangle is half the area of the parallelogram formed by vectors $\vec{PQ}$ and $\vec{PR}$.)

Problem 15: Finding a Normal Vector

Find a vector that is orthogonal to the plane containing the points $A(1,1,1)$, $B(2,3,4)$, and $C(3,0,1)$. (Hint: $\vec{AB} \times \vec{AC}$ is orthgognal to the plane containing the vectors $\vec{AB}$ and $\vec{AC}$ in standard position.)

Problem 16: Scalar Triple Product Calculation

Let $\vec{a} = \langle 1,2,1 \rangle$, $\vec{b} = \langle 0,3,4 \rangle$, and $\vec{c} = \langle 5,-1,0 \rangle$. Compute the scalar triple product $\vec{a} \cdot (\vec{b} \times \vec{c})$.

Problem 17: Alternative Scalar Triple Product Calculation

For the vectors in Problem 16, compute $(\vec{a} \times \vec{b}) \cdot \vec{c}$. What do you notice about your answer compared to the result from Problem 16?

Problem 18: Volume of a Parallelepiped

Find the volume of the parallelepiped with adjacent edges given by the vectors $\vec{u} = \vec{i} + \vec{j}$, $\vec{v} = \vec{j} + \vec{k}$, and $\vec{w} = \vec{i} + \vec{j} + \vec{k}$.

Problem 19: Checking for Coplanarity

Determine whether the vectors $\vec{a} = \langle 1,5,-2 \rangle$, $\vec{b} = \langle 3,-1,0 \rangle$, and $\vec{c} = \langle 5,9,-4 \rangle$ are coplanar (i.e., they lie on the same plane). Justify your answer with a calculation.

Problem 20: Application - Torque

A force of $\vec{F} = \langle -3,1,4 \rangle$ is applied at the point $P(2,3,1)$. Find the torque vector $\vec{\tau}$ about the pivot point $Q(1,1,-1)$. (Hint: The lever arm vector is $\vec{r} = \vec{QP}$.)

Quiz 3

Quiz 3

Name: _____

(Show all your work for full credit.)

Problem 1: Line of Intersection of Two Planes

Find the parametric equations for the line of intersection of the planes:

$$x + y + z = 3 \quad \text{and} \quad x + 2y + 3z = 6$$

Problem 2: Plane Containing Two Intersecting Lines

Find the equation of the plane that contains the following two intersecting lines:

$$\mathbf{r}(t) = \langle 1 + t, 2 - t, 3t \rangle \qquad \text{and} \qquad \mathbf{u}(t) = \langle 2 - t, 1 + t, 2t \rangle$$

Write your answer in the form $ax + by + cz = d$.

CHAPTER 4

Cycle 4: Lines and Planes in Space (Homework 4)

Overview

This cycle studies lines and planes in three-dimensional space. Vector and symmetric equations of lines, equations of planes, and their intersections are used to analyze relative positions of geometric objects (parallel, intersecting, skew) and to solve applied geometry problems.

Lecture Notes and Classwork

September 17: Interactions Between Lines and Planes

Calculus III: Interactions Between Lines and Planes

Lecture Notes

Now that we have developed equations for both lines and planes, we can explore how these objects interact with each other in 3D space. The key to solving these problems is to remember the fundamental "ingredients" needed for each object:

- To define a **Line**: You need a **point** P_0 and a parallel **direction vector** $\vec{v}$.
- To define a **Plane**: You need a **point** P_0 and an orthogonal **normal vector** $\vec{n}$.

Our primary tool for generating an orthogonal vector is the cross product. Let's work through the most common scenarios.

Learning Objectives

By the end of this lecture, you will be able to:

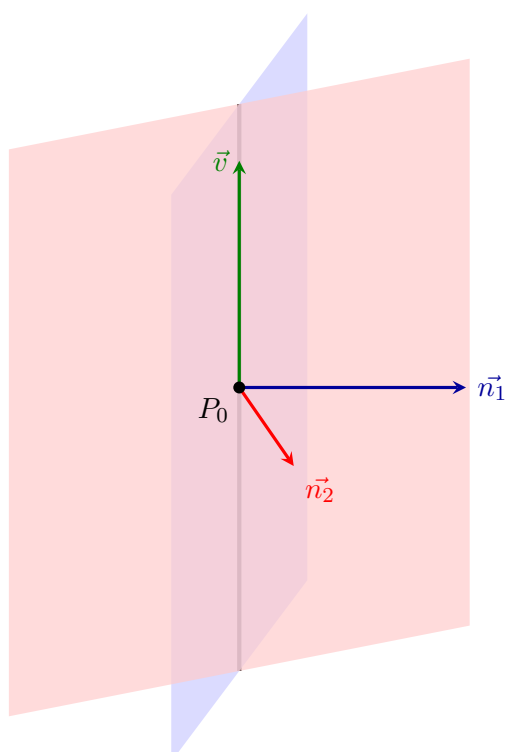
1. Find the point of intersection between a line and a plane.
2. Determine the equation of the line of intersection of two planes.
3. Determine the equation of a plane containing two intersecting lines.

Example 1: Finding the Line of Intersection of Two Planes

Find the parametric equations for the line of intersection of the planes $x - 2y + z = 1$ and $2x + y + z = 1$.

Thought Process:

- **Goal:** Find the equation of a line.
- **Ingredients needed:** A point P_0 on the line and a direction vector $\vec{v}$ for the line.
- **Strategy for $\vec{v}$:** The line of intersection lies in *both* planes. Therefore, the line's direction vector $\vec{v}$ must be orthogonal to *both* of the planes' normal vectors, $\vec{n_1}$ and $\vec{n_2}$. The cross product is the perfect tool for this: $\vec{v} = \vec{n_1} \times \vec{n_2}$.
- **Strategy for P_0:** The point must lie on *both* planes, satisfying both equations. The easiest way to find one such point is to set one variable to a constant (like $z = 0$) and solve the remaining system of two equations for the other two variables.



Step 1: Find the Direction Vector ($\vec{v}$). Extract the normal vectors from the plane equations:

$$\vec{n_1} = \langle 1, -2, 1 \rangle \quad \text{and} \quad \vec{n_2} = \langle 2, 1, 1 \rangle$$

Compute the cross product:

$$\vec{v} = \vec{n_1} \times \vec{n_2} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & -2 & 1 \\ 2 & 1 & 1 \end{vmatrix} = \vec{i}(-2-1) - \vec{j}(1-2) + \vec{k}(1-(-4)) = \langle -3, 1, 5 \rangle$$

Step 2: Find a Point on the Line (P_0). Set $z = 0$ in both plane equations:

$$\begin{cases} x - 2y = 1 \\ 2x + y = 1 \end{cases}$$

From the second equation, $y = 1 - 2x$. Substituting into the first: $x - 2(1 - 2x) = 1 \implies 5x - 2 = 1 \implies x = 3/5$. Then $y = 1 - 2(3/5) = -1/5$. A point on the line is $P_0(3/5, -1/5, 0)$.

Step 3: Assemble the Line Equation. Using point P_0 and direction vector $\vec{v}$, the parametric equations are:

$$x = \frac{3}{5} - 3t, \quad y = -\frac{1}{5} + t, \quad z = 5t$$

Example 2: Finding the Intersection Point of a Line and a Plane

Find the point where the line $x = 1 + 2t, y = -1 + t, z = t$ intersects the plane $2x + y - z = 7$.

Thought Process:

- **Goal:** Find a single point (x, y, z).
- **Strategy:** The coordinates of the intersection point must satisfy the equations for both the line and the plane simultaneously. We can represent a generic point on the line using the parameter t. By substituting this generic point into the plane equation, we find the specific value of t where the line "pierces" the plane.

Step 1: Substitute the line's parametric equations into the plane equation. Plug the expressions for x, y, and z from the line into the plane equation:

$$2(1 + 2t) + (-1 + t) - (t) = 7$$

Step 2: Solve for t.

$$2 + 4t - 1 + t - t = 7 \implies 1 + 4t = 7 \implies 4t = 6 \implies t = \frac{3}{2}$$

Step 3: Find the Intersection Point. Plug this specific value of t back into the original parametric equations for the line:

$$x = 1 + 2\left(\frac{3}{2}\right) = 4, \quad y = -1 + \frac{3}{2} = \frac{1}{2}, \quad z = t = \frac{3}{2}$$

The point of intersection is $(4, 1/2, 3/2)$.

Example 3: Finding the Plane Containing Two Intersecting Lines

Find the equation of the plane that contains the intersecting lines:

$$L_1: \quad x = 1 + t, \quad y = 2 - t, \quad z = 3t$$

$$L_2: \quad x = 2 - s, \quad y = 1 + 2s, \quad z = 3 + s$$

Thought Process:

- **Goal:** Find the equation of a plane.
- **Ingredients needed:** A point P_0 on the plane and a normal vector $\vec{n}$ for the plane.
- **Strategy for $\vec{n}$:** The plane contains both lines, which means the plane is parallel to both direction vectors $\vec{v_1}$ and $\vec{v_2}$. Therefore, the plane's normal vector must be orthogonal to both $\vec{v_1}$ and $\vec{v_2}$. We can find this normal vector using the cross product: $\vec{n} = \vec{v_1} \times \vec{v_2}$.
- **Strategy for P_0:** Since the plane contains both lines, any point on either line is a valid choice. The easiest point to find is usually by setting one of the parameters to zero.

Step 1: Find the Normal Vector ($\vec{n}$). Extract the direction vectors from the parametric equations:

$$\vec{v_1} = \langle 1, -1, 3 \rangle \quad \text{(from the coefficients of } t\text{)}$$
$$\vec{v_2} = \langle -1, 2, 1 \rangle \quad \text{(from the coefficients of } s\text{)}$$

Compute the cross product:

$$\vec{n} = \vec{v_1} \times \vec{v_2} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & -1 & 3 \\ -1 & 2 & 1 \end{vmatrix} = \vec{i}(-1-6) - \vec{j}(1-(-3)) + \vec{k}(2-1) = \langle -7, -4, 1 \rangle$$

Step 2: Find a Point on the Plane (P_0). Use line L_1 and set $t = 0$:

$$x = 1, \quad y = 2, \quad z = 0$$

A point on the plane is $P_0(1, 2, 0)$.

Step 3: Assembling the Plane Equation. Using point $P_0(1, 2, 0)$ and normal vector $\vec{n} = \langle -7, -4, 1 \rangle$:

$$-7(x-1) - 4(y-2) + 1(z-0) = 0$$
$$-7x + 7 - 4y + 8 + z = 0$$
$$-7x - 4y + z = -15 \quad \text{or} \quad 7x + 4y - z = 15$$

Classwork: Interactions Between Lines and Planes

Name: _____

Problem 1: Finding the Line of Intersection of Two Planes

Find the parametric equations for the line of intersection of the following two planes:

$$\text{Plane 1: } x - 2y + z = 1$$

$$\text{Plane 2: } 2x + y + z = 1$$

(a) Find the normal vectors $\vec{n_1}$ and $\vec{n_2}$ for each plane.

(b) Find the direction vector $\vec{v}$ of the line of intersection by computing $\vec{n_1} \times \vec{n_2}$.

(c) Find a point P_0 that lies on both planes by setting $z = 0$ and solving the resulting system of equations for x and y.

(d) Write the final parametric equations for the line.

Problem 2: Finding the Plane Containing Two Intersecting Lines

Find the equation of the plane that contains the following two intersecting lines:

$$L_1: \quad x = 1 + t, \quad y = 2 - t, \quad z = 3t$$

$$L_2: \quad x = 2 - s, \quad y = 1 + 2s, \quad z = 3 + s$$

(a) State the direction vectors $\vec{v_1}$ and $\vec{v_2}$ for each line.

(b) Find the normal vector $\vec{n}$ for the plane by computing $\vec{v_1} \times \vec{v_2}$.

(c) Find a point on the plane. (Hint: Any point on either line will work. The point $(1, 2, 0)$ is on L_1 when $t = 0$.)

(d) Write the final equation for the plane in the form $ax + by + cz = d$.

Problem 3: Finding a Plane Parallel to a Line

Find the equation of the plane that passes through the point $P(3, 0, -1)$ and contains the line given by $x = 2 + t, y = 3 - 2t, z = 4 + t$. (Hint: To find the normal vector $\vec{n}$, you need two different vectors that lie in the plane. One is the direction vector of the line. The other can be found by connecting a point on the line to the given point P.)

Problem 4: Finding the Intersection Point of a Line and a Plane

Find the point where the line $x = 1 + 2t, y = -1 + t, z = t$ intersects the plane $2x + y - z = 7$.

September 19: Calculus of Vector-Valued Functions

Calculus III: Calculus of Vector-Valued Functions

Lecture Notes

On Monday, we saw that the equation of a line, $\vec{r}(t) = \vec{r_0} + t\vec{v}$, can be thought of as a function that "draws" a line as the parameter t changes. This is our first example of a **vector-valued function**. In general, we can define any curve in space by a function of the form $\vec{r}(t) = \langle f(t), g(t), h(t) \rangle$.

If we think of the parameter t as time, then $\vec{r}(t)$ represents the **position vector** of a particle moving through space. This opens the door to describing motion using calculus: velocity, acceleration, and more.

Learning Objectives

By the end of this lecture, you will be able to:

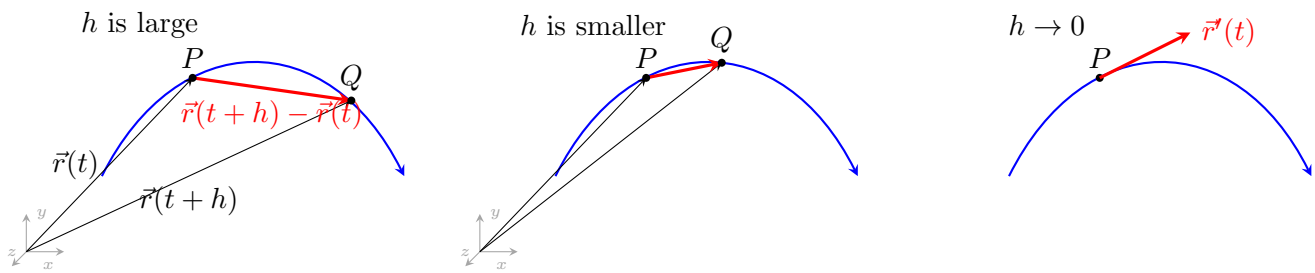
1. Find the derivative of a vector-valued function.
2. Find the parametric equation of the tangent line to a curve at a given point.
3. Find the unit tangent vector, $\vec{T}(t)$.
4. Find the arc length function for a vector-valued function.

1. The Derivative of a Vector-Valued Function

Recall from Calculus I that the derivative is the limit of the difference quotient. The same is true for vector-valued functions. The derivative of $\vec{r}(t)$ is defined as:

$$\vec{r}'(t) = \lim_{h \to 0} \frac{\vec{r}(t+h) - \vec{r}(t)}{h}$$

The vector $\vec{r}(t+h) - \vec{r}(t)$ is a **secant vector** connecting two points on the curve. Dividing by the scalar h scales this vector. As h approaches zero, the two points on the curve get closer, and the secant vector's direction approaches the direction of the **tangent vector** at the point $\vec{r}(t)$.



To compute this limit, we can analyze the components. Let $\vec{r}(t) = \langle f(t), g(t), h(t) \rangle$.

$$\begin{aligned}
\vec{r}'(t) &= \lim_{h \to 0} \frac{\langle f(t+h), g(t+h), h(t+h) \rangle - \langle f(t), g(t), h(t) \rangle}{h} \\
&= \lim_{h \to 0} \left\langle \frac{f(t+h) - f(t)}{h}, \frac{g(t+h) - g(t)}{h}, \frac{h(t+h) - h(t)}{h} \right\rangle \\
&= \left\langle \lim_{h \to 0} \frac{f(t+h) - f(t)}{h}, \lim_{h \to 0} \frac{g(t+h) - g(t)}{h}, \lim_{h \to 0} \frac{h(t+h) - h(t)}{h} \right\rangle \\
&= \langle f'(t), g'(t), h'(t) \rangle
\end{aligned}$$

This confirms that the derivative is found by simply differentiating each component.

Definition: The Derivative of a Vector-Valued Function

If $\vec{r}(t) = \langle f(t), g(t), h(t) \rangle$, where f, g, and h are differentiable functions, then its derivative is:

$$\vec{r}'(t) = \frac{d\vec{r}}{dt} = \langle f'(t), g'(t), h'(t) \rangle$$

If $\vec{r}(t)$ is position, then $\vec{r}'(t)$ is the **velocity vector**, and $||\vec{r}'(t)||$ is the **speed**.

2. The Tangent Line to a Curve

The tangent line to a curve at a point is the line that "just touches" the curve at that point and points in the same direction as the curve's motion. We now have all the ingredients to define it:

- **A point on the line:** This is the point on the curve itself, given by $\vec{r}(a)$.
- **A direction vector:** This is the tangent (velocity) vector at that point, $\vec{r}'(a)$.

Equation of the Tangent Line

The tangent line to the curve $\vec{r}(t)$ at the point where $t = a$ is given by the vector equation:

$$\vec{L}(t) = \vec{r}(a) + t \cdot \vec{r}'(a)$$

Example 1: Finding a Tangent Line

Find the parametric equations of the tangent line to the helix $\vec{r}(t) = \langle \cos(t), \sin(t), t \rangle$ at the point where $t = \pi/2$.

Solution: 1. Find the point on the curve, $\vec{r}(\pi/2)$:

$$\vec{r}(\pi/2) = \langle \cos(\pi/2), \sin(\pi/2), \pi/2 \rangle = \langle 0, 1, \pi/2 \rangle$$

This is our point $(x_0, y_0, z_0) = (0, 1, \pi/2)$.

2. Find the derivative (tangent vector), $\vec{r}'(t)$:

$$\vec{r}'(t) = \frac{d}{dt}\langle \cos(t), \sin(t), t \rangle = \langle -\sin(t), \cos(t), 1 \rangle$$

3. Find the specific tangent vector at $t = \pi/2$, $\vec{r}'(\pi/2)$:

$$\vec{r}'(\pi/2) = \langle -\sin(\pi/2), \cos(\pi/2), 1 \rangle = \langle -1, 0, 1 \rangle$$

This is our direction vector $\vec{v} = \langle a, b, c \rangle$.

4. Assemble the line equation: Using the point $(0, 1, \pi/2)$ and direction $\langle -1, 0, 1 \rangle$:

$$x = 0 - 1t, \quad y = 1 + 0t, \quad z = \pi/2 + 1t$$

The parametric equations are $x = -t, y = 1, z = \pi/2 + t$.

3. The Unit Tangent Vector

The velocity vector $\vec{r}'(t)$ tells us two things: the direction of motion and the speed. If we only care about the **direction**, we can normalize the velocity vector to have a magnitude of 1.

Definition: The Unit Tangent Vector

The **unit tangent vector**, denoted $\vec{T}(t)$, is a vector of length 1 that points in the direction of the tangent vector $\vec{r}'(t)$. It is found by:

$$\vec{T}(t) = \frac{\vec{r}'(t)}{||\vec{r}'(t)||}$$

Example 2: Finding the Unit Tangent Vector

For the helix $\vec{r}(t) = \langle \cos(t), \sin(t), t \rangle$, find the unit tangent vector $\vec{T}(t)$.

Solution: 1. Find the derivative, $\vec{r}'(t)$: We already found this:

$$\vec{r}'(t) = \langle -\sin(t), \cos(t), 1\rangle$$

2. Find the magnitude of the derivative, $||\vec{r}'(t)||$:

$$||\vec{r}'(t)|| = \sqrt{(-\sin t)^2 + (\cos t)^2 + 1^2} = \sqrt{\sin^2 t + \cos^2 t + 1} = \sqrt{1+1} = \sqrt{2}$$

This means the helix has a constant speed of $\sqrt{2}$.

3. Divide the derivative by its magnitude:

$$\vec{T}(t) = \frac{\vec{r}'(t)}{||\vec{r}'(t)||} = \frac{1}{\sqrt{2}}\langle -\sin(t), \cos(t), 1\rangle = \left\langle -\frac{\sin t}{\sqrt{2}}, \frac{\cos t}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right\rangle$$

This vector has a length of 1 for any value of t.

4. Arc Length and Speed

We know from Calculus II that the arc length of a parametric curve from $t = a$ to $t = b$ is found by integrating the speed. This concept extends perfectly to 3D.

Formula for Arc Length

The **arc length**, L, of a curve $\vec{r}(t)$ from $t = a$ to $t = b$ is the integral of its speed:

$$L = \int_a^b ||\vec{r}'(t)||\, dt = \int_a^b \sqrt{(f'(t))^2 + (g'(t))^2 + (h'(t))^2}\, dt$$

The Arc Length Function

Often, we want to know the distance traveled from a starting point (say, $t = 0$) up to some arbitrary time t. This defines the **arc length function**, $s(t)$.

The Arc Length Function

The distance traveled along a curve from a starting time a to any time t is:

$$s(t) = \int_a^t ||\vec{r}'(u)||\, du$$

(We use a dummy variable u for integration to avoid confusion with the upper limit t.)

The Relationship Between Arc Length and Speed

What is the derivative of the arc length function, $s'(t)$? We can apply the Fundamental Theorem of Calculus (Part 1) to the arc length function:

$$\frac{ds}{dt} = \frac{d}{dt}\left[\int_a^t ||\vec{r}'(u)||\, du\right] = ||\vec{r}'(t)||$$

This gives us a profound result:

The rate of change of distance traveled is the speed.

This confirms that our definition of speed as the magnitude of the velocity vector is consistent with our understanding of distance from single-variable calculus. Integrating the speed gives the total distance traveled.

Example: Arc Length of a Helix

Find the length of the helix $\vec{r}(t) = \langle \cos(t), \sin(t), t \rangle$ from $t = 0$ to $t = 2\pi$.

Solution: 1. Find the speed, $||\vec{r}'(t)||$: We already calculated this in the previous example and found that the speed is constant:

$$||\vec{r}'(t)|| = \sqrt{2}$$

2. Integrate the speed from $t = 0$ to $t = 2\pi$:

$$L = \int_0^{2\pi} \sqrt{2}\, dt = \sqrt{2}t\Big|_0^{2\pi} = \sqrt{2}(2\pi - 0) = 2\sqrt{2}\pi$$

The length of one full turn of the helix is $2\sqrt{2}\pi$.

Classwork: Calculus of Vector-Valued Functions

Name: _____

Problem 1: Finding a Tangent Line

Consider the curve described by the vector-valued function $\vec{r}(t) = \langle t^2, t^3, t \rangle$.

(a) Find the derivative, $\vec{r}'(t)$.

(b) Find the parametric equations for the tangent line to the curve at the point where $t = 2$.

Problem 2: Finding the Unit Tangent Vector

Consider the curve described by the vector-valued function $\vec{r}(t) = \langle 4\sqrt{t}, t^2, t\rangle$.

(a) Find the tangent vector, $\vec{r}'(t)$.

(b) Find the unit tangent vector $\vec{T}(t)$ at the point where $t = 1$.

Problem 3: Finding Arc Length

Find the total length of the curve (arc length) for the helix $\vec{r}(t) = \langle 5\cos(t), 5\sin(t), 12t\rangle$ from $t = 0$ to $t = 2\pi$.

(a) Find the derivative vector $\vec{r}'(t)$.

(b) Find the speed, $||\vec{r}'(t)||$.

(c) Set up and evaluate the integral for arc length.

September 22: Curvature and the Normal Vector

Calculus III: Curvature and the Normal Vector

Lecture Notes

In our last lecture, we explored the calculus of vector-valued functions, defining the velocity vector $\vec{v}(t) = \vec{r}'(t)$ and the unit tangent vector $\vec{T}(t)$. Today, we deepen our understanding of motion by introducing the acceleration vector. Then, we pivot from the time-dependent nature of motion to the intrinsic geometry of the curve itself.

The results of this lecture come from reparameterizing a curve by its **arc length**, s. This powerful idea strips away the speed of travel, leaving only the pure shape of the path. From this perspective, the definitions of **curvature** (κ) and the **principal unit normal vector** ($\vec{N}$) emerge naturally and elegantly. We will see that curvature measures how sharply a curve bends, and the normal vector points in the direction of that bend.

Finally, we will use the Chain Rule to translate these pure geometric concepts back into practical, computable formulas in terms of our original parameter, t. While the concepts are elegant, we will see that the computations can be algebraically challenging.

Learning Objectives

By the end of this lecture, you will be able to:

1. Compute the velocity and acceleration vectors of a particle in motion.
2. Compute the unit tangent vector, $\vec{T}(t)$.
3. Compute the principal unit normal vector, $\vec{N}(t)$.
4. Recall the precise definition of curvature, κ. (Note: We will find more convenient computational formulas for κ in a future lecture.)

1. Velocity and Acceleration

If $\vec{r}(t)$ is the position vector of a particle at time t, its velocity is the rate of change of position, and its acceleration is the rate of change of velocity.

Definition: Velocity and Acceleration

The **velocity vector** of a particle with position function $\vec{r}(t)$ is:

$$\vec{v}(t) = \vec{r}'(t)$$

The **acceleration vector** is the second derivative of the position function:

$$\vec{a}(t) = \vec{v}'(t) = \vec{r}''(t)$$

The velocity vector is tangent to the curve and its magnitude is the particle's speed. The acceleration vector, however, can point in any direction. It describes how the velocity vector is changing, which can mean a change in speed, a change in direction, or both.

2. The Geometry of Curves: Arc Length Parameterization

To study the intrinsic geometry of a curve, we want to separate its shape from the speed at which it is traced. We do this by parameterizing the curve with respect to its arc length, s.

Recall that the unit tangent vector is $\vec{T}(t) = \frac{\vec{r}'(t)}{||\vec{r}'(t)||} = \frac{d\vec{r}/dt}{ds/dt}$. By the Chain Rule, this simplifies to $\vec{T}(s) = \frac{d\vec{r}}{ds}$. This means that if you travel along the curve at a constant speed of 1 unit per second, your velocity vector is exactly the unit tangent vector.

Now for the key insight. Let's see how this unit tangent vector changes as we move along the curve. We examine its derivative with respect to arc length, $\frac{d\vec{T}}{ds}$. Since $\vec{T}$ is a unit vector, we know $||\vec{T}(s)||^2 = \vec{T}(s) \cdot \vec{T}(s) = 1$. Differentiating with respect to s:

$$\frac{d}{ds}(\vec{T}(s) \cdot \vec{T}(s)) = \frac{d}{ds}(1) \implies \vec{T}'(s) \cdot \vec{T}(s) + \vec{T}(s) \cdot \vec{T}'(s) = 0 \implies 2\vec{T}(s) \cdot \vec{T}'(s) = 0$$

This shows that $\vec{T}'(s)$ is always **orthogonal** to $\vec{T}(s)$. The vector $\vec{T}'(s)$ points in the direction that the tangent vector is turning, which is necessarily perpendicular to the tangent vector itself.

This leads us to our core geometric definitions.

Definition: Curvature and the Principal Unit Normal Vector

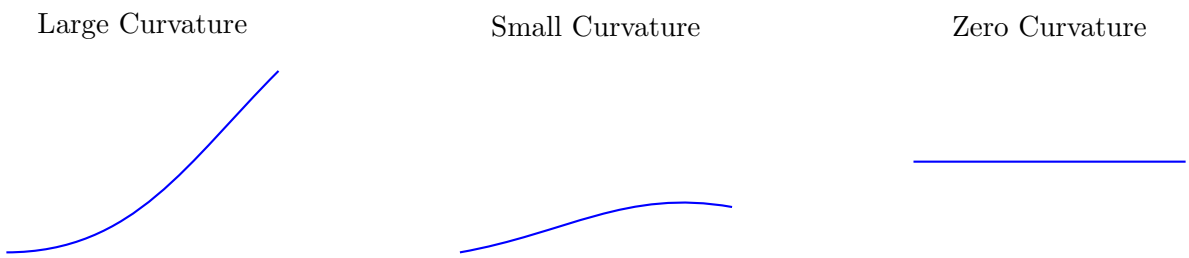
The **curvature**, $\kappa(s)$ (kappa), is the magnitude of the rate of change of the unit tangent vector with respect to arc length. It measures how sharply the curve bends.

$$\kappa(s) = \left|\left|\frac{d\vec{T}}{ds}\right|\right|$$

The **principal unit normal vector**, $\vec{N}(s)$, is the unit vector that gives the direction in which the tangent vector is changing.

$$\vec{N}(s) = \frac{d\vec{T}/ds}{||d\vec{T}/ds||} = \frac{1}{\kappa(s)}\frac{d\vec{T}}{ds}$$

From these definitions, we see the beautiful relationship $\frac{d\vec{T}}{ds} = \kappa(s)\vec{N}(s)$. The change in the tangent vector is broken down into its magnitude (how much it's changing - the curvature) and its direction (which way it's changing - the normal vector).



The diagrams above illustrate the concept of curvature. For large curvature, the unit tangent vectors (red arrows) change direction rapidly over a short distance. For small curvature, they change direction slowly. For zero curvature (a straight line), they do not change direction at all.

3. Computational Formulas for $\vec{T}(t)$ and $\vec{N}(t)$

Finding the arc length parameterization $s(t)$ explicitly is often difficult. We need formulas that work with our original parameter, t.

Computational Formula: Unit Tangent Vector

The **unit tangent vector** is found by normalizing the velocity vector:

$$\vec{T}(t) = \frac{\vec{r}'(t)}{||\vec{r}'(t)||}$$

To find $\vec{N}(t)$, we need to compute the derivative of $\vec{T}(t)$ with respect to t. Using the Chain Rule:

$$\frac{d\vec{T}}{dt} = \frac{d\vec{T}}{ds}\frac{ds}{dt}$$

We know that $\frac{ds}{dt} = ||\vec{r}'(t)||$. We also know from our geometric definition that $\frac{d\vec{T}}{ds}$ points in the same direction as $\vec{N}(s)$. Therefore, $\frac{d\vec{T}}{dt}$ must also point in the same direction as $\vec{N}(t)$. To find the unit vector $\vec{N}(t)$, we simply need to find the derivative $\vec{T}'(t)$ and then normalize it.

Computational Formula: Principal Unit Normal Vector

The **principal unit normal vector** is found by computing the derivative of the unit tangent vector, $\vec{T}'(t)$, and then normalizing it:

$$\vec{N}(t) = \frac{\vec{T}'(t)}{||\vec{T}'(t)||}$$

Warning: The computation of $\vec{T}'(t)$ often involves complex applications of the quotient or product rule and can be algebraically intensive.

4. Examples

Example 1: The Helix

Find the unit tangent vector $\vec{T}(t)$ and the principal unit normal vector $\vec{N}(t)$ for the helix $\vec{r}(t) = \langle \cos(t), \sin(t), t \rangle$.

Solution: 1. Find $\vec{r}'(t)$ and $||\vec{r}'(t)||$:

$$\vec{r}'(t) = \langle -\sin(t), \cos(t), 1 \rangle$$

$$||\vec{r}'(t)|| = \sqrt{(-\sin t)^2 + (\cos t)^2 + 1^2} = \sqrt{\sin^2 t + \cos^2 t + 1} = \sqrt{1+1} = \sqrt{2}$$

The speed is constant.

2. Find $\vec{T}(t)$:

$$\vec{T}(t) = \frac{\vec{r}'(t)}{||\vec{r}'(t)||} = \frac{\langle -\sin(t), \cos(t), 1 \rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}} \langle -\sin(t), \cos(t), 1 \rangle$$

3. Find $\vec{T}'(t)$:

$$\vec{T}'(t) = \frac{d}{dt}\left(\frac{1}{\sqrt{2}} \langle -\sin(t), \cos(t), 1 \rangle \right) = \frac{1}{\sqrt{2}} \langle -\cos(t), -\sin(t), 0 \rangle$$

4. Find $||\vec{T}'(t)||$:

$$||\vec{T}'(t)|| = \frac{1}{\sqrt{2}} \sqrt{(-\cos t)^2 + (-\sin t)^2 + 0^2} = \frac{1}{\sqrt{2}} \sqrt{\cos^2 t + \sin^2 t} = \frac{1}{\sqrt{2}}(1) = \frac{1}{\sqrt{2}}$$

5. Find $\vec{N}(t)$:

$$\vec{N}(t) = \frac{\vec{T}'(t)}{||\vec{T}'(t)||} = \frac{\frac{1}{\sqrt{2}} \langle -\cos(t), -\sin(t), 0 \rangle}{1/\sqrt{2}} = \langle -\cos(t), -\sin(t), 0 \rangle$$

This result is very geometric. The tangent vector $\vec{T}(t)$ has a constant z-component, pointing slightly upwards as it spirals. The normal vector $\vec{N}(t)$ has a zero z-component and points from the curve's position directly towards the z-axis, indicating the center of the circular motion.

Example 2: An Algebraically Complex Case

Find $\vec{T}(t)$ and $\vec{N}(t)$ for the curve $\vec{r}(t) = \langle e^t, e^{-t}, \sqrt{2}t \rangle$.

Solution: This example illustrates how the algebra can become challenging. **1. Find $\vec{r}'(t)$ and $||\vec{r}'(t)||$:**

$$\vec{r}'(t) = \langle e^t, -e^{-t}, \sqrt{2} \rangle$$

$$||\vec{r}'(t)|| = \sqrt{(e^t)^2 + (-e^{-t})^2 + (\sqrt{2})^2} = \sqrt{e^{2t} + e^{-2t} + 2} = \sqrt{(e^t + e^{-t})^2} = e^t + e^{-t}$$

2. Find $\vec{T}(t)$:

$$\vec{T}(t) = \frac{\langle e^t, -e^{-t}, \sqrt{2} \rangle}{e^t + e^{-t}} = \left\langle \frac{e^t}{e^t + e^{-t}}, \frac{-e^{-t}}{e^t + e^{-t}}, \frac{\sqrt{2}}{e^t + e^{-t}} \right\rangle$$

3. Find $\vec{T}'(t)$: We apply the quotient rule, $\frac{f'g-fg'}{g^2}$, to each component individually. Let $g(t) = e^t + e^{-t}$, so $g'(t) = e^t - e^{-t}$.

x-component: Let $f(t) = e^t$, so $f'(t) = e^t$.

$$\frac{d}{dt}\left(\frac{e^t}{e^t+e^{-t}}\right) = \frac{e^t(e^t+e^{-t}) - e^t(e^t-e^{-t})}{(e^t+e^{-t})^2} = \frac{(e^{2t}+1)-(e^{2t}-1)}{(e^t+e^{-t})^2} = \frac{2}{(e^t+e^{-t})^2}$$

y-component: Let $f(t) = -e^{-t}$, so $f'(t) = e^{-t}$.

$$\frac{d}{dt}\left(\frac{-e^{-t}}{e^t+e^{-t}}\right) = \frac{e^{-t}(e^t+e^{-t}) - (-e^{-t})(e^t-e^{-t})}{(e^t+e^{-t})^2} = \frac{(1+e^{-2t})+(1-e^{-2t})}{(e^t+e^{-t})^2} = \frac{2}{(e^t+e^{-t})^2}$$

z-component: Let $f(t) = \sqrt{2}$, so $f'(t) = 0$.

$$\frac{d}{dt}\left(\frac{\sqrt{2}}{e^t+e^{-t}}\right) = \frac{0(e^t+e^{-t}) - \sqrt{2}(e^t-e^{-t})}{(e^t+e^{-t})^2} = \frac{-\sqrt{2}(e^t-e^{-t})}{(e^t+e^{-t})^2}$$

Combining these results gives the derivative vector:

$$\vec{T}'(t) = \frac{1}{(e^t+e^{-t})^2}\langle 2, 2, -\sqrt{2}(e^t-e^{-t})\rangle$$

4. Find $||\vec{T}'(t)||$:

$$||\vec{T}'(t)|| = \frac{1}{(e^t+e^{-t})^2}||\langle 2, 2, -\sqrt{2}(e^t-e^{-t})\rangle||$$

The magnitude of the numerator vector is:

$$\begin{aligned}\sqrt{2^2+2^2+(-\sqrt{2}(e^t-e^{-t}))^2} &= \sqrt{8+2(e^{2t}-2+e^{-2t})} \\ &= \sqrt{4+2e^{2t}+2e^{-2t}} = \sqrt{2(2+e^{2t}+e^{-2t})} \\ &= \sqrt{2(e^t+e^{-t})^2} = \sqrt{2}(e^t+e^{-t})\end{aligned}$$

Thus, the magnitude of the full derivative is:

$$||\vec{T}'(t)|| = \frac{\sqrt{2}(e^t+e^{-t})}{(e^t+e^{-t})^2} = \frac{\sqrt{2}}{e^t+e^{-t}}$$

5. Find $\vec{N}(t)$:

$$\vec{N}(t) = \frac{\vec{T}'(t)}{||\vec{T}'(t)||} = \frac{\frac{1}{(e^t+e^{-t})^2}\langle 2, 2, -\sqrt{2}(e^t-e^{-t})\rangle}{\frac{\sqrt{2}}{e^t+e^{-t}}} = \frac{\langle 2, 2, -\sqrt{2}(e^t-e^{-t})\rangle}{\sqrt{2}(e^t+e^{-t})}$$

$$\vec{N}(t) = \frac{1}{e^t+e^{-t}}\langle \sqrt{2}, \sqrt{2}, -(e^t-e^{-t})\rangle$$

Classwork: The Unit Tangent and Normal Vectors

Name: _____

Problem 1: Velocity and Acceleration

Consider the curve described by the vector-valued function $\vec{r}(t) = \langle 4\cos(t), \sin(t), t^2 \rangle$.

(a) Find the velocity vector, $\vec{v}(t) = \vec{r}'(t)$.

(b) Find the acceleration vector, $\vec{a}(t) = \vec{r}''(t)$.

(c) Find the acceleration vector at the point where $t = \pi$.

Problem 2: Finding $\vec{T}(t)$ and $\vec{N}(t)$

Consider the curve traced by the function $\vec{r}(t) = \langle 1, t, t^2 \rangle$.

(a) Find the unit tangent vector $\vec{T}(t)$.

(b) Find the derivative of the unit tangent vector, $\vec{T}'(t)$. *(Hint: This is the most algebraically intensive step. Be careful with the chain rule and quotient/product rule.)*

(c) Find the magnitude of your result from the previous step, $||\vec{T}'(t)||$.

(d) Using your results from parts (b) and (c), find the principal unit normal vector $\vec{N}(t)$.

Homework 4

Homework 4 - Lines and Planes in Space

Work solutions on blank paper, showing all steps. Scan and submit by the due date on Blackboard.

Part 1: Equations of Planes

Problem 1: The Basics of Plane Equations

A plane in space passes through the point $P_0(1, -2, 3)$ and has a normal vector $\vec{n} = \langle 4, 5, -1 \rangle$.

(a) Write the equation of the plane in standard (point-normal) form.

(b) Write the equation of the plane in the general form: $ax + by + cz = d$.

(c) Determine if the point $Q(2, 0, 9)$ lies on this plane.

Problem 2: Finding the Equation of a Plane from Three Points

Find the equation of the plane containing the points $A(1, 1, -1)$, $B(2, 0, 2)$, and $C(0, -2, 1)$.

(a) Find two vectors that lie in the plane, for example $\vec{AB}$ and $\vec{AC}$.

(b) Calculate a normal vector $\vec{n}$ for the plane.

(c) Find the equation of the plane in general form.

Problem 3: Identifying Plane Relationships

Consider the planes P1: $3x - 2y + z = 4$, P2: $x + 2y - z = 3$, and P3: $-6x + 4y - 2z = 5$.

(a) State the normal vector for each of the three planes.

(b) Identify a pair of planes that are parallel. Justify your answer.

(c) Identify a pair of planes that are orthogonal. Justify your answer.

Problem 4: Finding the Angle Between Two Planes

Find the acute angle of intersection between the planes $2x + 2y - z = 3$ and $x - 3y + 2z = 1$.

Problem 5: Finding a Parallel Plane

Find the equation of the plane that passes through the point $(3, -1, 4)$ and is parallel to the plane $x - 2y + 5z = 10$.

Problem 6: Intercepts of a Plane

Consider the plane given by the equation $2x + 4y + z = 8$.

(a) Find the x, y, and z-intercepts of the plane.

(b) Using the intercepts, find the area of the triangle formed by the plane in the first octant.

Part 2: Equations of Lines

Problem 7: The Basics of Line Equations

A line passes through the point $P_0(1, 0, -3)$ and is parallel to the vector $\vec{v} = \langle 2, -4, 5 \rangle$.

(a) Write the parametric equations for the line.

(b) Write the symmetric equations for the line.

(c) Find another point on the line by choosing a value for t.

Problem 8: Finding the Equation of a Line Through Two Points

Find the parametric equations for the line that passes through the points $A(3, -1, 4)$ and $B(1, 5, -2)$.

Problem 9: Symmetric Equations with a Zero Component

A line passes through the point $(2, 1, 5)$ and has a direction vector $\vec{v} = \langle 3, 0, -1 \rangle$. Write the symmetric equations for this line.

Problem 10: Parallel Lines

Determine if the following two lines are parallel, and explain your reasoning.

$$L_1: \quad x = 1 + 2t, \quad y = 3t, \quad z = 2 - t$$

$$L_2: \quad x = -1 + 4s, \quad y = 5 + 6s, \quad z = 3 - 2s$$

Problem 11: Skew vs Intersecting Lines

Consider the lines:

$$L_1: \quad x = 5 + t, \quad y = 1 + 3t, \quad z = -2 - t$$

$$L_2: \quad x = 2 - s, \quad y = 4 + s, \quad z = 1 + 2s$$

(a) Are the lines parallel?

(b) Do the lines intersect? If so, find the point of intersection. If not, what is the relationship between them?

Problem 12: Finding a Perpendicular Line

Find the parametric equations for the line that passes through the point $(0, 1, 2)$ and is perpendicular to both of the vectors $\vec{a} = \langle 1, 1, 0 \rangle$ and $\vec{b} = \langle 0, 2, 3 \rangle$.

Part 3: Interactions Between Lines and Planes

Problem 13: Finding the Intersection Point of a Line and Plane

Find the point where the line $x = 2 - t, y = 1 + 3t, z = 4t$ intersects the plane $2x - y + z = 2$.

Problem 14: Finding the Intersection Point of a Line and Plane

Find the point where the line passing through $(1, 0, 1)$ and $(4, -2, 2)$ intersects the plane $x + y + z = 6$.

Problem 15: Finding the Line of Intersection of Two Planes

Find the parametric equations for the line of intersection of the planes $x + y + z = 1$ and $x - y + z = 1$.

Problem 16: Finding the Line of Intersection of Two Planes

Find the symmetric equations for the line of intersection of the planes $2x + 5z + 3 = 0$ and $x - 3y + z + 2 = 0$.

Problem 17: Finding the Plane Containing Two Intersecting Lines

Find the equation of the plane that contains the intersecting lines:

$$L_1: \frac{x-1}{2} = \frac{y}{3} = \frac{z+1}{-1} \quad \text{and} \quad L_2: \frac{x-1}{-1} = \frac{y}{2} = \frac{z+1}{1}$$

Problem 18: Finding a Plane Containing a Point and a Line

Find the equation of the plane that passes through the point $P(1, 2, 3)$ and contains the line $x = 3t, y = 1 + t, z = 2 - t$.

Problem 19: Finding a Perpendicular Plane

Find the equation of the plane that passes through the point $(6, 0, -2)$ and is perpendicular to the line $x = 4 - t, y = 3 + 2t, z = t$.

Problem 20: Conceptual Question

Consider a line L with direction vector $\vec{v}$ and a plane P with normal vector $\vec{n}$.

(a) What is the geometric relationship between the line and the plane if $\vec{v} \cdot \vec{n} = 0$?

(b) What is the geometric relationship between the line and the plane if $\vec{v} \times \vec{n} = \vec{0}$?

Quiz 4

Quiz 4: Calculus III

Name: _____

(Show all your work for full credit.)

Problem 1: Curvature

A particle travels along the path given by the vector function $\vec{r}(t) = \langle 2t, t^2, 5 \rangle$.

(a) Find the curvature, $\kappa(t)$, at the point where $t = 0$.

(b) In a sentence, explain what your answer from part (a) tells you about the curve at the point $P(0, 0, 5)$.

Problem 2: Tangential Component of Acceleration

A particle's position is given by the vector function $\vec{r}(t) = \langle e^t, t^2, 2\rangle$.

(a) Find the tangential component of acceleration, a_T, at the point where $t = 0$.

(b) What does the sign (positive, negative, or zero) of your answer from part (a) tell you about the motion of the particle at that instant?

CHAPTER 5

Cycle 5: Curvature, Acceleration, and the TNB Frame (Homework 5)

Overview

This cycle treats vector-valued functions as particle motion in space. The main ideas are curvature, unit tangent/normal/binormal vectors (the TNB frame), and the tangential and normal components of acceleration. These concepts describe how a curve bends and how motion decomposes along and across the path.

Lecture Notes and Classwork

September 24: The TNB Frame and Acceleration Components

Calculus III: The TNB Frame and Acceleration Components
Lecture Outline

Today's Goals

1. Derive and compute curvature κ using the cross product.
2. Derive and compute the tangential (a_T) and normal (a_N) components of acceleration.
3. Define the binormal vector $\vec{B}$ and the TNB Frame.
4. Connect the formulas to the physical meaning of motion.

1. Derivation of the Curvature Formula

Goal: Prove that $\kappa(t) = \frac{||\vec{r}'(t) \times \vec{r}''(t)||}{||\vec{r}'(t)||^3}$.

Step 1: Start with the definition of $\vec{T}(t)$.

$$\vec{r}'(t) = ||\vec{r}'(t)||\vec{T}(t)$$

Step 2: Differentiate using the product rule.

$$\vec{r}''(t) = \left(\frac{d}{dt}||\vec{r}'(t)||\right)\vec{T}(t) + ||\vec{r}'(t)||\vec{T}'(t)$$

Step 3: Cross $\vec{r}'(t)$ with $\vec{r}''(t)$.

$$\vec{r}'(t) \times \vec{r}''(t) = \left(||\vec{r}'||\vec{T}\right) \times \left[\left(\frac{d}{dt}||\vec{r}'||\right)\vec{T} + ||\vec{r}'||\vec{T}'\right]$$

$$= \left(||\vec{r}'||\vec{T} \times \left(\frac{d}{dt}||\vec{r}'||\right)\vec{T}\right) + \left(||\vec{r}'||\vec{T} \times ||\vec{r}'||\vec{T}'\right)$$
$$= \underbrace{\ldots(\vec{T} \times \vec{T})}_{\vec{0} \text{ since parallel}} + ||\vec{r}'||^2(\vec{T} \times \vec{T}')$$

$$\implies \vec{r}'(t) \times \vec{r}''(t) = ||\vec{r}'(t)||^2(\vec{T}(t) \times \vec{T}'(t))$$

Step 4: Take the magnitude of both sides.

$$||\vec{r}' \times \vec{r}''|| = ||\vec{r}'||^2||\vec{T} \times \vec{T}'||$$

Since $\vec{T} \perp \vec{T}'$ and $||\vec{T}|| = 1$, we have $||\vec{T} \times \vec{T}'|| = ||\vec{T}|| \cdot ||\vec{T}'|| \sin(90°) = ||\vec{T}'||$.

$$||\vec{r}' \times \vec{r}''|| = ||\vec{r}'||^2||\vec{T}'||$$

Step 5: Solve for κ. Recall the definition: $\kappa = \frac{||\vec{T}'||}{||\vec{r}'||}$. First, isolate $||\vec{T}'||$ from Step 4:

$$||\vec{T}'|| = \frac{||\vec{r}' \times \vec{r}''||}{||\vec{r}'||^2}$$

Now, substitute this into the definition of κ:

$$\kappa = \frac{1}{||\vec{r}'||}\left(\frac{||\vec{r}' \times \vec{r}''||}{||\vec{r}'||^2}\right) = \frac{||\vec{r}' \times \vec{r}''||}{||\vec{r}'||^3} \quad \blacksquare$$

2. Derivation of Acceleration Components

Goal: Find formulas for a_T and a_N.

Step 1: Decompose the acceleration vector. From the product rule in Step 2 of the previous proof, we have:

$$\vec{r}''(t) = \underbrace{\left(\frac{d}{dt}||\vec{r}'||\right)}_{a_T}\vec{T}(t) + \underbrace{||\vec{r}'|| \cdot ||\vec{T}'||}_{a_N}\vec{N}(t)$$

So we can write:

$$\vec{r}''(t) = a_T\vec{T}(t) + a_N\vec{N}(t)$$

Step 2: Derive the formula for a_T. To isolate a_T, dot both sides with $\vec{T}$.

$$\begin{aligned}\vec{r}'' \cdot \vec{T} &= (a_T\vec{T} + a_N\vec{N}) \cdot \vec{T} \\ &= a_T(\vec{T} \cdot \vec{T}) + a_N(\vec{N} \cdot \vec{T}) \quad (\text{since } \vec{T} \cdot \vec{T} = 1, \vec{N} \cdot \vec{T} = 0) \\ &= a_T\end{aligned}$$

Now substitute $\vec{T} = \frac{\vec{r}'}{||\vec{r}'||}$:

$$\boxed{a_T = \frac{\vec{r}' \cdot \vec{r}''}{||\vec{r}'||}}$$

Step 3: Derive the formula for a_N. To isolate a_N, cross both sides with $\vec{T}$.

$$\begin{aligned}\vec{r}'' \times \vec{T} &= (a_T\vec{T} + a_N\vec{N}) \times \vec{T} \\ &= a_T(\vec{T} \times \vec{T}) + a_N(\vec{N} \times \vec{T}) \quad (\text{since } \vec{T} \times \vec{T} = \vec{0}) \\ &= a_N(\vec{N} \times \vec{T})\end{aligned}$$

Take the magnitude of both sides. Since $\vec{N}$ and $\vec{T}$ are orthogonal unit vectors, $||\vec{N} \times \vec{T}|| = 1$.

$$||\vec{r}'' \times \vec{T}|| = |a_N| \cdot ||\vec{N} \times \vec{T}|| = a_N \quad (\text{since } a_N \geq 0)$$

Now substitute $\vec{T} = \frac{\vec{r}'}{||\vec{r}'||}$:

$$a_N = \left|\left|\vec{r}'' \times \frac{\vec{r}'}{||\vec{r}'||}\right|\right| = \frac{||\vec{r}'' \times \vec{r}'||}{||\vec{r}'||} = \frac{||\vec{r}' \times \vec{r}''||}{||\vec{r}'||}$$

$$\boxed{a_N = \frac{||\vec{r}' \times \vec{r}''||}{||\vec{r}'||}}$$

3. The TNB Frame & Summary

The TNB Frame

A moving, right-handed coordinate system on the curve.

- $\vec{T}(t)$: Unit Tangent Vector (direction of motion)
- $\vec{N}(t)$: Principal Unit Normal Vector (direction of turning)
- $\vec{B}(t) = \vec{T}(t) \times \vec{N}(t)$: Binormal Vector (orthogonal to the plane of motion)

Summary Physical Meaning

Let $\vec{v} = \vec{r}'$ (velocity), $v = ||\vec{v}||$ (speed), and $\vec{a} = \vec{r}''$ (acceleration).

- The acceleration vector is $\vec{a} = a_T\vec{T} + a_N\vec{N}$.
- $a_T = v' = \frac{\vec{v} \cdot \vec{a}}{v}$: Tangential component, measures the **rate of change of speed**.
- $a_N = \kappa v^2 = \frac{||\vec{v} \times \vec{a}||}{v}$: Normal component, measures the acceleration due to the **change in direction**.

4. Example Calculation

Let $\vec{r}(t) = \langle t, t^2, t^3 \rangle$. Find κ, a_T, and a_N at $t = 1$.

1. Derivatives at $t = 1$:

$$\vec{r}'(t) = \langle 1, 2t, 3t^2 \rangle \implies \vec{r}'(1) = \langle 1, 2, 3 \rangle$$
$$\vec{r}''(t) = \langle 0, 2, 6t \rangle \implies \vec{r}''(1) = \langle 0, 2, 6 \rangle$$

2. Necessary Products and Magnitudes:

- $||\vec{r}'(1)|| = \sqrt{1^2 + 2^2 + 3^2} = \sqrt{14}$
- $\vec{r}'(1) \cdot \vec{r}''(1) = (1)(0) + (2)(2) + (3)(6) = 22$
- $\vec{r}'(1) \times \vec{r}''(1) = \langle 6, -6, 2 \rangle$
- $||\vec{r}'(1) \times \vec{r}''(1)|| = \sqrt{36 + 36 + 4} = \sqrt{76} = 2\sqrt{19}$

3. Final Computations:

- $\kappa(1) = \frac{||\vec{r}' \times \vec{r}''||}{||\vec{r}'||^3} = \frac{2\sqrt{19}}{(\sqrt{14})^3} = \frac{2\sqrt{19}}{14\sqrt{14}} = \frac{\sqrt{19}}{7\sqrt{14}}$
- $a_T(1) = \frac{\vec{r}' \cdot \vec{r}''}{||\vec{r}'||} = \frac{22}{\sqrt{14}}$
- $a_N(1) = \frac{||\vec{r}' \times \vec{r}''||}{||\vec{r}'||} = \frac{2\sqrt{19}}{\sqrt{14}}$

5. Second Example: Analysis of a Helix

Example: General Formulas for a Helix

Let $\vec{r}(t) = \langle 2\cos(t), 2\sin(t), 3t \rangle$. Find the curvature $\kappa(t)$ and the tangential and normal components of acceleration, $a_T(t)$ and $a_N(t)$, as functions of t.

Solution: 1. Find the first and second derivatives:

$$\vec{r}'(t) = \langle -2\sin(t), 2\cos(t), 3 \rangle$$
$$\vec{r}''(t) = \langle -2\cos(t), -2\sin(t), 0 \rangle$$

2. Calculate necessary products and magnitudes:

- $||\vec{r}'(t)|| = \sqrt{(-2\sin t)^2 + (2\cos t)^2 + 3^2} = \sqrt{4\sin^2 t + 4\cos^2 t + 9} = \sqrt{4 + 9} = \sqrt{13}$
- $\vec{r}'(t) \cdot \vec{r}''(t) = (-2\sin t)(-2\cos t) + (2\cos t)(-2\sin t) + (3)(0) = 4\sin t\cos t - 4\sin t\cos t = 0$
- $\vec{r}'(t) \times \vec{r}''(t) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -2\sin t & 2\cos t & 3 \\ -2\cos t & -2\sin t & 0 \end{vmatrix} = \langle 6\sin t, -6\cos t, 4\sin^2 t + 4\cos^2 t \rangle = \langle 6\sin t, -6\cos t, 4 \rangle$
- $||\vec{r}'(t) \times \vec{r}''(t)|| = \sqrt{(6\sin t)^2 + (-6\cos t)^2 + 4^2} = \sqrt{36\sin^2 t + 36\cos^2 t + 16} = \sqrt{36 + 16} = \sqrt{52}$

3. Final Computations:

- $\kappa(t) = \frac{||\vec{r}' \times \vec{r}''||}{||\vec{r}'||^3} = \frac{\sqrt{52}}{(\sqrt{13})^3} = \frac{2\sqrt{13}}{13\sqrt{13}} = \frac{2}{13}$
- $a_T(t) = \frac{\vec{r}' \cdot \vec{r}''}{||\vec{r}'||} = \frac{0}{\sqrt{13}} = 0$
- $a_N(t) = \frac{||\vec{r}' \times \vec{r}''||}{||\vec{r}'||} = \frac{\sqrt{52}}{\sqrt{13}} = \sqrt{\frac{52}{13}} = \sqrt{4} = 2$

Interpretation: For this helix, the curvature is constant everywhere. The tangential acceleration is zero, which makes sense because the speed $||\vec{r}'(t)|| = \sqrt{13}$ is constant. The entire acceleration is directed towards turning the path.‘‘‘

Classwork: Curvature and Acceleration Components

Name: _____

Problem: A Comprehensive Calculation

Consider the path of a particle given by the vector function $\vec{r}(t) = \langle t^2, 2t, \ln(t) \rangle$. We will analyze its motion at the point where $t = 1$.

(a) Find the tangential component of acceleration, $a_T(1)$, using the formula $a_T = \frac{\vec{r}' \cdot \vec{r}''}{||\vec{r}'||}$.

(b) Find the normal component of acceleration, $a_N(1)$, using the formula $a_N = \frac{||\vec{r}' \times \vec{r}''||}{||\vec{r}'||}$.

(c) Based on your values for a_T and a_N, is the particle speeding up or slowing down at $t = 1$? Is its path curving at this point? Explain.

September 26: Functions of Several Variables and Contour Maps

Calculus III: Functions of Several Variables and Contour Maps

Lecture Notes: Visualizing Surfaces

In single-variable calculus, we studied functions of the form $y = f(x)$. The input was a single number, x, and the output was a single number, y. The graph of such a function is a **curve** in the 2D plane.

Now, we extend this idea to functions of two variables, written as $z = f(x, y)$.

- The **input** is an ordered pair (x, y), which represents a point in the xy-plane.
- The **output** is a single number, z.

The graph of a function of two variables is a **surface** in 3D space.

A Physical Analogy: The Heated Plate Imagine a thin metal plate lying on the xy-plane. If you heat the center of it with a torch, the temperature is not uniform. The temperature at any point depends on its (x, y) coordinates. We can describe this with a function $T(x, y)$, where the output T is the temperature at the point (x, y). The graph of this function, $z = T(x, y)$, would be a surface where the "height" z represents the temperature.

Learning Objectives

By the end of this lecture, you will be able to:

1. Define a function of two variables and understand its graph as a surface.
2. Define level curves and explain their relationship to the 3D surface.
3. Interpret a contour map to understand the shape of a surface (steepness, peaks, valleys).
4. Create a simple contour map from a given function $z = f(x, y)$.

1. The Challenge of 3D and the 2D Solution

Drawing surfaces in 3D is difficult and often doesn't fully capture the nature of the function. To simplify this, we can borrow an idea from map-making: topographical maps. A topographical map represents a 3D mountain on a 2D piece of paper by showing curves of constant elevation. We will do the same for our surfaces.

A **level curve** is the set of all points (x, y) in the domain where the function $f(x, y)$ has a constant value. That is, a level curve is a curve in the xy-plane defined by the equation:

$$f(x, y) = k$$

where k is a constant (the "height" or "level").

A **contour map** is a collection of level curves drawn for several different values of k.

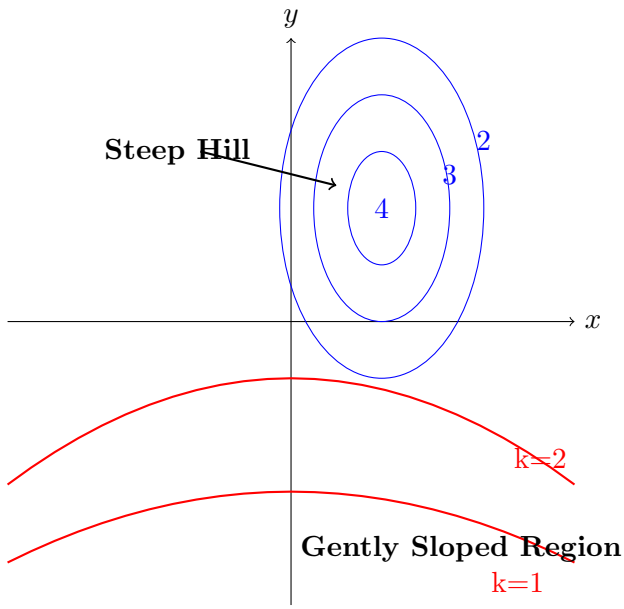
Back to the Heated Plate: If we draw curves on our metal plate connecting all points that have the same temperature, we are drawing level curves. For example, the curve defined by $T(x, y) = 100°$ shows all points with a temperature of exactly 100°. A collection of these curves for, say, $T = 50°, 75°, 100°, 125°$ would form a contour map. These specific level curves are often called *isotherms* (lines of constant temperature).

2. How to Read a Contour Map

A contour map encodes the 3D shape of the surface in a 2D plot. Here are the key rules for interpretation:

Interpreting Contours

- **Spacing of Curves $\implies$ Steepness:**
  - **Closely spaced** contour lines indicate a **steep** region on the surface. The function value is changing rapidly.
  - **Widely spaced** contour lines indicate a **flatter** or gently sloped region. The function value is changing slowly.
- **Shape of Curves $\implies$ Features:**
  - A set of **closed, concentric curves** usually indicates a **hill (local maximum)** or a **basin (local minimum)**. You must check the values of k to know which one it is. If the values increase as you move inward, it's a hill. If they decrease, it's a basin.


y
x
Steep Hill
2
3
4
k=2
Gently Sloped Region
k=1

3. Example: The Paraboloid

Let's create a contour map for the function $z = f(x, y) = x^2 + y^2$. The graph of this function is a 3D surface called a paraboloid.

Step 1: Set $f(x, y) = k$.
The equation for our level curves is $x^2 + y^2 = k$.

Step 2: Analyze the equation.
We recognize $x^2 + y^2 = k$ as the equation of a circle centered at the origin $(0, 0)$ with radius $r = \sqrt{k}$. This tells us two things:

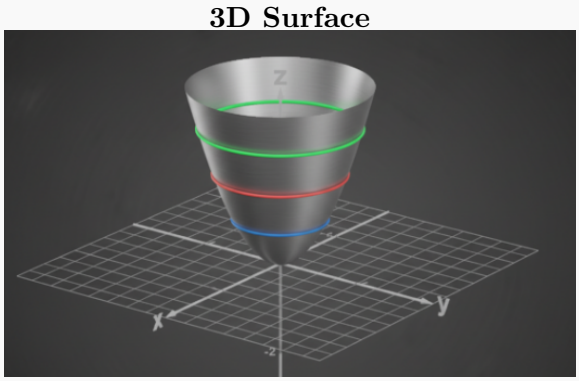
- The level curves are concentric circles.
- We must have $k \geq 0$, since the sum of squares cannot be negative.

Step 3: Choose values for k and sketch.
Let's pick some convenient values for k (perfect squares make the radius easy to calculate).

- If $k = 0$: $x^2 + y^2 = 0 \implies$ The origin $(0, 0)$.
- If $k = 1$: $x^2 + y^2 = 1 \implies$ A circle of radius 1.
- If $k = 4$: $x^2 + y^2 = 4 \implies$ A circle of radius 2.
- If $k = 9$: $x^2 + y^2 = 9 \implies$ A circle of radius 3.

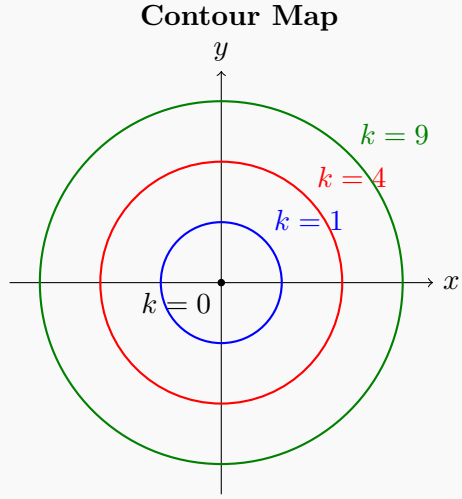
Surface vs. Contour Map for $z = x^2 + y^2$

3D Surface



The graph is a bowl-shaped surface opening upwards.

Contour Map



The level curves are circles.

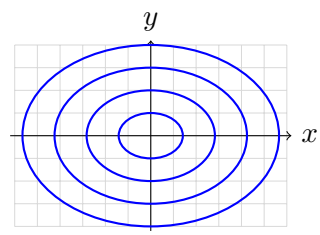
Classwork

Name: _____

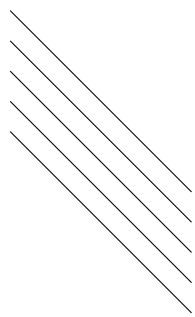
Instructions: For each equation in Column A, find the corresponding graph of its level curves in Column B. Draw a

Column A: Equations **Column B: Level Curve Graphs**

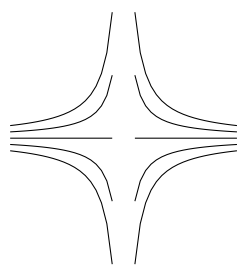
1. $f(x, y) = xy$



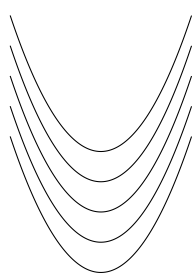
2. $f(x, y) = y - 0.5x^2$



3. $f(x, y) = \frac{x^2}{2} + y^2$



4. $f(x, y) = x + y$



September 29: Tangent Planes and Partial Derivatives

Calculus III: Tangent Planes and Partial Derivatives

Lecture Notes: The Geometry of Derivatives in 3D

In single-variable calculus, the derivative $f'(a)$ gave us the slope of the tangent line to the curve $y = f(x)$ at the point $(a, f(a))$. This tangent line is the best linear approximation of the function near that point.

Our goal today is to extend this idea to surfaces. For a surface given by $z = f(x, y)$, what is the equivalent of a tangent line? It's a **tangent plane**. This plane will be the best linear approximation

of the surface near a point. To find this plane, we first need to answer a more fundamental question: how do we even define "slope" on a surface? The slope depends on the direction you are moving!

Learning Objectives

By the end of this lecture, you will be able to:

1. Understand the geometric meaning of a partial derivative in terms of slopes of tangent lines on a surface.
2. Compute the partial derivatives f_x and f_y for a function of two variables.
3. Derive the equation of the plane tangent to a surface $z = f(x, y)$ at a given point.
4. Find the linearization $L(x, y)$ of a function at a point.

1. Finding the Tangent Plane by Slicing the Surface

To define a plane, we need a point and a normal vector. The point is easy: we want the plane to touch the surface at a point $P_0(a, b, f(a, b))$. To find the normal vector, we will first find two vectors that lie *within* the tangent plane.

The strategy is to slice the surface with planes parallel to the coordinate planes, creating curves. The tangent vectors to these curves at our point P_0 must lie in the tangent plane.

Slice 1: Hold x constant at $x = a$.
Imagine slicing our surface with the vertical plane $x = a$. This creates a curve, C_1, on the surface. We can describe this curve using a vector-valued function where the parameter is y:

$$\vec{r}_1(y) = \langle a, y, f(a, y)\rangle$$

The tangent vector to this curve is its derivative with respect to y.

$$\vec{r}_1'(y) = \left\langle 0, 1, \frac{d}{dy} f(a, y)\right\rangle$$

The derivative $\frac{d}{dy} f(a, y)$ represents the rate of change of f when x is held fixed. We call this the **partial derivative of f with respect to** y, denoted $f_y(a, y)$. At our specific point where $y = b$, the tangent vector is:

$$\vec{v}_1 = \vec{r}_1'(b) = \langle 0, 1, f_y(a, b)\rangle$$

Slice 2: Hold y constant at $y = b$.
Similarly, slicing the surface with the plane $y = b$ creates another curve, C_2, parameterized by x:

$$\vec{r}_2(x) = \langle x, b, f(x, b)\rangle$$

Its tangent vector involves the **partial derivative of f with respect to** x, $f_x(x, b)$. At our point where $x = a$, this second tangent vector is:

$$\vec{v}_2 = \vec{r}_2'(a) = \langle 1, 0, f_x(a, b)\rangle$$

$$xyz^2 - (y-1)^2 + 2.5 \quad P_0 \quad 3[thick, blue, domain = 0:2, samplesy = 0$$

Definition: Partial Derivatives

The **partial derivative of f with respect to** x at (x, y) is:

$$f_x(x, y) = \frac{\partial f}{\partial x} = \lim_{h \to 0} \frac{f(x + h, y) - f(x, y)}{h}$$

Working Rule: To find f_x, treat the variable y as if it were a constant and differentiate $f(x, y)$ with respect to x as usual.

The **partial derivative of f with respect to** y at (x, y) is:

$$f_y(x, y) = \frac{\partial f}{\partial y} = \lim_{h \to 0} \frac{f(x, y + h) - f(x, y)}{h}$$

Working Rule: To find f_y, treat the variable x as if it were a constant and differentiate $f(x, y)$ with respect to y as usual.

Constructing the Plane

Since both vectors $\vec{v}_1$ and $\vec{v}_2$ lie in the tangent plane, their cross product will give us a normal vector $\vec{n}$ to the plane.

$$\vec{n} = \vec{v}_2 \times \vec{v}_1 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & f_x(a, b) \\ 0 & 1 & f_y(a, b) \end{vmatrix}$$

$$\vec{n} = \vec{i}(0 - f_x(a, b)) - \vec{j}(f_y(a, b) - 0) + \vec{k}(1 - 0)$$

$$\vec{n} = \langle -f_x(a, b), -f_y(a, b), 1 \rangle$$

Now we use the standard equation of a plane with point $P_0(a, b, f(a, b))$ and our normal vector $\vec{n}$.

$$-f_x(a, b)(x - a) - f_y(a, b)(y - b) + 1(z - f(a, b)) = 0$$

Solving for z gives us the standard equation for the tangent plane.

The Tangent Plane and Linearization

The equation of the **tangent plane** to the surface $z = f(x, y)$ at the point (a, b) is:

$$z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

This plane is the graph of a linear function that best approximates $f(x, y)$ for points near (a, b). We call this function the **linearization** of f at (a, b):

$$L(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

2. Example Calculation

Example: Finding the Tangent Plane

Find the equation of the plane tangent to the surface $f(x, y) = \sin(x)\cos(y)$ at the point $(a, b) = (\pi/2, 0)$.

Step 1: Find the point of tangency, $P_0(a, b, f(a, b))$.

$$a = \pi/2, \quad b = 0$$

$$f(\pi/2, 0) = \sin(\pi/2)\cos(0) = (1)(1) = 1$$

So the point is $P_0(\pi/2, 0, 1)$.

Step 2: Find the partial derivatives. To find f_x, treat y (and thus $\cos(y)$) as a constant:

$$f_x(x, y) = \frac{\partial}{\partial x}(\sin(x)\cos(y)) = \cos(x)\cos(y)$$

To find f_y, treat x (and thus $\sin(x)$) as a constant:

$$f_y(x, y) = \frac{\partial}{\partial y}(\sin(x)\cos(y)) = \sin(x)(-\sin(y)) = -\sin(x)\sin(y)$$

Step 3: Evaluate the partial derivatives at the point $(\pi/2, 0)$.

$$f_x(\pi/2, 0) = \cos(\pi/2)\cos(0) = (0)(1) = 0$$

$$f_y(\pi/2, 0) = -\sin(\pi/2)\sin(0) = -(1)(0) = 0$$

The "slopes" in both the x and y directions are zero at this point. This means the tangent plane is horizontal.

Step 4: Assemble the tangent plane equation. Using the formula $z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$:

$$z = 1 + (0)(x - \pi/2) + (0)(y - 0)$$

Step 5: Simplify.

$$z = 1$$

The equation of the tangent plane at this point is simply $z = 1$. This makes sense, as the point $(\pi/2, 0)$ is a local maximum (a peak) of the wavy surface, and the plane touching the top of a smooth peak is horizontal. You can graph both $f(x, y) = \sin(x)\cos(y)$ and the plane $z = 1$ in Desmos or another 3D plotter to see this!

3. Example: Finding the Linearization

Example: Finding the Linearization of a Paraboloid

Find the linearization, $L(x, y)$, of the function $f(x, y) = x^2 + y^2$ at the point $(a, b) = (1, 2)$.

Step 1: Find the function value $f(a, b)$.

$$f(1, 2) = 1^2 + 2^2 = 1 + 4 = 5$$

Our point of tangency on the surface is $(1, 2, 5)$.

Step 2: Find the partial derivatives. To find f_x, treat y as a constant:

$$f_x(x, y) = 2x$$

To find f_y, treat x as a constant:

$$f_y(x, y) = 2y$$

Step 3: Evaluate the partial derivatives at the point $(1, 2)$**.**

$$f_x(1, 2) = 2(1) = 2$$

$$f_y(1, 2) = 2(2) = 4$$

Step 4: Assemble the linearization formula. The formula is $L(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$.

$$L(x, y) = 5 + 2(x - 1) + 4(y - 2)$$

Step 5: Simplify the expression.

$$L(x, y) = 5 + 2x - 2 + 4y - 8$$

$$L(x, y) = 2x + 4y - 5$$

This is the linearization of $f(x, y)$ near the point $(1, 2)$. The equation of the tangent plane to the surface at this point is simply $z = 2x + 4y - 5$.

Homework 5

Homework 5 - Calculus of Vector-Valued Functions

Work solutions on blank paper, showing all steps. Scan and submit by the due date on Blackboard.

Problem 1: Derivatives and Tangent Vectors

Consider the vector-valued function $\vec{r}(t) = \langle t^3 - t, \sqrt{2t + 1}, \ln(t) \rangle$.

(a) Find the derivative of the vector function, $\vec{r}'(t)$.

(b) Find the tangent vector at the point where $t = 1$.

Problem 2: Finding the Tangent Line

Find the parametric equations for the tangent line to the curve with the given vector equation at the specified point.

$$\vec{r}(t) = \langle e^{-t}, t^2 + 1, \sin(2t) \rangle; \quad t = 0$$

(a) Find the point on the curve corresponding to $t = 0$.

(b) Find the tangent vector (direction vector) at $t = 0$.

(c) Write the final parametric equations for the tangent line.

Problem 3: Finding the Unit Tangent Vector

Consider the curve $\vec{r}(t) = \langle 2t, t^2, \frac{1}{3}t^3 \rangle$.

(a) Find the velocity vector, $\vec{r}'(t)$.

(b) Find the speed of the particle, $||\vec{r}'(t)||$. (Hint: The expression under the square root should be a perfect square).

(c) Find the unit tangent vector, $\vec{T}(t)$.

Problem 4: Finding the Unit Tangent Vector at a Point

Find the unit tangent vector $\vec{T}(t)$ for the curve $\vec{r}(t) = \langle t, 2\cos(t), 2\sin(t)\rangle$ at the point where $t = \pi/3$.

Problem 5: Arc Length

Find the length of the curve given by the vector function $\vec{r}(t) = \langle 2t, \frac{4}{3}t^{3/2}, \frac{1}{2}t^2 \rangle$ from $t = 1$ to $t = 2$.

(a) Find the derivative vector $\vec{r}'(t)$.

(b) Find the magnitude of the derivative (the speed), $||\vec{r}'(t)||$. (Hint: Again, look for a perfect square).

(c) Set up and evaluate the definite integral for the arc length.

Problem 6: Velocity and Acceleration

A particle's position is given by the vector function $\vec{r}(t) = \langle t\sin(t), t\cos(t), t^2 \rangle$.

(a) Find the velocity vector $\vec{v}(t)$.

(b) Find the acceleration vector $\vec{a}(t)$.

Problem 7: Finding the Principal Unit Normal Vector

Consider the circular path $\vec{r}(t) = \langle 5\cos(t), 5\sin(t), 4 \rangle$.

(a) Find the unit tangent vector $\vec{T}(t)$.

(b) Find the derivative of the unit tangent vector, $\vec{T}'(t)$.

(c) Find the principal unit normal vector $\vec{N}(t)$.

Problem 8: Finding $\vec{N}(t)$ at a Point

Consider the curve $\vec{r}(t) = \langle t^2, \sin(t) - t\cos(t), \cos(t) + t\sin(t) \rangle$. Find the principal unit normal vector $\vec{N}(t)$ at the point where $t = \pi$.

Problem 9: Orthogonality of $\vec{T}$ and $\vec{N}$

For the circular path in Problem 7, $\vec{r}(t) = \langle 5\cos(t), 5\sin(t), 4\rangle$, use your computed vectors $\vec{T}(t)$ and $\vec{N}(t)$ to show explicitly that they are orthogonal for all values of t by computing their dot product, $\vec{T}(t) \cdot \vec{N}(t)$.

Problem 10: Curvature

For the helix $\vec{r}(t) = \langle \cos(t), \sin(t), t\rangle$ calculate the curvature $\kappa(t)$.

Problem 11: Arc Length of a Spiral

Find the length of the curve $\vec{r}(t) = \langle e^t\cos(t), e^t\sin(t), e^t\rangle$ from $t = 0$ to $t = 1$.

Problem 12: Integration and Projectile Motion

A projectile is fired from a height of 100 meters with an initial velocity vector of $\vec{v}(0) = \langle 10, 20, 30\rangle$ m/s. The only force acting on it is gravity, so its acceleration is given by $\vec{a}(t) = \langle 0, 0, -9.8\rangle$ m/s^2.

(a) Find the velocity vector of the projectile, $\vec{v}(t)$, at any time t.

(b) Find the position vector of the projectile, $\vec{r}(t)$, at any time t. (Don't forget the initial position is $\langle 0, 0, 100\rangle$).

Problem 13: A Complete Calculation

Find both the unit tangent vector $\vec{T}(t)$ and the principal unit normal vector $\vec{N}(t)$ for the curve $\vec{r}(t) = \langle 3, 3t, t^2\rangle$.

Problem 14: Curvature with the Cross Product Formula

Find $\kappa(t)$ the curvature of the curve $\vec{r}(t) = \langle e^t, e^{-t}, \sqrt{2}t\rangle$.

Problem 15: The TNB Frame

For the helix $\vec{r}(t) = \langle \cos(t), \sin(t), t\rangle$.

(a) Find the binormal vector $\vec{B}(t) = \vec{T}(t) \times \vec{N}(t)$.

(b) Evaluate all three vectors, $\vec{T}$, $\vec{N}$, and $\vec{B}$, at the point $t = \pi/2$.

Problem 16: Components of Acceleration

Find the tangential and normal components of acceleration, a_T and a_N, for the curve $\vec{r}(t) = \langle 2t, t^2, \frac{1}{3}t^3\rangle$.

Problem 17: Motion with Constant Speed

Consider the motion of a particle along the helix $\vec{r}(t) = \langle 3\cos(t), 3\sin(t), 4t\rangle$. This problem explores the relationship between constant speed and acceleration.

(a) Find the velocity vector $\vec{v}(t)$ and show that the speed, $||\vec{v}(t)||$, is a constant value.

(b) Find the acceleration vector $\vec{a}(t)$.

(c) Explicitly calculate the tangential component of acceleration, a_T, using the formula $a_T = \frac{\vec{v}(t)\cdot\vec{a}(t)}{||\vec{v}(t)||}$.

Problem 18: Acceleration Components at a Point

Find the tangential and normal components of acceleration, a_T and a_N, for the motion of a particle with position vector $\vec{r}(t) = \langle \cos(t) + t\sin(t), \sin(t) - t\cos(t), t^2\rangle$ at the point $t = \pi$.

Problem 19: Finding the Binormal Vector

Find the binormal vector $\vec{B}(t)$ for the curve $\vec{r}(t) = \langle 1, t, t^2\rangle$ from homework problem 13. You will first need to find $\vec{T}(t)$ and $\vec{N}(t)$ if you have not already.

Problem 20: Comprehensive Analysis

A particle's position is given by $\vec{r}(t) = \langle e^t\cos(t), e^t\sin(t), 2\rangle$.

(a) Find the curvature $\kappa(t)$.

(b) Find the tangential component of acceleration, $a_T(t)$.

(c) Find the normal component of acceleration, $a_N(t)$.

Quiz 5

Quiz 5

Name: _____

Problem 1: Partial Derivatives and the Gradient

Consider the function $f(x, y) = x^4y^2 - 3x$.

(a) Find the gradient of f.

(b) Find the second-order partial derivative f_{xy}.

Problem 2: Steepest Ascent and Descent

Let $g(x, y) = 20 - x^2 - 2y^2$. You are at the point $P(2, 1)$.

(a) Find a vector pointing in the direction of steepest ascent.

(b) What is the rate of change in the direction of steepest **descent**?

Problem 3: Tangent Planes

Find the equation of the plane tangent to the surface $f(x, y) = x^3 - 4y^2$ at the point $(2, 1)$.

Problem 4: Linearization

Consider the function $f(x, y) = \sqrt{2x + y}$ at the point $(4, 1)$.

(a) Find the linearization $L(x, y)$ of the function at the point $(4, 1)$.

(b) Use your linearization to approximate the value of $f(4.1, 0.9)$.

CHAPTER 6

Cycle 6: Functions of Several Variables and the Gradient (Homework 6)

Overview

This cycle introduces functions of several variables, their graphs and contour maps, and partial derivatives. The gradient vector appears as the collection of first-order rates of change and as the direction of steepest ascent.

Lecture Notes and Classwork

October 6: Partial Derivatives and the Gradient Vector

Calculus III: Partial Derivatives and the Gradient Vector

Lecture Notes: Formalizing Derivatives and Introducing the Gradient

In our last lecture, we developed the idea of a tangent plane to a surface $z = f(x, y)$. The key insight was to find the "slope" of the surface by moving in two specific directions: parallel to the x-axis and parallel to the y-axis. These slopes were given by the partial derivatives, f_x and f_y.

Today, we will formalize the definition of the partial derivative, moving from a purely geometric idea to a rigorous limit-based definition, just as we did for single-variable derivatives. We will also explore higher-order partial derivatives and introduce a new, powerful object called the **gradient vector**, which packages all the first-order partial derivative information into a single vector.

Learning Objectives

By the end of this lecture, you will be able to:

1. **Define** the partial derivative of a multivariable function using the formal limit definition.
2. **Apply** differentiation rules to compute first and second-order partial derivatives for functions of two and three variables.
3. **Analyze** the meaning of mixed partial derivatives and apply Clairaut's Theorem.

4. **Formulate** the gradient vector for a given multivariable function.

1. Partial Derivatives from First Principles

Our geometric intuition of slicing a surface is formalized by the limit definition of a derivative, where we hold one variable constant and find the rate of change with respect to the other.

Definition: Partial Derivatives

The **partial derivative of f with respect to** x at (x, y) is:

$$f_x(x, y) = \frac{\partial f}{\partial x} = \lim_{h \to 0} \frac{f(x + h, y) - f(x, y)}{h}$$

Working Rule: To find f_x, treat the variable y as if it were a constant and differentiate $f(x, y)$ with respect to x as usual.

The **partial derivative of f with respect to** y at (x, y) is:

$$f_y(x, y) = \frac{\partial f}{\partial y} = \lim_{h \to 0} \frac{f(x, y + h) - f(x, y)}{h}$$

Working Rule: To find f_y, treat the variable x as if it were a constant and differentiate $f(x, y)$ with respect to y as usual.

Example 1: Calculating First-Order Partial Derivatives

Find f_x and f_y for the function $f(x, y) = x^2 \sin(xy)$.

Step 1: Find $f_x(x, y)$. Treat y as a constant. Notice that x appears in two places, so we must use the Product Rule: $d/dx(uv) = u'v + uv'$. Let $u = x^2$ and $v = \sin(xy)$.

$$\frac{\partial}{\partial x}(x^2) = 2x$$

$$\frac{\partial}{\partial x}(\sin(xy)) = \cos(xy) \cdot \frac{\partial}{\partial x}(xy) = \cos(xy) \cdot y$$

Applying the product rule:

$$f_x(x, y) = (2x)(\sin(xy)) + (x^2)(y \cos(xy)) = 2x \sin(xy) + x^2 y \cos(xy)$$

Step 2: Find $f_y(x, y)$. Treat x as a constant. The term x^2 is now just a constant multiplier. We only need to differentiate $\sin(xy)$ with respect to y using the Chain Rule.

$$f_y(x, y) = x^2 \cdot \frac{\partial}{\partial y}(\sin(xy))$$

$$\frac{\partial}{\partial y}(\sin(xy)) = \cos(xy) \cdot \frac{\partial}{\partial y}(xy) = \cos(xy) \cdot x$$

So,

$$f_y(x, y) = x^2(x \cos(xy)) = x^3 \cos(xy)$$

2. Higher-Order Partial Derivatives

Since f_x and f_y are themselves functions of x and y, we can differentiate them again to find second-order partial derivatives. This gives us information about the concavity of the surface.

Second-Order Partial Derivatives

- **Concavity in x-direction:** $f_{xx} = \frac{\partial}{\partial x}(f_x) = \frac{\partial^2 f}{\partial x^2}$
- **Concavity in y-direction:** $f_{yy} = \frac{\partial}{\partial y}(f_y) = \frac{\partial^2 f}{\partial y^2}$
- **Mixed partial (x then y):** $f_{xy} = \frac{\partial}{\partial y}(f_x) = \frac{\partial^2 f}{\partial y \partial x}$
- **Mixed partial (y then x):** $f_{yx} = \frac{\partial}{\partial x}(f_y) = \frac{\partial^2 f}{\partial x \partial y}$

The derivatives f_{xy} and f_{yx} are called the **mixed partial derivatives** and measure how the slope in one direction changes as you move in the perpendicular direction (a sense of "twist" in the surface).

Example 2: Calculating Second-Order Partial Derivatives

Let $f(x, y) = e^{2x}\cos(y)$. Find all four second-order partial derivatives.

Step 1: Find the first-order partials (if not already known).

$$f_x = \frac{\partial}{\partial x}(e^{2x}\cos(y)) = 2e^{2x}\cos(y)$$

$$f_y = \frac{\partial}{\partial y}(e^{2x}\cos(y)) = -e^{2x}\sin(y)$$

Step 2: Calculate f_{xx} and f_{yy}.

$$f_{xx} = \frac{\partial}{\partial x}(f_x) = \frac{\partial}{\partial x}(2e^{2x}\cos(y)) = 4e^{2x}\cos(y)$$

$$f_{yy} = \frac{\partial}{\partial y}(f_y) = \frac{\partial}{\partial y}(-e^{2x}\sin(y)) = -e^{2x}\cos(y)$$

Step 3: Calculate the mixed partials f_{xy} and f_{yx}.

$$f_{xy} = \frac{\partial}{\partial y}(f_x) = \frac{\partial}{\partial y}(2e^{2x}\cos(y)) = -2e^{2x}\sin(y)$$

$$f_{yx} = \frac{\partial}{\partial x}(f_y) = \frac{\partial}{\partial x}(-e^{2x}\sin(y)) = -2e^{2x}\sin(y)$$

In the example above, notice that $f_{xy} = f_{yx}$. This is not a coincidence! For most well-behaved functions, the order of differentiation does not matter.

Clairaut's Theorem

If the mixed partial derivatives f_{xy} and f_{yx} are continuous on an open disk containing the point (a, b), then

$$f_{xy}(a, b) = f_{yx}(a, b)$$

For nearly all functions we encounter in this course (polynomials, sines, cosines, exponentials, rational functions away from zero in the denominator), this theorem holds.

3. The Gradient Vector

We now introduce a new object that conveniently packages the first-order partial derivatives of a function into a single vector. This vector will become central to our study of multivariable calculus.

The Gradient Vector (2 Variables)

The **gradient** of a function $f(x, y)$ is the vector-valued function ∇f, defined as:

$$\nabla f(x, y) = \langle f_x(x, y), f_y(x, y) \rangle = \frac{\partial f}{\partial x}\vec{i} + \frac{\partial f}{\partial y}\vec{j}$$

The symbol ∇ is called "del" or "nabla."

Example 3: Computing the 2D Gradient

Find the gradient of $f(x, y) = x^2 - y^2$ and evaluate it at the point $(2, 1)$.

Step 1: Find the partial derivatives.

$$f_x(x, y) = \frac{\partial}{\partial x}(x^2 - y^2) = 2x$$

$$f_y(x, y) = \frac{\partial}{\partial y}(x^2 - y^2) = -2y$$

Step 2: Assemble the gradient vector. The gradient function is:

$$\nabla f(x, y) = \langle f_x, f_y \rangle = \langle 2x, -2y \rangle$$

Step 3: Evaluate the gradient at the specific point $(2, 1)$**.**

$$\nabla f(2, 1) = \langle 2(2), -2(1) \rangle = \langle 4, -1 \rangle$$

The concept extends naturally to functions of three or more variables.

The Gradient Vector (3 Variables)

The **gradient** of a function $f(x, y, z)$ is defined as:

$$\nabla f(x, y, z) = \langle f_x(x, y, z), f_y(x, y, z), f_z(x, y, z) \rangle$$

Example 4: Computing the 3D Gradient

Find the gradient of $f(x, y, z) = x \ln(yz)$.

Step 1: Find the partial derivative with respect to x. Treat y and z as constants, so $\ln(yz)$ is a constant multiplier.

$$f_x(x, y, z) = \ln(yz)$$

Step 2: Find the partial derivative with respect to y. Treat x and z as constants. Use the chain rule for the logarithm.

$$f_y(x, y, z) = x \cdot \frac{1}{yz} \cdot \frac{\partial}{\partial y}(yz) = x \cdot \frac{1}{yz} \cdot z = \frac{xz}{yz} = \frac{x}{y}$$

Step 3: Find the partial derivative with respect to z. Treat x and y as constants. Use the chain rule for the logarithm.

$$f_z(x, y, z) = x \cdot \frac{1}{yz} \cdot \frac{\partial}{\partial z}(yz) = x \cdot \frac{1}{yz} \cdot y = \frac{xy}{yz} = \frac{x}{z}$$

Step 4: Assemble the gradient vector.

$$\nabla f(x, y, z) = \left\langle \ln(yz), \frac{x}{y}, \frac{x}{z} \right\rangle$$

Looking Ahead: The Geometric Meaning of the Gradient

For now, we have focused on the computation of the gradient. In our upcoming lessons, we will give this vector a profound geometric interpretation. The gradient vector ∇f at a point (a, b) is incredibly important because:

1. It points in the direction of the **steepest ascent** on the surface at that point.
2. Its magnitude, $\|\nabla f(a, b)\|$, is the **rate of change** in that steepest direction.
3. It is always **orthogonal (normal) to the level curves** (or level surfaces for 3D functions) that pass through that point.

This makes the gradient one of the most fundamental tools in multivariable calculus, with applications ranging from optimization in machine learning to modeling heat flow and fluid dynamics.

Classwork: Partial Derivatives and Gradients

Name: _____

Instructions: Solve the following problems, showing all of your work. For partial derivatives, be careful to treat the other variables as constants.

Problem 1: First-Order Partial Derivatives

Find the first-order partial derivatives, f_x and f_y, for the function below.

$$f(x, y) = 5x^4y^3 - 3x^2y^2 + 2x - 7y + 1$$

Problem 2: First-Order Partials with Chain Rule

Find the first-order partial derivatives, $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$, for the function below.

$$f(x, y) = e^{xy} + \ln(x + y)$$

Problem 3: Second-Order Partial Derivatives

Find all four second-order partial derivatives (f_{xx}, f_{yy}, f_{xy}, and f_{yx}) for the following function. Confirm that $f_{xy} = f_{yx}$.

$$f(x, y) = x^3 + xy - 2y^3$$

Problem 4: Partials for a Function of Three Variables

Find the specified partial derivatives for the function $g(x, y, z) = z\sin(3xy)$.

(a) $g_x(x, y, z)$

(b) $g_y(x, y, z)$

(c) $g_z(x, y, z)$

Problem 5: Finding the Gradient in 2D

Find the gradient vector, $\nabla f(x, y)$, for the function:

$$f(x, y) = y^3 \cos(x)$$

Problem 6: Evaluating the Gradient in 2D

Find the gradient of the function $f(x, y) = \sqrt{x^2 + y^2}$ and then evaluate it at the point $P(3, 4)$.

Problem 7: Finding the Gradient in 3D

Find the gradient vector, $\nabla g(x, y, z)$, for the function:

$$g(x, y, z) = xy + yz + xz$$

Problem 8: Evaluating the Gradient in 3D

Find the gradient of the function $h(x, y, z) = xe^{yz}$ and then evaluate it at the point $P(2, 0, -1)$.

October 8: The Geometric Meaning of the Gradient (Direction and Rate of Steepest Ascent)

Calculus III: The Geometric Meaning of the Gradient

Lecture Notes: Direction and Rate of Steepest Ascent

In our last class, we learned the mechanics of computing the gradient vector, ∇f. Today, we will discover its profound geometric meaning. We know that on a surface, the "slope" changes depending on the direction you move. Today we will prove that the gradient vector elegantly tells us both the direction of the steepest slope and the value of that slope. To do this, we will start from the fundamental definition of the derivative.

Learning Objectives

By the end of this lecture, you will be able to:

1. **Derive** the formula for the rate of change of a function along a path using the limit definition of a derivative.
2. **Calculate** the direction of steepest ascent and descent for a function at a given point.
3. **Calculate** the maximum and minimum rate of change (steepest slope) of a function at a given point.
4. **Apply** these concepts to solve problems for functions of both two and three variables.

1. Deriving the Rate of Change Along a Path

$z = f(x, y)$ $xyz^2 + (y - 1.5)^2) + 3[domain = 130 : 320, samples = 50, variable$

Imagine a surface $z = f(x, y)$ and a path in the domain $\vec{r}(t) = \langle x(t), y(t) \rangle$. As you walk along this path, your height changes. We want to find the rate of change of this height, $\frac{dz}{dt}$, using only the definition of the derivative.

$$\frac{dz}{dt} = \lim_{h \to 0} \frac{z(t+h) - z(t)}{h} = \lim_{h \to 0} \frac{f(x(t+h), y(t+h)) - f(x(t), y(t))}{h}$$

The key is to add and subtract the term $f(x(t), y(t+h))$ in the numerator. This allows us to isolate the change in x and the change in y separately.

$$\frac{dz}{dt} = \lim_{h \to 0} \frac{1}{h} \left[f(x(t+h), y(t+h)) - f(x(t), y(t+h)) + f(x(t), y(t+h)) - f(x(t), y(t)) \right]$$

Now, split this into two separate limits and rearrange each one to look like a derivative.

$$\begin{aligned}
\frac{dz}{dt} &= \lim_{h \to 0} \left[\frac{f(x(t+h), y(t+h)) - f(x(t), y(t+h))}{h} \right] + \lim_{h \to 0} \left[\frac{f(x(t), y(t+h)) - f(x(t), y(t))}{h} \right] \\
&= \lim_{h \to 0} \left[\frac{f(x(t+h), y(t+h)) - f(x(t), y(t+h))}{x(t+h) - x(t)} \cdot \frac{x(t+h) - x(t)}{h} \right] \\
&\quad + \lim_{h \to 0} \left[\frac{f(x(t), y(t+h)) - f(x(t), y(t))}{y(t+h) - y(t)} \cdot \frac{y(t+h) - y(t)}{h} \right]
\end{aligned}$$

As $h \to 0$, each piece of this expression converges to a derivative we know:

- $\lim_{h \to 0} \frac{f(x(t+h), y(t+h)) - f(x(t), y(t+h))}{x(t+h) - x(t)}$ becomes $\frac{\partial f}{\partial x}$ at (x, y).
- $\lim_{h \to 0} \frac{x(t+h) - x(t)}{h}$ becomes $\frac{dx}{dt}$.
- $\lim_{h \to 0} \frac{f(x(t), y(t+h)) - f(x(t), y(t))}{y(t+h) - y(t)}$ becomes $\frac{\partial f}{\partial y}$ at (x, y).
- $\lim_{h \to 0} \frac{y(t+h) - y(t)}{h}$ becomes $\frac{dy}{dt}$.

Combining these gives a result, which can be expressed perfectly as a dot product:

$$\frac{dz}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} = \langle f_x, f_y \rangle \cdot \langle x', y' \rangle = \nabla f(x, y) \cdot \vec{r}'(t)$$

2. The Meaning of the Gradient

Now we can use our derived formula to understand the gradient.

$$\frac{dz}{dt} = \nabla f \cdot \vec{r}'(t) = \|\nabla f\| \|\vec{r}'(t)\| \cos(\theta)$$

Here, θ is the angle between the gradient vector ∇f and the direction of motion $\vec{r}'(t)$. We want to find the direction that maximizes $\frac{dz}{dt}$. The value of $\cos(\theta)$ is at its maximum of 1 when $\theta = 0$. This means that $\vec{r}'(t)$ must point in the **same direction** as ∇f.

When we talk about the "slope" of the surface, we mean the rate of change per unit of distance traveled. This corresponds to setting our speed $\|\vec{r}'(t)\|$ to 1. In this case, the rate of change is:

$$\frac{dz}{dt} = \|\nabla f\| \cos(\theta)$$

This rate of change is maximized when $\cos(\theta) = 1$, and the maximum value is precisely $\|\nabla f\|$.

The Two Key Properties of the Gradient Vector

1. **Direction of Steepest Ascent:** The vector $\nabla f(a, b)$ points in the direction of the maximum possible rate of increase of the function f at the point (a, b).

2. **Rate of Steepest Ascent:** The magnitude of the gradient, $\|\nabla f(a, b)\|$, is the value of this maximum rate of increase (the "steepest slope").

The direction of steepest **descent** is $-\nabla f$, and its rate is $-\|\nabla f\|$.

3. Examples and Applications

Example 1: Steepest Ascent on a Paraboloid (2D)

A surface is described by the function $f(x, y) = 100 - x^2 - y^2$. You are standing at the point $(6, 8)$.

(a) Find the direction of steepest ascent.

(b) Find the rate of change (slope) in that direction.

Step 1: Find the gradient vector.

$$\nabla f(x, y) = \langle \frac{\partial}{\partial x}(100 - x^2 - y^2), \frac{\partial}{\partial y}(100 - x^2 - y^2) \rangle = \langle -2x, -2y \rangle$$

Step 2: Evaluate the gradient at the point $(6, 8)$**.**

$$\nabla f(6, 8) = \langle -2(6), -2(8) \rangle = \langle -12, -16 \rangle$$

Answer (a): The direction of steepest ascent is $\langle -12, -16 \rangle$.

Step 3: Find the magnitude of this gradient vector.

$$\begin{aligned} \|\nabla f(6, 8)\| &= \sqrt{(-12)^2 + (-16)^2} \\ &= \sqrt{144 + 256} \\ &= \sqrt{400} \\ &= 20 \end{aligned}$$

Answer (b): The maximum rate of increase is 20. The steepest slope is 20.

Example 2: Steepest Descent on a Surface (2D)

Find the direction and rate of steepest **descent** for $f(x, y) = e^x \sin(y)$ at the point $P(0, \pi/4)$.

Step 1: Find the gradient.

$$\nabla f(x, y) = \langle e^x \sin(y), e^x \cos(y) \rangle$$

Step 2: Evaluate the gradient at $P(0, \pi/4)$.

$$\begin{aligned}\nabla f(0, \pi/4) &= \langle e^0 \sin(\pi/4), e^0 \cos(\pi/4)\rangle \\ &= \left\langle (1) \cdot \frac{\sqrt{2}}{2}, (1) \cdot \frac{\sqrt{2}}{2} \right\rangle \\ &= \left\langle \frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} \right\rangle\end{aligned}$$

Step 3: Find the direction and rate of steepest descent. The direction of steepest descent is $-\nabla f(0, \pi/4) = \left\langle -\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2} \right\rangle$. The rate of steepest ascent is:

$$\begin{aligned}\|\nabla f(0, \pi/4)\| &= \sqrt{\left(\frac{\sqrt{2}}{2}\right)^2 + \left(\frac{\sqrt{2}}{2}\right)^2} \\ &= \sqrt{\frac{2}{4} + \frac{2}{4}} = \sqrt{\frac{4}{4}} = \sqrt{1} = 1\end{aligned}$$

The rate of steepest descent is $-\|\nabla f\| = -1$.

4. Extension to Three Variables

The entire argument we made holds perfectly for a function of three variables, $w = f(x, y, z)$. The derivation is identical, just with one extra variable to track. Let a path in 3D be given by $\vec{r}(t) = \langle x(t), y(t), z(t)\rangle$. We want to find $\frac{dw}{dt}$.

We start with the limit definition:

$$\frac{dw}{dt} = \lim_{h \to 0} \frac{f(x(t+h), y(t+h), z(t+h)) - f(x(t), y(t), z(t))}{h}$$

We add and subtract two intermediate terms this time to isolate the change in each variable:

$$\begin{aligned}\frac{dw}{dt} = \lim_{h \to 0} \frac{1}{h}[&f(x(t+h), \dots) - f(x(t), y(t+h), \dots) \\ &+ f(x(t), y(t+h), \dots) - f(x(t), y(t), z(t+h)) \\ &+ f(x(t), y(t), z(t+h)) - f(x(t), y(t), z(t))]\end{aligned}$$

Separating into three limits and multiplying/dividing by the change in each variable (just as before) yields:

$$\frac{dw}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt} + \frac{\partial f}{\partial z}\frac{dz}{dt}$$

This is precisely the dot product in 3D:

$$\frac{dw}{dt} = \langle f_x, f_y, f_z\rangle \cdot \langle x', y', z'\rangle = \nabla f(x, y, z) \cdot \vec{r}'(t)$$

Since the final formula $\frac{dw}{dt} = \nabla f \cdot \vec{r}'(t)$ has the exact same structure, our entire geometric argument applies. The gradient $\nabla f(x, y, z)$ still points in the direction of fastest increase, and its magnitude $\|\nabla f\|$ is that maximum rate of increase.

Example 3: Heat Flow in a Room (3D)

The temperature in a room is given by $T(x, y, z) = 90 - x^2 - 2y^2 - 3z^2$. A fly is at the point $(1, 1, 1)$. In what direction should the fly move to cool off as quickly as possible? What is the rate of temperature change in that direction?

Step 1: Find the gradient of the temperature function.

$$\nabla T(x, y, z) = \langle -2x, -4y, -6z \rangle$$

Step 2: Evaluate the gradient at the fly's position $(1, 1, 1)$**.**

$$\nabla T(1, 1, 1) = \langle -2(1), -4(1), -6(1) \rangle = \langle -2, -4, -6 \rangle$$

This vector gives the direction of the fastest *increase* in temperature. **Step 3: Determine the direction to cool off fastest.** To cool off, the fly must move in the direction of steepest descent, which is $-\nabla T$. Direction to cool off: $-\langle -2, -4, -6 \rangle = \langle 2, 4, 6 \rangle$.

Step 4: Find the rate of change. The maximum rate of *increase* is $\|\nabla T(1, 1, 1)\|$.

$$\begin{aligned} \|\nabla T(1, 1, 1)\| &= \sqrt{(-2)^2 + (-4)^2 + (-6)^2} \\ &= \sqrt{4 + 16 + 36} \\ &= \sqrt{56} = \sqrt{4 \cdot 14} = 2\sqrt{14} \end{aligned}$$

The maximum rate of *decrease* (cooling) is $-\|\nabla T\| = -2\sqrt{14}$. The temperature will drop at a rate of $2\sqrt{14}$ degrees per unit distance.

October 10: The Directional Derivative

Calculus III: The Directional Derivative

Lecture Notes: The Gradient, Directional Derivatives, and Level Sets

In our last lecture, we established that the gradient vector, ∇f, points in the direction of steepest ascent and its magnitude, $\|\nabla f\|$, is the rate of that ascent. Today, we will generalize the idea of a "rate of change" to any arbitrary direction. This new concept, the **directional derivative**, will not only complete our understanding of the gradient but also reveal a beautiful geometric property: the gradient vector is always orthogonal to the level sets of a function.

Learning Objectives

By the end of this lecture, you will be able to:

1. **Define** the directional derivative as a rate of change in an arbitrary direction.
2. **Calculate** the directional derivative using the gradient vector.
3. **Explain** why the gradient vector is orthogonal to level curves and level surfaces.
4. **Apply** this orthogonality to find tangent lines to level curves and tangent planes to level surfaces.

1. Rate of Change in an Arbitrary Direction

The partial derivatives f_x and f_y tell us the rate of change of f in the directions of the standard basis vectors, $\vec{i} = \langle 1, 0 \rangle$ and $\vec{j} = \langle 0, 1 \rangle$. But what if we want to move in a different direction?

Let $\vec{u} = \langle a, b \rangle$ be a **unit vector** that specifies a direction. We want to find the rate of change of $f(x, y)$ at a point $P(x_0, y_0)$ in the direction of $\vec{u}$. We call this the **directional derivative** and denote it by $D_{\vec{u}} f(x_0, y_0)$.

$$xyz^2 - y^2) * 3; 3[domain = -2.5 : 1.5, samples = 50, variable =$$
$$(x_0, y_0)$$

Geometrically, we slice the surface $z = f(x, y)$ with a vertical plane that passes through $(x_0, y_0, 0)$ and points in the direction of $\vec{u}$. The directional derivative is the slope of the tangent line to the resulting curve at the point $(x_0, y_0, f(x_0, y_0))$.

The formal definition uses a limit, just like in single-variable calculus.

$$D_{\vec{u}} f(x, y) = \lim_{h \to 0} \frac{f(x + ha, y + hb) - f(x, y)}{h} \quad \text{where } \vec{u} = \langle a, b \rangle$$

Fortunately, there is a much simpler way to compute this. Recall the multivariable chain rule from our first lecture: if $z = f(x, y)$ and $x = x(t), y = y(t)$, then $\frac{dz}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}$.

Let's define a path $\vec{r}(t) = \langle x_0 + at, y_0 + bt \rangle$. This is a line passing through (x_0, y_0) with direction vector $\vec{u} = \langle a, b \rangle$. The derivative $\frac{dz}{dt}$ along this path is precisely the directional derivative.

$$\frac{dz}{dt} = \nabla f \cdot \vec{r}'(t) = \langle f_x, f_y \rangle \cdot \langle a, b \rangle = \nabla f \cdot \vec{u}$$

This gives us a simple and powerful formula.

Directional Derivative Formula

The directional derivative of f at (x, y) in the direction of the **unit vector** $\vec{u} = \langle a, b \rangle$ is:

$$D_{\vec{u}} f(x, y) = \nabla f(x, y) \cdot \vec{u}$$

The same formula applies for functions of three variables: $D_{\vec{u}} f(x, y, z) = \nabla f(x, y, z) \cdot \vec{u}$.

Important Note: The direction vector $\vec{v}$ must be a **unit vector**. If you are given a direction vector $\vec{v}$ that is not a unit vector, you must normalize it first by setting $\vec{u} = \frac{\vec{v}}{\|\vec{v}\|}$.

2. The Gradient's Relationship to Level Sets

One of the most elegant properties of the gradient is its relationship to the level curves (in 2D) and level surfaces (in 3D) of a function.

2.1 The 2D Case: Orthogonality to Level Curves

A level curve of a function $f(x, y)$ is a curve in the xy-plane defined by the equation $f(x, y) = k$ for some constant k. Imagine the contour map of a mountain; each contour line is a level curve.

Let's consider a point $P(a, b)$ on a level curve $f(x, y) = k$. Let this curve be parameterized by a vector function $\vec{r}(t) = \langle x(t), y(t)\rangle$ such that $\vec{r}(t_0) = \langle a, b\rangle$.

Since the curve lies on the level set, for any value of t, we have:

$$f(x(t), y(t)) = k$$

Now, let's differentiate both sides of this equation with respect to t.

$$\frac{d}{dt}[f(x(t), y(t))] = \frac{d}{dt}[k]$$

Using the chain rule on the left side, we get:

$$\frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt} = 0$$

This is just a dot product!

$$\langle f_x, f_y\rangle \cdot \langle x'(t), y'(t)\rangle = 0 \implies \nabla f(\vec{r}(t)) \cdot \vec{r}'(t) = 0$$

The vector $\vec{r}'(t)$ is the **tangent vector** to the level curve. This equation tells us that the dot product of the gradient vector and the tangent vector is zero. Therefore, they are orthogonal.

2.2 The 3D Case: Orthogonality to Level Surfaces

The exact same logic applies to a function of three variables, $F(x, y, z)$. A level surface is defined by the equation $F(x, y, z) = k$.

Let $\vec{r}(t) = \langle x(t), y(t), z(t)\rangle$ be any smooth curve that lies on the level surface. Then $F(x(t), y(t), z(t)) = k$. Differentiating with respect to t gives:

$$\frac{\partial F}{\partial x}\frac{dx}{dt} + \frac{\partial F}{\partial y}\frac{dy}{dt} + \frac{\partial F}{\partial z}\frac{dz}{dt} = 0$$

$$\nabla F \cdot \vec{r}'(t) = 0$$

Since $\vec{r}'(t)$ is a tangent vector to a curve on the surface, this means that the gradient vector ∇F is orthogonal to every tangent vector at that point. This implies that ∇F is the **normal vector** to the tangent plane of the level surface.

The Gradient as a Normal Vector

1. For a function $f(x, y)$, the gradient vector $\nabla f(a, b)$ is orthogonal to the level curve $f(x, y) = k$ at the point (a, b).
2. For a function $F(x, y, z)$, the gradient vector $\nabla F(a, b, c)$ is orthogonal (normal) to the level surface $F(x, y, z) = k$ at the point (a, b, c).

This is a crucial link between the gradient and geometry. Since the gradient vector is always **normal** (orthogonal) to the level set at any given point, we can use it to define the tangent line or plane at that point.

Tangent Line to a Level Curve (2D)

Let $f(x, y) = k$ be a level curve, and let $P(a, b)$ be a point on this curve. The tangent line to the curve at P is the line that passes through P and is perpendicular to the normal vector.

- **Normal Vector:** The gradient vector $\vec{n} = \nabla f(a, b)$.
- **Vector on the Line:** Let $Q(x, y)$ be any other point on the tangent line. The vector from P to Q, given by $\vec{v} = \langle x - a, y - b\rangle$, lies flat along the tangent line.

Since the normal vector $\vec{n}$ must be orthogonal to the vector $\vec{v}$ on the line, their dot product must be zero.

$$\nabla f(a, b) \cdot \langle x - a, y - b\rangle = 0$$

Expanding this dot product gives the standard point-normal equation for the tangent line:

$$f_x(a, b)(x - a) + f_y(a, b)(y - b) = 0$$

Tangent Plane to a Level Surface (3D)

The logic extends perfectly to three dimensions. Let $F(x, y, z) = k$ be a level surface, and let $P(a, b, c)$ be a point on this surface. The tangent plane to the surface at P is the plane that passes through P and is perpendicular to the normal vector.

- **Normal Vector:** The gradient vector $\vec{n} = \nabla F(a, b, c)$.
- **Vector in the Plane:** Let $Q(x, y, z)$ be any other point on the tangent plane. The vector from P to Q, given by $\vec{v} = \langle x - a, y - b, z - c\rangle$, lies flat within the tangent plane.

Since the normal vector $\vec{n}$ is orthogonal to any vector $\vec{v}$ in the plane, their dot product must be zero.

$$\nabla F(a, b, c) \cdot \langle x - a, y - b, z - c\rangle = 0$$

Expanding this dot product gives the standard point-normal equation for the tangent plane:

$$F_x(a, b, c)(x - a) + F_y(a, b, c)(y - b) + F_z(a, b, c)(z - c) = 0$$

Example 1: Directional Derivative

Let $g(x, y, z) = y\cos(x) + z^2$. Find the directional derivative at $P(\pi/2, 1, 2)$ in the direction of the vector $\vec{v} = \langle 2, -1, 2\rangle$.

Step 1: Find the gradient vector of g.

$$\nabla g(x, y, z) = \langle -y\sin(x), \cos(x), 2z\rangle$$

Step 2: Evaluate the gradient at the point $P(\pi/2, 1, 2)$.

$$\begin{aligned}\nabla g(\pi/2, 1, 2) &= \langle -(1)\sin(\pi/2), \cos(\pi/2), 2(2)\rangle \\ &= \langle -1, 0, 4\rangle\end{aligned}$$

Step 3: Normalize the direction vector $\vec{v}$. The formula requires a unit vector.

$$\|\vec{v}\| = \sqrt{2^2 + (-1)^2 + 2^2} = \sqrt{4+1+4} = \sqrt{9} = 3$$

The unit vector $\vec{u}$ is:

$$\vec{u} = \frac{\vec{v}}{\|\vec{v}\|} = \left\langle \frac{2}{3}, -\frac{1}{3}, \frac{2}{3} \right\rangle$$

Step 4: Compute the dot product $\nabla g \cdot \vec{u}$.

$$\begin{aligned}D_{\vec{u}} g(\pi/2, 1, 2) &= \langle -1, 0, 4\rangle \cdot \left\langle \frac{2}{3}, -\frac{1}{3}, \frac{2}{3} \right\rangle \\ &= (-1)\left(\frac{2}{3}\right) + (0)\left(-\frac{1}{3}\right) + (4)\left(\frac{2}{3}\right) \\ &= -\frac{2}{3} + 0 + \frac{8}{3} = \frac{6}{3} = 2\end{aligned}$$

The directional derivative at the given point and direction is 2.

Example 2: Steepest Ascent and Descent

Consider the function $f(x, y) = x^2y - y^3$. Find the direction and rate of steepest ascent at the point $P(3, 1)$.

Step 1: Find the gradient vector.

$$\nabla f(x, y) = \langle 2xy, x^2 - 3y^2\rangle$$

Step 2: Evaluate the gradient at the point $P(3, 1)$.

$$\begin{aligned}\nabla f(3, 1) &= \langle 2(3)(1), (3)^2 - 3(1)^2\rangle \\ &= \langle 6, 9 - 3\rangle = \langle 6, 6\rangle\end{aligned}$$

Answer (Direction): The direction of steepest ascent is given by the gradient vector $\langle 6, 6\rangle$.

Step 3: Find the magnitude of this gradient vector.

$$\begin{aligned}\|\nabla f(3, 1)\| &= \sqrt{6^2 + 6^2} \\ &= \sqrt{36 + 36} = \sqrt{72} = \sqrt{36 \cdot 2} = 6\sqrt{2}\end{aligned}$$

The rate of steepest ascent is $6\sqrt{2}$. The rate of steepest descent would be $-6\sqrt{2}$.

Example 3: Tangent Line to a Level Curve

Find the equation of the line tangent to the ellipse defined by $f(x, y) = 4x^2 + y^2 = 8$ at the point $(1, 2)$.

Step 1: Find the gradient of $f(x, y)$. The gradient at $(1, 2)$ will serve as the normal vector to our tangent line.

$$\nabla f(x, y) = \langle 8x, 2y \rangle$$

Step 2: Evaluate the gradient at the point $(1, 2)$.

$$\nabla f(1, 2) = \langle 8(1), 2(2) \rangle = \langle 8, 4 \rangle$$

This is the normal vector $\vec{n}$ to the tangent line.

Step 3: Use the point-normal form of a line. The equation for a line with normal vector $\vec{n} = \langle A, B \rangle$ passing through a point (x_0, y_0) is $A(x - x_0) + B(y - y_0) = 0$.

$$8(x - 1) + 4(y - 2) = 0$$

Example 4: Tangent Plane to a Level Surface

Find the equation of the plane tangent to the level surface $F(x, y, z) = \ln(xy) + z = 2$ at the point $P(1, e, 1)$.

Step 1: Find the gradient of $F(x, y, z)$. This will be the normal vector to the tangent plane. Recall that $\frac{d}{dx} \ln(u) = \frac{1}{u} \frac{du}{dx}$.

$$\nabla F(x, y, z) = \left\langle \frac{1}{xy} \cdot y, \frac{1}{xy} \cdot x, 1 \right\rangle = \left\langle \frac{1}{x}, \frac{1}{y}, 1 \right\rangle$$

Step 2: Evaluate the gradient at the point $P(1, e, 1)$.

$$\nabla F(1, e, 1) = \left\langle \frac{1}{1}, \frac{1}{e}, 1 \right\rangle = \langle 1, e^{-1}, 1 \rangle$$

This is our normal vector $\vec{n}$.

Step 3: Use the point-normal form for a plane. The equation for a plane with normal $\vec{n} = \langle A, B, C \rangle$ through (x_0, y_0, z_0) is $A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$.

$$1(x - 1) + \frac{1}{e}(y - e) + 1(z - 1) = 0$$

Classwork: Directional Derivatives and Tangent Planes

Name: _____

Instructions: Work through the following problems. Show your steps clearly.

Problem 1: Directional Derivative

Let $f(x, y, z) = xe^y - z$. Find the directional derivative at $P(0, 0, 1)$ in the direction of $\vec{v} = \langle 1, 2, -2 \rangle$.

Problem 2: Steepest Ascent

Let $f(x, y) = 3x - 4y + 2$.

(a) In what direction does f have its steepest ascent?

(b) What is the rate of change in that direction?

Problem 3: Tangent Line

Consider the function $f(x, y) = x^2 + xy + y^2$. Find the equation of the tangent line to the level curve $f(x, y) = 3$ at the point $(1, 1)$.

Problem 4: More Steepest Ascent/Descent

For $f(x, y) = \ln(x^2 + y^2)$ at the point $(1, 0)$, find:

(a) The direction of steepest descent.

(b) The directional derivative in the direction $\vec{u} = \langle 0, 1 \rangle$.

Problem 5: Tangent Plane

Consider the function

$$F(x, y, z) = xy + z^3.$$

Find the equation of the plane tangent to the level surface $F(x, y, z) = 5$ at the point $(1, 2, \sqrt[3]{3})$.

Homework 6

Homework: Partial Derivatives, Gradients, and Tangent Planes

Work solutions on blank paper, showing all steps. Scan and submit by the due date on Blackboard.

Part 1: Partial Derivatives

Find the requested partial derivatives for each function.

Problem 1: First-Order Partials

Find f_x and f_y for $f(x, y) = 3x^5y^2 - 7x^3y + x - 5y^2 + 9$.

Problem 2: First-Order Partials with Chain Rule

Find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ for $f(x, y) = \cos(x^2y^3)$.

Problem 3: First-Order Partials with Product Rule

Find g_x and g_y for $g(x, y) = y^2e^{xy}$.

Problem 4: Second-Order Partials

Find all four second-order partial derivatives $(f_{xx}, f_{yy}, f_{xy}, f_{yx})$ for $f(x, y) = \ln(x^2 + y)$. Verify that $f_{xy} = f_{yx}$.

Problem 5: Higher-Order Partials

Find the partial derivative f_{yyy} for $f(x, y) = x^3e^{2y}$.

Problem 6: Partials of Three Variables

Find f_x, f_y, and f_z for $f(x, y, z) = \frac{xy}{z}$.

Part 2: Tangent Planes and Linear Approximations

Problem 7: Finding a Tangent Plane

Find the equation of the plane tangent to the surface $z = x^2 + 3y^2$ at the point $(3, -1)$.

Problem 8: Finding a Tangent Plane

Find the equation of the plane tangent to the surface $z = \arctan(y/x)$ at the point $(1, 1)$.

Problem 9: Finding a Linearization

Find the linearization $L(x, y)$ of the function $f(x, y) = \sqrt{x + e^{4y}}$ at the point $(3, 0)$.

Problem 10: Using Linearization to Approximate

Use the linearization you found in Problem 9 to approximate the value of $f(3.05, -0.01)$.

Problem 11: Working Backwards

The equation of the tangent plane to the surface $z = f(x, y)$ at the point $(2, -1)$ is given by $z = 4 - 3(x - 2) + 5(y + 1)$. Determine the values of $f(2, -1)$, $f_x(2, -1)$, and $f_y(2, -1)$.

Part 3: The Gradient and Steepest Ascent/Descent

Problem 12: Finding a Gradient (2D)

Find the gradient of the function $f(x, y) = x^3 - 2xy + 3y^2$.

Problem 13: Finding a Gradient (3D)

Find the gradient of the function $f(x, y, z) = z^2 \ln(x^2 + y^2)$.

Problem 14: Steepest Ascent

Let $f(x, y) = xy^2$. Find the direction of steepest ascent and the maximum rate of change of f at the point $(2, 3)$.

Problem 15: Steepest Descent

The altitude of a mountain is given by $H(x, y) = 2000 - 0.01(x^2 + y^2)$. If a hiker is standing at the point $(10, 10)$, in which direction should they move to descend the most rapidly? What is the slope of their path in that direction?

Problem 16: Steepest Ascent (3D)

The temperature at a point (x, y, z) is given by $T(x, y, z) = 100e^{-x^2-2y^2-3z^2}$. Find the direction of the greatest rate of temperature increase at the point $(1, 1, 1)$.

Problem 17: Rate of Change

For the temperature function in Problem 16, what is the value of the maximum rate of change at the point $(1, 1, 1)$?

Part 4: Linear Approximation in Three Variables

Note on Linear Approximation for $f(x, y, z)$***:*** Just as a tangent plane approximates a function $f(x, y)$ near a point, we can approximate a function of three variables, $w = f(x, y, z)$, near a point $P_0(a, b, c)$. The graph of this function exists in 4D space, so we can't visualize the "tangent hyperplane," but the formula is a direct extension of the 2D case. The **linearization of** f **at** (a, b, c) is:

$$L(x, y, z) = f(a, b, c) + f_x(a, b, c)(x - a) + f_y(a, b, c)(y - b) + f_z(a, b, c)(z - c)$$

We can use this to approximate function values: $f(x, y, z) \approx L(x, y, z)$ for points (x, y, z) close to (a, b, c).

Problem 18: Finding a Linearization (3D)

Find the linearization $L(x, y, z)$ of the function $f(x, y, z) = x\sqrt{y + z}$ at the point $(2, 1, 3)$.

Problem 19: Finding a Linearization (3D)

Find the linearization $L(x, y, z)$ of the function $f(x, y, z) = \sin(x+2y+3z)$ at the point $(0, 0, 0)$.

Problem 20: Using Linearization to Approximate (3D)

Use the linearization you found in Problem 18 to approximate the value of $f(1.9, 1.1, 3.05)$.

Quiz 6

Quiz 6

Name: _____

Problem 1: Directional Derivative

Find the directional derivative of the function $f(x, y) = e^x \sin(y)$ at the point $P(0, \pi/3)$ in the direction of the vector $\vec{v} = \langle -6, 8 \rangle$.

Problem 2: Tangent Plane to a Level Surface

Find the equation of the tangent plane to the ellipsoid (a level surface) defined by $x^2+2y^2+z^2 = 7$ at the point $(2, -1, 1)$.

Problem 3: Lagrange Multipliers

Find the extreme values (maximum and minimum) of the function $f(x, y) = 2xy$ subject to the constraint of the circle $x^2 + y^2 = 25$.

CHAPTER 7

Cycle 7: Directional Derivatives and Constrained Optimization (Homework 7)

Overview

This cycle develops directional derivatives and tangent planes to level surfaces, then applies these ideas to constrained optimization via the method of Lagrange multipliers. The emphasis is on interpreting gradients and constraints geometrically.

Lecture Notes and Classwork

October 15: Constrained Optimization (Lagrange Multipliers)

Calculus III: Constrained Optimization
Lecture Notes: The Method of Lagrange Multipliers

Many real-world optimization problems require us to find the maximum or minimum value of a function, not over its entire domain, but only along a specific path, boundary, or surface. For example, we might want to find the hottest point on a metal ring, where the temperature is a function $T(x, y)$ and the ring is our boundary, or "constraint." This is called **constrained optimization**.

Today, we develop a powerful and elegant technique called the method of Lagrange multipliers to solve exactly these kinds of problems. The method relies on a beautiful geometric insight involving gradient vectors.

Learning Objective

By the end of this lecture, you will be able to:
- **Use** the method of Lagrange multipliers to solve constrained optimization problems.

1. The Geometric Derivation

Let's say we want to find the extreme values of a function $z = f(x, y)$ subject to a constraint $g(x, y) = k$. The constraint is a curve in the xy-plane. As we travel along this path, our "altitude"

on the surface $z = f(x, y)$ will change. We want to find the points on the path that correspond to the highest and lowest altitudes.

$$xyz^2 - ^2; 3[surf, opacity = 0.5, colormap/viridis, zbuffer = sort$$

Let's parameterize the constraint curve by a vector function $\vec{r}(t) = \langle x(t), y(t) \rangle$.

First Insight: The Constraint Gradient

Since every point on the path satisfies the constraint, we have $g(x(t), y(t)) = k$. If we differentiate this equation with respect to t, we get:

$$\frac{d}{dt}\left[g(x(t), y(t))\right] = \frac{d}{dt}[k]$$

By the multivariable chain rule, this becomes:

$$\frac{\partial g}{\partial x}\frac{dx}{dt} + \frac{\partial g}{\partial y}\frac{dy}{dt} = 0$$

This is just a dot product: $\langle g_x, g_y \rangle \cdot \langle x'(t), y'(t) \rangle = 0$. In vector notation, this is:

$$\nabla g \cdot \vec{r}'(t) = 0$$

This confirms a fact we already know: the gradient of a function, ∇g, is always orthogonal to its level curves (here, the constraint curve).

Second Insight: The Objective Function Gradient at an Extremum

The value of our function f along the constraint path is given by the composite function $h(t) = f(x(t), y(t))$. From single-variable calculus, an extremum (max or min) of $h(t)$ can only occur when its derivative is zero, i.e., $h'(t) = 0$. Let's calculate this derivative:

$$\frac{dh}{dt} = \frac{d}{dt}\left[f(x(t), y(t))\right] = 0$$

Using the chain rule again:

$$\frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt} = 0$$

This is also a dot product: $\langle f_x, f_y \rangle \cdot \langle x'(t), y'(t) \rangle = 0$. In vector notation:

$$\nabla f \cdot \vec{r}'(t) = 0$$

This means that at a point of maximum or minimum value along the path, the gradient of the objective function, ∇f, must also be orthogonal to the tangent vector of the path.

The Main Conclusion

At an extreme point on the constraint curve, we have two facts:

- ∇g is orthogonal to the curve's tangent vector, $\vec{r}'(t)$.
- ∇f is also orthogonal to the curve's tangent vector, $\vec{r}'(t)$.

If both gradient vectors are orthogonal to the same tangent vector, they must be **parallel to each other**.

If two vectors are parallel, one must be a scalar multiple of the other. We call this scalar multiplier λ (lambda). Assuming $\nabla g \neq \vec{0}$, we arrive at the central equation of this method:

The Lagrange Condition

At an extreme point of f subject to the constraint $g(x, y) = k$, there must exist a scalar λ such that:

$$\nabla f(x, y) = \lambda \nabla g(x, y)$$

This tells us exactly where to look for the critical points along the curve. The rest is just algebra.

2. The Method of Lagrange Multipliers

Method for Constrained Optimization

To find the maximum and minimum values of $f(x, y)$ subject to the constraint $g(x, y) = k$:

1. **Set up the system of equations.** There are three equations and three unknowns (x, y, λ). The first two come from the gradient condition and the third is the constraint itself.

$$\begin{aligned} f_x(x, y) &= \lambda \cdot g_x(x, y) \\ f_y(x, y) &= \lambda \cdot g_y(x, y) \\ g(x, y) &= k \end{aligned}$$

2. **Solve the system** for all points (x, y) that are solutions. This is often the most challenging part.

3. **Evaluate f at all candidate points** from Step 2.

4. **Compare the values.** The largest value is the maximum, and the smallest value is the minimum.

The same method applies to functions of three variables, $f(x, y, z)$, with a constraint $g(x, y, z) = k$. This simply adds a fourth equation, $f_z = \lambda g_z$, and a fourth unknown, z.

3. Examples

Example 1: Finding Extrema on a Circle

Find the maximum and minimum values of $f(x, y) = xy$ subject to the constraint $x^2 + y^2 = 8$.

Step 1: Set up the system. First, identify the functions.

- Objective function: $f(x, y) = xy$
- Constraint function: $g(x, y) = x^2 + y^2 = 8$

Next, find the gradients.

$$\nabla f = \langle y, x \rangle \quad \text{and} \quad \nabla g = \langle 2x, 2y \rangle$$

Now, set up the three equations from the Lagrange condition $\nabla f = \lambda \nabla g$ and the constraint.

$$\begin{aligned} (1) \quad & y = \lambda(2x) \\ (2) \quad & x = \lambda(2y) \\ (3) \quad & x^2 + y^2 = 8 \end{aligned}$$

Step 2: Solve the system. Substitute (1) into (2):

$$x = \lambda(2(\lambda(2x))) \implies x = 4\lambda^2 x \implies x(1 - 4\lambda^2) = 0$$

This gives two possibilities: $x = 0$ or $1 - 4\lambda^2 = 0$.

If $x = 0$, equation (1) implies $y = 0$. But the point $(0, 0)$ does not satisfy the constraint $x^2 + y^2 = 8$. Therefore, we can conclude that $x \neq 0$. This means we must be in the second case.

Case: $1 - 4\lambda^2 = 0$**.** This implies $4\lambda^2 = 1$, so $\lambda = \pm\frac{1}{2}$.

- If $\lambda = 1/2$: From equation (1), $y = \frac{1}{2}(2x) \implies y = x$. Substitute this into the constraint (3): $x^2 + (x)^2 = 8 \implies 2x^2 = 8 \implies x^2 = 4 \implies x = \pm 2$. This gives two candidate points: $(\mathbf{2}, \mathbf{2})$ and $(-\mathbf{2}, -\mathbf{2})$.
- If $\lambda = -1/2$: From equation (1), $y = -\frac{1}{2}(2x) \implies y = -x$. Substitute into constraint (3): $x^2 + (-x)^2 = 8 \implies 2x^2 = 8 \implies x^2 = 4 \implies x = \pm 2$. This gives two more candidate points: $(\mathbf{2}, -\mathbf{2})$ and $(-\mathbf{2}, \mathbf{2})$.

Step 3: Evaluate f at the candidate points.

$$\begin{aligned} f(2,2) &= (2)(2) = 4 \\ f(-2,-2) &= (-2)(-2) = 4 \\ f(2,-2) &= (2)(-2) = -4 \\ f(-2,2) &= (-2)(2) = -4 \end{aligned}$$

Step 4: Compare the values. The maximum value is 4, which occurs at $(2, 2)$ and $(-2, -2)$. The minimum value is -4, which occurs at $(2, -2)$ and $(-2, 2)$.

Example 2: Optimizing the Dimensions of a Soda Can

A manufacturer wants to produce a cylindrical can that holds 1 liter (1000 cm^3) of liquid. Find the dimensions (radius r and height h) that will minimize the amount of metal used to make the can.

Step 1: Set up the system. First, identify the objective and constraint functions.

- **Objective:** We want to minimize the surface area, A. The area of the top and bottom circles is $2(\pi r^2)$, and the area of the side is $(2\pi r)h$.

$$A(r, h) = 2\pi r^2 + 2\pi rh$$

- **Constraint:** The volume must be 1000 cm^3.

$$V(r, h) = \pi r^2 h = 1000$$

Next, find the gradients of A and V.

$$\nabla A = \left\langle \frac{\partial A}{\partial r}, \frac{\partial A}{\partial h} \right\rangle = \langle 4\pi r + 2\pi h, 2\pi r \rangle$$
$$\nabla V = \left\langle \frac{\partial V}{\partial r}, \frac{\partial V}{\partial h} \right\rangle = \langle 2\pi rh, \pi r^2 \rangle$$

Now, set up the Lagrange system $\nabla A = \lambda \nabla V$ and the constraint.

$$\begin{aligned} (1) \quad & 4\pi r + 2\pi h = \lambda(2\pi rh) \\ (2) \quad & 2\pi r = \lambda(\pi r^2) \\ (3) \quad & \pi r^2 h = 1000 \end{aligned}$$

Step 2: Solve the system. From (2), we can simplify by dividing by πr (since r cannot be zero for a can to exist):

$$2 = \lambda r \implies \lambda = \frac{2}{r}$$

Now, simplify equation (1) by dividing everything by 2π:

$$2r + h = \lambda(rh)$$

Substitute our expression for λ into this simplified equation:

$$2r + h = \left(\frac{2}{r}\right)(rh) \implies 2r + h = 2h \implies \mathbf{h = 2r}$$

This is a famous result: the most efficient can has a height equal to its diameter!

Now, use this relationship in our constraint equation (3) to find the actual dimensions.

$$\begin{aligned} \pi r^2(2r) &= 1000 \\ 2\pi r^3 &= 1000 \\ r^3 &= \frac{500}{\pi} \\ r &= \sqrt[3]{\frac{500}{\pi}} \approx 5.42 \text{ cm} \end{aligned}$$

Since $h = 2r$, the optimal height is:

$$h = 2\sqrt[3]{\frac{500}{\pi}} \approx 10.84 \text{ cm}$$

Step 3 & 4: Evaluate and Conclude. Since we only found one candidate solution, and a can must have a minimum surface area (it can't be zero) and can have an infinitely large surface area (a very wide, short can), this single point must be the minimum.

The dimensions that minimize the cost of metal for a 1-liter can are a radius of $r = \sqrt[3]{500/\pi}$ cm and a height of $h = 2r$.

4. What Does λ Mean?

The Lagrange multiplier λ is not just an intermediate variable; it has a useful physical and economic interpretation.

The Meaning of the Lagrange Multiplier

The value of λ at an optimal point is the rate of change of the maximum (or minimum) value of the objective function f with respect to a change in the constraint constant k.

$$\lambda = \frac{df_{optimal}}{dk}$$

In economics, if f is a profit function and k is a budget constraint, λ represents the **marginal utility** of the budget. It tells you approximately how much the maximum profit would increase if you increased your budget by one unit.

Applying this to our example: We found optimal points when $\lambda = 1/2$ (for the max) and $\lambda = -1/2$ (for the min). Our constraint was $x^2 + y^2 = 8$, so $k = 8$. The value $\lambda = 1/2$ means that if we were to increase the constraint constant from $k = 8$ to $k = 9$ (making the circle slightly larger), the maximum value of f would increase by approximately 1/2. The current maximum is 4, so the new maximum would be about 4.5. This is a powerful tool for sensitivity analysis.

Classwork: Lagrange Multipliers

Name: _____

Problem 1: Extrema on a Circle

Find the maximum and minimum values of the function $f(x, y) = 3x + 4y$ subject to the constraint $x^2 + y^2 = 25$.

October 17: Local Extrema and the Second Derivative Test

Local Extrema and the Second Derivative Test

Lecture Notes: Classifying Critical Points

Learning Objectives

By the end of this lesson, you will be able to:

- Calculate the critical points of a function of two variables by setting its gradient equal to zero and solving the resulting system of equations.
- Apply the Second Derivative Test to classify a critical point as a local maximum, local minimum, or a saddle point by evaluating the discriminant and the second partial derivative f_{xx}.

In Calculus 1, we found local maxima and minima by first finding critical points where $f'(x) = 0$, and then using the Second Derivative Test to classify them. If $f''(x) > 0$, the function was concave up, indicating a local minimum. If $f''(x) < 0$, it was concave down, indicating a local maximum.

Today, we extend this idea to functions of two variables, $z = f(x, y)$. A critical point still occurs where the tangent plane is horizontal, which means $\nabla f(a, b) = \vec{0}$ (i.e., both f_x and f_y are zero). However, classifying these points is more complex. A surface can be concave up in one direction (e.g., along the x-axis) but concave down in another (e.g., along the y-axis). This would give us a "saddle point," which is neither a maximum nor a minimum.

To handle this, we need a tool that considers the concavity in all directions at once. This tool is the **Second Derivative Test for Multivariable Functions**.

1. The Discriminant and the Hessian Matrix

The test relies on a quantity called the **discriminant**, D, which is calculated from the second-order partial derivatives.

The Discriminant $D(x, y)$

The discriminant $D(x, y)$ is the determinant of a matrix of the second partial derivatives, known as the **Hessian Matrix**.

$$D(x, y) = \begin{vmatrix} f_{xx}(x, y) & f_{xy}(x, y) \\ f_{yx}(x, y) & f_{yy}(x, y) \end{vmatrix} = f_{xx}f_{yy} - (f_{xy})^2$$

(Recall that by Clairaut's Theorem, $f_{xy} = f_{yx}$ for the continuous functions we will be working with.)

Idea: The sign of the discriminant tells us about the concavity. If $D(a, b) > 0$, it means the concavity is "consistent" in all directions (either all up or all down). If $D(a, b) < 0$, the concavity changes depending on the direction, which signals a saddle point.

2. The Second Derivative Test

The Second Derivative Test

Suppose that $\nabla f(a,b) = \vec{0}$ (so (a,b) is a critical point) and all second partial derivatives are continuous. Let $D = D(a,b) = f_{xx}(a,b)f_{yy}(a,b) - (f_{xy}(a,b))^2$.

1. If $\boldsymbol{D > 0}$ and $\boldsymbol{f_{xx}(a,b) > 0}$, then $f(a,b)$ is a **local minimum**.
2. If $\boldsymbol{D > 0}$ and $\boldsymbol{f_{xx}(a,b) < 0}$, then $f(a,b)$ is a **local maximum**.
3. If $\boldsymbol{D < 0}$, then $f(a,b)$ is a **saddle point**.
4. If $\boldsymbol{D = 0}$, the test is **inconclusive**.

3. Algorithm and Examples

Algorithm for Finding Local Extrema

1. Find all critical points by solving the system of equations $\nabla f(x,y) = \vec{0}$, which means solving both $f_x(x,y) = 0$ and $f_y(x,y) = 0$.
2. Compute all second partial derivatives: f_{xx}, f_{yy}, and f_{xy}.
3. Construct the discriminant: $D(x,y) = f_{xx}f_{yy} - (f_{xy})^2$.
4. For each critical point (a,b), evaluate $D(a,b)$ and $f_{xx}(a,b)$.
5. Use the Second Derivative Test to classify each critical point.

Example: Find and classify all critical points of $f(x,y) = x^3 - 6xy + y^3$

Step 1: Find Critical Points First, we compute the gradient and set it to zero.

$$f_x = 3x^2 - 6y$$

$$f_y = -6x + 3y^2$$

Set both to zero to form a system of equations:

$$(1) \quad 3x^2 - 6y = 0 \implies y = \frac{1}{2}x^2$$

$$(2) \quad -6x + 3y^2 = 0 \implies 3y^2 = 6x \implies y^2 = 2x$$

Now, substitute equation (1) into equation (2):

$$\left(\frac{1}{2}x^2\right)^2 = 2x$$

$$\frac{1}{4}x^4 = 2x$$

$$x^4 = 8x$$

$$x^4 - 8x = 0$$

$$x(x^3 - 8) = 0$$

This gives two possible solutions for x: $x = 0$ and $x = 2$. Now we find the corresponding y values using $y = \frac{1}{2}x^2$. If $x = 0$, then $y = \frac{1}{2}(0)^2 = 0$. Critical point: $\mathbf{(0, 0)}$. If $x = 2$, then $y = \frac{1}{2}(2)^2 = 2$. Critical point: $\mathbf{(2, 2)}$.

Step 2 & 3: Compute Second Partials and the Discriminant

$$f_{xx} = \frac{\partial}{\partial x}(3x^2 - 6y) = 6x$$

$$f_{yy} = \frac{\partial}{\partial y}(-6x + 3y^2) = 6y$$

$$f_{xy} = \frac{\partial}{\partial y}(3x^2 - 6y) = -6$$

The discriminant is:

$$D(x, y) = (6x)(6y) - (-6)^2 = 36xy - 36$$

Step 4 & 5: Classify Each Critical Point Now we test each point. For the point $(0, 0)$:

$$D(0, 0) = 36(0)(0) - 36 = -36$$

Since $D < 0$, the point $\mathbf{(0, 0)}$ is a **saddle point**.

For the point $(2, 2)$:

$$D(2, 2) = 36(2)(2) - 36 = 144 - 36 = 108$$

Since $D > 0$, it is either a local max or min. We check the sign of f_{xx}:

$$f_{xx}(2, 2) = 6(2) = 12$$

Since $f_{xx} > 0$, the point $\mathbf{(2, 2)}$ is a **local minimum**.

Example 2: A Function with Maxima and a Saddle Point

Find and classify all critical points of $f(x, y) = 4xy - x^4 - y^4$.

Solution:

Step 1: Find Critical Points We compute the gradient and set it to zero.

$$f_x = \frac{\partial}{\partial x}(4xy - x^4 - y^4) = 4y - 4x^3$$

$$f_y = \frac{\partial}{\partial y}(4xy - x^4 - y^4) = 4x - 4y^3$$

Now, we set both partial derivatives to zero to create a system of equations:

$$(1) \quad 4y - 4x^3 = 0 \implies y = x^3$$

$$(2) \quad 4x - 4y^3 = 0 \implies x = y^3$$

This is a non-linear system. We can solve it by substituting the expression for y from equation (1) into equation (2).

$$x = (x^3)^3$$

$$x = x^9$$

Bring all terms to one side:

$$x^9 - x = 0$$

Factor out an x:

$$x(x^8 - 1) = 0$$

This gives us solutions for x. We have $x = 0$ or $x^8 - 1 = 0$. The second equation means $x^8 = 1$, which has real solutions $x = 1$ and $x = -1$. Now we find the corresponding y-values for each x using the relation $y = x^3$.

- If $x = 0$, then $y = 0^3 = 0$. Critical point: $\mathbf{(0, 0)}$.
- If $x = 1$, then $y = 1^3 = 1$. Critical point: $\mathbf{(1, 1)}$.
- If $x = -1$, then $y = (-1)^3 = -1$. Critical point: $\mathbf{(-1, -1)}$.

We have three critical points to test.

Step 2 & 3: Compute Second Partials and the Discriminant Next, we find the second-order partial derivatives.

$$f_{xx} = \frac{\partial}{\partial x}(4y - 4x^3) = -12x^2$$

$$f_{yy} = \frac{\partial}{\partial y}(4x - 4y^3) = -12y^2$$

$$f_{xy} = \frac{\partial}{\partial y}(4y - 4x^3) = 4$$

Now we construct the discriminant, $D(x, y)$.

$$D(x, y) = f_{xx}f_{yy} - (f_{xy})^2$$

$$D(x, y) = (-12x^2)(-12y^2) - (4)^2$$

$$D(x, y) = 144x^2y^2 - 16$$

Step 4 & 5: Classify Each Critical Point We apply the Second Derivative Test to each of our three critical points.

<u>For the point $(0, 0)$:</u>

$$D(0, 0) = 144(0)^2(0)^2 - 16 = -16$$

Since $D < 0$, the point $\mathbf{(0, 0)}$ is a **saddle point**.

<u>For the point $(1, 1)$:</u>

$$D(1, 1) = 144(1)^2(1)^2 - 16 = 144 - 16 = 128$$

Since $D > 0$, we have a local extremum. We must check the sign of f_{xx} to classify it.

$$f_{xx}(1, 1) = -12(1)^2 = -12$$

Since $f_{xx} < 0$, the point $\mathbf{(1,1)}$ is a **local maximum**.

For the point $(-1,-1)$:

$$D(-1,-1) = 144(-1)^2(-1)^2 - 16 = 144 - 16 = 128$$

Since $D > 0$, we have a local extremum. We again check the sign of f_{xx}.

$$f_{xx}(-1,-1) = -12(-1)^2 = -12$$

Since $f_{xx} < 0$, the point $\mathbf{(-1,-1)}$ is also a **local maximum**.

Classwork: The Second Derivative Test

Name: _____

Instructions: Find all critical points of the function below and classify each one as a local maximum, a local minimum, or a saddle point using the Second Derivative Test. Show all of your steps clearly.

Problem 1: A Basic Example

Find and classify the critical point(s) for the function:

$$f(x,y) = x^2 + 2y^2 - 6x + 8y - 1$$

October 20: Double Integrals as Volume

An Introduction to Multiple Integration: Day 1

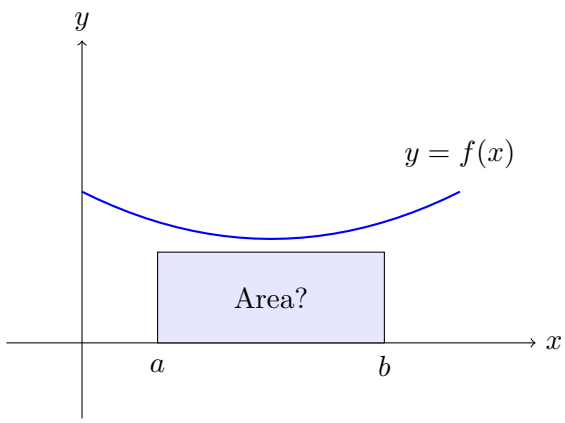
Lecture Notes: Double Integrals as Volume

By the end of this lesson, you will be able to:

- Connect the concept of a definite integral in 1D (area) to a double integral in 2D (volume).
- Understand how a double integral is defined as the limit of a double Riemann sum.
- Evaluate a double integral over a rectangular region by computing an iterated integral.

1. Motivation: From Area to Volume

In Calculus 1, we spent a great deal of time on the definite integral, $\int_a^b f(x)\,dx$. The central question was, "What is the net signed area between the curve $y = f(x)$ and the x-axis over the interval $[a, b]$?"



Today, we move to a higher dimension. Our functions are no longer curves, but **surfaces** of the form $z = f(x, y)$. Our domain is no longer an interval on a line, but a **region** R in the xy-plane. The analogous question is:

"What is the net signed volume between the surface $z = f(x, y)$ and the xy-plane over the region R?"

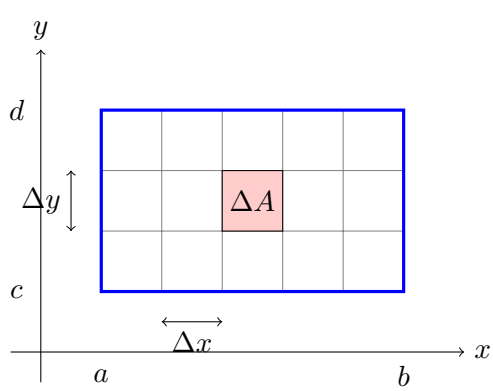
Imagine you are a landscape architect with a digital map of a client's property. The height of the land is given by $z = f(x, y)$. You need to build a rectangular patio over a region R. To do this, you must level the ground. This requires calculating the volume of earth you need to remove or add. This is precisely the problem the double integral solves.

2. Building the Double Integral from the Ground Up

Let's replicate the process we used to invent the definite integral in Calculus 1, but now for a function of two variables over a rectangular domain $R = [a, b] \times [c, d]$.

Step 1: Partition the Domain

In 1D, we chopped the interval $[a, b]$ into n small segments of width Δx. In 2D, we chop the rectangle R into a grid of $m \times n$ smaller sub-rectangles. Each sub-rectangle has an area $\Delta A = \Delta x \cdot \Delta y$.

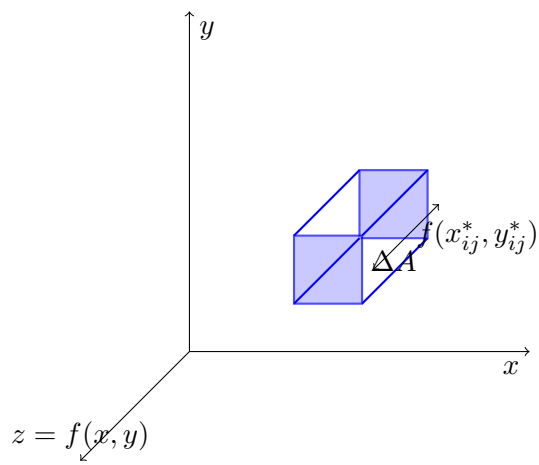


Step 2: Approximate with Prisms

In 1D, we built a rectangle on each segment. The height was $f(x_i^*)$ for some sample point x_i^* in the segment. In 2D, we build a rectangular prism (a box) on each sub-rectangle. We pick a sample point (x_{ij}^*, y_{ij}^*) in the sub-rectangle, and the height of the prism is the value of the function there: $z = f(x_{ij}^*, y_{ij}^*)$.

The volume of one of these small prisms is:

$$V_{ij} = (\text{height}) \times (\text{base area}) = f(x_{ij}^*, y_{ij}^*) \cdot \Delta A$$



Step 3: Sum and Take the Limit

In 1D, we summed the areas of all the rectangles: $\sum f(x_i^*)\Delta x$. In 2D, we sum the volumes of all the prisms. Since our grid has two indices, i for the x-direction and j for the y-direction, this is a **double summation**:

$$V \approx \sum_{i=1}^{m} \sum_{j=1}^{n} f(x_{ij}^*, y_{ij}^*)\Delta A$$

To get the exact volume, we let the number of prisms go to infinity in both directions ($m \to \infty, n \to \infty$). This limit is the definition of the double integral.

The Double Integral over a Rectangle

The **double integral** of f over the rectangle R is defined as the limit of the double Riemann sum:

$$\iint_R f(x, y)\, dA = \lim_{m,n\to\infty} \sum_{i=1}^{m} \sum_{j=1}^{n} f(x_{ij}^*, y_{ij}^*)\Delta A$$

This definition is theoretically perfect, but computationally a nightmare. We need a better way.

3. Iterated Integrals: The Practical Method

Let's go back to the volume analogy. Imagine you want to find the volume of a loaf of bread. You don't need to add up molecules. You can simply slice the loaf, find the area of each slice, and then "add up" the areas of all the slices. This is the idea behind iterated integrals.

1. **Slice the Volume:** Pick a fixed value of x. This creates a 2D cross-section of our solid parallel to the yz-plane. The area of this slice depends on which x we chose, so we call it $A(x)$.

2. **Find the Area of a Slice:** The area $A(x)$ is the area under the curve $z = f(x, y)$, where y goes from c to d and x is held constant. This is a standard Calc 1 integral!

$$A(x) = \int_c^d f(x, y)\, dy$$

3. **Integrate the Areas:** To find the total volume, we integrate these cross-sectional areas as x sweeps from a to b.

$$V = \int_a^b A(x)\, dx = \int_a^b \left[\int_c^d f(x, y)\, dy \right] dx$$

This is an **iterated integral**. We always evaluate it from the **inside out**.

Example 1: The Volume of a Box

Find the volume of the solid under the plane $z = 5$ and above the rectangle $R = [0, 4] \times [0, 2]$.

Solution: Geometric Check: This is a simple box. Its dimensions are length=4, width=2, and height=5. The volume should be $V = 4 \times 2 \times 5 = 40$. Let's see if the calculus gives the same result.

Calculus Method: We need to evaluate $\iint_R 5\, dA$. We set up the iterated integral. The outer variable is x, with bounds $[0, 4]$. The inner variable is y, with bounds $[0, 2]$.

$$V = \int_0^4 \int_0^2 5\, dy\, dx$$

Step 1 (Inner Integral): Integrate with respect to y, treating x (and 5) as constants.

$$\begin{aligned} \int_0^2 5\, dy &= [5y]_{y=0}^{y=2} \\ &= (5(2)) - (5(0)) \\ &= 10 - 0 = 10 \end{aligned}$$

This makes sense: the area of any cross-section of the box is a rectangle of height 5 and width 2, so its area is 10.

Step 2 (Outer Integral): Integrate the result from Step 1.

$$\begin{aligned} \int_0^4 10\, dx &= [10x]_{x=0}^{x=4} \\ &= (10(4)) - (10(0)) \\ &= 40 - 0 = 40 \end{aligned}$$

The results match! This shows that the method works.

What does $\iint_R 1\,dA$ mean?

If we integrate the function $f(x, y) = 1$, we are finding the volume of a solid with height 1 over the region R. The volume of such a solid is (base area) $\times$ 1. Therefore:

$$\iint_R 1\,dA = \text{Area}(R)$$

This is a very useful property.

Example 2: A More Complex Volume

Evaluate $\iint_R (x - 3y^2)\,dA$ where $R = [0, 2] \times [1, 2]$.

Solution: We set up the iterated integral in the order $dy\,dx$. The outer bounds for x are $[0, 2]$ and the inner bounds for y are $[1, 2]$.

$$V = \int_0^2 \left[\int_1^2 (x - 3y^2)\,dy\right] dx$$

Step 1: Compute the Inner Integral. We integrate with respect to y. For this step, we must treat the variable x as if it were a constant number.

$$\begin{aligned}
\int_1^2 (x - 3y^2)\,dy &= [\text{The integral of the constant } x \text{ is } xy] - \left[\text{The integral of } 3y^2 \text{ is } y^3\right] \\
&= \left[xy - y^3\right]_{y=1}^{y=2} \\
&\text{Now, substitute the top and bottom bounds for } y. \\
&= (x(2) - (2)^3) - (x(1) - (1)^3) \\
&= (2x - 8) - (x - 1) \\
&= 2x - 8 - x + 1 \\
&= x - 7
\end{aligned}$$

This result is the function $A(x)$ that gives the area of a cross-section at any value of x.

Step 2: Compute the Outer Integral. Now we integrate the much simpler function that resulted from Step 1.

$$\begin{aligned}
\int_0^2 (x - 7)\,dx &= \left[\frac{1}{2}x^2 - 7x\right]_{x=0}^{x=2} \\
&\text{Substitute the bounds for } x. \\
&= \left(\frac{1}{2}(2)^2 - 7(2)\right) - \left(\frac{1}{2}(0)^2 - 7(0)\right) \\
&= \left(\frac{1}{2}(4) - 14\right) - (0) \\
&= (2 - 14) \\
&= -12
\end{aligned}$$

Interpreting the Result: What does a volume of -12 mean? This is a **net volume**. It means that over the region R, the part of the surface that lies *below* the xy-plane has a greater volume than the part that lies *above* it. The double integral automatically subtracts the volume below from the volume above, just like a 1D integral subtracts area below the x-axis.

October 22: Fubini's Theorem and Triple Integrals

An Introduction to Multiple Integration: Day 2

Lecture Notes: Fubini's Theorem and Triple Integrals

Learning Objectives

By the end of this lesson, you will be able to:

- Apply Fubini's Theorem to change the order of integration for a double integral over a rectangle.
- Recognize when one order of integration is strategically better than the other.
- Set up and evaluate a triple integral over a rectangular box.
- Interpret a triple integral as mass or volume.

1. Fubini's Theorem: A Powerful Tool

Recap: Yesterday, we calculated the double integral $\iint_R (x - 3y^2)\, dA$ over $R = [0, 2] \times [1, 2]$ by computing the iterated integral $\int_0^2 \int_1^2 (x - 3y^2)\, dy\, dx$. Our calculation method was based on slicing the volume with planes of constant x.

Motivating Question: What if we had sliced the volume the other way, with planes of constant y? This would correspond to integrating with respect to x first, then y. Would we get the same answer?

Common sense says yes. It shouldn't matter how you slice an object to find its volume. The mathematician Guido Fubini proved that this intuition is correct for continuous functions over rectangles.

Fubini's Theorem

If $f(x, y)$ is continuous on the rectangle $R = [a, b] \times [c, d]$, then

$$\iint_R f(x, y)\, dA = \int_a^b \int_c^d f(x, y)\, dy\, dx = \int_c^d \int_a^b f(x, y)\, dx\, dy$$

Visual Intuition: Imagine finding the volume of a stack of paper.

- $dy\, dx$ **order:** Find the area of a vertical sheet of paper (integrating in y), and then add up all the sheets (integrating in x).
- $dx\, dy$ **order:** Find the area of a horizontal strip of paper (integrating in x), and then add up all the strips from bottom to top (integrating in y).

The total amount of paper is the same regardless of how you count it.

Example 1: Verifying Fubini's Theorem

Evaluate $\iint_R (x - 3y^2)\, dA$ where $R = [0, 2] \times [1, 2]$ using the order $dx\, dy$.

Solution: Yesterday, we found the answer to be -12 using the $dy\,dx$ order. Let's confirm this. The setup is now:

$$V = \int_1^2 \left[\int_0^2 (x - 3y^2)\,dx \right] dy$$

Step 1: Inner Integral (wrt x). Treat y as a constant.

$$\begin{aligned}
\int_0^2 (x - 3y^2)\,dx &= \left[\frac{1}{2}x^2 - (3y^2)x \right]_{x=0}^{x=2} \\
&= \left(\frac{1}{2}(2)^2 - 3y^2(2) \right) - \left(\frac{1}{2}(0)^2 - 3y^2(0) \right) \\
&= (2 - 6y^2) - (0) = 2 - 6y^2
\end{aligned}$$

Step 2: Outer Integral (wrt y).

$$\begin{aligned}
\int_1^2 (2 - 6y^2)\,dy &= \left[2y - 2y^3 \right]_{y=1}^{y=2} \\
&= (2(2) - 2(2)^3) - (2(1) - 2(1)^3) \\
&= (4 - 16) - (2 - 2) = -12 - 0 = -12
\end{aligned}$$

It works! The order did not change the final answer.

The Strategic Importance of Fubini's Theorem

Fubini's Theorem isn't just a neat trick; it's a vital problem-solving strategy. Sometimes an integral that is difficult or even impossible to solve in one order becomes simple in the other.

Example 2: A Strategic Choice

Evaluate $\iint_R y\sin(xy)\,dA$ over the region $R = [0,2] \times [0,\pi/2]$.

Solution: We have two options. Let's analyze the inner integral for each.

Option 1 (The Hard Way): $dy\,dx$ The inner integral would be $\int_0^{\pi/2} y\sin(xy)\,dy$. Since both y and $\sin(xy)$ contain the integration variable y, this requires **integration by parts**. This is doable, but it's a lot of work and prone to error.

Option 2 (The Easy Way): $dx\,dy$ The inner integral would be $\int_0^2 y\sin(xy)\,dx$. Here, y is treated as a constant. The integral is of the form $\int C\sin(Cx)\,dx$, which is a basic integral. Let's proceed with this order.

$$I = \int_0^{\pi/2} \left[\int_0^2 y\sin(xy)\,dx\right] dy$$

Step 1: Inner Integral (wrt x). Treat y as constant. The antiderivative of $\sin(ax)$ is $-\frac{1}{a}\cos(ax)$. Here, $a = y$.

$$\begin{aligned}
\int_0^2 y\sin(xy)\,dx &= y\int_0^2 \sin(xy)\,dx \\
&= y\left[-\frac{1}{y}\cos(xy)\right]_{x=0}^{x=2} \\
&= [-\cos(xy)]_{x=0}^{x=2} \\
&= (-\cos(y\cdot 2)) - (-\cos(y\cdot 0)) \\
&= -\cos(2y) + \cos(0) \\
&= 1 - \cos(2y)
\end{aligned}$$

Step 2: Outer Integral (wrt y).

$$\begin{aligned}
\int_0^{\pi/2} (1-\cos(2y))\,dy &= \left[y - \frac{1}{2}\sin(2y)\right]_{y=0}^{y=\pi/2} \\
&= \left(\frac{\pi}{2} - \frac{1}{2}\sin\left(2\cdot\frac{\pi}{2}\right)\right) - \left(0 - \frac{1}{2}\sin(0)\right) \\
&= \left(\frac{\pi}{2} - \frac{1}{2}\sin(\pi)\right) - (0) \\
&= \frac{\pi}{2} - \frac{1}{2}(0) = \frac{\pi}{2}
\end{aligned}$$

By choosing the correct order, we transformed a difficult integration-by-parts problem into a sequence of two elementary integrals.

2. Triple Integrals

The entire concept extends seamlessly to three dimensions. We now want to integrate a function of three variables, $w = f(x, y, z)$, over a 3D rectangular box, $B = [a, b] \times [c, d] \times [e, f]$.

What could this mean?

- **Mass:** If $f(x, y, z)$ represents the density of a substance at point (x, y, z) (in, say, kg/m^3), then the triple integral gives the total mass of the substance in the box. This is the most common application.
- **Hypervolume:** Mathematically, it is the 4-dimensional volume of the region that is "under" the 3D hypersurface $w = f(x, y, z)$ and "above" our 3D box B. This is hard to visualize but is mathematically sound.

The **triple integral** is written as $\iiint_B f(x, y, z)\, dV$, where $dV = dx\, dy\, dz$ represents an infinitesimal element of volume. We evaluate it as an iterated integral with three layers. Fubini's Theorem extends here as well: for a rectangular box, any of the $3! = 6$ possible orders of integration ($dz\, dy\, dx$, $dy\, dx\, dz$, etc.) will give the same answer.

Example 3: Volume of a Box

Find the volume of the box $B = [0, 1] \times [-1, 1] \times [0, 2]$ using a triple integral.

Solution: To find volume, we integrate the function $f(x, y, z) = 1$.

$$V = \iiint_B 1\, dV = \int_0^1 \int_{-1}^1 \int_0^2 1\, dz\, dy\, dx$$

Step 1: Innermost Integral (wrt z):

$$\int_0^2 1\, dz = [z]_{z=0}^{z=2} = 2 - 0 = 2$$

Step 2: Middle Integral (wrt y):

$$\int_{-1}^1 2\, dy = [2y]_{y=-1}^{y=1} = (2(1)) - (2(-1)) = 2 - (-2) = 4$$

Step 3: Outermost Integral (wrt x):

$$\int_0^1 4\, dx = [4x]_{x=0}^{x=1} = 4(1) - 4(0) = 4$$

The volume is 4. Let's check: length (x-range) is 1. Width (y-range) is $1 - (-1) = 2$. Height (z-range) is 2. Volume $= 1 \times 2 \times 2 = 4$. It matches.

Example 4: Mass of a Non-Uniform Box

Find the total mass of the box $B = [0,1] \times [0,2] \times [0,3]$ if its density at any point is given by the function $\rho(x,y,z) = x + y + z$.

Solution: The total mass is given by the triple integral of the density function. We'll use the order $dz\,dy\,dx$.

$$M = \int_0^1 \int_0^2 \int_0^3 (x+y+z)\,dz\,dy\,dx$$

Step 1: Innermost Integral (wrt z). Treat x, y as constants.

$$\begin{aligned}\int_0^3 (x+y+z)\,dz &= \left[(x+y)z + \frac{1}{2}z^2\right]_{z=0}^{z=3} \\ &= \left((x+y)(3) + \frac{1}{2}(3)^2\right) - (0) \\ &= 3x + 3y + \frac{9}{2}\end{aligned}$$

Step 2: Middle Integral (wrt y). Treat x as a constant.

$$\begin{aligned}\int_0^2 \left(3x + 3y + \frac{9}{2}\right)\,dy &= \left[3xy + \frac{3}{2}y^2 + \frac{9}{2}y\right]_{y=0}^{y=2} \\ &= \left(3x(2) + \frac{3}{2}(2)^2 + \frac{9}{2}(2)\right) - (0) \\ &= 6x + \frac{3}{2}(4) + 9 \\ &= 6x + 6 + 9 = 6x + 15\end{aligned}$$

Step 3: Outermost Integral (wrt x).

$$\begin{aligned}\int_0^1 (6x+15)\,dx &= \left[3x^2 + 15x\right]_{x=0}^{x=1} \\ &= (3(1)^2 + 15(1)) - (0) \\ &= 3 + 15 = 18\end{aligned}$$

The total mass of the box is 18.

Classwork - Multiple Integration over Rectangles

Name: _____

Problem 1: Verifying Fubini's Theorem

Evaluate the double integral $\iint_R x^2 y\,dA$ over the region $R = [0,2] \times [1,3]$ in two different ways.

(a) By setting up the iterated integral in the order $dy\,dx$.

(b) By setting up the iterated integral in the order $dx\,dy$. Confirm you get the same answer.

Problem 2: Choosing the Order of Integration

Evaluate the integral $\iint_R y\cos(xy)\,dA$ over the region $R = [0,1] \times [0,\pi]$.

(Hint: One order of integration is significantly easier than the other. Choose wisely!)

Problem 3: Triple Integral over a Box

Evaluate the triple integral $\iiint_B (6xz - y^2)\, dV$ where the box B is given by

$$B = \{(x, y, z) \mid 0 \leq x \leq 1, -1 \leq y \leq 2, 1 \leq z \leq 2\}$$

Homework 7

Homework 7: Gradients, Directional Derivatives, and Lagrange Multipliers

Work solutions on blank paper, showing all steps. Scan and submit by the due date on Blackboard.

Part 1: Gradients, Directional Derivatives, and Tangent Geometry

Problem 1: Directional Derivative in 2D

Find the directional derivative of the function $f(x, y) = x^3 - 4xy^2$ at the point $P(2, 1)$ in the direction of the vector $\vec{v} = \langle 1, 1\rangle$.

Problem 2: Directional Derivative in 3D

Find the directional derivative of the function $f(x, y, z) = xe^{yz}$ at the point $P(3, 0, 2)$ in the direction of the vector $\vec{v} = \langle 2, -1, 2\rangle$.

Problem 3: Directional Derivative

Find the directional derivative of $f(x, y) = x^2y^3$ at the point $P(2, 1)$ in the direction of the vector $\vec{v} = \langle 4, 3\rangle$.

Problem 4: Steepest Ascent

For the function $f(x, y) = \ln(x^2 + y^2)$, find the direction of steepest ascent at the point $P(1, 2)$. What is the rate of change in this direction?

Problem 5: Steepest Descent

For the function $f(x, y, z) = x^2 + y^2 - z^2$, find the direction of steepest descent at the point $P(1, 1, 1)$. What is the rate of change in this direction?

Problem 6: Tangent Line to a Level Curve

Find the equation of the tangent line to the level curve $x^2 + 4y^2 = 8$ at the point $(2, 1)$.

Problem 7: Another Tangent Line

Find the equation of the tangent line to the curve $y = \cos(x)$ at the point $(\pi/3, 1/2)$. (Hint: Treat the curve as a level curve of $f(x, y) = y - \cos(x)$.)

Problem 8: Tangent Plane to a Level Surface

Find the equation of the tangent plane to the level surface $xy + yz + xz = 11$ at the point $(1, 2, 3)$.

Problem 9: Tangent Plane to a Function Graph

Find the equation of the tangent plane to the surface $z = x^2 + 5y^2$ at the point $(2, 1, 9)$.

Problem 10: Tangent Plane to a Sphere

Find the equation of the tangent plane to the sphere $x^2 + y^2 + z^2 = 14$ at the point $(1, 2, 3)$.

Problem 11: Tangent Plane with Exponential Function

Find the equation of the tangent plane to the surface $e^{xyz} = 1$ at the point $(1, 1, 0)$.

Part 2: The Method of Lagrange Multipliers

For each problem, use the method of Lagrange multipliers to find the maximum and minimum values of the function f subject to the given constraint.

Problem 12

Find the extreme values of $f(x, y) = x + 2y$ subject to the constraint $x^2 + y^2 = 5$.

Problem 13

Find the extreme values of $f(x, y) = xy$ subject to the constraint $x^2 + y^2 = 18$.

Problem 14

Find the extreme values of $f(x, y) = x^2 + y^2$ subject to the constraint $x^2 + 4y^2 = 16$.

Problem 15

Find the extreme values of $f(x, y) = 4x^2 + y^2$ subject to the constraint $x^2 + y^2 = 1$.

Problem 16

Find the extreme values of $f(x, y) = 2x + 6y$ subject to the constraint $x^2 + y^2 = 10$.

Problem 17

Find the extreme values of $f(x, y, z) = x + y + z$ subject to the constraint $x^2 + y^2 + z^2 = 12$.

Problem 18

Find the maximum value of $f(x, y, z) = xyz$ subject to the constraint $x + y + z = 6$, assuming $x, y, z > 0$.

Problem 19: Application

You have 100 feet of fencing to build a rectangular enclosure. What is the maximum area you can enclose? (Objective function is $A(x, y) = xy$, constraint is the perimeter).

Quiz 7

Quiz 7

Name: _____

Problem 1: Second Derivative Test

Find the critical point of the function $f(x, y) = \frac{1}{2}(x - 2)^2 + \frac{1}{2}(y - 3)^2 + 5$ and use the Second Derivative Test to classify it as a local maximum, local minimum, or saddle point.

Problem 2: Double Integral over a Rectangle

Evaluate the double integral $\iint_R (6xy^2)\, dA$ where R is the rectangle defined by $R = [1, 2] \times [0, 3]$.

Problem 3: Triple Integral over a Box

Evaluate the triple integral $\iiint_B (x+yz)\,dV$ where the box B is given by $B = [0,1]\times[0,2]\times[0,3]$.

CHAPTER 8

Cycle 8: Second Derivatives and the Shape of Surfaces (Homework 8)

Overview

This cycle focuses on second-order partial derivatives, Hessians, and classification of critical points for functions of several variables. The goal is to understand how second derivatives control the local shape of surfaces (maxima, minima, and saddles).

Lecture Notes and Classwork

October 24: Double Integrals Over General Regions

Double Integrals Over General Regions
Lecture Notes: Beyond Rectangles

Learning Objectives

By the end of this lesson, you will be able to:

- Use a double integral to calculate the mass of a flat plate (lamina) with variable density.
- Identify and describe Type I and Type II regions in the plane.
- Set up and evaluate iterated double integrals over Type I and Type II regions.

1. Application: Mass of a Lamina

So far, we have only integrated over rectangular regions. Before we learn how to integrate over more complex shapes, let's introduce a key physical application: finding the mass of a flat object.

Imagine a thin, flat sheet of material, which we call a **lamina**. We can model this lamina as a region D in the xy-plane. If the material is not uniform, its density will vary from point to point. We describe this with a **area density function**, $\rho(x, y)$, which gives the density at point (x, y) in units of mass per unit area (e.g., kg/m^2).

To find the total mass, we can follow the logic of integration:

1. Chop the lamina D into tiny rectangular pieces, each with area dA.
2. The mass of a single piece at (x, y) is approximately mass $\approx \rho(x, y)\, dA$.
3. To get the total mass, we "sum up" the masses of all the pieces by integrating over the entire region D.

Mass from Density

The total mass M of a lamina with shape D and density function $\rho(x, y)$ is given by the double integral:

$$M = \iint_D \rho(x, y)\, dA$$

Example 1: Mass of a Rectangular Lamina

A rectangular plate is defined by $R = [0, 2] \times [0, 1]$. The density of the plate is given by $\rho(x, y) = x + y^2$. Find the total mass of the plate.

Solution: We simply integrate the density function over the rectangular region.

$$\begin{aligned}
M = \iint_R (x + y^2)\, dA &= \int_0^2 \int_0^1 (x + y^2)\, dy\, dx \\
&= \int_0^2 \left[xy + \frac{1}{3}y^3 \right]_{y=0}^{y=1} dx \\
&= \int_0^2 \left((x(1) + \frac{1}{3}(1)^3) - (0) \right) dx \\
&= \int_0^2 \left(x + \frac{1}{3} \right) dx \\
&= \left[\frac{1}{2}x^2 + \frac{1}{3}x \right]_{x=0}^{x=2} \\
&= \left(\frac{1}{2}(2)^2 + \frac{1}{3}(2) \right) - (0) = 2 + \frac{2}{3} = \frac{8}{3}
\end{aligned}$$

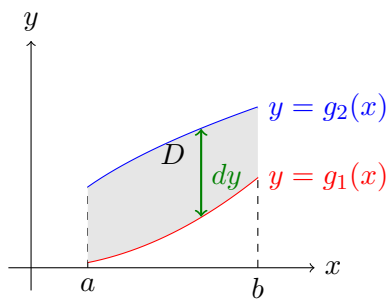
The total mass is 8/3.

2. Double Integrals over General Regions

Real-world objects are rarely perfect rectangles. To integrate over more complex shapes like triangles, circles, or regions bounded by curves, we define two general types of regions.

Type I Regions (Vertically Simple)

A Type I region is bounded below and above by the graphs of two continuous functions of x, and on the sides by vertical lines.



A region D is **Type I** if it can be described as:

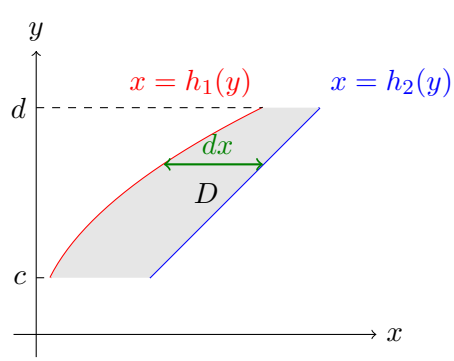
$$D = \{(x, y) \mid a \leq x \leq b, \quad g_1(x) \leq y \leq g_2(x)\}$$

To integrate over a Type I region, we use the order $dy\,dx$. The outer integral has constant limits (a to b), while the inner integral has function limits ($g_1(x)$ to $g_2(x)$).

$$\iint_D f(x, y)\,dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y)\,dy\,dx$$

Type II Regions (Horizontally Simple)

A Type II region is bounded on the left and right by the graphs of two continuous functions of y, and on the top and bottom by horizontal lines.



A region D is **Type II** if it can be described as:

$$D = \{(x, y) \mid c \leq y \leq d, \quad h_1(y) \leq x \leq h_2(y)\}$$

To integrate over a Type II region, we use the order $dx\,dy$. The outer integral has constant limits (c to d), while the inner integral has function limits ($h_1(y)$ to $h_2(y)$).

$$\iint_D f(x,y)\,dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x,y)\,dx\,dy$$

Crucial Rule: The limits of the outermost integral must always be constants.

3. Comprehensive Example: Mass of a Lamina

Example 2: Mass of a Non-Rectangular Lamina

Find the mass of a lamina that occupies the region D bounded by the parabola $y = x^2$ and the line $y = 2x$. The density of the lamina at any point (x, y) is given by $\rho(x, y) = x^2y$.

Solution: First, let's sketch the region D and determine the bounds of integration. To find the intersection points, we set the functions equal:

$$x^2 = 2x \implies x^2 - 2x = 0 \implies x(x-2) = 0$$

The curves intersect at $x = 0$ and $x = 2$. For x between 0 and 2, the line $y = 2x$ is above the parabola $y = x^2$.

The region is naturally a Type I region. We can describe it using set-builder notation:

$$D = \{(x,y) \mid 0 \le x \le 2, \quad x^2 \le y \le 2x\}$$

The mass M is the integral of the density function $\rho(x, y) = x^2y$ over this region.

$$M = \iint_D x^2y\,dA = \int_0^2 \int_{x^2}^{2x} x^2y\,dy\,dx$$

Step 1: Inner Integral (wrt y). Treat x as a constant.

$$\begin{aligned}
\int_{x^2}^{2x} x^2y\,dy &= x^2 \int_{x^2}^{2x} y\,dy \\
&= x^2 \left[\frac{1}{2}y^2\right]_{y=x^2}^{y=2x} \\
&= \frac{1}{2}x^2\left((2x)^2 - (x^2)^2\right) \\
&= \frac{1}{2}x^2(4x^2 - x^4) \\
&= 2x^4 - \frac{1}{2}x^6
\end{aligned}$$

Step 2: Outer Integral (wrt x).

$$
\begin{aligned}
M = \int_0^2 \left(2x^4 - \frac{1}{2}x^6\right) dx &= \left[\frac{2}{5}x^5 - \frac{1}{2\cdot 7}x^7\right]_{x=0}^{x=2} \\
&= \left[\frac{2}{5}x^5 - \frac{1}{14}x^7\right]_{x=0}^{x=2} \\
&= \left(\frac{2}{5}(2)^5 - \frac{1}{14}(2)^7\right) - (0) \\
&= \frac{2}{5}(32) - \frac{1}{14}(128) \\
&= \frac{64}{5} - \frac{64}{7} \\
&= 64\left(\frac{1}{5} - \frac{1}{7}\right) = 64\left(\frac{7-5}{35}\right) = 64\left(\frac{2}{35}\right) = \frac{128}{35}
\end{aligned}
$$

The total mass of the lamina is $\frac{128}{35}$.

Classwork: Integrals over General Regions

Name: _____

Instructions: For each problem, first sketch the region of integration. Then, set up and evaluate the double integral. Show your work clearly.

Problem 1: A Type I Region

Evaluate the double integral $\iint_D (x+y)\,dA$ where D is the region bounded by the parabola $y = x^2$ and the line $y = \sqrt{x}$.

Problem 2: A Type II Region

Find the mass of a lamina that occupies the region D bounded by the graphs of $x = y^2$ and the line $x = y + 2$. The density of the lamina is given by $\rho(x, y) = y^2$.

October 27: Applications of Double Integrals: Center of Mass

Applications of Double Integrals: Center of Mass

Lecture Notes: Finding the Balancing Point

Learning Objectives

By the end of this lesson, you will be able to:

- Understand the physical concept of center of mass.
- Use double integrals to calculate the moments of a lamina about the x and y axes.
- Derive and apply the formulas to find the center of mass of a lamina with variable density.

1. Moments and Center of Mass

In the previous lecture, we learned how to find the total mass of a lamina (a thin flat plate) with a variable density function $\rho(x, y)$. The **center of mass** is the point $(\bar{x}, \bar{y})$ where the lamina would be perfectly balanced if it were placed on a pinpoint.

To find this balancing point, we introduce the concept of **moments**. The moment of an object measures its tendency to rotate about an axis.

- The **moment about the x-axis**, denoted M_x, measures the tendency to rotate about the x-axis. A small piece of mass $\rho(x, y)\, dA$ at a distance y from the x-axis contributes $y \cdot (\rho(x, y)\, dA)$ to this moment.
- The **moment about the y-axis**, denoted M_y, measures the tendency to rotate about the y-axis. A small piece of mass at a distance x from the y-axis contributes $x \cdot (\rho(x, y)\, dA)$ to this moment.

To find the total moments, we integrate these contributions over the entire lamina D.

Formulas for Moments and Center of Mass

For a lamina occupying a region D with density function $\rho(x, y)$:

1. **Mass:** $m = \iint_D \rho(x, y)\, dA$
2. **Moment about the x-axis:** $M_x = \iint_D y \cdot \rho(x, y)\, dA$
3. **Moment about the y-axis:** $M_y = \iint_D x \cdot \rho(x, y)\, dA$
4. **Center of Mass $(\bar{x}, \bar{y})$:**
$$\bar{x} = \frac{M_y}{m} \quad \text{and} \quad \bar{y} = \frac{M_x}{m}$$

Think of the center of mass formulas as weighted averages. $\bar{x}$ is the average x-coordinate weighted by mass, and $\bar{y}$ is the average y-coordinate weighted by mass.

2. Comprehensive Example: Center of Mass

Example 1: Center of Mass Calculation

A lamina occupies the region D in the first quadrant bounded by the parabola $y = x^2$ and the line $y = 4$. The density of the lamina is proportional to the distance from the x-axis, so $\rho(x, y) = ky$ for some constant k. Find the center of mass $(\bar{x}, \bar{y})$.

Solution: First, let's sketch the region D. The region is bounded by $y = x^2$, $y = 4$, and the y-axis ($x = 0$). We can treat this as either a Type I or Type II region. Let's use Type II, as the y-limits are constant. The intersection of $y = x^2$ and $y = 4$ is at $x = 2$.

$$D = \{(x, y) \mid 0 \leq y \leq 4, \quad 0 \leq x \leq \sqrt{y}\}$$

The density is $\rho(x, y) = ky$. We need to calculate m, M_x, and M_y.

Step 1: Calculate the Mass (m)

$$m = \iint_D \rho(x, y)\, dA = \int_0^4 \int_0^{\sqrt{y}} ky\, dx\, dy$$

First, integrate with respect to x:

$$\begin{aligned}\int_0^{\sqrt{y}} ky\, dx &= ky \int_0^{\sqrt{y}} 1\, dx \\ &= ky[x]_{x=0}^{x=\sqrt{y}} \\ &= ky(\sqrt{y} - 0) = ky^{3/2}\end{aligned}$$

Now, integrate the result with respect to y:

$$\begin{aligned}m = \int_0^4 ky^{3/2}\, dy &= k\left[\frac{y^{5/2}}{5/2}\right]_{y=0}^{y=4} \\ &= \frac{2k}{5}[y^{5/2}]_{y=0}^{y=4} \\ &= \frac{2k}{5}(4^{5/2} - 0^{5/2}) \\ &= \frac{2k}{5}((4^{1/2})^5) = \frac{2k}{5}(2^5) = \frac{2k}{5}(32) = \frac{64k}{5}\end{aligned}$$

Step 2: Calculate Moment about the x-axis (M_x)

$$M_x = \iint_D y \cdot \rho(x, y)\, dA = \int_0^4 \int_0^{\sqrt{y}} y(ky)\, dx\, dy = \int_0^4 \int_0^{\sqrt{y}} ky^2\, dx\, dy$$

First, integrate with respect to x:

$$\begin{aligned}\int_0^{\sqrt{y}} ky^2\, dx &= ky^2[x]_{x=0}^{x=\sqrt{y}} \\ &= ky^2(\sqrt{y} - 0) = ky^{5/2}\end{aligned}$$

Now, integrate the result with respect to y:

$$\begin{aligned} M_x = \int_0^4 ky^{5/2}\,dy &= k\left[\frac{y^{7/2}}{7/2}\right]_{y=0}^{y=4} \\ &= \frac{2k}{7}[y^{7/2}]_{y=0}^{y=4} \\ &= \frac{2k}{7}(4^{7/2}-0) = \frac{2k}{7}(2^7) = \frac{2k}{7}(128) = \frac{256k}{7} \end{aligned}$$

Step 3: Calculate Moment about the y-axis (M_y)

$$M_y = \iint_D x\cdot\rho(x,y)\,dA = \int_0^4\int_0^{\sqrt{y}} x(ky)\,dx\,dy = \int_0^4\int_0^{\sqrt{y}} kxy\,dx\,dy$$

First, integrate with respect to x:

$$\begin{aligned} \int_0^{\sqrt{y}} kxy\,dx &= ky\int_0^{\sqrt{y}} x\,dx \\ &= ky\left[\frac{1}{2}x^2\right]_{x=0}^{x=\sqrt{y}} \\ &= \frac{ky}{2}((\sqrt{y})^2-0^2) = \frac{ky}{2}(y) = \frac{k}{2}y^2 \end{aligned}$$

Now, integrate the result with respect to y:

$$\begin{aligned} M_y = \int_0^4 \frac{k}{2}y^2\,dy &= \frac{k}{2}\left[\frac{1}{3}y^3\right]_{y=0}^{y=4} \\ &= \frac{k}{6}[y^3]_{y=0}^{y=4} \\ &= \frac{k}{6}(4^3-0^3) = \frac{k}{6}(64) = \frac{32k}{3} \end{aligned}$$

Step 4: Find the Center of Mass $(\bar{x},\bar{y})$

$$\bar{x} = \frac{M_y}{m} = \frac{32k/3}{64k/5} = \frac{32k}{3}\cdot\frac{5}{64k} = \frac{32}{64}\cdot\frac{5}{3} = \frac{1}{2}\cdot\frac{5}{3} = \frac{5}{6}$$

$$\bar{y} = \frac{M_x}{m} = \frac{256k/7}{64k/5} = \frac{256k}{7}\cdot\frac{5}{64k} = \frac{256}{64}\cdot\frac{5}{7} = 4\cdot\frac{5}{7} = \frac{20}{7}$$

The center of mass is at $(\frac{5}{6},\frac{20}{7})$. Notice that the constant of proportionality k cancelled out, which will always happen.

Classwork: Center of Mass

Name: _____

Instructions: For each problem, find the mass and the center of mass $(\bar{x},\bar{y})$ of the lamina. Sketch the region of integration and show all your work clearly.

Problem 1: A Rectangular Lamina

A lamina occupies the rectangular region $D = [0, 4] \times [0, 2]$. The density of the lamina at any point (x, y) is given by $\rho(x, y) = y$.

Problem 2: A Triangular Lamina

A lamina occupies the triangular region D bounded by the lines $y = x$, $y = 0$, and $x = 2$. The density of the lamina is given by $\rho(x, y) = x$.

October 29: Integration in Polar Coordinates

Integration in Polar Coordinates
Lecture Notes: Circles, Sectors, and Beyond

Learning Objectives

By the end of this lesson, you will be able to:

- Convert double integrals from rectangular to polar coordinates.
- Set up and evaluate double integrals over regions described by polar curves.
- Apply polar integration to find physical quantities like mass and volume.

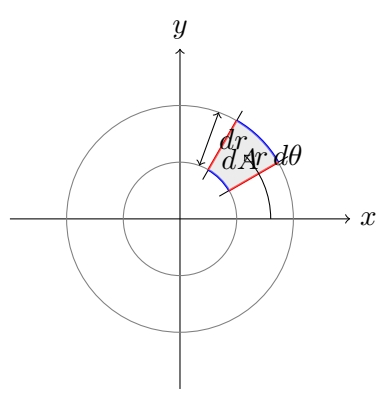
1. Double Integrals in Polar Coordinates

Some regions, like circles, sectors, or cardioids, are very difficult to describe with rectangular coordinates but are quite simple in polar coordinates. A polar rectangle is a region where r and θ are bounded by constants: $R = \{(r, \theta) \mid a \leq r \leq b, \alpha \leq \theta \leq \beta\}$.

When we switch from Cartesian coordinates (x, y) to polar coordinates (r, θ), we must change three things: the function, the limits of integration, and the area element dA. The coordinate conversion is:

$$x = r\cos\theta \quad \text{and} \quad y = r\sin\theta$$

A crucial difference is how we calculate the area element. A small rectangle in the Cartesian plane has area $dA = dx\,dy$. In the polar plane, a small "polar rectangle" has a different area.



The area of this small polar region is approximately that of a rectangle with side lengths dr and $r\,d\theta$. So, we have $dA \approx (dr)(r\,d\theta) = r\,dr\,d\theta$.

The extra factor of r is a geometric scaling factor that accounts for the fact that the area of a polar sector changes as you move away from the origin. The formal reason for this factor comes from a concept called the **Jacobian**, which we will cover in a future lecture. For now, it is critical to remember to include it.

Change of Variables to Polar Coordinates

To change a double integral to polar coordinates over a region D:

$$\iint_D f(x,y)\,dA = \iint_D f(r\cos\theta, r\sin\theta)\,r\,dr\,d\theta$$

Checklist:

- Replace x with $r\cos\theta$ and y with $r\sin\theta$ in the function f.
- Replace dA with $r\,dr\,d\theta$. **Do not forget the r!**
- Change the limits of integration to describe the region D in terms of r and θ.

2. Examples

Example 1: Volume of a Paraboloid

Find the volume of the solid that lies under the circular paraboloid $z = 9 - x^2 - y^2$ and above the xy-plane.

Solution: The volume is the double integral of the height function $f(x,y) = z = 9 - x^2 - y^2$ over the region D on the xy-plane. The solid intersects the xy-plane when $z = 0$, so $x^2 + y^2 = 9$. The region of integration D is a circular disk of radius 3 centered at the origin.

Step 1: Convert the function and region to polar coordinates. The function is $f(x,y) = 9 - (x^2 + y^2)$. Since $r^2 = x^2 + y^2$, this becomes $f(r,\theta) = 9 - r^2$. The region D is a disk of radius 3, described by constant bounds:

$$D = \{(r,\theta) \mid 0 \le r \le 3, \quad 0 \le \theta \le 2\pi\}$$

Step 2: Set up the integral. The volume V is given by:

$$V = \iint_D (9 - x^2 - y^2)\,dA = \int_0^{2\pi}\int_0^3 (9 - r^2)\cdot(r\,dr\,d\theta)$$

$$V = \int_0^{2\pi}\int_0^3 (9r - r^3)\,dr\,d\theta$$

Step 3: Evaluate the integral. First, the inner integral with respect to r:

$$\begin{aligned}\int_0^3 (9r - r^3)\,dr &= \left[9\frac{r^2}{2} - \frac{r^4}{4}\right]_{r=0}^{r=3} \\ &= \left(\frac{81}{2} - \frac{81}{4}\right) - (0) = \frac{81}{4}\end{aligned}$$

Now, the outer integral with respect to θ:

$$V = \int_0^{2\pi} \frac{81}{4}\,d\theta = \frac{81}{4}[\theta]_0^{2\pi} = \frac{81}{4}(2\pi - 0) = \frac{81\pi}{2}$$

Example 2: Mass of a Lamina on a Sector

A thin plate covers the region D inside the circle $x^2+y^2=4$ in the first quadrant. The density of the plate is given by $\rho(x,y)=y$. Find the total mass of the plate.

Solution: The mass m is the double integral of the density function $\rho(x,y)=y$ over the specified region D.

Step 1: Convert the function and region to polar coordinates. The density function is $\rho(x,y)=y$, which in polar coordinates becomes $\rho(r,\theta)=r\sin\theta$. The region D is a quarter-disk of radius 2 in the first quadrant. This is described by:

$$D=\{(r,\theta)\mid 0\le r\le 2,\quad 0\le\theta\le\pi/2\}$$

Step 2: Set up the integral. The mass m is given by the integral of the density function.

$$m=\iint_D y\,dA=\int_0^{\pi/2}\int_0^2(r\sin\theta)\cdot(r\,dr\,d\theta)$$

$$m=\int_0^{\pi/2}\int_0^2 r^2\sin\theta\,dr\,d\theta$$

Step 3: Evaluate the integral. First, we evaluate the inner integral with respect to r. We treat $\sin\theta$ as a constant.

$$\begin{aligned}\int_0^2 r^2\sin\theta\,dr&=\sin\theta\int_0^2 r^2\,dr\\&=\sin\theta\left[\frac{r^3}{3}\right]_{r=0}^{r=2}\\&=\sin\theta\left(\frac{2^3}{3}-0\right)=\frac{8}{3}\sin\theta\end{aligned}$$

Now, we evaluate the outer integral with respect to θ:

$$\begin{aligned}m=\int_0^{\pi/2}\frac{8}{3}\sin\theta\,d\theta&=\frac{8}{3}[-\cos\theta]_0^{\pi/2}\\&=\frac{8}{3}\left((-\cos(\pi/2))-(-\cos(0))\right)\\&=\frac{8}{3}\left((0)-(-1)\right)=\frac{8}{3}\end{aligned}$$

The total mass of the lamina is $\frac{8}{3}$.

Classwork: Double Integrals in Polar Coordinates

Name: _____

Instructions: For each problem, sketch the region of integration. Then, set up the integral in polar coordinates and evaluate it. Show your work clearly.

Problem 1: Integral over a Disk

Evaluate the integral $\iint_D(x^2+y^2)\,dA$ where D is the disk of radius 2 centered at the origin (i.e., $x^2+y^2\le 4$).

Problem 2: Mass on a Semicircle

Find the mass of a lamina that occupies the upper half of the disk $x^2 + y^2 \leq 9$ (i.e., $y \geq 0$). The density of the lamina is constant, $\rho(x, y) = 5$.

Problem 3: Volume over an Annulus

Find the volume of the solid under the cone $z = \sqrt{x^2 + y^2}$ and above the annular region D given by $1 \leq x^2 + y^2 \leq 16$. (An annulus is the region between two circles).

Problem 4: Integral over a Sector

Evaluate the integral $\iint_D x\,dA$ where D is the region in the first quadrant bounded by the circle $x^2 + y^2 = 4$ and the lines $y = 0$ and $y = x$.

October 31: Change of Variables and the Jacobian Determinant

Change of Variables and the Jacobian Determinant

Lecture Notes: The Geometry of Transformations

Learning Objectives

By the end of this lesson, you will be able to:

- Understand the geometric motivation for the Jacobian determinant.
- Define the Jacobian matrix and calculate its determinant for transformations in 2D and 3D.
- Use the Jacobian to change variables in double and triple integrals.
- Apply a change of variables to simplify the evaluation of integrals over complex regions.

1. The Jacobian in Two Dimensions

When we evaluate a double integral $\iint_R f(x, y)\,dA$, the process can be difficult if either the function $f(x, y)$ or the region R is complicated. A change of variables, or transformation, can often simplify both.

Consider a transformation T from a uv-coordinate system to an xy-coordinate system, defined by the functions:

$$x = x(u, v) \quad \text{and} \quad y = y(u, v)$$

This transformation maps a region S in the uv-plane to a region R in the xy-plane. Our goal is to relate the area element $dA = dx\,dy$ to an area element $du\,dv$.

Geometric Intuition

An infinitesimal rectangle in the uv-plane with area $du\,dv$ is not, in general, mapped to a rectangle in the xy-plane. Instead, it is mapped to a small curvilinear shape that can be approximated by a parallelogram.

Let $\vec{r}(u, v) = \langle x(u, v), y(u, v)\rangle$. The sides of the infinitesimal rectangle in the uv-plane can be represented by the vectors $\langle du, 0\rangle$ and $\langle 0, dv\rangle$. The transformation T maps these sides to curves in the xy-plane. The vectors tangent to these curves are:

- $\vec{r}_u = \frac{\partial \vec{r}}{\partial u} = \left\langle \frac{\partial x}{\partial u}, \frac{\partial y}{\partial u} \right\rangle$ (tangent to a curve of constant v)
- $\vec{r}_v = \frac{\partial \vec{r}}{\partial v} = \left\langle \frac{\partial x}{\partial v}, \frac{\partial y}{\partial v} \right\rangle$ (tangent to a curve of constant u)

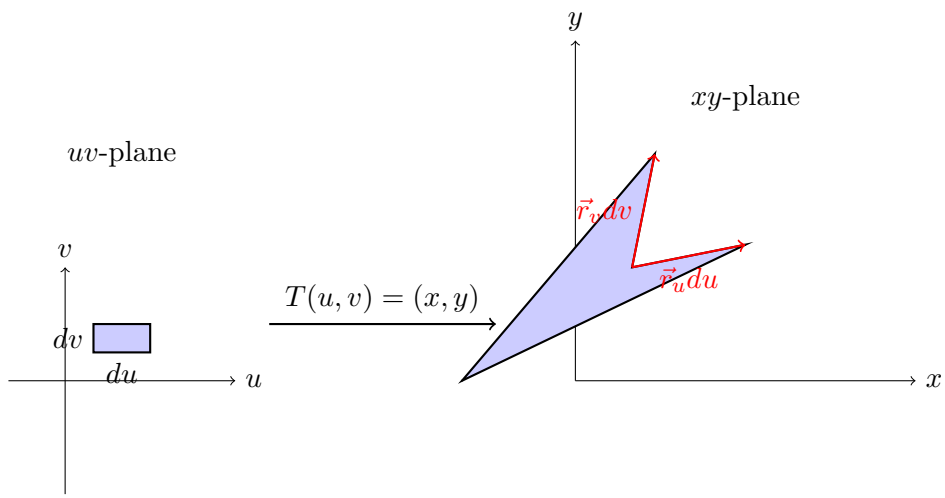
The sides of the parallelogram in the xy-plane are approximated by the vectors $\vec{r}_u du$ and $\vec{r}_v dv$. The area of this parallelogram, dA, is given by the magnitude of the cross product of these vectors (if we embed them in 3D space with $z = 0$).

$$dA \approx |(\vec{r}_u du) \times (\vec{r}_v dv)| = |\vec{r}_u \times \vec{r}_v|\,du\,dv$$

The cross product is:

$$\vec{r}_u \times \vec{r}_v = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} & 0 \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} & 0 \end{vmatrix} = \left(\frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u} \right) \vec{k}$$

The magnitude of this vector is the absolute value of the scalar component. This expression is precisely the determinant of the **Jacobian matrix**.



The **Jacobian determinant** of the transformation is denoted $\frac{\partial(x,y)}{\partial(u,v)}$ and is defined as:

$$\frac{\partial(x,y)}{\partial(u,v)} = \det \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix} = \frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u}$$

The change of the area element is given by the absolute value of this determinant:

$$dA = dx\,dy = \left| \frac{\partial(x,y)}{\partial(u,v)} \right| du\,dv$$

Example: Polar Coordinates

For the transformation from polar (r,θ) to Cartesian (x,y) coordinates, we have $u = r, v = \theta$:

$$x = r\cos\theta \quad \text{and} \quad y = r\sin\theta$$

The Jacobian determinant is:

$$\frac{\partial(x,y)}{\partial(r,\theta)} = \det \begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{pmatrix} = \det \begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix} = (r\cos^2\theta) - (-r\sin^2\theta) = r(\cos^2\theta + \sin^2\theta) = r$$

Thus, the area element is $dx\,dy = |r|\,dr\,d\theta$. Since we typically take $r \geq 0$, this becomes the familiar $dx\,dy = r\,dr\,d\theta$.

2. The Jacobian in Three Dimensions

The concept extends directly to triple integrals. Consider a transformation from a uvw-coordinate system to an xyz-system:

$$x = x(u, v, w), \quad y = y(u, v, w), \quad z = z(u, v, w)$$

This transformation maps an infinitesimal box in uvw-space to a small parallelepiped in xyz-space. The volume of this parallelepiped, dV, is given by the absolute value of the scalar triple product of the tangent vectors that form its edges: $|(\vec{r}_u du) \cdot ((\vec{r}_v dv) \times (\vec{r}_w dw))|$. This corresponds to the absolute value of the determinant of the 3x3 matrix formed by these vectors.

The **Jacobian determinant** for a 3D transformation is:

$$\frac{\partial(x, y, z)}{\partial(u, v, w)} = \det \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{pmatrix}$$

The volume element transforms as:

$$dV = dx\, dy\, dz = \left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| du\, dv\, dw$$

Example: Spherical Coordinates

The transformation from spherical (ρ, θ, ϕ) to Cartesian (x, y, z) coordinates is:

$$x = \rho \sin\phi \cos\theta, \quad y = \rho \sin\phi \sin\theta, \quad z = \rho \cos\phi$$

The Jacobian determinant is (after a lengthy but straightforward calculation):

$$\frac{\partial(x, y, z)}{\partial(\rho, \theta, \phi)} = \det \begin{pmatrix} \sin\phi\cos\theta & -\rho\sin\phi\sin\theta & \rho\cos\phi\cos\theta \\ \sin\phi\sin\theta & \rho\sin\phi\cos\theta & \rho\cos\phi\sin\theta \\ \cos\phi & 0 & -\rho\sin\phi \end{pmatrix} = -\rho^2 \sin\phi$$

The volume element is $dV = |-\rho^2 \sin\phi|\, d\rho\, d\theta\, d\phi$. Since $\rho \geq 0$ and $\phi \in [0, \pi]$ (so $\sin\phi \geq 0$), this simplifies to $dV = \rho^2 \sin\phi\, d\rho\, d\theta\, d\phi$.

3. The Change of Variables Formula

Change of Variables in Multiple Integrals

Let T be a transformation that maps a region S in the uv-plane to a region R in the xy-plane. Then:

$$\iint_R f(x, y)\, dx\, dy = \iint_S f(x(u, v), y(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du\, dv$$

Similarly, for a transformation from uvw-space to xyz-space:

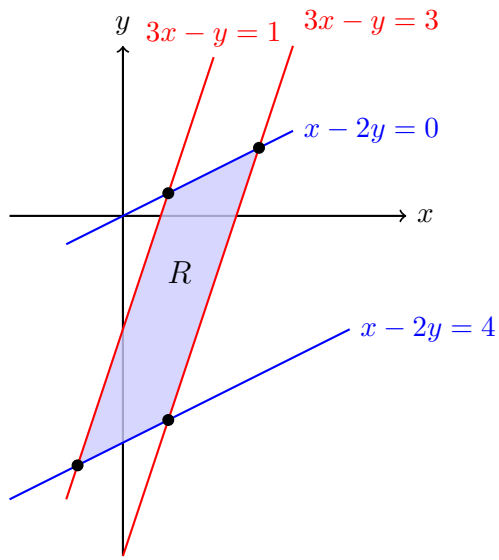
$$\iiint_R f(x, y, z)\, dx\, dy\, dz = \iiint_S f(x(u, v, w), \dots) \left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| du\, dv\, dw$$

4. Worked Example

Example: Integral over a Parallelogram

Evaluate $\iint_R (x-2y)^2\,dA$ where R is the parallelogram enclosed by the lines $x-2y=0$, $x-2y=4$, $3x-y=1$, and $3x-y=3$.

Solution: The region of integration R is a parallelogram in the xy-plane.



The region R is complicated to integrate over using Cartesian coordinates. We can simplify it with the transformation:

$$u = x-2y \quad \text{and} \quad v = 3x-y$$

In the uv-plane, the region of integration S is a simple rectangle defined by the transformed boundaries: $0 \le u \le 4$ and $1 \le v \le 3$.

To find the Jacobian $\frac{\partial(x,y)}{\partial(u,v)}$, we must first express x and y in terms of u and v. We solve the system of linear equations:

$$\begin{aligned} u &= x-2y \\ v &= 3x-y \end{aligned}$$

From the second equation, we have $y = 3x-v$. Substituting this into the first equation:

$$u = x-2(3x-v) = x-6x+2v = -5x+2v \implies 5x = 2v-u \implies x = -\frac{1}{5}u+\frac{2}{5}v$$

Now substitute x back into the expression for y:

$$y = 3\left(-\frac{1}{5}u+\frac{2}{5}v\right)-v = -\frac{3}{5}u+\frac{6}{5}v-v = -\frac{3}{5}u+\frac{1}{5}v$$

So our transformation from the uv-plane to the xy-plane is:

$$x(u,v) = -\frac{1}{5}u+\frac{2}{5}v \quad \text{and} \quad y(u,v) = -\frac{3}{5}u+\frac{1}{5}v$$

Now we can compute the partial derivatives and find the Jacobian determinant:

$$\frac{\partial(x,y)}{\partial(u,v)} = \det\begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix} = \det\begin{pmatrix} -1/5 & 2/5 \\ -3/5 & 1/5 \end{pmatrix} = \left(-\frac{1}{5}\right)\left(\frac{1}{5}\right)-\left(\frac{2}{5}\right)\left(-\frac{3}{5}\right) = -\frac{1}{25}+\frac{6}{25} = \frac{5}{25} = \frac{1}{5}$$

The absolute value is $\left|\frac{1}{5}\right| = \frac{1}{5}$. Now we can transform the integral. The integrand $(x-2y)^2$ becomes simply u^2.

$$\begin{aligned}
\iint_R (x-2y)^2\,dA &= \iint_S u^2 \left|\frac{\partial(x,y)}{\partial(u,v)}\right| du\,dv \\
&= \int_1^3 \int_0^4 u^2 \left(\frac{1}{5}\right) du\,dv \\
&= \frac{1}{5}\int_1^3 \left[\frac{u^3}{3}\right]_{u=0}^{u=4} dv \\
&= \frac{1}{5}\int_1^3 \left(\frac{64}{3} - 0\right) dv \\
&= \frac{64}{15}\int_1^3 dv = \frac{64}{15}[v]_1^3 = \frac{64}{15}(3-1) = \frac{128}{15}
\end{aligned}$$

Classwork: Calculating the Jacobian Determinant

Name: _____

Instructions: For each transformation, compute the Jacobian determinant $\dfrac{\partial(x,y)}{\partial(u,v)}$. *Show your work clearly on a separate sheet of paper.*

Problem 1: A Linear Transformation

Consider the transformation given by:

$$x = 3u + 2v \quad \text{and} \quad y = u - 4v$$

Compute the Jacobian determinant $\dfrac{\partial(x,y)}{\partial(u,v)}$.

Problem 2: Polar Coordinates

For the transformation from polar coordinates (r,θ) to Cartesian coordinates (x,y), we have:

$$x = r\cos\theta \quad \text{and} \quad y = r\sin\theta$$

Compute the Jacobian determinant $\dfrac{\partial(x,y)}{\partial(r,\theta)}$.

Problem 3: A Non-linear Transformation

Consider the transformation given by:

$$x = u^2 - v^2 \quad \text{and} \quad y = 2uv$$

Compute the Jacobian determinant $\dfrac{\partial(x, y)}{\partial(u, v)}$.

Problem 4: A Scaling Transformation

Consider the transformation given by:

$$x = 5u \quad \text{and} \quad y = -2v$$

Compute the Jacobian determinant $\dfrac{\partial(x, y)}{\partial(u, v)}$.

Homework 8

Homework 8: Second Derivative Test & Multiple Integrals

Problem 1

Find all critical points of the function $f(x, y) = x^2 + y^2 - 6x + 2y + 7$ and classify each as a local maximum, local minimum, or saddle point.

Problem 2

Find all critical points of the function $f(x, y) = xy - x^2 - y^2$ and classify each as a local maximum, local minimum, or saddle point.

Problem 3

Find all critical points of the function $f(x, y) = x^3 - 12xy + 8y^3$ and classify each as a local maximum, local minimum, or saddle point.

Problem 4

Find all critical points of the function $f(x, y) = e^{2x}\cos(y)$ and classify each as a local maximum, local minimum, or saddle point. (Note: there are infinitely many critical points; describe the general form and test one.)

Problem 5

Find all critical points of the function $f(x, y) = y\ln(x) - x - 2y$ and classify each as a local maximum, local minimum, or saddle point.

Problem 6

Find all critical points of the function $f(x, y) = x^3 - 3x + y^3 - 12y$ and classify each as a local maximum, local minimum, or saddle point. (This function has four critical points.)

Problem 7

Find all critical points of the function $f(x, y) = x^2y^2 - 4x^2 - y^2$ and classify each as a local maximum, local minimum, or saddle point. (This function has five critical points.)

Problem 8

Construct a function $f(x, y)$ that has a single critical point at $(1, 2)$ which is a local minimum. Verify using the Second Derivative Test that your function satisfies these conditions.

Problem 9

Evaluate the double integral $\iint_R 7\,dA$ where the region R is the rectangle $R = [-1, 3] \times [2, 5]$.

Problem 10

Evaluate the iterated integral $\int_0^2 \int_1^3 (x^2 + y)\,dy\,dx$.

Problem 11

Evaluate the double integral $\iint_R (2x - 4y^3)\,dA$ where $R = [0, 1] \times [0, 2]$.

Problem 12

Evaluate the double integral $\iint_R \frac{xy^2}{x^2+1}\,dA$ where $R = [0, 1] \times [-3, 3]$.

Problem 13

Evaluate $\iint_R 2xy\,dA$ where $R = [0, 1] \times [0, 2]$ by computing the iterated integral in **both** possible orders ($dy\,dx$ and $dx\,dy$) to explicitly verify Fubini's Theorem.

Problem 14

Evaluate the double integral $\iint_R y\cos(xy)\,dA$ where $R = [0, 1] \times [0, \pi]$. Choose the easier order of integration.

Problem 15

Evaluate the double integral $\iint_R xe^{xy}\,dA$ where $R = [0,2] \times [0,1]$. Choose the easier order of integration.

Problem 16

Evaluate the double integral $\iint_R y\sec^2(xy)\,dA$ where $R = [0,1] \times [0,\pi/4]$. Choose the easier order of integration.

Problem 17

Evaluate the iterated integral $\int_0^1 \int_{-1}^0 \int_1^2 (x + 2y - z)\, dz\, dy\, dx$.

Problem 18

Evaluate the triple integral $\iiint_B (y+z)^2\, dV$ where the box B is given by $B = [0,2] \times [0,1] \times [0,1]$.

Problem 19

Evaluate the triple integral $\iiint_B z e^{x+y}\, dV$ where $B = [0,1] \times [0,1] \times [0,1]$.

Problem 20

A solid box is bounded by the planes $x = 0, x = 2, y = 0, y = 1, z = 0, z = 1$. The density of the box at any point (x, y, z) is given by the function $\rho(x, y, z) = 1 + 2y$. Find the total mass of the box by setting up and evaluating a triple integral.

Quiz 8

Quiz 9: Double Integrals

Name: _____

Problem 1: Double Integral over a General Region

Find the volume of the solid that lies under the plane $z = x + 1$ and above the region D in the xy-plane bounded by the parabola $y = x^2$ and the line $y = 4$.

Problem 2: Double Integral in Polar Coordinates

Evaluate the integral $\iint_D (x^2 + y^2)^{3/2}\, dA$ where D is the region in the first quadrant bounded by the circle $x^2 + y^2 = 4$.

Problem 3: Mass of a Lamina

Find the mass of the lamina that occupies the triangular region D with vertices $(0, 0)$, $(2, 0)$, and $(0, 4)$, if the density function is given by $\rho(x, y) = y$.

CHAPTER 9

Cycle 9: Double Integrals over Planar Regions (Homework 9)

Overview

This cycle introduces double integrals over rectangles and more general planar regions. Iterated integrals and region descriptions are used to compute areas, masses, and other quantities depending on two variables.

Lecture Notes and Classwork

November 3: Line Integrals (Work & Flux in 2D)

Line Integrals (Work & Flux in 2D)
Lecture Notes: Vector Calculus

Learning Objectives

By the end of this lesson, you will be able to:

- Evaluate the line integral for work, $\int_C \vec{F} \cdot d\vec{r}$, by parameterizing the curve C.
- Compute the circulation of a vector field $\vec{F}$ around a closed loop.
- Understand the concept of flux and compute the flux of a vector field $\vec{F}$ across a curve C.

Introduction to Vector Fields

A **vector field** is a function that assigns a vector to each point in its domain.

This function can represent forces or flows. For example:

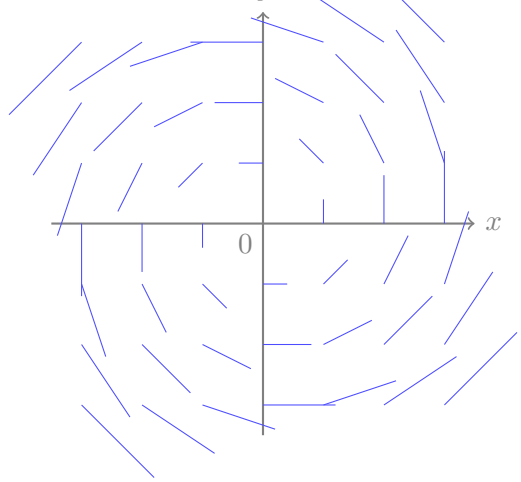
- The velocity of water at every point in a river.
- The direction and strength of wind at every point in the atmosphere.

- The gravitational or electric force experienced by a particle at any point in space.

In two dimensions, a vector field $\vec{F}$ is a function $\vec{F}(x, y) = \langle P(x, y), Q(x, y)\rangle$.

We visualize a vector field by drawing the vector $\vec{F}(x, y)$ based at the point (x, y) for a selection of points.

Vector Field: $\vec{F}(x, y) = \langle -y, x\rangle$



1. Line Integrals for Work

The line integral for work measures how much a vector field $\vec{F}$ helps or hinders movement along an oriented curve C.

On each infinitesimally small piece of the curve, represented by the displacement vector $d\vec{r}$, the work done is the component of the force in the direction of that displacement.

This small amount of work is $dW = \vec{F} \cdot d\vec{r}$.

To find the total work, we sum these contributions by integrating over the curve C.

Definition: Line Integral for Work

The work done by a continuous vector field $\vec{F}$ along a smooth, oriented curve C parameterized by $\vec{r}(t)$ from $t = a$ to $t = b$ is:

$$W = \int_C \vec{F} \cdot d\vec{r} = \int_a^b \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t)\, dt$$

To compute the line integral for work, we follow these steps line-by-line:

1. **Parameterize the curve C**: Find a vector function $\vec{r}(t) = \langle x(t), y(t)\rangle$ for $a \le t \le b$.
2. **Find the derivative $\vec{r}'(t)$**: Compute $\vec{r}'(t) = \langle x'(t), y'(t)\rangle$. This is the tangent vector.
3. **Express $\vec{F}$ in terms of t**: Substitute $x(t)$ and $y(t)$ into $\vec{F}$ to get $\vec{F}(\vec{r}(t))$.
4. **Compute the dot product**: Calculate the scalar function $\vec{F}(\vec{r}(t)) \cdot \vec{r}'(t)$.
5. **Integrate**: Evaluate the definite integral of the resulting scalar function with respect to t from a to b.

Example: Work along a Parabola

Calculate the work done by the vector field $\vec{F}(x, y) = \langle -y, x\rangle$ in moving a particle along the parabola $y = x^2$ from $(0, 0)$ to $(2, 4)$.

Solution:

First, we parameterize the curve C. Let $x = t$, which implies $y = t^2$. The starting point $(0, 0)$ corresponds to $t = 0$. The ending point $(2, 4)$ corresponds to $t = 2$. So, our parameterization is $\vec{r}(t) = \langle t, t^2\rangle$ for $0 \le t \le 2$.

Next, we find the derivative of the parameterization. $\vec{r}'(t) = \langle \frac{d}{dt}(t), \frac{d}{dt}(t^2)\rangle = \langle 1, 2t\rangle$.

Now, we evaluate the vector field $\vec{F}$ on the curve. $\vec{F}(\vec{r}(t)) = \vec{F}(t, t^2) = \langle -t^2, t\rangle$.

We then compute the dot product of the field and the tangent vector. $\vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) = \langle -t^2, t\rangle \cdot \langle 1, 2t\rangle = (-t^2)(1) + (t)(2t) = -t^2 + 2t^2 = t^2$.

Finally, we integrate this expression from $t = 0$ to $t = 2$. $W = \int_0^2 t^2\, dt = \left[\frac{t^3}{3}\right]_0^2 = \frac{2^3}{3} - \frac{0^3}{3} = \frac{8}{3}$.

2. Circulation

Definition: Circulation

When the curve C is a closed loop, the line integral for work is called the **circulation** of $\vec{F}$ around C. It is denoted using a circle on the integral sign:

$$\text{Circulation} = \oint_C \vec{F} \cdot d\vec{r}$$

Circulation measures the net tendency of the vector field to flow along the curve.

Example: Circulation on a Piecewise Path

Calculate the circulation of $\vec{F}(x, y) = \langle xy, y^2\rangle$ around the triangle C with vertices $(0, 0)$, $(1, 0)$, and $(1, 2)$, traversed counter-clockwise.

Solution: The curve C is made of three segments, C_1, C_2, and C_3. The total circulation is the sum of the integrals over each segment.

$$\oint_C \vec{F} \cdot d\vec{r} = \int_{C_1} \vec{F} \cdot d\vec{r} + \int_{C_2} \vec{F} \cdot d\vec{r} + \int_{C_3} \vec{F} \cdot d\vec{r}$$

Path C_1: From $(0, 0)$ to $(1, 0)$. $\vec{r}_1(t) = \langle t, 0\rangle$ for $0 \le t \le 1$. $\vec{r}_1'(t) = \langle 1, 0\rangle$. $\vec{F}(\vec{r}_1(t)) = \langle (t)(0), 0^2\rangle = \langle 0, 0\rangle$. $\int_{C_1} \vec{F} \cdot d\vec{r} = \int_0^1 \langle 0, 0\rangle \cdot \langle 1, 0\rangle\, dt = 0$.

Path C_2: From $(1, 0)$ to $(1, 2)$. $\vec{r}_2(t) = \langle 1, t\rangle$ for $0 \le t \le 2$. $\vec{r}_2'(t) = \langle 0, 1\rangle$. $\vec{F}(\vec{r}_2(t)) = \langle (1)(t), t^2\rangle = \langle t, t^2\rangle$. $\int_{C_2} \vec{F} \cdot d\vec{r} = \int_0^2 \langle t, t^2\rangle \cdot \langle 0, 1\rangle\, dt = \int_0^2 t^2\, dt = \left[\frac{t^3}{3}\right]_0^2 = \frac{8}{3}$.

Path C_3: From $(1, 2)$ to $(0, 0)$. $\vec{r}_3(t) = (1 - t)\langle 1, 2\rangle + t\langle 0, 0\rangle = \langle 1 - t, 2 - 2t\rangle$ for $0 \le t \le 1$. $\vec{r}_3'(t) = \langle -1, -2\rangle$. $\vec{F}(\vec{r}_3(t)) = \langle (1-t)(2-2t), (2-2t)^2\rangle = \langle 2(1-t)^2, 4(1-t)^2\rangle$. $\vec{F} \cdot \vec{r}_3' = (2(1-t)^2)(-1) + (4(1-t)^2)(-2) = -10(1-t)^2$. $\int_{C_3} \vec{F} \cdot d\vec{r} = \int_0^1 -10(1-t)^2\, dt = -10\left[-\frac{(1-t)^3}{3}\right]_0^1 = \frac{10}{3}(0 - 1) = -\frac{10}{3}$.

Total Circulation: $0 + \frac{8}{3} - \frac{10}{3} = -\frac{2}{3}$.

3. Flux in 2D

While work measures the component of a field *along* a curve, **flux** measures the component of a field *across* a curve.

Definition: Flux Across a Curve

The flux of a vector field $\vec{F}$ across an oriented curve C is:

$$\text{Flux} = \int_C \vec{F} \cdot \vec{n}\, ds$$

Here, $\vec{n}$ is the unit normal vector (perpendicular to the curve, pointing to the right of its orientation) and ds is the differential element of arc length.

Deriving the Computational Formula for Flux

We need a way to compute the flux integral using a parameterization $\vec{r}(t) = \langle x(t), y(t)\rangle$.

Step 1: **Find the Normal Vector**. The tangent vector is $\vec{r}'(t) = \langle x'(t), y'(t)\rangle$. The outward normal vector is $\vec{N}(t) = \langle y'(t), -x'(t)\rangle$.

Step 2: **Find the Unit Normal** $\vec{n}$. $\vec{n}(t) = \frac{\vec{N}(t)}{|\vec{N}(t)|} = \frac{\langle y'(t), -x'(t)\rangle}{|\vec{r}'(t)|}$.

Step 3: **Relate Arc Length** ds **to** dt. $ds = |\vec{r}'(t)|\, dt$.

Step 4: **Assemble the Integral**. Flux $= \int_a^b \vec{F}(\vec{r}(t)) \cdot \left(\frac{\langle y'(t), -x'(t)\rangle}{|\vec{r}'(t)|}\right) (|\vec{r}'(t)|\, dt)$ The $|\vec{r}'(t)|$ terms cancel, leading to: Flux $= \int_a^b \vec{F}(\vec{r}(t)) \cdot \langle y'(t), -x'(t)\rangle\, dt$

Flux Computation Formula

For a curve C parameterized by $\vec{r}(t) = \langle x(t), y(t)\rangle$ and a vector field $\vec{F} = \langle P, Q\rangle$:

$$\text{Flux} = \int_a^b \left(P(x(t), y(t))\frac{dy}{dt} - Q(x(t), y(t))\frac{dx}{dt} \right) dt$$

This is often written in the compact differential form: Flux $= \int_C P\, dy - Q\, dx$.

Example: Flux across a Parabolic Arc

Find the flux of the vector field $\vec{F}(x, y) = \langle y, x\rangle$ across the parabolic arc C given by $y = x^2$ from $(0, 0)$ to $(2, 4)$.

Solution:

First, we parameterize the curve. Let $x = t$, which means $y = t^2$. The path from $(0, 0)$ to $(2, 4)$ corresponds to the interval $0 \le t \le 2$. The parameterization is $\vec{r}(t) = \langle t, t^2\rangle$.

Next, we identify the components and their derivatives. $x(t) = t \implies \frac{dx}{dt} = 1$. $y(t) = t^2 \implies \frac{dy}{dt} = 2t$.

Now, we identify the component functions of the vector field $\vec{F} = \langle P, Q\rangle = \langle y, x\rangle$. $P(x, y) = y$. $Q(x, y) = x$.

We evaluate these component functions along the parameterized curve. $P(x(t), y(t)) = P(t, t^2) = t^2$. $Q(x(t), y(t)) = Q(t, t^2) = t$.

Now, assemble the integrand for the flux formula: $P\frac{dy}{dt} - Q\frac{dx}{dt}$. Integrand $= (t^2)(2t) - (t)(1) = 2t^3 - t$.

Finally, we integrate this polynomial from $t = 0$ to $t = 2$. Flux $= \int_0^2 (2t^3 - t)\,dt = \left[2\frac{t^4}{4} - \frac{t^2}{2}\right]_0^2$ $= \left[\frac{t^4}{2} - \frac{t^2}{2}\right]_0^2 = \left(\frac{2^4}{2} - \frac{2^2}{2}\right) - \left(\frac{0^4}{2} - \frac{0^2}{2}\right) = \left(\frac{16}{2} - \frac{4}{2}\right) - 0 = 8 - 2 = 6.$

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November 10: Conservative Fields and the Fundamental Theorem of Line Integrals

Continuation: 2D Flux (From Nov 3)

3. Flux in 2D (Cont.)

Key Contrast: Work vs. Flux

The two line integrals we have studied are fundamentally different:

- **Work/Circulation** measures flow **ALONG** the curve ($d\vec{r}$ is the tangent vector).

$$\text{Work} = \int_C \vec{F} \cdot d\vec{r} = \int_C P\,dx + Q\,dy$$

- **Flux** measures flow **ACROSS** the curve ($\vec{N}ds$ is the normal vector).

$$\text{Flux} = \int_C \vec{F} \cdot \vec{N}\,ds = \int_C P\,dy - Q\,dx$$

The computational key is the normal vector $\vec{N} = \langle y'(t), -x'(t)\rangle$, which is derived from the tangent vector $\vec{r}'(t) = \langle x'(t), y'(t)\rangle$ by rotating it 90° clockwise (i.e., pointing outward/to the right of the path).

Example: Flux vs. Circulation on a Closed Loop

Let $\vec{F} = \langle -y, x\rangle$ and C be the unit circle $x^2 + y^2 = 1$, traversed counterclockwise.

1. Find the **Circulation** ($\oint_C \vec{F} \cdot d\vec{r}$).
2. Find the **Flux** ($\oint_C \vec{F} \cdot \vec{N}\,ds$).

Solution: Parameterization: $\vec{r}(t) = \langle \cos t, \sin t\rangle$, $0 \le t \le 2\pi$. Derivatives: $x'(t) = -\sin t$, $y'(t) = \cos t$. Field along C: $\vec{F}(\vec{r}(t)) = \langle -\sin t, \cos t\rangle$.

1. **Circulation (Work Form: $\int_C P\,dx + Q\,dy$)**

$$\oint_C \vec{F} \cdot d\vec{r} = \int_0^{2\pi} \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t)\,dt$$

$$\vec{F} \cdot \vec{r}' = \langle -\sin t, \cos t\rangle \cdot \langle -\sin t, \cos t\rangle = \sin^2 t + \cos^2 t = 1$$

$$\text{Circulation} = \int_0^{2\pi} 1\,dt = [t]_0^{2\pi} = 2\pi$$

(The field constantly flows in the direction of the path.)

2. **Flux (Flux Form: $\int_C P\,dy - Q\,dx$)**

$$\oint_C \vec{F}\cdot\vec{N}\,ds = \int_0^{2\pi}\left(P\frac{dy}{dt} - Q\frac{dx}{dt}\right)\,dt$$

$$P = -y = -\sin t, \quad Q = x = \cos t$$

$$\text{Integrand} = (-\sin t)(\cos t) - (\cos t)(-\sin t) = -\sin t\cos t + \sin t\cos t = 0$$

$$\text{Flux} = \int_0^{2\pi} 0\,dt = 0$$

(The field is purely rotational, so there is no net flow across the boundary.)

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Conservative Fields and the Fundamental Theorem of Line Integrals

Lecture Notes: Vector Calculus (Nov 10)

Learning Objectives

By the end of this lesson, you will be able to:

- Determine if a vector field $\vec{F}$ is **conservative**.
- Find the **scalar potential function** f for a conservative field.
- Use the **Fundamental Theorem of Line Integrals (FTC)** to evaluate line integrals.

1. Conservative Vector Fields

Definition: Conservative Field and Potential Function

A vector field $\vec{F}$ is called **conservative** if it is the gradient of some scalar function f. That is,

$$\vec{F} = \nabla f.$$

The scalar function f is called the **potential function** for $\vec{F}$.

In physics, a force field is conservative if the work done by the force in moving an object depends only on the start and end points (e.g., gravity, spring force), not the path taken.

The Cross-Partials Test for Conservatism

The gradient of f is

$$\nabla f = \langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\rangle.$$

If $\vec{F} = \langle P, Q, R\rangle = \nabla f$, then we must have:

$$P = \frac{\partial f}{\partial x}, \quad Q = \frac{\partial f}{\partial y}, \quad R = \frac{\partial f}{\partial z}$$

Since mixed second partial derivatives must be equal (e.g., $f_{xy} = f_{yx}$), we derive the necessary conditions for a field to be conservative.

Theorem: Test for Conservative Vector Field (I.2)

Let $\vec{F} = \langle P, Q \rangle$ be a 2D vector field. If $\vec{F}$ is conservative, then:

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$$

Let $\vec{F} = \langle P, Q, R \rangle$ be a 3D vector field. If $\vec{F}$ is conservative, then the following three cross-partial conditions must hold:

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} \quad \text{and} \quad \frac{\partial P}{\partial z} = \frac{\partial R}{\partial x} \quad \text{and} \quad \frac{\partial Q}{\partial z} = \frac{\partial R}{\partial y}$$

Note: This condition is necessary, and also sufficient if the domain is simply connected (no holes).

Example: Testing for Conservatism

Test if the field $\vec{F} = \langle 2xy, x^2 + 3y^2 \rangle$ is conservative.

Solution: We have $P = 2xy$ and $Q = x^2 + 3y^2$. We check the cross-partials:

$$\frac{\partial P}{\partial y} = \frac{\partial}{\partial y}(2xy) = 2x$$

$$\frac{\partial Q}{\partial x} = \frac{\partial}{\partial x}(x^2 + 3y^2) = 2x$$

Since $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$, the field is **conservative**.

Finding the Potential Function

If $\vec{F} = \langle P, Q \rangle$ is conservative, we find $f(x, y)$ by integrating its partial derivatives.

We use the fact that $\frac{\partial f}{\partial x} = P$ and $\frac{\partial f}{\partial y} = Q$.

Example: Finding a Potential Function

Find the potential function $f(x, y)$ for the conservative field $\vec{F} = \langle 2xy, x^2 + 3y^2 \rangle$.

Solution: 1. **Integrate P with respect to x:**

$$f(x, y) = \int P\,dx = \int 2xy\,dx = x^2y + C(y)$$

(The constant of integration must be a function of y, $C(y)$, since it vanishes when we take $\frac{\partial}{\partial x}$.)

2. **Take $\frac{\partial f}{\partial y}$ and compare to Q:**

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}(x^2y + C(y)) = x^2 + C'(y)$$

We know that $Q = x^2 + 3y^2$. Therefore, we set them equal:

$$x^2 + C'(y) = x^2 + 3y^2$$

3. **Solve for $C(y)$:**

$$C'(y) = 3y^2$$

$$C(y) = \int 3y^2\,dy = y^3 + K$$

(where K is the final constant of integration).

4. Final Potential Function:

$$f(x, y) = x^2y + y^3 + K$$

(Check: $\nabla f = \langle 2xy, x^2 + 3y^2 \rangle = \vec{F}$.)

2. The Fundamental Theorem of Line Integrals (FTC)

The defining property of conservative fields is that their line integrals are path independent.

Theorem: Fundamental Theorem of Line Integrals

Let $\vec{F}$ be a continuous conservative vector field with potential function f (i.e. $\vec{F} = \nabla f$). Let C be a smooth curve parameterized by $\vec{r}(t)$ from $t = a$ (initial point A) to $t = b$ (terminal point B). Then:

$$\int_C \vec{F} \cdot d\vec{r} = \int_C \nabla f \cdot d\vec{r} = f(B) - f(A)$$

This theorem allows us to bypass complex parameterizations and line integral calculations entirely, requiring only the potential function and the endpoints.

Consequences of the FTC

1. **Path Independence:** If $\vec{F}$ is conservative, the integral $\int_C \vec{F} \cdot d\vec{r}$ depends only on the endpoints of C, not the path taken.

2. **Closed Loop Integral:** If $\vec{F}$ is conservative and C is a closed loop (so $A = B$), then the circulation is always zero:

$$\oint_C \vec{F} \cdot d\vec{r} = f(B) - f(A) = 0$$

Example: Applying the FTC

Evaluate $\int_C \vec{F} \cdot d\vec{r}$, where $\vec{F} = \langle 2xy, x^2 + 3y^2 \rangle$ and C is a path from $(1, 1)$ to $(2, 4)$.

Solution:

1. **Confirm Conservative and Find** f: We already confirmed this field is conservative, and found the potential function (using $K = 0$):

$$f(x, y) = x^2y + y^3$$

2. **Identify Endpoints**: Initial Point $P = (1, 1)$. Terminal Point $Q = (2, 4)$.
3. **Apply FTC**:

$$\int_C \vec{F} \cdot d\vec{r} = f(2, 4) - f(1, 1)$$

$$f(2, 4) = (2^2)(4) + (4^3) = 16 + 64 = 80$$

$$f(1, 1) = (1^2)(1) + (1^3) = 1 + 1 = 2$$

$$\int_C \vec{F} \cdot d\vec{r} = 80 - 2 = 78$$

(Notice how this single subtraction replaces a potentially complicated line integral!)

Classwork: Conservative Fields and the Fundamental Theorem

Name: _____

Problem 1: Conservative Field Test

Consider the vector field:

$$\vec{F}(x, y) = \left\langle \frac{1}{y} - 2x, -\frac{x}{y^2} + 2y \right\rangle$$

Determine if $\vec{F}$ is conservative by checking the cross-partials $\frac{\partial P}{\partial y}$ and $\frac{\partial Q}{\partial x}$.

Problem 2: Find Potential Function

Consider the vector field:

$$\vec{F}(x, y) = \left\langle \frac{1}{y} - 2x, -\frac{x}{y^2} + 2y \right\rangle$$

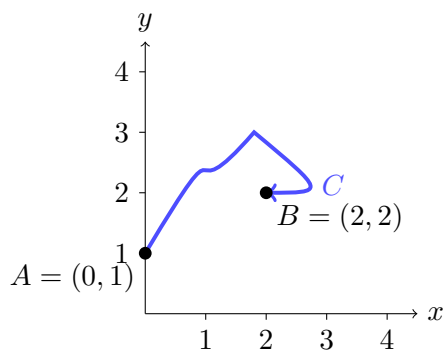
If $\vec{F}$ is conservative (use your result from Problem 1), find the scalar potential function $f(x, y)$ such that $\vec{F} = \nabla f$. (Let the constant of integration be $K = 0$).

Problem 3: Apply Fundamental Theorem (Objective II.2)

Consider the vector field:

$$\vec{F}(x,y) = \left\langle \frac{1}{y} - 2x, -\frac{x}{y^2} + 2y \right\rangle$$

Use the Fundamental Theorem of Line Integrals (and your result from Problem 2) to evaluate the integral $\int_C \vec{F} \cdot d\vec{r}$, where C is the path shown below from $A = (0,1)$ to $B = (2,2)$.



November 12: Green's Theorem and Differential Operators (Curl & Divergence)

Green's Theorem and Differential Operators (Curl & Divergence)

Lecture Notes: Vector Calculus

Learning Objectives (Nov 12)

By the end of this lesson, you will be able to:

- Use **Green's theorem** to simplify the evaluation of circulation integrals.
- Compute the **Divergence** ($\nabla \cdot \vec{F}$) and **Curl** ($\nabla \times \vec{F}$) of a 3D vector field.
- Use Green's theorem for the Flux integral.

1. Green's Theorem

Green's Theorem provides the fundamental link between a line integral around a closed boundary C and a double integral over the flat region R enclosed by C.

Green's Theorem (Circulation Form)

Let C be a positively oriented (counter-clockwise), piecewise smooth, simple closed curve that bounds a simply connected region R in the plane. Let $\vec{F} = \langle P, Q \rangle$ be a vector field with continuous partial derivatives on R. Then:

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

Example: Green's for Circulation

Use Green's Theorem to evaluate the circulation $\oint_C (x^2 y\,dx + x^3\,dy)$, where C is the circle $x^2 + y^2 = 4$, oriented counterclockwise.

Solution: Direct integration would require a long parameterization and integration of trigonometric functions. Green's Theorem simplifies this significantly.

1. **Identify P and Q**: $P = x^2 y$ and $Q = x^3$. 2. **Calculate the difference of cross-partials**:

$$\frac{\partial Q}{\partial x} = \frac{\partial}{\partial x}(x^3) = 3x^2$$

$$\frac{\partial P}{\partial y} = \frac{\partial}{\partial y}(x^2 y) = x^2$$

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 3x^2 - x^2 = 2x^2$$

3. **Set up the Double Integral**: The region R is the disk $x^2 + y^2 \leq 4$. We use Polar Coordinates.

$$\iint_R 2x^2\,dA = \int_0^{2\pi} \int_0^2 2(r\cos\theta)^2 \cdot r\,dr\,d\theta$$

$$= \int_0^{2\pi} \int_0^2 2r^3 \cos^2\theta\,dr\,d\theta$$

4. **Evaluate the integral**:

$$= \left(\int_0^2 2r^3\,dr\right)\left(\int_0^{2\pi} \cos^2\theta\,d\theta\right)$$

$$= \left[\frac{2r^4}{4}\right]_0^2 \cdot \left[\frac{1}{2}\left(\theta + \frac{1}{2}\sin(2\theta)\right)\right]_0^{2\pi}$$

$$= \left(\frac{16}{2}\right) \cdot \left(\frac{1}{2}(2\pi)\right) = 8 \cdot \pi = 8\pi$$

2. Differential Operators in 3D: Divergence and Curl

These two operators are fundamental to the rest of vector calculus (Stokes' and Divergence Theorems). They are often defined using the ∇ (nabla or del) operator:

$$\nabla = \langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \rangle$$

A. The Divergence Operator

The divergence measures the local rate of expansion (outward flow) or contraction (inward flow) of a vector field at a point.

Definition: Divergence ($\nabla \cdot \vec{F}$)

Let $\vec{F} = \langle P, Q, R \rangle$. The **divergence** of $\vec{F}$ is the scalar function given by the dot product of ∇

and $\vec{F}$:

$$\text{div}\vec{F} = \nabla \cdot \vec{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$$

- If $\nabla \cdot \vec{F} > 0$, the point is a **source** (outward flow).
- If $\nabla \cdot \vec{F} < 0$, the point is a **sink** (inward flow).
- If $\nabla \cdot \vec{F} = 0$, the field is called **incompressible** or

Example: Computing Divergence (HW 2, Q5)

Compute the divergence for the vector field $\vec{F} = \langle 3x, 5y^2, -2z \rangle$.

Solution:

$$\begin{aligned}\nabla \cdot \vec{F} &= \frac{\partial}{\partial x}(3x) + \frac{\partial}{\partial y}(5y^2) + \frac{\partial}{\partial z}(-2z) \\ &= 3 + 10y - 2 = 10y + 1\end{aligned}$$

B. The Curl Operator (III.2)

The curl measures the local rotational tendency (swirling or circulation) of a vector field at a point.

Definition: Curl ($\nabla \times \vec{F}$) (III.2)

Let $\vec{F} = \langle P, Q, R \rangle$. The **curl** of $\vec{F}$ is the vector function given by the cross product of ∇ and $\vec{F}$:

$$\vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ P & Q & R \end{vmatrix}$$

Expanding the determinant gives the curl vector:

$$\nabla \times \vec{F} = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}\right)\mathbf{i} - \left(\frac{\partial R}{\partial x} - \frac{\partial P}{\partial z}\right)\mathbf{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\mathbf{k}$$

- If $\nabla \times \vec{F} = \vec{0}$, the field is called *irrotational*.
- Connection to Conservatism: Recall the cross-partials test for conservatism: the conditions are exactly the components of the $\nabla \times \vec{F}$ vector.

A simply connected vector field is conservative if and only if its curl is zero: $\nabla \times \vec{F} = \vec{0}$.

Example: Computing Curl (HW 2, Q5 cont.)

Compute the curl for the vector field $\vec{F} = \langle 3x, 5y^2, -2z \rangle$.

Solution:

$$\nabla \times \vec{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 3x & 5y^2 & -2z \end{vmatrix}$$

$$= \mathbf{i}\left(\frac{\partial(-2z)}{\partial y} - \frac{\partial(5y^2)}{\partial z}\right) - \mathbf{j}\left(\frac{\partial(-2z)}{\partial x} - \frac{\partial(3x)}{\partial z}\right) + \mathbf{k}\left(\frac{\partial(5y^2)}{\partial x} - \frac{\partial(3x)}{\partial y}\right)$$

$$= \mathbf{i}(0-0) - \mathbf{j}(0-0) + \mathbf{k}(0-0) = \langle 0, 0, 0 \rangle = \vec{0}$$

3. Green's Theorem (Flux Form)

The divergence operator is what links the Flux line integral to a double integral.

Green's Theorem (Flux Form)

Let $\vec{F} = \langle P, Q \rangle$. The total outward flux of $\vec{F}$ across the closed boundary C is equal to the integral of the divergence over the enclosed region R:

$$\oint_C \vec{F} \cdot \vec{N}\, ds = \iint_R \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) dA = \iint_R \text{div}\vec{F}\, dA$$

(This shows that the total flux is the sum of the local outward flow (divergence) over the entire region.)

Example: Green's for Flux

Use the flux form of Green's theorem to find the flux of $\vec{F} = \langle \sin(y) + 3x, e^x - 5y \rangle$ across the unit circle $x^2 + y^2 = 1$.

Solution: 1. **Calculate the Divergence**:

$$\nabla \cdot \vec{F} = \frac{\partial}{\partial x}(\sin(y) + 3x) + \frac{\partial}{\partial y}(e^x - 5y)$$
$$= 3 + (-5) = -2$$

2. **Apply Green's Flux Theorem**:

$$\text{Flux} = \iint_R \text{div}\vec{F}\, dA = \iint_R -2\, dA$$
$$= -2 \iint_R 1\, dA = -2 \cdot (\text{Area of } R)$$

3. **Evaluate**: The region R is the unit disk, which has Area $A = \pi r^2 = \pi(1)^2 = \pi$.

$$\text{Flux} = -2 \cdot \pi = -2\pi$$

November 14: Surface Parameterization and Flux Integrals

Surface Parameterization and Flux Integrals

Lecture Notes: Vector Calculus

Learning Objectives

By the end of this lesson, you will be able to:

- Parameterize a surface.
- Compute the vector product $\vec{r}_u \times \vec{r}_v$ for a parameterized surface.
- Evaluate the flux integral $\iint_S \vec{F} \cdot d\vec{S}$ for a simple parameterized surface (Objective IV.2).

1. Parameterized Surfaces

Just as a curve C is described by one parameter t, a surface S requires two parameters, typically u and v.

$$\text{Surface } S: \quad \vec{r}(u, v) = \langle x(u, v), y(u, v), z(u, v) \rangle$$

Case: Surface defined by $z = g(x, y)$

If the surface S is the graph of a function $z = g(x, y)$ over a region R in the xy-plane, we can use the x and y coordinates as the parameters.

$$\vec{r}(x, y) = \langle x, y, g(x, y)\rangle \quad \text{over the region } R$$

Example: Parameterize a Plane Segment

Parameterize the portion of the plane $z = 4-x-2y$ that lies over the rectangle $R = [0, 2]\times[0, 1]$ in the xy-plane.

Solution: Let the parameters be $u = x$ and $v = y$. Since $z = g(x, y) = 4 - x - 2y$, the parameterization is:

$$\vec{r}(u, v) = \langle u, v, 4 - u - 2v\rangle$$

The domain of the parameters is $0 \leq u \leq 2$ and $0 \leq v \leq 1$.

2. The Surface Normal Vector

To compute integrals over a surface, we need a small surface area element, $d\vec{S}$, which is a vector parallel to the normal direction.

Definition: The Surface Differential Vector $d\vec{S}$

The vector element of surface area (the normal vector) is defined by the cross product of the partial derivative vectors:

$$d\vec{S} = \vec{N}\, dA = (\vec{r}_u \times \vec{r}_v)\, dA$$

Where $\vec{r}_u = \frac{\partial \vec{r}}{\partial u}$ and $\vec{r}_v = \frac{\partial \vec{r}}{\partial v}$. The direction of $\vec{N}$ gives the orientation.

Example: Computing the Normal Vector

For the plane segment $\vec{r}(u, v) = \langle u, v, 4 - u - 2v\rangle$, compute the normal vector $\vec{r}_u \times \vec{r}_v$.

Solution: 1. **Compute Partial Derivatives**:

$$\vec{r}_u = \langle \frac{\partial x}{\partial u}, \frac{\partial y}{\partial u}, \frac{\partial z}{\partial u}\rangle = \langle 1, 0, -1\rangle$$

$$\vec{r}_v = \langle \frac{\partial x}{\partial v}, \frac{\partial y}{\partial v}, \frac{\partial z}{\partial v}\rangle = \langle 0, 1, -2\rangle$$

2. **Compute Cross Product**:

$$\vec{r}_u \times \vec{r}_v = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & -1 \\ 0 & 1 & -2 \end{vmatrix} = \mathbf{i}(0 - (-1)) - \mathbf{j}(-2 - 0) + \mathbf{k}(1 - 0) = \langle 1, 2, 1\rangle$$

The vector $\vec{N} = \langle 1, 2, 1\rangle$ is the normal vector. Since the z-component is positive, this vector points upward from the plane.

3. Flux Across a Surface

The flux integral $\iint_S \vec{F} \cdot d\vec{S}$ measures the rate of flow (flux) of the vector field $\vec{F}$ across the surface S.

Definition: Flux Integral

The flux of a vector field $\vec{F}$ across the oriented surface S is given by:

$$\iint_S \vec{F} \cdot d\vec{S} = \iint_R \vec{F}(\vec{r}(u,v)) \cdot (\vec{r}_u \times \vec{r}_v)\, dA$$

Example: Computing Surface Flux

Compute the flux of $\vec{F} = \langle x, y, z \rangle$ across the upward-oriented plane segment S defined by $\vec{r}(u,v) = \langle u, v, 4 - u - 2v \rangle$, where $R : 0 \le u \le 2, 0 \le v \le 1$.

Solution: 1. **Identify Components**:

$$\vec{F}(\vec{r}(u,v)) = \langle u, v, 4 - u - 2v \rangle.$$

$$\vec{r}_u \times \vec{r}_v = \langle 1, 2, 1 \rangle$$

(upward-oriented, matching the problem).

$$dA = dv\, du.$$

2. **Compute the Dot Product** $\vec{F} \cdot (\vec{r}_u \times \vec{r}_v)$:

$$\vec{F} \cdot (\vec{r}_u \times \vec{r}_v) = u(1) + v(2) + (4 - u - 2v)(1)$$

$$= u + 2v + 4 - u - 2v = 4$$

3. **Set up and Evaluate the Double Integral**:

$$\iint_S \vec{F} \cdot d\vec{S} = \int_0^2 \int_0^1 4\, dv\, du$$

$$= \int_0^2 [4v]_0^1\, du = \int_0^2 4\, du = [4u]_0^2 = 8$$

The total flux of the vector field $\vec{F} = \langle x, y, z \rangle$ across the plane is 8.

Classwork: Green's Theorem & Surface Integrals

Name: _____

Problem 1: Green's Theorem (Circulation)

Use the circulation form of Green's Theorem to evaluate the line integral $\oint_C (2y\, dx + (3x - y)\, dy)$, where C is the boundary of the square with vertices $(0,0), (2,0), (2,2), (0,2)$, oriented counterclockwise.

Problem 2: Curl and Divergence

Consider the 3D vector field $\vec{F}(x, y, z) = \langle xy, yz, xz \rangle$.

1. Compute the **Divergence** $(\nabla \cdot \vec{F})$.
2. Compute the **Curl** $(\nabla \times \vec{F})$.

Problem 3: Surface Parameterization

Let S be the part of the paraboloid $z = 4 - x^2 - y^2$ that lies above the xy-plane.

1. Parameterize the surface S using the parameters u and v (let $x = u, y = v$). State the domain R for your parameters.
2. Compute the normal vector $\vec{N} = \vec{r}_u \times \vec{r}_v$.

Problem 4: Surface Flux Integral

Use the results from Problem 3 to set up (DO NOT EVALUATE) the flux integral $\iint_S \vec{F} \cdot d\vec{S}$ for the vector field $\vec{F} = \langle z, x, y \rangle$ across the surface S, oriented outward (upward).

Homework 9

Homework 9: Double Integrals over General & Polar Regions

Problem 1: Type I Integral

Evaluate the iterated integral $\int_0^2 \int_0^{x^2} xy \, dy \, dx$.

Problem 2: Type II Integral

Evaluate the iterated integral $\int_0^1 \int_0^y (x + y) \, dx \, dy$.

Problem 3: Volume over a General Region

Find the volume of the solid that lies under the plane $z = x$ and above the region D bounded by the parabola $y = 2x^2$ and the line $y = x + 1$.

Problem 4: Volume under a Surface

Find the volume of the solid that lies under the plane $z = 2x + 1$ and above the triangular region D in the xy-plane with vertices $(0, 0)$, $(1, 0)$, and $(1, 1)$.

Problem 5: Integral over a Type II Region

Evaluate the double integral $\iint_D x^2 y \, dA$ where D is the region bounded by $x = y^2$, $x = 4$, and the x-axis ($y = 0$).

Problem 6: Volume over a Triangle

Find the volume of the solid under the surface $z = xy$ and above the triangular region with vertices (0,0), (2,0), and (2,4).

Problem 7: Mass of a Lamina

Find the mass of the lamina that occupies the region D bounded by $y = \sqrt{x}$, $y = 0$, and $x = 4$, with density function $\rho(x, y) = y$.

Problem 8: Center of Mass

Find the center of mass $(\bar{x}, \bar{y})$ of the lamina from Problem 7.

Problem 9: Mass of a Lamina

Find the mass of the lamina that occupies the triangular region D with vertices $(0, 0)$, $(0, 3)$, and $(1, 0)$, with density function $\rho(x, y) = x$.

Problem 10: Center of Mass

Find the center of mass $(\bar{x}, \bar{y})$ of the lamina from Problem 9.

Problem 11: Converting to Polar Coordinates

Evaluate the integral $\iint_D (x^2 + y^2 + 1)\, dA$ where D is the disk $x^2 + y^2 \leq 4$.
Hint: *The region is a full disk of radius 2. The polar bounds are* $0 \leq r \leq 2$ *and* $0 \leq \theta \leq 2\pi$.

Problem 12: Polar Integral

Evaluate the integral $\iint_D e^{-x^2-y^2}\, dA$ where D is the region bounded by the semicircle $x = \sqrt{4 - y^2}$ and the y-axis.
Hint: *The region is the right half of a disk of radius 2. The polar bounds are* $0 \leq r \leq 2$ *and* $-\pi/2 \leq \theta \leq \pi/2$.

Problem 13: Volume in Polar Coordinates

Use a polar double integral to find the volume of the solid that lies under the paraboloid $z = x^2 + y^2$ and above the disk $x^2 + y^2 \leq 9$.
Hint: *The region of integration is a full disk of radius 3. The polar bounds are* $0 \leq r \leq 3$ *and* $0 \leq \theta \leq 2\pi$.

Problem 14: Volume in Polar Coordinates

Find the volume of the solid bounded by the paraboloid $z = 16 - x^2 - y^2$ and the xy-plane, for the portion that lies in the first octant.
Hint: *The base of the solid is a quarter-disk of radius 4 in the first quadrant. The polar bounds are* $0 \leq r \leq 4$ *and* $0 \leq \theta \leq \pi/2$.

Problem 15: Mass of a Polar Lamina

Find the mass of a lamina that covers the annular region D given by $1 \leq x^2 + y^2 \leq 9$. The density is $\rho(x, y) = \frac{k}{\sqrt{x^2+y^2}}$.
Hint: *The region is an annulus (a ring) between circles of radius 1 and 3. The polar bounds are* $1 \leq r \leq 3$ *and* $0 \leq \theta \leq 2\pi$.

Problem 16: Center of Mass of a Polar Lamina

Find the center of mass of a lamina that occupies the region D in the first quadrant bounded by the circle $x^2 + y^2 = 1$. The density is $\rho(x, y) = k\sqrt{x^2 + y^2}$.
Hint: *The region is a quarter-disk of radius 1 in the first quadrant. The polar bounds are* $0 \leq r \leq 1$ *and* $0 \leq \theta \leq \pi/2$.

Problem 17: Integral over a Polar Sector

Evaluate $\iint_D y\, dA$ where D is the region in the first quadrant between the circles $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$.
Hint: *This region is a sector of an annulus. The polar bounds are* $1 \leq r \leq 2$ *and* $0 \leq \theta \leq \pi/2$.

Problem 18: Integral over a Semicircle

Evaluate the double integral $\iint_D x^2\, dA$ where D is the top half of the disk centered at the origin with radius 2.
Hint: *The region is a semicircle of radius 2 above the x-axis. The polar bounds are* $0 \leq r \leq 2$ *and* $0 \leq \theta \leq \pi$.

Problem 19: Mass of a Solid Box

Find the total mass of the solid box $B = [0, 1] \times [0, 2] \times [0, 3]$ if the density at any point (x, y, z) is given by the function $\rho(x, y, z) = x^2 y$.

Problem 20: Mass of a Solid Wedge

Find the total mass of the solid tetrahedron (a wedge in the first octant) bounded by the coordinate planes $(x = 0, y = 0, z = 0)$ and the plane $x + y + z = 1$. The density of the solid is given by $\rho(x, y, z) = y$.

Quiz 9

Quiz 9: Line Integrals and Conservative Fields

Name: _____

Problem 1: Line Integral of a Vector Field

Evaluate the line integral $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = \langle y, x^2 \rangle$ and C is the curve along the parabola $y = x^2$ from the point $(0, 0)$ to $(2, 4)$.

Problem 2: Conservative Vector Fields

Consider the vector field $\vec{F} = \langle ye^x + \sin(y), e^x + x\cos(y)\rangle$.

(a) Show that the vector field $\vec{F}$ is conservative.

(b) Find a scalar potential function $f(x, y)$ such that $\nabla f = \vec{F}$.

Problem 3: Fundamental Theorem of Line Integrals

Using the potential function $f(x, y)$ you found in Problem 2, evaluate the line integral

$$\int_C \vec{F} \cdot d\vec{r}$$

where C is any path from the point $(0, \pi)$ to the point $(1, 0)$.

CHAPTER 10

Cycle 10: Double Integrals in Polar Coordinates (Homework 10)

Overview

This cycle adapts double integrals to polar coordinates, emphasizing regions with radial symmetry and the geometric meaning of the Jacobian factor. Many classical area and mass computations become simpler in this coordinate system.

Lecture Notes and Classwork

November 17: Stokes' Theorem

Stokes' Theorem
Lecture Notes: Vector Calculus

Learning Objectives

By the end of this lesson, you will be able to:

- State Stokes' Theorem and understand its relationship to Green's Theorem.
- Define the curl of a vector field as the integrand in Stokes' Theorem.
- Use Stokes' Theorem to evaluate a line integral by computing a flux integral (or vice versa).

1. From Green's Theorem to Stokes' Theorem

Recall the circulation form of Green's Theorem:

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

This theorem relates the circulation (a line integral) around a 2D region's boundary curve C to the "microscopic rotation" inside the region D.

Stokes' Theorem is the generalization of this idea to a surface S in 3D space. It states that the total circulation of a vector field around the boundary curve C of a surface is equal to the flux of the "rotation" of the field through the surface itself.

To quantify this "rotation," we need the **curl**.

2. Curl of a Vector Field

The curl of a vector field $\vec{F} = \langle P, Q, R\rangle$ is a new vector field that measures the rotation at every point.

Definition: Curl

The curl of $\vec{F}$ is defined as the formal cross product of the del operator ∇ and $\vec{F}$:

$$(\vec{F}) = \nabla \times \vec{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}$$

Notice that the **k**-component of this vector is $\left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)$, which is exactly the integrand from Green's Theorem!

3. Stokes' Theorem

Stokes' Theorem connects the flux of the curl through a surface S to the line integral of the field $\vec{F}$ around the boundary curve C of that surface.

Stokes' Theorem

Let S be an oriented surface with a boundary curve C. The orientation of C must be consistent with the orientation of S via the right-hand rule (if your fingers curl in the direction of C, your thumb points in the direction of the surface normal $\vec{n}$). Then:

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot d\vec{S}$$

In words: The work done on the boundary equals the flux of the curl through the surface.

Why is this useful? It allows us to choose the easier of two integrals. If a line integral is difficult to parameterize or compute, we can compute a (potentially easier) surface integral instead, and vice-versa.

Example: Verifying Stokes' Theorem

Let $\vec{F} = \langle -y^2, x, z^2\rangle$ and let S be the part of the paraboloid $z = x^2 + y^2$ below the plane $z = 1$, oriented upward. Verify Stokes' Theorem.

Solution: The boundary curve C is the circle $x^2 + y^2 = 1$ in the plane $z = 1$.

Part 1: The Line Integral $\oint_C \vec{F} \cdot d\vec{r}$ 1. Parameterize C: $\vec{r}(t) = \langle \cos(t), \sin(t), 1\rangle$ for $0 \leq t \leq 2\pi$. Then $\vec{r}'(t) = \langle -\sin(t), \cos(t), 0\rangle$. 2. Evaluate $\vec{F}$ on C: $\vec{F}(\vec{r}(t)) = \langle -(\sin t)^2, \cos t, 1^2\rangle = \langle -\sin^2 t, \cos t, 1\rangle$. 3. Compute the integral:

$$\oint_C \vec{F} \cdot d\vec{r} = \int_0^{2\pi} \langle -\sin^2 t, \cos t, 1\rangle \cdot \langle -\sin t, \cos t, 0\rangle \, dt$$

$$= \int_0^{2\pi} (\sin^3 t + \cos^2 t)\, dt = \int_0^{2\pi} \sin t(1 - \cos^2 t)\, dt + \int_0^{2\pi} \frac{1 + \cos(2t)}{2}\, dt$$

The first integral is 0. The second evaluates to $\frac{1}{2}[t + \frac{1}{2}\sin(2t)]_0^{2\pi} = \pi$. So, $\oint_C \vec{F} \cdot d\vec{r} = \pi$.

Part 2: The Surface Integral $\iint_S (\nabla \times \vec{F}) \cdot d\vec{S}$ 1. **Compute the Curl**:

$$\nabla \times \vec{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial_x & \partial_y & \partial_z \\ -y^2 & x & z^2 \end{vmatrix} = \langle 0 - 0, 0 - 0, 1 - (-2y) \rangle = \langle 0, 0, 1 + 2y \rangle$$

2. **Find the Normal Vector**: Since the surface is given as a function $z = g(u, v)$, we use u and v as our parameters, where $u = x$ and $v = y$. The surface is $z = u^2 + v^2$. The parameterization is:

$$\vec{r}(u, v) = \langle u, v, u^2 + v^2 \rangle$$

First, we find the partial derivative vectors with respect to u and v:

$$\vec{r}_u = \langle 1, 0, 2u \rangle$$

$$\vec{r}_v = \langle 0, 1, 2v \rangle$$

Next, we compute the cross product. The problem specifies an upward orientation, so we check the **k**-component to ensure it is positive.

$$\vec{r}_u \times \vec{r}_v = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & 2u \\ 0 & 1 & 2v \end{vmatrix} = \mathbf{i}(0 - 2u) - \mathbf{j}(2v - 0) + \mathbf{k}(1 - 0) = \langle -2u, -2v, 1 \rangle$$

Since the **k**-component is positive (+1), this is the correct upward-pointing normal vector. Therefore, the vector surface element is:

$$d\vec{S} = \langle -2u, -2v, 1 \rangle\, dA$$

3. **Compute the dot product**: On the surface, the curl $\langle 0, 0, 1 + 2y \rangle$ becomes $\langle 0, 0, 1 + 2v \rangle$.

$$(\nabla \times \vec{F}) \cdot d\vec{S} = \langle 0, 0, 1 + 2v \rangle \cdot \langle -2u, -2v, 1 \rangle\, dA = (1 + 2v)\, dA$$

4. **Evaluate the integral over the projection disk** $D : u^2 + v^2 \leq 1$. The integral to compute is $\iint_D (1 + 2v)\, dA$. We split this into two parts:

$$\iint_D (1 + 2v)\, dA = \iint_D 1\, dA + \iint_D 2v\, dA$$

To evaluate, we switch to polar coordinates, where $u = r\cos\theta$, $v = r\sin\theta$, and $dA = r\, dr\, d\theta$. The domain D is described by $0 \leq r \leq 1$ and $0 \leq \theta \leq 2\pi$.

First integral:

$$\iint_D 1\, dA = \int_0^{2\pi} \int_0^1 r\, dr\, d\theta = \int_0^{2\pi} \left[\frac{r^2}{2}\right]_0^1 d\theta = \int_0^{2\pi} \frac{1}{2}\, d\theta = \frac{1}{2}[\theta]_0^{2\pi} = \pi$$

Second integral:

$$\iint_D 2v\, dA = \int_0^{2\pi} \int_0^1 2(r\sin\theta) r\, dr\, d\theta == 0$$

Combining the results:

$$\iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = \pi + 0 = \pi$$

The results match, verifying Stokes' Theorem.

November 19: The Divergence Theorem

The Divergence Theorem
Lecture Notes: Vector Calculus

Learning Objectives

By the end of this lesson, you will be able to:

- State the Divergence Theorem and understand its relationship to the flux form of Green's Theorem.
- Understand the physical intuition behind the Divergence Theorem.
- Use the Divergence Theorem to evaluate a flux integral over a closed surface by computing a triple integral.

1. Motivation: From 2D Flux to 3D Flux

Recall the flux form of Green's Theorem, which relates the flux across a closed curve C in the plane to a double integral over the region D it encloses:

$$\oint_C \vec{F} \cdot \vec{n}\, ds = \iint_D \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) dA$$

The term on the right is the integral of the 2D divergence of $\vec{F}$ over the region. This theorem tells us that the total outward flow across the boundary is equal to the sum of all the "sources" and "sinks" inside the region.

The Divergence Theorem is the direct generalization of this idea to three dimensions. It states that the total flux of a vector field out of a **closed surface** S is equal to the triple integral of the divergence of the field over the solid volume E that the surface encloses.

2. Review: Divergence

The key ingredient for this theorem is the divergence of a 3D vector field.

Definition: Divergence

For a vector field $\vec{F} = \langle P, Q, R \rangle$, the divergence is a scalar function defined as:

$$\text{div}(\vec{F}) = \nabla \cdot \vec{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$$

Physically, the divergence at a point measures the rate of expansion (if positive, a "source") or compression (if negative, a "sink") of the vector field at that point.

3. The Divergence Theorem (Gauss's Theorem)

The Divergence Theorem

Let E be a solid region in $\mathbb{R}^3$ whose boundary is a closed surface S, and let the orientation of

S be outward. Let $\vec{F}$ be a vector field with continuous partial derivatives. Then:

$$\iint_S \vec{F} \cdot d\vec{S} = \iiint_E (\nabla \cdot \vec{F})\, dV$$

In words: The total outward flux through a closed surface is equal to the sum of all the divergence (sources and sinks) within the volume enclosed by the surface.

Intuition: Imagine the volume E is filled with a huge number of tiny boxes. The divergence of $\vec{F}$ tells you the net flux out of each tiny box. For any two boxes that touch, the flux out of one box's face is exactly the flux into the adjacent box's face. Therefore, all the internal fluxes cancel each other out. The only flux that remains is from the faces on the outer boundary of the solid, which make up the surface S.

Why is this useful? Calculating the flux over a closed surface can be very difficult. For example, the flux out of a cylinder would require three separate surface integrals: one for the top disk, one for the bottom disk, and one for the cylindrical side. The Divergence Theorem allows us to replace this work with a single, often much easier, triple integral.

Example: Flux through a Sphere

Let $\vec{F} = \langle 2x, y^2, z^2 \rangle$. Find the flux of $\vec{F}$ out of the unit sphere S defined by $x^2 + y^2 + z^2 = 1$.

Solution: Calculating this flux directly would be a long process. The surface S is closed, so we can apply the Divergence Theorem.

Step 1: Compute the divergence of $\vec{F}$.

$$\nabla \cdot \vec{F} = \frac{\partial}{\partial x}(2x) + \frac{\partial}{\partial y}(y^2) + \frac{\partial}{\partial z}(z^2) = 2 + 2y + 2z$$

Step 2: Apply the Divergence Theorem. The flux is equal to the triple integral of the divergence over the solid unit ball E.

$$\iint_S \vec{F} \cdot d\vec{S} = \iiint_E (2 + 2y + 2z)\, dV$$

Step 3: Evaluate the triple integral. We can split the integral:

$$\iiint_E 2\, dV + \iiint_E 2y\, dV + \iiint_E 2z\, dV$$

The integral $\iiint_E 2\, dV = 2 \times (\text{Volume of } E)$. The volume of a sphere of radius 1 is $\frac{4}{3}\pi(1)^3 = \frac{4}{3}\pi$. So, this part is $2 \cdot \frac{4}{3}\pi = \frac{8\pi}{3}$.

For the integrals $\iiint_E 2y\, dV$ and $\iiint_E 2z\, dV$, the region of integration (the unit ball) is symmetric with respect to the planes $y = 0$ and $z = 0$, but the integrands ($2y$ and $2z$) are odd functions. By symmetry, both of these integrals are 0.

Therefore, the total flux is:

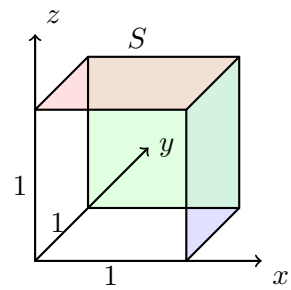
$$\text{Flux} = \frac{8\pi}{3} + 0 + 0 = \frac{8\pi}{3}$$

Classwork: Flux through a Cube

Name:

Problem 1

Let $\vec{F} = \langle x^2, 3y, z\rangle$. Use the Divergence Theorem to find the outward flux of $\vec{F}$ across the surface S of the cube defined by $0 \le x \le 1$, $0 \le y \le 1$, and $0 \le z \le 1$. *(Think about how difficult it would be to compute the flux across all six faces of the cube separately!)*



Homework 10

Homework 10: Line Integrals, Conservative Fields, and FTC
Due: Sunday, November 16

Problem 1: Work (Line Segment 1)

Evaluate the line integral $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = \langle 3xy, x^2 + y\rangle$ and C is the line segment from $(0, 0)$ to $(1, 2)$.

Problem 2: Work (Line Segment 2)

Evaluate the line integral $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = \langle z, x, y\rangle$ and C is the line segment from $(0, 0, 0)$ to $(1, 3, -1)$.

Problem 3: Work (Parabola)

Evaluate the line integral $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = \langle y^2, -x\rangle$ and C is the parabola $y = x^2$ from $(-1, 1)$ to $(1, 1)$.

Problem 4: Work (Exponential Curve)

Evaluate the line integral $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = \langle ye^x, x^2\rangle$ and C is the curve $\vec{r}(t) = \langle t, e^t\rangle$ for $0 \le t \le 1$.

Problem 5: Work (Closed Triangle)

Evaluate the line integral $\oint_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = \langle x+y, x-y \rangle$ and C is the triangle with vertices $(0,0)$, $(1,0)$, and $(0,1)$, oriented counterclockwise.

Problem 6: Work (Closed Box Boundary)

Evaluate the line integral $\oint_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = \langle y, -x, z^2 \rangle$ and C is the boundary of the square in the plane $z = 1$ with vertices $(0,0,1)$, $(1,0,1)$, $(1,1,1)$, and $(0,1,1)$, oriented counterclockwise when viewed from above.

Problem 7: Work (Helix)

Evaluate the line integral $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = \langle z, y, x \rangle$ and C is the helix $\vec{r}(t) = \langle \cos t, \sin t, t \rangle$ for $0 \leq t \leq \pi$.

Problem 8: Line Integral (Scalar Function)

Evaluate the line integral with respect to arc length $\int_C (x + y^2)\, ds$, where C is the upper half of the circle $x^2 + y^2 = 4$.

Problem 9: Flux (Line Segment)

Compute the flux of the vector field $\vec{F} = \langle x, y \rangle$ across the line segment C from $(0,0)$ to $(2,1)$.

Problem 10: Flux (Semicircle)

Compute the flux of the vector field $\vec{F} = \langle y, -x \rangle$ across the upper semicircle of $x^2 + y^2 = 1$, oriented clockwise.

Problem 11: Flux (Rectangle Boundary)

Compute the flux of $\vec{F} = \langle 2x + y, x - 3y \rangle$ across the boundary of the rectangle C defined by $0 \leq x \leq 1$, $0 \leq y \leq 2$, oriented counterclockwise.

Problem 12: Conservative Test (2D)

Determine if the vector field $\vec{F} = \langle 2xy, x^2 \rangle$ is conservative.

Problem 13: Find Potential Function (2D)

If the vector field from Problem 12 is conservative, find its scalar potential function $f(x, y)$.

Problem 14: Conservative Test (2D)

Determine if the vector field $\vec{F} = \langle y^2, x^2 \rangle$ is conservative.

Problem 15: Conservative Test (3D)

Determine if the vector field $\vec{F} = \langle yz, xz, xy \rangle$ is conservative.

Problem 16: Find Potential Function (3D)

If the vector field from Problem 15 is conservative, find its scalar potential function $f(x, y, z)$.

Problem 17: Conservative Test (3D)

Determine if the vector field $\vec{F} = \langle e^z, xe^z, xe^y \rangle$ is conservative.

Problem 18: FTC for Line Integrals

First, confirm the vector field $\vec{F} = \langle 2x+3y, 3x-2y \rangle$ is conservative. Then, use the Fundamental Theorem of Line Integrals to evaluate $\int_C \vec{F} \cdot d\vec{r}$ where C is any path from $(-1, 1)$ to $(2, -1)$.

Problem 19: FTC with Given Potential

Evaluate the integral $\int_C \nabla f \cdot d\vec{r}$, where $f(x, y, z) = x \ln(y) + z^2$ and C is any path from $(1, e, 0)$ to $(3, 1, 2)$.

Problem 20: FTC Over a Closed Loop

Evaluate $\oint_C \vec{F} \cdot d\vec{r}$ for the vector field $\vec{F} = \langle \frac{x}{x^2+y^2}, \frac{y}{x^2+y^2} \rangle$, where C is the boundary of the square with vertices $(1, 1)$, $(-1, 1)$, $(-1, -1)$, and $(1, -1)$ oriented counterclockwise. (Assume the field is conservative on a suitable domain).

Problem 21: The Jacobian Determinant

For a transformation from the uv-plane to the xy-plane, given by $x = g(u, v)$ and $y = h(u, v)$, Compute the Jacobian $\frac{\partial(x,y)}{\partial(u,v)}$ determinant for each of the following transformations.

(a) $x = 5u - 2v, \quad y = 3u + v$

(b) $x = u^2 - v^2, \quad y = 2uv$

(c) $x = u^2 \cos(v), \quad y = u^2 \sin(v)$

(d) $x = e^u \cos(v), \quad y = e^u \sin(v)$

CHAPTER 11

Cycle 11: Triple Integrals and Mass in Three Dimensions (Homework 11)

Overview

This cycle extends integration to three dimensions via triple integrals over boxes and more general regions. Applications include mass and center-of-mass computations in three-dimensional settings.

Lecture Notes and Classwork

November 21: Vector Calculus Surface Integrals (Prep for Homework 11)

Classwork: Vector Calculus Surface Integrals

Preparation for Homework 11

Name: _____

Problem 1: Mechanics of Div and Curl

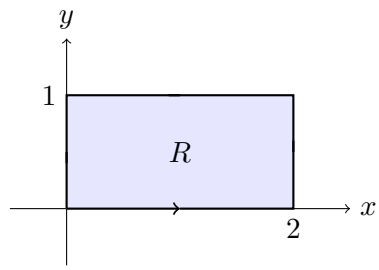
Given the vector field $\vec{F} = \langle x^2 z,\, \cos(y),\, yz^2 \rangle$:

(a) Compute the divergence $(\nabla \cdot \vec{F})$.

(b) Compute the curl $(\nabla \times \vec{F})$.

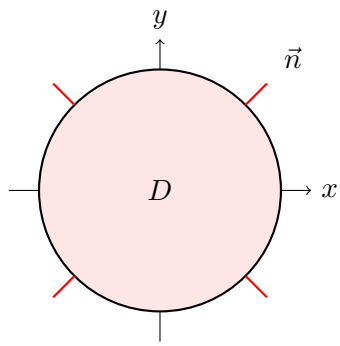
Problem 2: Green's Theorem (Circulation)

Use Green's Theorem to evaluate the line integral $\oint_C (3y\,dx + 2x\,dy)$, where C is the boundary of the rectangle defined by $0 \le x \le 2$ and $0 \le y \le 1$, oriented counterclockwise.



Problem 3: Green's Theorem (Flux)

Use the **flux form** of Green's Theorem ($\oint_C \vec{F} \cdot \vec{n}\, ds = \iint_D \vec{F}\, dA$) to find the flux of $\vec{F} = \langle 2x, 3y \rangle$ across the circle $x^2 + y^2 = 4$.



Problem 4: Parameterizing Surfaces

(a) A Plane: Parameterize the part of the plane $z = 6 - 2x - 3y$ that lies above the square $[0, 1] \times [0, 1]$. Find the normal vector $\vec{r}_u \times \vec{r}_v$.

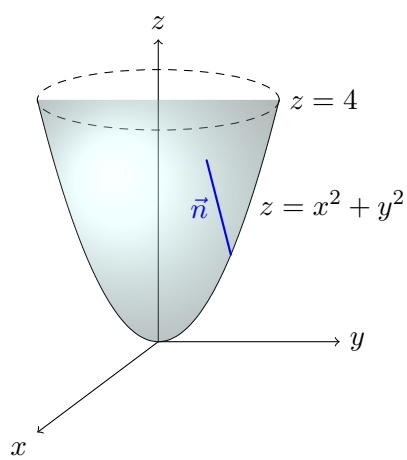
(b) A Cylinder: Parameterize the cylinder $x^2 + y^2 = 9$ for $0 \leq z \leq 4$. Find the outward normal vector $\vec{r}_u \times \vec{r}_v$.

Problem 5: Surface Flux (Paraboloid)

Evaluate the surface integral $\iint_S \vec{F} \cdot d\vec{S}$ where $\vec{F} = \langle 0, 0, z \rangle$ and S is the portion of the paraboloid $z = x^2 + y^2$ that lies below $z = 4$, oriented with the normal vector pointing **upward** (inward).

Steps to solve:

1. Parameterize S using $x = u, y = v$.
2. Find $\vec{r}_u$ and $\vec{r}_v$.
3. Compute the normal vector $\vec{n} = \vec{r}_u \times \vec{r}_v$. Check the orientation!
4. Calculate $\vec{F}(\vec{r}(u, v)) \cdot \vec{n}$.
5. Integrate over the region D (the shadow in the uv-plane, convert to polar coordinates $u = r\sin(\theta)$ and $v = r\cos(\theta)$)



Homework 11

Homework 11: Green's Theorem, Curl/Divergence, and Surfaces
Due: Sunday, November 30

Problem 1: Divergence and Curl

Compute the divergence ($\nabla \cdot \vec{F}$) and curl ($\nabla \times \vec{F}$) for the vector field $\vec{F} = \langle x^2y, yz^2, zx^2 \rangle$.

Problem 2: Divergence and Curl

Compute the divergence and curl for the vector field $\vec{F} = \langle e^{xy}, \cos(yz), z^3 \rangle$.

Problem 3: Divergence and Curl

Compute the divergence and curl for the vector field $\vec{F} = \langle y\ln(z), x\ln(z), xy/z \rangle$.

Problem 4: Divergence and Curl

Compute the divergence and curl for the vector field $\vec{F} = \langle \cos(xy), -\sin(xy), z \rangle$.

Problem 5: Divergence and Curl

Compute the divergence and curl for the vector field $\vec{F} = \langle 3x, 5y^2, -2z \rangle$.

Problem 6: Divergence and Curl

Compute the divergence and curl for the vector field $\vec{F} = \langle \frac{x}{x^2+y^2+z^2}, \frac{y}{x^2+y^2+z^2}, \frac{z}{x^2+y^2+z^2} \rangle$.

Problem 7: Green's Theorem (Circulation)

Use Green's theorem to evaluate the line integral $\oint_C (3y\,dx - 5x\,dy)$, where C is the boundary of the rectangle with vertices $(0,0), (4,0), (4,2), (0,2)$, oriented counterclockwise.

Problem 8: Green's Theorem (Circulation)

Use Green's theorem to evaluate $\oint_C (x^2y\,dx + x^3\,dy)$, where C is the ellipse $x^2 + 4y^2 = 4$, oriented counterclockwise.

Problem 9: Area with Green's Theorem

Use a line integral to find the area of the region bounded by the curves $y = x^2$ and $y = x$.
Hint: *Use the formula Area* $= \frac{1}{2}\oint_C x\,dy - y\,dx$.

Problem 10: Green's Theorem (Flux)

Use the flux form of Green's theorem to find the flux of $\vec{F} = \langle x^2, y^2\rangle$ out of the triangular region with vertices $(0,0), (2,0)$, and $(0,3)$.

Problem 11: Green's Theorem (Flux)

Use the flux form of Green's theorem to find the flux of $\vec{F} = \langle \sin(y) + 3x, e^x - 5y\rangle$ across the unit circle $x^2 + y^2 = 1$.

Problem 12: Parameterize a Plane

Parameterize the surface S which is the part of the plane $z = 4 - x - 2y$ that lies over the rectangle $D = [0,2] \times [0,1]$. Then, compute the normal vector $\vec{r}_x \times \vec{r}_y$.

Problem 13: Parameterize a Cylinder

Parameterize the surface S which is the part of the cylinder $x^2 + y^2 = 9$ for $0 \le z \le 5$. Then, compute the outward-pointing normal vector $\vec{r}_\theta \times \vec{r}_z$.

Problem 14: Parameterize a Paraboloid

Parameterize the surface S which is the part of the paraboloid $z = x^2 + y^2$ that lies below the plane $z = 4$. Then, compute the upward-pointing normal vector.

Problem 15: Surface Flux (Plane)

Evaluate $\iint_S \vec{F} \cdot d\vec{S}$ where $\vec{F} = \langle 1, 1, 1\rangle$ and S is the surface from **Problem 12**, oriented upward.

Problem 16: Surface Flux (Cylinder)

Evaluate $\iint_S \vec{F} \cdot d\vec{S}$ where $\vec{F} = \langle x, y, z\rangle$ and S is the lateral surface of the cylinder from **Problem 13**, oriented outward.

Problem 17: Surface Flux (Paraboloid)

Evaluate $\iint_S \vec{F} \cdot d\vec{S}$ where $\vec{F} = \langle 0, 0, z\rangle$ and S is the paraboloid surface from **Problem 14**, oriented upward.

Problem 18: Surface Flux (Hemisphere)

Evaluate $\iint_S \vec{F} \cdot d\vec{S}$ where $\vec{F} = \langle x, y, 0 \rangle$ and S is the upper hemisphere of radius 1, $x^2+y^2+z^2 = 1$ for $z \geq 0$, oriented outward (upward).

Problem 19: Surface Flux (Simple Surface)

Evaluate $\iint_S \vec{F} \cdot d\vec{S}$ where $\vec{F} = \langle z, x, y \rangle$ and S is the part of the plane $z = x$ that lies over the square $D = [0, 1] \times [0, 1]$, oriented upward.

Problem 20: Surface Flux (Cone)

Evaluate $\iint_S \vec{F} \cdot d\vec{S}$ where $\vec{F} = \langle x, y, -2z \rangle$ and S is the part of the cone $z = \sqrt{x^2 + y^2}$ between the planes $z = 1$ and $z = 2$, oriented upward.

CHAPTER 12

Cycle 12: Vector Fields, Divergence, Curl, and Line Integrals (Homework 12)

Overview

This cycle introduces vector fields and the differential operators grad, div, and curl. Line integrals, flux, and circulation provide a unifying viewpoint that leads toward the fundamental theorem for line integrals and the classical vector calculus theorems.

Lecture Notes and Classwork

November 24: Stokes' Theorem & Divergence Theorem (Prep for Homework 12)

Classwork: Stokes' Theorem & Divergence Theorem

Preparation for Homework 12

Name: _____

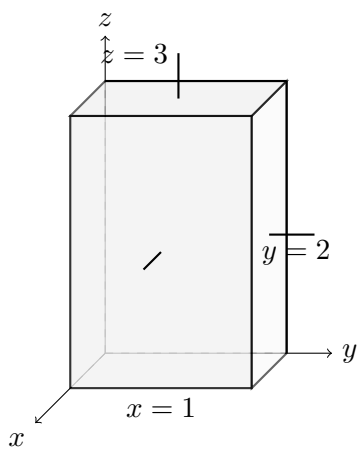
Problem 1: Stokes' Theorem (Line to Surface)

Use Stokes' Theorem to evaluate the line integral $\oint_C \vec{F} \cdot d\vec{r}$, where $\vec{F} = \langle -y, x, z^2 \rangle$ and C is the boundary of the paraboloid $z = 4 - x^2 - y^2$ that lies above the xy-plane, oriented upward.

$^2); 3[domain = 0 : 360, samples = 60, linewidth = 1.5pt, decorati$

Problem 2: Divergence Theorem (Rectangular Box)

Use the Divergence Theorem to calculate the outward flux of the vector field $\vec{F} = \langle x^2, 3y, z \rangle$ across the surface of the rectangular box bounded by the planes $x = 0, x = 1$, $y = 0, y = 2$, and $z = 0, z = 3$.



Problem 3: Divergence Theorem (Closed Surface)

Find the total outward flux of $\vec{F} = \langle x^3, y^3, z^3 \rangle$ across the **closed** cylindrical surface S bounded by $x^2 + y^2 = 4$, $z = 0$, and $z = 2$. The surface includes the top disk, bottom disk, and the side wall.

E

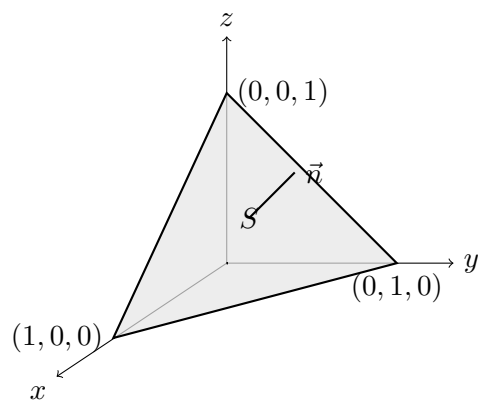
Problem 4: Stokes' Theorem (Surface Independence)

Evaluate the line integral $\oint_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = \langle xz, yz, xy \rangle$ and C is the circle $x^2 + y^2 = 1$ in the plane $z = 3$, oriented counterclockwise when viewed from above. Use the simplest possible surface for your calculation.

$\mathcal{S}$

Problem 5: Stokes' Theorem (Planar Surface)

Evaluate $\oint_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = \langle z, x, y \rangle$ and C is the boundary of the triangle with vertices $(1,0,0)$, $(0,1,0)$, and $(0,0,1)$, oriented counterclockwise when viewed from the first octant. Use Stokes' Theorem to evaluate this by parameterizing the planar surface S enclosed by C.



Homework 12

Homework 12: Stokes' Theorem and the Divergence Theorem
Due: Day of Final Exam

Problem 1: Stokes' Theorem (Line to Surface)

Use Stokes' Theorem to evaluate $\oint_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = \langle z, x, y \rangle$ and C is the boundary of the portion of the plane $x + y + z = 1$ in the first octant, oriented counterclockwise when viewed from above.

Problem 2: Stokes' Theorem (Line to Surface)

Use Stokes' Theorem to evaluate $\oint_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = \langle 2y, 3z, x \rangle$ and C is the circle $x^2 + y^2 = 4$ in the plane $z = 1$, oriented counterclockwise.

Problem 3: Stokes' Theorem (Line to Surface)

Use Stokes' Theorem to evaluate $\oint_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = \langle xz, xy, yz \rangle$ and C is the boundary of the part of the paraboloid $z = 1 - x^2 - y^2$ with $z \geq 0$, oriented upward.

Problem 4: Stokes' Theorem (Line to Surface)

Use Stokes' Theorem to evaluate $\oint_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = \langle -y^2, x, z^2 \rangle$ and C is the boundary of the cylindrical surface $x^2 + y^2 = 1$ between $z = 0$ and $z = 1$, oriented counter-clockwise when

viewed from above.

Problem 5: Stokes' Theorem (Surface to Line)

Use Stokes' Theorem to evaluate $\iint_S \text{curl}\vec{F} \cdot d\vec{S}$ where $\vec{F} = \langle y^2, x, z^2 \rangle$ and S is the hemisphere $x^2 + y^2 + z^2 = 4$ with $z \geq 0$, oriented upward.

Problem 6: Stokes' Theorem (Surface to Line)

Use Stokes' Theorem to evaluate $\iint_S \text{curl}\vec{F} \cdot d\vec{S}$ where $\vec{F} = \langle \cos(z), x^2, e^y \rangle$ and S is the square $[0, 1] \times [0, 1]$ in the xy-plane, oriented upward.

Problem 7: Divergence Theorem

Use the Divergence Theorem to find the outward flux of $\vec{F} = \langle xy, yz, xz \rangle$ across the surface S of the cube defined by $0 \leq x \leq 1$, $0 \leq y \leq 1$, and $0 \leq z \leq 1$.

Problem 8: Divergence Theorem

Use the Divergence Theorem to find the outward flux of $\vec{F} = \langle x^3, y^3, z^3 \rangle$ across the surface S of the cylinder bounded by $x^2 + y^2 = 4$, $z = 0$, and $z = 2$.

Problem 9: Divergence Theorem

Use the Divergence Theorem to find the outward flux of $\vec{F} = \langle x, 2y, 3z \rangle$ across the surface S of the solid hemisphere bounded by $x^2 + y^2 + z^2 = 1$ with $z \geq 0$ and the plane $z = 0$.

Problem 10: Divergence Theorem

Use the Divergence Theorem to find the outward flux of $\vec{F} = \langle x^2, y^2, z^2 \rangle$ over the surface of the sphere of radius a centered at the origin.

Problem 11: Flux Across One Side

Find the flux of $\vec{F} = \langle 3x^2, xy, 2xz \rangle$ across the top face (S_{top} at $z = 1$) of the unit cube.
Hint: *Use the Divergence Theorem to find the total flux across the entire cube, then directly calculate and subtract the flux across the other five faces.*

Problem 12: Flux Across One Side

Find the flux of $\vec{F} = \langle x, y, z^2 \rangle$ across the slanted surface S of the cone $z = 1 - \sqrt{x^2 + y^2}$ for $z \geq 0$.
Hint: *Apply the Divergence Theorem to the entire solid cone (including its base), and then subtract the flux across the circular base at $z = 0$.*

Problem 13: Theorem Identification

State the most efficient method to evaluate $\oint_C (x^2y\,dx + y^2x\,dy)$, where C is the boundary of the region defined by $x^2 + y^2 \leq 9$, and justify your choice. **Do NOT evaluate the integral.**

Problem 14: Theorem Identification

State the most efficient method to evaluate $\int_C \langle e^x \cos y, -e^x \sin y\rangle \cdot d\vec{r}$, where C is a path from $(0,0)$ to $(1, \pi/2)$, and justify your choice. **Do NOT evaluate the integral.**

Problem 15: Theorem Identification

State the most efficient method to evaluate $\iint_S \operatorname{curl}(\vec{F}) \cdot d\vec{S}$, where $\vec{F} = \langle y, x^2, z^3\rangle$ and S is the portion of the sphere $x^2 + y^2 + z^2 = 4$ above the plane $z = 1$, and justify your choice. **Do NOT evaluate the integral.**

Problem 16: Theorem Identification

State the most efficient method to evaluate $_S\langle x^2, 3y, 2z\rangle \cdot d\vec{S}$, where S is the closed surface of the region bounded by the paraboloid $z = 4 - x^2 - y^2$ and the plane $z = 0$, and justify your choice. **Do NOT evaluate the integral.**

Problem 17: Theorem Identification

State the most efficient method to evaluate $\oint_C \langle \sin(x^2), x^2\rangle \cdot d\vec{r}$, where C is the boundary of the square $[0,1] \times [0,1]$ oriented counterclockwise, and justify your choice. **Do NOT evaluate the integral.**

Problem 18: Theorem Identification

State the most efficient method to evaluate $\iint_S \langle z, 0, 0\rangle \cdot d\vec{S}$, where S is the part of the plane $z = x$ over the triangle with vertices $(0,0), (1,0), (0,1)$, oriented downward, and justify your choice. **Do NOT evaluate the integral.**

Problem 19: Theorem Identification

State the most efficient method to evaluate $\int_C (2xy^3\,dx + 3x^2y^2\,dy)$, where C is the curve $\vec{r}(t) = \langle t^2, \sin t\rangle$ for $0 \leq t \leq 2\pi$, and justify your choice. **Do NOT evaluate the integral.**

Problem 20: Theorem Identification

State the most efficient method to evaluate $\iint_S \operatorname{curl}(\vec{F}) \cdot d\vec{S}$, where $\vec{F} = \langle x + y, y + z, z + x\rangle$ and S is the closed surface of the rectangular box $0 \leq x, y, z \leq 1$, and justify your choice. **Do NOT evaluate the integral.**

Part II

Exams and Review Material

CHAPTER 13

Exam 1

Overview

Exam 1 primarily assesses material from Cycles 1–5: vector geometry, parametric curves, motion in space, and the basic three-dimensional tools needed for later multivariable calculus.

Learning Objectives

Calculus III Workbook

Dr. [Author's Name]
Sam Houston State University

Fall 2025

Learning Objectives

The following objectives reflect the foundational concepts and specific computational skills developed in the course materials, organized by topic area.

I. Polar Coordinates, 3D Coordinates, and Area

0. **Convert** coordinates between the standard cartesian system (x, y) and the polar coordinate system (r, θ).
1. **Convert** coordinates between the 3D rectangular system (x, y, z) and the cylindrical system (r, θ, z).
2. **Determine** the area of a region bounded by a single polar curve (e.g., circle, cardioid, rose petal) by setting up and evaluating the appropriate integral $A = \int \frac{1}{2} r^2 d\theta$.
3. **Set up and calculate** the area of a region lying between two polar curves r_{outer} and r_{inner} by finding the correct bounds of integration.

II. Vectors and the Dot Product

4. **Represent** vectors in component form $\langle a, b, c \rangle$ and as a linear combination of the standard unit vectors $a\vec{i} + b\vec{j} + c\vec{k}$.
5. **Perform** vector arithmetic (addition, subtraction, scalar multiplication, and linear combinations) and **interpret** vector addition graphically using the Head-to-Tail Method or the Parallelogram Law.
6. **Calculate** the magnitude $||\vec{v}||$ of a vector and **normalize** a vector to find the corresponding unit vector.
7. **Apply** the dot product $(\vec{a} \cdot \vec{b})$ to **determine** the angle (θ) between two vectors and **test** for vector orthogonality $(\vec{a} \cdot \vec{b} = 0)$.
8. **Compute** the scalar projection $(\text{comp}_{\vec{a}}\vec{b})$ and the vector projection $(\text{proj}_{\vec{a}}\vec{b})$ of one vector onto another.

III. The Cross Product and Determinants

9. **Calculate** the cross product $(\vec{a} \times \vec{b})$ of two vectors using the determinant formula.
10. **Analyze** and **apply** the algebraic and geometric properties of the cross product, specifically demonstrating orthogonality ($\vec{a} \times \vec{b}$ is perpendicular to $\vec{a}$ and $\vec{b}$) and anticommutativity ($\vec{a} \times \vec{b} = -(\vec{b} \times \vec{a})$).
11. **Determine** the area of a parallelogram or triangle spanned by two vectors using the magnitude of the cross product: Area $= ||\vec{a} \times \vec{b}||$.
12. **Calculate** the scalar triple product $\vec{a} \cdot (\vec{b} \times \vec{c})$ to **find** the volume of a parallelepiped and **determine** if three vectors are coplanar (Volume $= 0$).
13. **Apply** the cross product to **calculate** the torque vector $(\vec{\tau})$ and its magnitude, given a force $(\vec{F})$ applied at a position vector $(\vec{r})$ relative to a pivot: $\vec{\tau} = \vec{r} \times \vec{F}$.

IV. Equations of Lines and Planes in Space

14. **Formulate** the standard (point-normal) and general equations of a plane given a point and a normal vector.

15. **Construct** the equation of a plane given three non-collinear points by using the cross product to find the normal vector.

16. **Analyze** the normal vectors of two planes to **determine** if they are parallel or orthogonal, or to **find** the angle between them.

17. **Formulate** the parametric and symmetric equations of a line given a point and a direction vector, or given two points.

18. **Calculate** the point of intersection between a line and a plane by substituting the line's parametric equations into the plane's general equation.

19. **Determine** the parametric and symmetric equations of the line of intersection of two planes by finding a direction vector (using the cross product of the normals) and a common point.

20. **Construct** the equation of a plane that satisfies specific geometric constraints (e.g., containing two intersecting lines, parallel to another plane, perpendicular to a line).

V. Calculus of Vector-Valued Functions and Motion

21. **Compute** the derivative $\vec{r}'(t)$, velocity $\vec{v}(t)$, and acceleration $\vec{a}(t)$ vectors of a vector-valued function by differentiating each component.

22. **Find** the parametric equations for the tangent line to a curve at a specified point $\vec{r}(a)$, using $\vec{r}'(a)$ as the direction vector.

23. **Calculate** the unit tangent vector $\vec{T}(t)$ by normalizing the velocity vector $\vec{r}'(t)$.

24. **Calculate** the principal unit normal vector $\vec{N}(t)$ by normalizing the derivative of the unit tangent vector $\vec{T}'(t)$.

25. **Verify** the orthogonality of the unit tangent and normal vectors, $\vec{T}(t)$ and $\vec{N}(t)$, via their dot product.

26. **Calculate** the arc length (total distance traveled) of a curve over a given interval by integrating the speed ($||\vec{r}'(t)||$), applying appropriate integration techniques.

27. **Determine** the curvature $\kappa(t)$ using the cross-product formula $\kappa(t) = \frac{||\vec{r}'(t) \times \vec{r}''(t)||}{||\vec{r}'(t)||^3}$.

28. **Decompose** the acceleration vector into its tangential (a_T) and normal (a_N) components using the formulas $a_T = \frac{\vec{r}' \cdot \vec{r}''}{||\vec{r}'||}$ and $a_N = \frac{||\vec{r}' \times \vec{r}''||}{||\vec{r}'||}$.

29. **Interpret** the physical meaning of a_T (rate of change of speed) and a_N (acceleration due to turning) based on their computed values.

Exam 1

Calculus III - Exam 1

Instructions:

1. Show all your work. An answer without justification will receive little to no credit.
2. No graphing calculators are permitted.
3. You have 75 minutes to complete this exam.
4. A formula sheet is provided on the last page. You may detach it for ease of use.

Name: _____

1. Consider the vectors $\vec{a} = \langle 1, 1, 0 \rangle$ and $\vec{b} = \langle 1, 0, 1 \rangle$.

   (a) Find the angle θ between vectors $\vec{a}$ and $\vec{b}$.

   (b) Find the vector projection of $\vec{b}$ onto $\vec{a}$.

2. Consider the triangle with vertices at the points $P(1,1,1)$, $Q(2,3,2)$, and $R(3,1,0)$.

   (a) Find the vectors $\vec{PQ}$ and $\vec{PR}$.

   (b) Find the area of the triangle.

3. Find the parametric equations for the line of intersection of the two planes:

$$x + y + z = 3$$

$$x - y + 2z = 1$$

4. Find the equation of the plane that contains the two intersecting lines given by:

$$\vec{r}_1(t) = \langle 3t,\ 1+t,\ 2-t \rangle$$

$$\vec{r}_2(t) = \langle t,\ 1+t,\ 2+t \rangle$$

5. Consider the vectors $\vec{a} = \langle 1, 1, 0 \rangle$, $\vec{b} = \langle 0, 1, 1 \rangle$, and $\vec{c} = \langle 1, 2, 1 \rangle$.

   (a) Compute the scalar triple product $\vec{a} \cdot (\vec{b} \times \vec{c})$.

   (b) Does your result from part (a) prove that the three vectors are coplanar? Explain why or why not.

6. A particle's position is described by the vector function $\vec{r}(t) = \langle 2\cos(t), 2\sin(t), 3t\rangle$.

   (a) Find the particle's velocity vector and its acceleration vector.

   (b) Find the tangential component of the acceleration.

7. Find the parametric equations for the tangent line to the curve given by the vector function $\vec{r}(t) = \langle t^3 - t,\ t^2 + 1,\ \ln(t) \rangle$ at the point where $t = 1$.

8. A particle's position is given by the vector function

$$\vec{r}(t) = \langle t,\ t\cos(t),\ t\sin(t)\rangle$$

Find the curvature $\kappa(t)$ at the point where $t = 0$.

9. Perform the following coordinate conversions.

   (a) Convert the polar coordinates $(r, \theta) = (4, \frac{2\pi}{3})$ to rectangular coordinates (x, y).

   (b) Convert the rectangular coordinates $(x, y, z) = (2\sqrt{3}, 2, -5)$ to cylindrical coordinates (r, θ, z).

10. A particle is moving through space. For each scenario below, circle all of the statements that **must** be true. A correct response may have one or more statements circled.

    (a) If the tangential component of acceleration, a_T, is positive at time t_0.

        (i) The particle's speed is increasing at t_0.

        (ii) The particle is moving away from the origin at t_0.

        (iii) The particle's velocity vector, $\vec{v}(t_0)$, is parallel to its acceleration vector, $\vec{a}(t_0)$.

        (iv) None of the above statements must be true.

    (b) If the tangential component of acceleration, a_T, is negative at time t_0.

        (i) The particle's speed is decreasing at t_0.

        (ii) The particle is moving towards the origin at t_0.

        (iii) The curvature of the particle's path must be zero at t_0.

        (iv) None of the above statements must be true.

CHAPTER 14

Exam 2

Overview

Exam 2 primarily assesses material from Cycles 6–10: functions of several variables, gradients, optimization, and double integrals in both Cartesian and polar coordinates.

Learning Objectives

The following objectives reflect the foundational concepts and specific computational skills developed in the course materials, organized by topic area.

I. Polar Coordinates, 3D Coordinates, and Area

0. **Convert** coordinates between the standard cartesian system (x, y) and the polar coordinate system (r, θ).

1. **Convert** coordinates between the 3D rectangular system (x, y, z) and the cylindrical system (r, θ, z).

2. **Determine** the area of a region bounded by a single polar curve (e.g., circle, cardioid, rose petal) by setting up and evaluating the appropriate integral $A = \int \frac{1}{2} r^2 d\theta$.

3. **Set up and calculate** the area of a region lying between two polar curves r_{outer} and r_{inner} by finding the correct bounds of integration.

II. Vectors and the Dot Product

4. **Represent** vectors in component form $\langle a, b, c \rangle$ and as a linear combination of the standard unit vectors $a\vec{i} + b\vec{j} + c\vec{k}$.

5. **Perform** vector arithmetic (addition, subtraction, scalar multiplication, and linear combinations) and **interpret** vector addition graphically using the Head-to-Tail Method or the Parallelogram Law.

6. **Calculate** the magnitude $||\vec{v}||$ of a vector and **normalize** a vector to find the corresponding unit vector.

7. **Apply** the dot product ($\vec{a} \cdot \vec{b}$) to **determine** the angle (θ) between two vectors and **test** for vector orthogonality ($\vec{a} \cdot \vec{b} = 0$).

8. **Compute** the scalar projection ($\text{comp}_{\vec{a}}\vec{b}$) and the vector projection ($\text{proj}_{\vec{a}}\vec{b}$) of one vector onto another.

III. The Cross Product and Determinants

9. **Calculate** the cross product ($\vec{a} \times \vec{b}$) of two vectors using the determinant formula.

10. **Analyze** and **apply** the algebraic and geometric properties of the cross product, specifically demonstrating orthogonality ($\vec{a} \times \vec{b}$ is perpendicular to $\vec{a}$ and $\vec{b}$) and anticommutativity ($\vec{a} \times \vec{b} = -(\vec{b} \times \vec{a})$).

11. **Determine** the area of a parallelogram or triangle spanned by two vectors using the magnitude of the cross product: Area $= ||\vec{a} \times \vec{b}||$.

12. **Calculate** the scalar triple product $\vec{a} \cdot (\vec{b} \times \vec{c})$ to **find** the volume of a parallelepiped and **determine** if three vectors are coplanar (Volume $= 0$).

13. **Apply** the cross product to **calculate** the torque vector ($\vec{\tau}$) and its magnitude, given a force ($\vec{F}$) applied at a position vector ($\vec{r}$) relative to a pivot: $\vec{\tau} = \vec{r} \times \vec{F}$.

IV. Equations of Lines and Planes in Space

14. **Formulate** the standard (point-normal) and general equations of a plane given a point and a normal vector.

15. **Construct** the equation of a plane given three non-collinear points by using the cross product to find the normal vector.

16. **Analyze** the normal vectors of two planes to **determine** if they are parallel or orthogonal, or to **find** the angle between them.

17. **Formulate** the parametric and symmetric equations of a line given a point and a direction vector, or given two points.

18. **Calculate** the point of intersection between a line and a plane by substituting the line's parametric equations into the plane's general equation.

19. **Determine** the parametric and symmetric equations of the line of intersection of two planes by finding a direction vector (using the cross product of the normals) and a common point.

20. **Construct** the equation of a plane that satisfies specific geometric constraints (e.g., containing two intersecting lines, parallel to another plane, perpendicular to a line).

V. Calculus of Vector-Valued Functions and Motion

21. **Compute** the derivative $\vec{r}'(t)$, velocity $\vec{v}(t)$, and acceleration $\vec{a}(t)$ vectors of a vector-valued function by differentiating each component.

22. **Find** the parametric equations for the tangent line to a curve at a specified point $\vec{r}(a)$, using $\vec{r}'(a)$ as the direction vector.

23. **Calculate** the unit tangent vector $\vec{T}(t)$ by normalizing the velocity vector $\vec{r}'(t)$.

24. **Calculate** the principal unit normal vector $\vec{N}(t)$ by normalizing the derivative of the unit tangent vector $\vec{T}'(t)$.

25. **Verify** the orthogonality of the unit tangent and normal vectors, $\vec{T}(t)$ and $\vec{N}(t)$, via their dot product.

26. **Calculate** the arc length (total distance traveled) of a curve over a given interval by integrating the speed ($||\vec{r}'(t)||$), applying appropriate integration techniques.

27. **Determine** the curvature $\kappa(t)$ using the cross-product formula $\kappa(t) = \frac{||\vec{r}'(t)\times\vec{r}''(t)||}{||\vec{r}'(t)||^3}$.

28. **Decompose** the acceleration vector into its tangential (a_T) and normal (a_N) components using the formulas $a_T = \frac{\vec{r}'\cdot\vec{r}''}{||\vec{r}'||}$ and $a_N = \frac{||\vec{r}'\times\vec{r}''||}{||\vec{r}'||}$.

29. **Interpret** the physical meaning of a_T (rate of change of speed) and a_N (acceleration due to turning) based on their computed values.

VI. Functions of Several Variables and Contour Maps

30. **Define** a function of two variables and **understand** its graph as a surface in 3D space.

31. **Define** level curves for a function $f(x, y)$ and **explain** their relationship to its 3D surface.

32. **Interpret** a contour map to **understand** the shape of a surface, including identifying steepness, peaks, and valleys.

33. **Create** a simple contour map from a given function $z = f(x, y)$.

VII. Partial Derivatives, Gradients, and Directional Derivatives

34. **Understand** the geometric meaning of partial derivatives as slopes of tangent lines on a surface.

35. **Compute** first and second-order partial derivatives ($f_x, f_y, f_{xx}, f_{yy}, f_{xy}$) and apply Clairaut's Theorem.

36. **Formulate** the gradient vector ∇f and **evaluate** it at a specific point.

37. **Derive** and **apply** the equation of the tangent plane and the linearization $L(x, y)$ of a function at a point.

38. **Calculate** the directional derivative of a function at a point in the direction of a given vector.

39. **Determine** the direction of steepest ascent/descent and the value of the maximum/minimum rate of change of a function.

40. **Explain** why the gradient vector is orthogonal to level curves and level surfaces and **apply** this property to find tangent lines and planes.

VIII. Multivariable Optimization

41. **Calculate** the critical points of a function of two variables by setting its gradient equal to zero and solving the system of equations.
42. **Apply** the Second Derivative Test to **classify** each critical point as a local maximum, local minimum, or a saddle point.
43. **Use** the method of Lagrange multipliers to **solve** constrained optimization problems with one constraint.

IX. Double Integrals

44. **Connect** the concept of a definite integral in 1D (area) to a double integral in 2D (volume).
45. **Evaluate** a double integral over a rectangular region by computing an iterated integral.
46. **Apply** Fubini's Theorem to **change** the order of integration for a double integral.
47. **Identify** and **describe** Type I (vertically simple) and Type II (horizontally simple) regions in the plane.
48. **Set up and evaluate** iterated double integrals over general Type I and Type II regions.

X. Applications and Advanced Integration Topics

49. **Use** a double integral to **calculate** the mass of a flat plate (lamina) with a variable density function $\rho(x, y)$.
50. **Calculate** the moments and the center of mass of a lamina using double integrals.
51. **Convert** double integrals from rectangular to polar coordinates, including the function and the area element ($dA = r\,dr\,d\theta$).
52. **Set up and evaluate** double integrals over regions described by polar curves.
53. **Set up and evaluate** a triple integral over a rectangular box to find mass or volume.

Exam 2

Alternate Version

Calculus III - Exam 2 Alternate

Instructions:

1. Show all your work. An answer without justification will receive little to no credit.
2. No graphing calculators are permitted.
3. You have 75 minutes to complete this exam.
4. A formula sheet is provided on the last page. You may detach it for ease of use.

Name: _____

1. Consider the function $f(x, y) = e^{2x-y} - 4xy^2$ at the point $P(1, -1)$.

   (a) Find the gradient $\nabla f(P)$.

   (b) Find the directional derivative of f at P in the direction of $\vec{v} = \langle -6,\ 8 \rangle$.

2. Consider $g(x, y) = \ln(55 - 4x^2 - 9y^2)$.

   (a) Find the linear approximation $L(x, y)$ at $(x_0, y_0) = (2, 1)$.

   (b) Use your result to approximate $g(2.1,\ 0.9)$.

3. Find the equation of the tangent plane to the level surface

$$x^2 + 2y^2 - 3z - 10 = 0$$

at the point $Q(2, 1, -2)$.

4. Consider $p(x, y) = x^3 - 3xy^2 + 5x$.

   (a) Find all critical points of p.

   (b) Classify each critical point as local minimum, local maximum, or saddle point.

5. Use Lagrange multipliers to find the extrema of $f(x, y) = x^2 + y^2$ subject to the constraint $x + 2y = 8$. State all extreme values and where they occur.

6. Evaluate the double integral

$$\iint_R 7(x^2 + y^2)\, dA,$$

where R is the region inside of the circle $x^2 + y^2 = 16$ in the second quadrant.

7. Evaluate the triple integral

$$\iiint_E yx^2 + yz^2 \, dV,$$

where E is the region

$$[0,2] \times [1,2], \times [0,3].$$

8. Perform the following coordinate conversions.

   (a) Convert the polar coordinates $(r, \theta) = \left(5,\ \dfrac{5\pi}{6}\right)$ to rectangular coordinates (x, y).

   (b) Convert the rectangular coordinates $(x, y, z) = \left(-3,\ 3\sqrt{3},\ -4\right)$ to cylindrical coordinates (r, θ, z).

9. Let D be the region bounded by $y = x^2$ and $y = 3x$ with $x \geq 0$. Evaluate

$$\iint_D (2x + y)\, dA.$$

10. Consider the vectors $\vec{a} = \langle 2, -1, 3 \rangle$, $\vec{b} = \langle 1, 4, 0 \rangle$, and $\vec{c} = \langle -1, 2, 2 \rangle$.

    (a) Compute the scalar triple product $\vec{a} \cdot (\vec{b} \times \vec{c})$.

    (b) Find the volume of the parallelepiped determined by $\vec{a}, \vec{b}, \vec{c}$.

11. (Extra Credit) Consider the change of variables $u = x + y$ and $v = x - y$ with inverse $x = \dfrac{u+v}{2}$, $y = \dfrac{u-v}{2}$.

    (a) Find the Jacobian $\dfrac{\partial(x,y)}{\partial(u,v)}$.

    (b) Evaluate $\displaystyle\iint_D (x^2 - y^2)\, dA$ where $D = \{(x,y) : |x| + |y| \leq 2\}$ by transforming to the uv–plane.

Part III

Course Reference

Formula Sheets

Exam 1 Formula Sheet

FORMULA SHEET

2

$$\|\langle x_1, y_1, z_1\rangle\| = \sqrt{x_1^2 + y_1^2 + z_1^2}$$

$$\frac{\mathbf{u}}{\|\mathbf{u}\|}$$

$$\langle x_1, y_1, z_1\rangle = x_1\mathbf{i} + y_1\mathbf{j} + z_1\mathbf{k}$$

$$\langle x_1, y_1, z_1\rangle \cdot \langle x_2, y_2, z_2\rangle = x_1x_2 + y_1y_2 + z_1z_2$$

$$\cos(\theta) = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|}$$

$$\mathrm{Proj}_{\mathbf{v}}(\mathbf{u}) = \left(\frac{\mathbf{u}\cdot\mathbf{v}}{\mathbf{v}\cdot\mathbf{v}}\right)\mathbf{v}$$

$$\mathrm{Comp}_{\mathbf{v}}(\mathbf{u}) = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{v}\|}$$

$$\langle x_1, y_1, z_1\rangle \times \langle x_2, y_2, z_2\rangle = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \end{vmatrix}$$

$$\|\mathbf{u}\times\mathbf{v}\| = \|\mathbf{u}\|\|\mathbf{v}\|\sin(\theta)$$

$$\mathbf{u}\cdot(\mathbf{v}\times\mathbf{w})$$

$$\mathbf{r}(t) = \langle x_0, y_0, z_0\rangle + t\langle a, b, c\rangle$$

$$\langle a, b, c\rangle \cdot \langle x - x_0, y - y_0, z - z_0\rangle = 0$$

$$\cos(\theta) = \frac{\mathbf{n}_1 \cdot \mathbf{n}_2}{\|\mathbf{n}_1\|\|\mathbf{n}_2\|}$$

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|}$$

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$$

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t)$$

$$\int_a^b \|\mathbf{r}'(t)\|\, dt$$

$$s(t) = \int_a^t \|\mathbf{r}'(u)\|\, du$$

$$\kappa = \left\| \frac{d\mathbf{T}}{ds} \right\|$$

$$\kappa = \frac{\|\mathbf{T}'(t)\|}{\|\mathbf{r}'(t)\|}$$

$$\kappa = \frac{\|\mathbf{r}'(t) \times \mathbf{r}''(t)\|}{\|\mathbf{r}'(t)\|^3}$$

$$\mathbf{v}(t) = \mathbf{r}'(t)$$

$$\mathbf{a}(t) = \mathbf{r}''(t)$$

$$\frac{ds}{dt} = \|\mathbf{r}'(t)\|$$

$$a_N = \frac{\|\mathbf{v}(t) \times \mathbf{a}(t)\|}{\|\mathbf{v}(t)\|}$$

$$a_T = \frac{\mathbf{a}(t) \cdot \mathbf{v}(t)}{\|\mathbf{v}(t)\|}$$

$$\mathbf{a}(t) = a_T\mathbf{T}(t) + a_N\mathbf{N}(t)$$

Exam 2 Formula Sheet

FORMULA SHEET

2

$$\|\langle x_1, y_1, z_1\rangle\| = \sqrt{x_1^2 + y_1^2 + z_1^2}$$

$$\frac{\mathbf{u}}{\|\mathbf{u}\|}$$

$$\langle x_1, y_1, z_1\rangle = x_1\mathbf{i} + y_1\mathbf{j} + z_1\mathbf{k}$$

$$\langle x_1, y_1, z_1\rangle \cdot \langle x_2, y_2, z_2\rangle = x_1x_2 + y_1y_2 + z_1z_2$$

$$\cos(\theta) = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|}$$

$$\text{proj}_{\mathbf{v}}(\mathbf{u}) = \left(\frac{\mathbf{u}\cdot\mathbf{v}}{\mathbf{v}\cdot\mathbf{v}}\right)\mathbf{v}$$

$$\text{comp}_{\mathbf{v}}(\mathbf{u}) = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{v}\|}$$

$$\langle x_1, y_1, z_1\rangle \times \langle x_2, y_2, z_2\rangle = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \end{vmatrix}$$

$$\|\mathbf{u}\times\mathbf{v}\| = \|\mathbf{u}\|\|\mathbf{v}\|\sin(\theta)$$

$$\mathbf{u}\cdot(\mathbf{v}\times\mathbf{w})$$

$$\mathbf{r}(t) = \langle x_0, y_0, z_0\rangle + t\langle a, b, c\rangle$$

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

$$\cos(\theta) = \frac{|\mathbf{n}_1\cdot\mathbf{n}_2|}{\|\mathbf{n}_1\|\|\mathbf{n}_2\|}$$

$$\nabla f(x, y) = \langle f_x, f_y\rangle$$

$$D_{\mathbf{u}}f(x, y) = \nabla f\cdot\mathbf{u}$$

$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

$$L(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

$$D = f_{xx} f_{yy} - (f_{xy})^2$$

$$\nabla f = \lambda \nabla g, \quad g(x, y, z) = k$$

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|}$$

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$$

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t)$$

$$\int_a^b \|\mathbf{r}'(t)\| \, dt$$

$$\kappa = \frac{\|\mathbf{T}'(t)\|}{\|\mathbf{r}'(t)\|} = \frac{\|\mathbf{r}'(t) \times \mathbf{r}''(t)\|}{\|\mathbf{r}'(t)\|^3}$$

$$\mathbf{v}(t) = \mathbf{r}'(t), \quad \mathbf{a}(t) = \mathbf{r}''(t)$$

$$\frac{ds}{dt} = \|\mathbf{r}'(t)\|$$

$$a_T = \frac{\mathbf{r}'(t) \cdot \mathbf{r}''(t)}{\|\mathbf{r}'(t)\|}$$

$$a_N = \frac{\|\mathbf{r}'(t) \times \mathbf{r}''(t)\|}{\|\mathbf{r}'(t)\|}$$

$$\mathbf{a}(t) = a_T \mathbf{T}(t) + a_N \mathbf{N}(t)$$

$$m = \iint_D \rho(x, y) \, dA$$

$$M_x = \iint_D y\rho(x, y) \, dA$$

$$M_y = \iint_D x\rho(x, y) \, dA$$

$$\bar{x} = \frac{M_y}{m}, \quad \bar{y} = \frac{M_x}{m}$$

$$x = r\cos\theta, \quad y = r\sin\theta$$

$$x^2 + y^2 = r^2, \quad \tan\theta = \frac{y}{x}$$

$$\iint_D f(x,y)\,dA = \iint_D f(r\cos\theta, r\sin\theta)\,r\,dr\,d\theta$$

Part IV

Homework Solutions

CHAPTER 15

Homework Solutions

Homework 1 Solutions

Homework - 1

Work solutions on blank paper, showing all steps. Scan and submit by Sunday at 11:59 pm on Blackboard. Solutions will be posted 1–2 days before the deadline.

Problem 1

We are given the parametric equations for an ellipse:

$$x = 2\cos t, \quad y = 3\sin t, \quad 0 \le t \le 2\pi$$

1. Eliminate the parameter to find the Cartesian equation.
2. Sketch the curve and indicate orientation.

Solution to Problem 1

1. **Eliminate the parameter:** The strategy for curves involving $\sin t$ and $\cos t$ is to use the Pythagorean identity $\cos^2 t + \sin^2 t = 1$. First, we solve the given parametric equations for $\cos t$ and $\sin t$:

$$x = 2\cos t \implies \cos t = \frac{x}{2}$$
$$y = 3\sin t \implies \sin t = \frac{y}{3}$$

Now, we substitute these expressions into the identity:

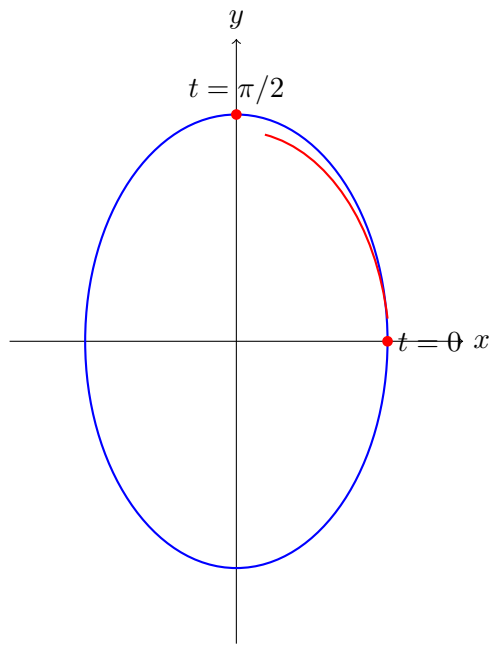
$$(\cos t)^2 + (\sin t)^2 = 1 \implies \left(\frac{x}{2}\right)^2 + \left(\frac{y}{3}\right)^2 = 1$$

The Cartesian equation is $\frac{x^2}{4} + \frac{y^2}{9} = 1$.

2. **Sketch and Orientation:** This is the equation of an ellipse centered at $(0, 0)$ with a horizontal semi-axis of $\sqrt{4} = 2$ and a vertical semi-axis of $\sqrt{9} = 3$. To find the orientation, we test points as t increases:

- At $t = 0$: $(x, y) = (2\cos 0, 3\sin 0) = (2, 0)$ (Starting point).
- At $t = \pi/2$: $(x, y) = (2\cos(\pi/2), 3\sin(\pi/2)) = (0, 3)$.

The curve starts at $(2, 0)$ and moves up to $(0, 3)$, so the orientation is **counter-clockwise**.



Problem 2

We are given the parametric equations for a line segment:

$$x = 1 + 2t, \quad y = t, \quad 0 \leq t \leq 3$$

1. Eliminate t to find the Cartesian equation.
2. Find the start and end points of the segment.

Solution to Problem 2

1. **Eliminate the parameter:** The strategy for polynomial equations is to solve for t in the simpler equation and substitute into the other. Here, the second equation is $y = t$. Substitute $t = y$ into the first equation:

$$x = 1 + 2y$$

   This is the Cartesian equation of the line on which the segment lies.

2. **Find endpoints:** We evaluate the parametric equations at the boundaries of the t interval.
   - **Start point (t=0):** $x(0) = 1 + 2(0) = 1$, $y(0) = 0$. The starting point is $(1, 0)$.
   - **End point (t=3):** $x(3) = 1 + 2(3) = 7$, $y(3) = 3$. The ending point is $(7, 3)$.

Problem 3

We are given:

$$x = t^3 - t, \quad y = t^2$$

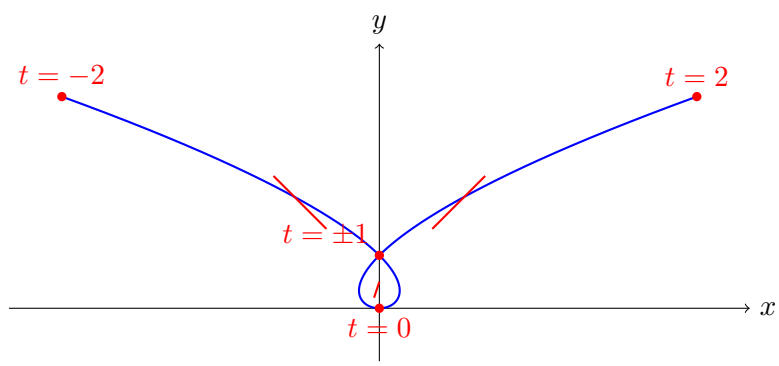
1. Sketch the curve for $t \in [-2, 2]$, showing direction.
2. Find the coordinates of the self-intersection point.

Solution to Problem 3

1. **Sketch:** We build a table of values and plot them to see the shape and orientation.

t	$x = t^3 - t$	$y = t^2$	Point
-2	$(-2)^3 - (-2) = -6$	$(-2)^2 = 4$	$(-6, 4)$
-1	$(-1)^3 - (-1) = 0$	$(-1)^2 = 1$	$(0, 1)$
0	$0^3 - 0 = 0$	$0^2 = 0$	$(0, 0)$
1	$1^3 - 1 = 0$	$1^2 = 1$	$(0, 1)$
2	$2^3 - 2 = 6$	$2^2 = 4$	$(6, 4)$

The curve comes from the left, creates a loop at the origin that passes through $(0, 1)$, and then goes out to the right.



2. **Self-intersection:** A self-intersection occurs when two **different** times, $t_1 \neq t_2$, produce the same (x, y) point. This requires solving the system:

$$\begin{array}{ll} x(t_1) = x(t_2) & y(t_1) = y(t_2) \\ t_1^3 - t_1 = t_2^3 - t_2 & t_1^2 = t_2^2 \end{array}$$

We start with the simpler equation, $y(t_1) = y(t_2)$:

$$t_1^2 = t_2^2 \implies t_1 = \pm t_2$$

Since we require $t_1 \neq t_2$, the only possibility is $t_1 = -t_2$. Now we substitute this into the x equation:

$$t_1^3 - t_1 = (-t_1)^3 - (-t_1)$$

$$t_1^3 - t_1 = -t_1^3 + t_1$$

$$2t_1^3 - 2t_1 = 0$$

$$2t_1(t_1^2 - 1) = 0$$
$$2t_1(t_1 - 1)(t_1 + 1) = 0$$

This gives potential values for t_1 as $0, 1, -1$.

- If $t_1 = 0$, then $t_2 = -0 = 0$. This is not a solution because $t_1 = t_2$.
- If $t_1 = 1$, then $t_2 = -1$. These are different times, so this is the solution.

To find the point, we plug either $t = 1$ or $t = -1$ into the original equations:

$$x(1) = 1^3 - 1 = 0, \quad y(1) = 1^2 = 1$$

The self-intersection occurs at the point $(0, 1)$.

Problem 4

We are given:

$$x = 4 + \cos t, \quad y = -2 + \sin t$$

1. Eliminate t to find the Cartesian equation.
2. Find the center and radius of the circle.

Solution to Problem 4

1. **Eliminate the parameter:** Again, we use the identity $\cos^2 t + \sin^2 t = 1$. First, isolate the trigonometric terms:

$$x - 4 = \cos t \qquad y - (-2) = \sin t \implies y + 2 = \sin t$$

Substitute these into the identity:

$$(x-4)^2 + (y+2)^2 = 1$$

2. **Center and Radius:** The standard form of a circle is $(x-h)^2 + (y-k)^2 = r^2$. By comparing our equation to the standard form, we can identify the center (h, k) and the radius r. The center is $(4, -2)$ and the radius is $r = \sqrt{1} = 1$.

Problem 5

The unit circle is parametrized by $x = \cos t, y = \sin t$. Provide two **different** parametrizations. For each, describe orientation and speed.

Solution to Problem 5

There are infinite possibilities. Here are two common examples:

- **Parametrization 1 (Clockwise):** To reverse the orientation, we can negate the y-component (or the t variable). Using the y-component is common.

$$x = \cos t, \quad y = -\sin t, \quad 0 \le t \le 2\pi$$

Orientation: At $t = 0$, point is $(1, 0)$. At $t = \pi/2$, point is $(0, -1)$. This moves clockwise. **Speed:** The speed is the same as the original, as it completes one revolution in 2π units of time.

- **Parametrization 2 (Double Speed):** To change the speed, we multiply the parameter t by a constant. To double the speed, we use $2t$.

$$x = \cos(2t), \quad y = \sin(2t)$$

Orientation: Counter-clockwise, same as the original. **Speed:** This curve completes a full revolution when $2t = 2\pi$, which means $t = \pi$. Since it completes the same path in half the time, it is traveling at **double the speed**. The domain would be $0 \le t \le \pi$.

Problem 6

We are given:

$$x = \sin t, \quad y = \cos t, \quad 0 \le t \le 2\pi$$

1. Show this is the unit circle.
2. Determine its orientation by checking points at $t = 0, \pi/2$.

Solution to Problem 6

1. **Show it is the unit circle:** We use the Pythagorean identity.

$$x^2 + y^2 = (\sin t)^2 + (\cos t)^2 = \sin^2 t + \cos^2 t = 1$$

   The equation $x^2 + y^2 = 1$ is the definition of the unit circle.

2. **Determine Orientation:**
   - At $t = 0$: $(x, y) = (\sin 0, \cos 0) = (0, 1)$.
   - At $t = \pi/2$: $(x, y) = (\sin(\pi/2), \cos(\pi/2)) = (1, 0)$.

   The curve starts at the top of the circle and moves to the right. This is a **clockwise** orientation.

Problem 7 (Desmos Required)

Use Desmos to graph:

- Curve 1: $(\cos t, \sin t)$, $0 \le t \le 2\pi$
- Curve 2: $(\cos(3t), \sin(3t))$, $0 \le t \le 2\pi$

Describe the motion and how many times each particle travels around the circle.

Solution to Problem 7

- **Curve 1:** As t increases from 0 to 2π, a particle on this path travels around the unit circle **once** in a counter-clockwise direction.
- **Curve 2:** The term $3t$ means the angle is increasing three times as fast. As t increases from 0 to 2π, the angle $3t$ increases from 0 to 6π. This means a particle on this path travels around the unit circle **three times**, also in a counter-clockwise direction.

Problem 8

We are given:

$$x = \sqrt{t}, \quad y = 1 - t, \quad t \ge 0$$

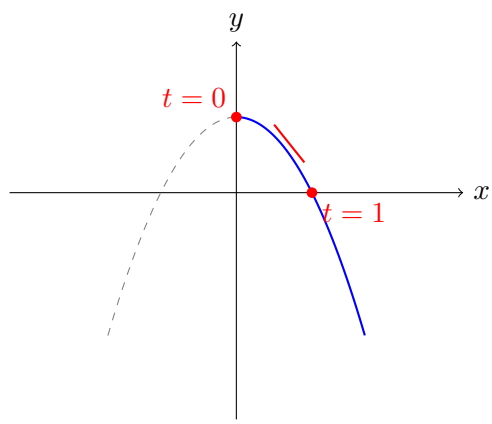
1. Eliminate t to relate x, y.

2. Sketch the curve. Why is it only part of the parabola?

Solution to Problem 8

1. **Eliminate the parameter:** From $x = \sqrt{t}$, we can square both sides to get $t = x^2$. Substitute this into the y-equation:
$$y = 1 - x^2$$

2. **Sketch and explanation:** The equation $y = 1 - x^2$ is a downward-opening parabola with its vertex at $(0, 1)$. However, the parametric equation for x is $x = \sqrt{t}$. The square root function's range is non-negative, so we must have $x \geq 0$. This domain restriction means we only trace the **right half** of the parabola.



Problem 9

Projectile motion:
$$x = 10t, \quad y = 40t - 5t^2$$

1. Eliminate t to express y in terms of x.
2. What is the shape of the path?

Solution to Problem 9

1. **Eliminate the parameter:** Solve the simpler linear equation for t.
$$x = 10t \implies t = \frac{x}{10}$$
Substitute this into the y-equation:
$$y = 40\left(\frac{x}{10}\right) - 5\left(\frac{x}{10}\right)^2 = 4x - 5\left(\frac{x^2}{100}\right) = 4x - \frac{x^2}{20}$$

2. **Shape of the path:** The equation $y = 4x - \frac{1}{20}x^2$ is a quadratic function of x. Because the x^2 term is negative, the shape is a **downward-opening parabola**.

Problem 10

We are given:

$$x = \sec t, \quad y = \tan t, \quad -\pi/2 < t < \pi/2$$

Eliminate t using $1 + \tan^2 t = \sec^2 t$.

Solution to Problem 10

We are given the relevant Pythagorean identity. We substitute $x = \sec t$ and $y = \tan t$ directly into it:

$$1 + (\tan t)^2 = (\sec t)^2 \implies 1 + y^2 = x^2$$

Rearranging gives the standard form $x^2 - y^2 = 1$. This is the equation of a **hyperbola**. The restriction $-\pi/2 < t < \pi/2$ ensures $\sec t \geq 1$, so we only get the right branch of the hyperbola.

Problem 11

Find the equation of the tangent line to the parametric curve $x(t) = t^2 - 3, y(t) = 3t - t^3$ at the point $(1, 2)$.

Solution to Problem 11

1. **Find the value of** t: We need to find t such that $x(t) = 1$ and $y(t) = 2$. $x(t) = t^2 - 3 = 1 \implies t^2 = 4 \implies t = \pm 2$. Now we check which t-value gives $y = 2$: $y(2) = 3(2) - 2^3 = 6 - 8 = -2$ (Incorrect). $y(-2) = 3(-2) - (-2)^3 = -6 + 8 = 2$ (Correct). The point $(1, 2)$ corresponds to $t = -2$.

2. **Find the slope** dy/dx: First, find the derivatives with respect to t.

$$\frac{dx}{dt} = 2t, \quad \frac{dy}{dt} = 3 - 3t^2 \implies \frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{3 - 3t^2}{2t}$$

3. **Evaluate the slope at** $t = -2$:

$$m = \left.\frac{dy}{dx}\right|_{t=-2} = \frac{3 - 3(-2)^2}{2(-2)} = \frac{3 - 12}{-4} = \frac{-9}{-4} = \frac{9}{4}$$

4. **Write the line equation** using point-slope form $y - y_1 = m(x - x_1)$:

$$y - 2 = \frac{9}{4}(x - 1)$$

Problem 12

Find the equation of the tangent line to the parametric curve $x(t) = \cos t, y(t) = \sin(2t)$ at $t = \pi/6$.

Solution to Problem 12

1. **Find the point** (x, y): $x(\pi/6) = \cos(\pi/6) = \sqrt{3}/2$. $y(\pi/6) = \sin(2 \cdot \pi/6) = \sin(\pi/3) = \sqrt{3}/2$. The point is $(\sqrt{3}/2, \sqrt{3}/2)$.

2. **Find the slope** dy/dx:

$$\frac{dx}{dt} = -\sin t, \quad \frac{dy}{dt} = 2\cos(2t) \implies \frac{dy}{dx} = \frac{2\cos(2t)}{-\sin t}$$

3. **Evaluate the slope at** $t = \pi/6$:

$$m = \frac{2\cos(\pi/3)}{-\sin(\pi/6)} = \frac{2(1/2)}{-1/2} = -2$$

4. **Write the line equation**:

$$y - \frac{\sqrt{3}}{2} = -2\left(x - \frac{\sqrt{3}}{2}\right)$$

Problem 13

Find all points on the parametric curve $x(t) = t^2 - 2t, y(t) = t^3 - 12t$ where the tangent line is horizontal.

Solution to Problem 13

A horizontal tangent occurs when the slope $dy/dx = 0$. This happens when $dy/dt = 0$ and $dx/dt \neq 0$.

$$\frac{dy}{dt} = 3t^2 - 12 = 3(t^2 - 4) = 3(t-2)(t+2)$$

Set $dy/dt = 0$: $3(t-2)(t+2) = 0 \implies t = 2$ or $t = -2$. We must check that $dx/dt = 2t - 2$ is not zero at these values. At $t = 2$, $dx/dt = 2(2) - 2 = 2 \neq 0$. At $t = -2$, $dx/dt = 2(-2) - 2 = -6 \neq 0$. Both are valid. Now we find the (x, y) points:

- At $t = 2$: $x(2) = 2^2 - 2(2) = 0$, $y(2) = 2^3 - 12(2) = -16$. Point: $(0, -16)$.
- At $t = -2$: $x(-2) = (-2)^2 - 2(-2) = 8$, $y(-2) = (-2)^3 - 12(-2) = 16$. Point: $(8, 16)$.

Problem 14

Find all points on the parametric curve $x(t) = 2\cos t, y(t) = \sin(2t)$ where the tangent line is vertical.

Solution to Problem 14

A vertical tangent occurs when the slope dy/dx is undefined. This happens when $dx/dt = 0$ and $dy/dt \neq 0$.

$$\frac{dx}{dt} = -2\sin t$$

Set $dx/dt = 0$: $-2\sin t = 0 \implies t = 0, \pi$ (within a 2π interval). We must check that $dy/dt = 2\cos(2t)$ is not zero at these values. At $t = 0$, $dy/dt = 2\cos(0) = 2 \neq 0$. At $t = \pi$, $dy/dt = 2\cos(2\pi) = 2 \neq 0$. Both are valid. Now we find the (x, y) points:

- At $t = 0$: $x(0) = 2\cos 0 = 2$, $y(0) = \sin 0 = 0$. Point: $(2, 0)$.
- At $t = \pi$: $x(\pi) = 2\cos\pi = -2$, $y(\pi) = \sin(2\pi) = 0$. Point: $(-2, 0)$.

Problem 15

For the parametric curve $x(t) = t^2$ and $y(t) = t^3 - 3t$, find the second derivative d^2y/dx^2 and determine for which values of t the curve is concave upward.

Solution to Problem 15

1. **Find the first derivative,** dy/dx:

$$\frac{dx}{dt} = 2t, \quad \frac{dy}{dt} = 3t^2 - 3 \implies \frac{dy}{dx} = \frac{3t^2 - 3}{2t} = \frac{3}{2}t - \frac{3}{2t}$$

2. **Differentiate** dy/dx **with respect to** t:

$$\frac{d}{dt}\left(\frac{dy}{dx}\right) = \frac{d}{dt}\left(\frac{3}{2}t - \frac{3}{2}t^{-1}\right) = \frac{3}{2} - \frac{3}{2}(-1)t^{-2} = \frac{3}{2} + \frac{3}{2t^2}$$

3. **Find the second derivative,** d^2y/dx^2:

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}(dy/dx)}{dx/dt} = \frac{\frac{3}{2} + \frac{3}{2t^2}}{2t} = \frac{\frac{3t^2+3}{2t^2}}{2t} = \frac{3(t^2+1)}{4t^3}$$

4. **Determine Concavity**: The curve is concave upward when $d^2y/dx^2 > 0$. The numerator $3(t^2 + 1)$ is always positive. The sign is determined by the denominator $4t^3$. $4t^3 > 0$ when $t > 0$. The curve is concave upward for all $t > 0$.

Problem 16

For the parametric curve $x(t) = te^{-t}$ and $y(t) = t^2e^{-t}$, find the second derivative d^2y/dx^2.

Solution to Problem 16

1. **Find** dx/dt **and** dy/dt using the product rule:

$$\frac{dx}{dt} = (1)e^{-t} + t(-e^{-t}) = e^{-t}(1 - t)$$

$$\frac{dy}{dt} = (2t)e^{-t} + t^2(-e^{-t}) = te^{-t}(2 - t)$$

2. **Find** dy/dx:

$$\frac{dy}{dx} = \frac{te^{-t}(2-t)}{e^{-t}(1-t)} = \frac{t(2-t)}{1-t} = \frac{2t - t^2}{1-t}$$

3. **Differentiate** dy/dx **with respect to** t using the quotient rule:

$$\frac{d}{dt}\left(\frac{dy}{dx}\right) = \frac{(2-2t)(1-t) - (2t-t^2)(-1)}{(1-t)^2} = \frac{2 - 2t - 2t + 2t^2 + 2t - t^2}{(1-t)^2} = \frac{t^2 - 2t + 2}{(1-t)^2}$$

4. **Find** d^2y/dx^2:

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}(dy/dx)}{dx/dt} = \frac{\frac{t^2-2t+2}{(1-t)^2}}{e^{-t}(1-t)} = \frac{t^2-2t+2}{e^{-t}(1-t)^3}$$

Problem 17

Find the length of the parametric curve $x(t) = t^2$ and $y(t) = t^3$ for $0 \le t \le 1$.

Solution to Problem 17

1. **State the arc length formula**: $L = \int_\alpha^\beta \sqrt{(dx/dt)^2 + (dy/dt)^2}\, dt$.
2. **Find derivatives**: $dx/dt = 2t$ and $dy/dt = 3t^2$.
3. **Set up the integral**:

$$L = \int_0^1 \sqrt{(2t)^2 + (3t^2)^2}\, dt = \int_0^1 \sqrt{4t^2 + 9t^4}\, dt = \int_0^1 \sqrt{t^2(4+9t^2)}\, dt = \int_0^1 t\sqrt{4+9t^2}\, dt$$

4. **Evaluate the integral** using u-substitution. Let $u = 4 + 9t^2$, then $du = 18t\, dt$, so $t\, dt = du/18$. Change bounds: When $t = 0, u = 4$. When $t = 1, u = 13$.

$$L = \int_4^{13} \sqrt{u}\frac{du}{18} = \frac{1}{18}\left[\frac{2}{3}u^{3/2}\right]_4^{13} = \frac{1}{27}[u^{3/2}]_4^{13} = \frac{1}{27}(13^{3/2} - 4^{3/2}) = \frac{1}{27}(13\sqrt{13} - 8)$$

Problem 18

Find the length of the parametric curve $x(t) = \cos t + t\sin t$ and $y(t) = \sin t - t\cos t$ for $0 \le t \le 2\pi$.

Solution to Problem 18

1. **Find derivatives**: $dx/dt = -\sin t + (1 \cdot \sin t + t\cos t) = t\cos t$. $dy/dt = \cos t - (1 \cdot \cos t + t(-\sin t)) = t\sin t$.
2. **Simplify the integrand's radical**:

$$\sqrt{(dx/dt)^2 + (dy/dt)^2} = \sqrt{(t\cos t)^2 + (t\sin t)^2} = \sqrt{t^2\cos^2 t + t^2\sin^2 t} = \sqrt{t^2(1)} = |t|$$

   Since $0 \le t \le 2\pi$, $|t| = t$.

3. **Evaluate the integral**:

$$L = \int_0^{2\pi} t\, dt = \left[\frac{1}{2}t^2\right]_0^{2\pi} = \frac{1}{2}(2\pi)^2 - 0 = 2\pi^2$$

Problem 19

Find the area of the surface generated by revolving the curve $x(t) = t^3$ and $y(t) = t^2$ for $0 \le t \le 1$ about the x-axis.

Solution to Problem 19

1. **State the surface area formula** (revolving about x-axis): $A = \int_\alpha^\beta 2\pi y\, ds$, where $ds = \sqrt{(dx/dt)^2 + (dy/dt)^2}\, dt$.
2. **Find derivatives**: $dx/dt = 3t^2$ and $dy/dt = 2t$.
3. **Calculate** ds:
$$ds = \sqrt{(3t^2)^2 + (2t)^2}\, dt = \sqrt{9t^4 + 4t^2}\, dt = t\sqrt{9t^2 + 4}\, dt$$
4. **Set up the integral**:
$$A = \int_0^1 2\pi(t^2)(t\sqrt{9t^2 + 4})\, dt = 2\pi \int_0^1 t^3\sqrt{9t^2 + 4}\, dt$$
This integral is complex and typically beyond the scope of a standard problem without further hints. The setup is the key part of the exercise.

Problem 20

Find the area of the surface generated by revolving the curve $x(t) = \cos^2 t$ and $y(t) = \sin^2 t$ for $0 \le t \le \pi/2$ about the y-axis.

Solution to Problem 20

1. **State the surface area formula** (revolving about y-axis): $A = \int_\alpha^\beta 2\pi x\, ds$.
2. **Find derivatives**: $dx/dt = 2\cos t(-\sin t) = -2\sin t\cos t = -\sin(2t)$. $dy/dt = 2\sin t(\cos t) = 2\sin t\cos t = \sin(2t)$.
3. **Calculate** ds:
$$ds = \sqrt{(-\sin(2t))^2 + (\sin(2t))^2}\, dt = \sqrt{2\sin^2(2t)}\, dt = \sqrt{2}|\sin(2t)|\, dt$$
For $0 \le t \le \pi/2$, $\sin(2t)$ is non-negative, so $|\sin(2t)| = \sin(2t)$.
4. **Set up and evaluate the integral**:
$$A = \int_0^{\pi/2} 2\pi(\cos^2 t)(\sqrt{2}\sin(2t))\, dt = 2\sqrt{2}\pi \int_0^{\pi/2} \cos^2 t(2\sin t\cos t)\, dt = 4\sqrt{2}\pi \int_0^{\pi/2} \cos^3 t\sin t\, dt$$
Let $u = \cos t$, then $du = -\sin t\, dt$. Bounds: $t = 0 \to u = 1$, $t = \pi/2 \to u = 0$.
$$A = -4\sqrt{2}\pi \int_1^0 u^3\, du = 4\sqrt{2}\pi \int_0^1 u^3\, du = 4\sqrt{2}\pi \left[\frac{u^4}{4}\right]_0^1 = 4\sqrt{2}\pi\left(\frac{1}{4}\right) = \sqrt{2}\pi$$

Homework 2 Solutions

Homework - 2 - Solution Key

Problem 1: Coordinate Conversion (Polar to Rectangular)

Find the rectangular coordinates (x, y) for the point given in polar coordinates by $(r, \theta) = (6, 5\pi/6)$.

Solution to Problem 1

We use the conversion formulas $x = r\cos\theta$ and $y = r\sin\theta$. Given $(r, \theta) = (6, 5\pi/6)$:

$$x = 6\cos\left(\frac{5\pi}{6}\right) = 6\left(-\frac{\sqrt{3}}{2}\right) = -3\sqrt{3}$$

$$y = 6\sin\left(\frac{5\pi}{6}\right) = 6\left(\frac{1}{2}\right) = 3$$

The rectangular coordinates are $(x, y) = (-3\sqrt{3}, 3)$.

Problem 2: Coordinate Conversion (Rectangular to Polar)

Find the polar coordinates (r, θ) for the point given in rectangular coordinates by $(x, y) = (-\sqrt{3}, 1)$. Provide an answer where $r > 0$ and $0 \leq \theta < 2\pi$.

Solution to Problem 2

We use the conversion formulas $r = \sqrt{x^2 + y^2}$ and $\tan\theta = \frac{y}{x}$. Given $(x, y) = (-\sqrt{3}, 1)$:

$$r = \sqrt{(-\sqrt{3})^2 + 1^2} = \sqrt{3+1} = \sqrt{4} = 2$$

$$\tan\theta = \frac{1}{-\sqrt{3}} = -\frac{1}{\sqrt{3}}$$

The reference angle is $\tan^{-1}(1/\sqrt{3}) = \pi/6$. Since the point $(-\sqrt{3}, 1)$ is in Quadrant II ($x < 0, y > 0$), the angle is $\theta = \pi - \pi/6 = 5\pi/6$. The polar coordinates are $(r, \theta) = (2, 5\pi/6)$.

Problem 3: Sketching a Polar Curve

Sketch the graph of the polar equation $r = 4\sin\theta$. What shape is the curve?

Solution to Problem 3

The curve is a **circle**. To verify this, we can convert the equation to rectangular coordinates:

$$r = 4\sin\theta$$

Multiply both sides by r:

$$r^2 = 4r\sin\theta$$

Substitute $r^2 = x^2 + y^2$ and $y = r\sin\theta$:

$$x^2 + y^2 = 4y$$

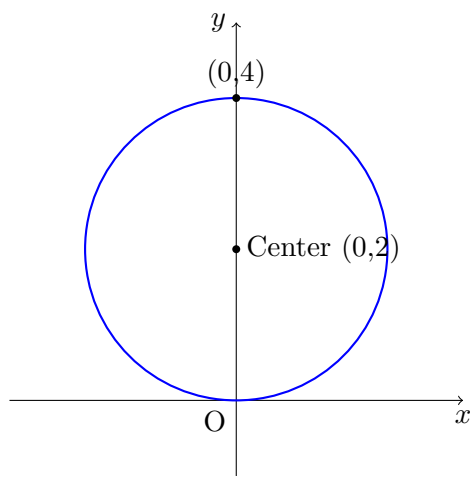
Rearrange and complete the square for y:

$$x^2 + y^2 - 4y = 0$$

$$x^2 + (y^2 - 4y + 4) = 4$$

$$x^2 + (y - 2)^2 = 2^2$$

This is the equation of a circle with center $(0, 2)$ and radius 2. The curve is traced once as θ goes from 0 to π.



Problem 4: Area of a Simple Polar Region

Set up and evaluate an integral to find the area of the region enclosed by the polar curve $r = 2\sin\theta$.

Solution to Problem 4

The curve $r = 2\sin\theta$ is a circle traced once for $0 \leq \theta \leq \pi$. The formula for the area of a polar region is $A = \frac{1}{2}\int_\alpha^\beta r^2 d\theta$.

$$A = \frac{1}{2}\int_0^\pi (2\sin\theta)^2 d\theta = \frac{1}{2}\int_0^\pi 4\sin^2\theta d\theta = 2\int_0^\pi \sin^2\theta d\theta$$

Using the half-angle identity $\sin^2\theta = \frac{1-\cos(2\theta)}{2}$:

$$A = 2\int_0^\pi \frac{1-\cos(2\theta)}{2} d\theta = \int_0^\pi (1 - \cos(2\theta)) d\theta$$

$$A = \left[\theta - \frac{1}{2}\sin(2\theta)\right]_0^\pi = \left(\pi - \frac{1}{2}\sin(2\pi)\right) - \left(0 - \frac{1}{2}\sin(0)\right) = (\pi - 0) - (0 - 0) = \pi$$

The area is π.

Problem 5: Area of a Cardioid

Find the total area of the region enclosed by the cardioid $r = 3 - 3\cos\theta$.

Solution to Problem 5
The cardioid is traced once for $0 \leq \theta \leq 2\pi$.

$$A = \frac{1}{2}\int_0^{2\pi}(3-3\cos\theta)^2 d\theta = \frac{1}{2}\int_0^{2\pi} 9(1-\cos\theta)^2 d\theta = \frac{9}{2}\int_0^{2\pi}(1-2\cos\theta+\cos^2\theta)d\theta$$

Using the half-angle identity $\cos^2\theta = \frac{1+\cos(2\theta)}{2}$:

$$A = \frac{9}{2}\int_0^{2\pi}\left(1-2\cos\theta+\frac{1+\cos(2\theta)}{2}\right)d\theta = \frac{9}{2}\int_0^{2\pi}\left(\frac{3}{2}-2\cos\theta+\frac{1}{2}\cos(2\theta)\right)d\theta$$

$$A = \frac{9}{2}\left[\frac{3}{2}\theta - 2\sin\theta + \frac{1}{4}\sin(2\theta)\right]_0^{2\pi} = \frac{9}{2}\left(\left(\frac{3}{2}(2\pi)-0+0\right)-(0-0+0)\right) = \frac{9}{2}(3\pi) = \frac{27\pi}{2}$$

The area is $\frac{27\pi}{2}$.

Problem 6: Area of a Rose Petal
Find the area of a single petal of the rose curve $r = 2\sin(3\theta)$.

Solution to Problem 6
To find the limits for one petal, we find consecutive values of θ for which $r = 0$. $2\sin(3\theta) = 0 \implies 3\theta = k\pi \implies \theta = k\pi/3$. One petal is traced between $\theta = 0$ and $\theta = \pi/3$.

$$A = \frac{1}{2}\int_0^{\pi/3}(2\sin(3\theta))^2 d\theta = \frac{1}{2}\int_0^{\pi/3} 4\sin^2(3\theta)d\theta = 2\int_0^{\pi/3}\sin^2(3\theta)d\theta$$

Using the half-angle identity $\sin^2(u) = \frac{1-\cos(2u)}{2}$:

$$A = 2\int_0^{\pi/3}\frac{1-\cos(6\theta)}{2}d\theta = \int_0^{\pi/3}(1-\cos(6\theta))d\theta$$

$$A = \left[\theta - \frac{1}{6}\sin(6\theta)\right]_0^{\pi/3} = \left(\frac{\pi}{3}-\frac{1}{6}\sin(2\pi)\right)-(0-0) = \frac{\pi}{3}$$

The area of one petal is $\frac{\pi}{3}$.

Problem 7: Area Between Curves
Find the area of the region that lies inside the circle $r = 3\cos\theta$ and outside the cardioid $r = 1+\cos\theta$.

Solution to Problem 7
First, find the intersection points by setting the radii equal: $3\cos\theta = 1+\cos\theta \implies 2\cos\theta = 1 \implies \cos\theta = 1/2$. The intersections occur at $\theta = -\pi/3$ and $\theta = \pi/3$. In this region, the circle $r = 3\cos\theta$ is the outer curve. The area is $A = \frac{1}{2}\int_{-\pi/3}^{\pi/3}(r_{outer}^2 - r_{inner}^2)d\theta$.

$$A = \frac{1}{2}\int_{-\pi/3}^{\pi/3}((3\cos\theta)^2-(1+\cos\theta)^2)d\theta = \frac{1}{2}\int_{-\pi/3}^{\pi/3}(9\cos^2\theta-(1+2\cos\theta+\cos^2\theta))d\theta$$

$$A = \frac{1}{2}\int_{-\pi/3}^{\pi/3}(8\cos^2\theta-2\cos\theta-1)d\theta$$

Using $\cos^2\theta = \frac{1+\cos(2\theta)}{2}$:

$$A = \frac{1}{2}\int_{-\pi/3}^{\pi/3}\left(8\left(\frac{1+\cos(2\theta)}{2}\right) - 2\cos\theta - 1\right)d\theta = \frac{1}{2}\int_{-\pi/3}^{\pi/3}(4 + 4\cos(2\theta) - 2\cos\theta - 1)d\theta$$

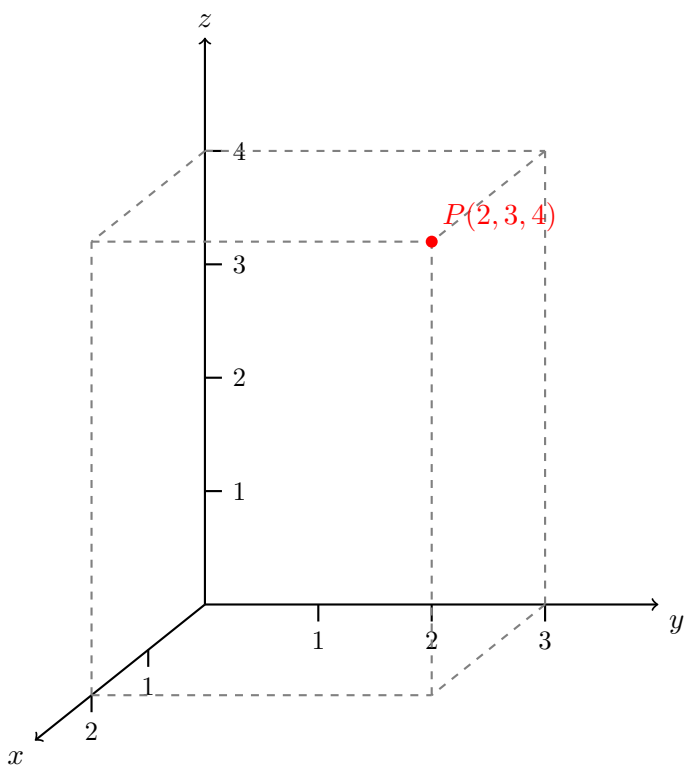
$$A = \frac{1}{2}\int_{-\pi/3}^{\pi/3}(3 + 4\cos(2\theta) - 2\cos\theta)d\theta = \frac{1}{2}\left[3\theta + 2\sin(2\theta) - 2\sin\theta\right]\Big|_{\pi/3}^{-\pi/3} = \pi$$

The area is π.

Problem 8: Plotting in 3D Rectangular Coordinates

Draw a 3D rectangular coordinate system and plot the point $P(2, 3, 4)$.

Solution to Problem 8



Problem 9: Cylindrical to Rectangular Conversion

Convert the cylindrical coordinates $(r, \theta, z) = (4, 4\pi/3, -2)$ to rectangular coordinates (x, y, z).

Solution to Problem 9

Using $x = r\cos\theta$, $y = r\sin\theta$, and $z = z$:

$$x = 4\cos(4\pi/3) = 4(-1/2) = -2$$

$$y = 4\sin(4\pi/3) = 4(-\sqrt{3}/2) = -2\sqrt{3}$$

$$z = -2$$

The rectangular coordinates are $(-2, -2\sqrt{3}, -2)$.

Problem 10: Rectangular to Cylindrical Conversion

Convert the rectangular coordinates $(4, -4, 3)$ to cylindrical coordinates (r, θ, z).

Solution to Problem 10

Using $r = \sqrt{x^2 + y^2}$, $\tan\theta = y/x$, and $z = z$:

$$r = \sqrt{4^2 + (-4)^2} = \sqrt{16 + 16} = \sqrt{32} = 4\sqrt{2}$$

$$\tan\theta = \frac{-4}{4} = -1$$

The point $(4, -4)$ is in Quadrant IV. The reference angle is $\pi/4$. So, $\theta = 2\pi - \pi/4 = 7\pi/4$.

$$z = 3$$

The cylindrical coordinates are $(4\sqrt{2}, 7\pi/4, 3)$.

Problem 11: Vectors Between Points

Find the vector $\vec{AB}$ that starts at point $A(1, 4, -2)$ and ends at point $B(3, 1, 1)$.

Solution to Problem 11

The vector is found by subtracting the coordinates of the initial point from the terminal point: $\vec{AB} = B - A$.

$$\vec{AB} = \langle 3 - 1, 1 - 4, 1 - (-2) \rangle = \langle 2, -3, 3 \rangle$$

(a) In component form: $\langle 2, -3, 3 \rangle$. (b) As a linear combination of unit vectors: $2\vec{i} - 3\vec{j} + 3\vec{k}$.

Problem 12: Vector Operations

Given $\vec{u} = \langle 2, -5, 1 \rangle$ and $\vec{v} = \langle -3, 0, 4 \rangle$, compute: (a) $\vec{u} + \vec{v}$, (b) $\vec{u} - \vec{v}$, (c) $4\vec{u}$.

Solution to Problem 12

(a) $\vec{u}+\vec{v} = \langle 2+(-3), -5+0, 1+4 \rangle = \langle -1, -5, 5 \rangle$. (b) $\vec{u}-\vec{v} = \langle 2-(-3), -5-0, 1-4 \rangle = \langle 5, -5, -3 \rangle$. (c) $4\vec{u} = 4\langle 2, -5, 1 \rangle = \langle 4(2), 4(-5), 4(1) \rangle = \langle 8, -20, 4 \rangle$.

Problem 13: Linear Combinations of Vectors

Let $\vec{a} = 2\vec{i} + 3\vec{j} - \vec{k}$ and $\vec{b} = \vec{i} - 4\vec{j} + 5\vec{k}$. Find $2\vec{a} - 3\vec{b}$.

Solution to Problem 13

First, find $2\vec{a}$ and $3\vec{b}$:

$$2\vec{a} = 2(2\vec{i} + 3\vec{j} - \vec{k}) = 4\vec{i} + 6\vec{j} - 2\vec{k} = \langle 4, 6, -2 \rangle$$

$$3\vec{b} = 3(\vec{i} - 4\vec{j} + 5\vec{k}) = 3\vec{i} - 12\vec{j} + 15\vec{k} = \langle 3, -12, 15 \rangle$$

Now, subtract:

$$2\vec{a} - 3\vec{b} = \langle 4 - 3, 6 - (-12), -2 - 15 \rangle = \langle 1, 18, -17 \rangle$$

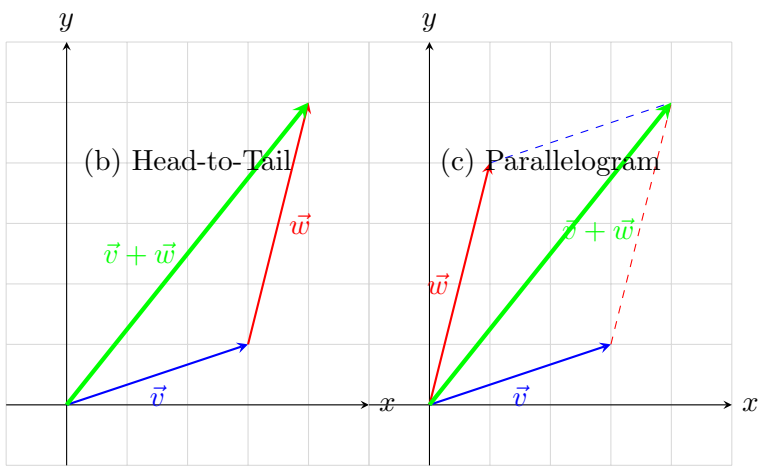
The vector is $\langle 1, 18, -17 \rangle$.

Problem 14: Graphical Vector Addition

Given $\vec{v} = \langle 3, 1\rangle$ and $\vec{w} = \langle 1, 4\rangle$, draw the resultant vector $\vec{v} + \vec{w}$.

Solution to Problem 14

The resultant vector is $\vec{v} + \vec{w} = \langle 3+1, 1+4\rangle = \langle 4, 5\rangle$.



Problem 15: Magnitude of a Vector

Find the magnitude of the vectors $\vec{w} = \langle 4, -2, 3\rangle$, $\vec{u} = \langle 4, -2, 5\rangle$, $\vec{v} = \langle -1, -2, 5\rangle$.

Solution to Problem 15

The magnitude $||\vec{a}||$ of a vector $\vec{a} = \langle a_1, a_2, a_3\rangle$ is $\sqrt{a_1^2 + a_2^2 + a_3^2}$.

$$||\vec{w}|| = \sqrt{4^2 + (-2)^2 + 3^2} = \sqrt{16+4+9} = \sqrt{29}$$

$$||\vec{u}|| = \sqrt{4^2 + (-2)^2 + 5^2} = \sqrt{16+4+25} = \sqrt{45} = 3\sqrt{5}$$

$$||\vec{v}|| = \sqrt{(-1)^2 + (-2)^2 + 5^2} = \sqrt{1+4+25} = \sqrt{30}$$

Problem 16: Finding a Unit Vector

Find the unit vector that has the same direction as the vectors $\vec{v} = \langle 1, -2, 2\rangle$, $\vec{w} = \langle 4, -2, 5\rangle$, and $\vec{u} = \langle -1, -2, 2\rangle$.

Solution to Problem 16

The unit vector $\hat{a}$ is found by $\vec{a}/||\vec{a}||$. For $\vec{v} = \langle 1, -2, 2\rangle$:

$$||\vec{v}|| = \sqrt{1^2 + (-2)^2 + 2^2} = \sqrt{1+4+4} = \sqrt{9} = 3$$

$$\hat{v} = \frac{\vec{v}}{||\vec{v}||} = \frac{1}{3}\langle 1, -2, 2\rangle = \left\langle \frac{1}{3}, -\frac{2}{3}, \frac{2}{3}\right\rangle$$

For $\vec{w} = \langle 4, -2, 5\rangle$:

$$||\vec{w}|| = \sqrt{4^2 + (-2)^2 + 5^2} = \sqrt{16+4+25} = \sqrt{45} = 3\sqrt{5}$$

$$\hat{w} = \frac{\vec{w}}{||\vec{w}||} = \frac{1}{3\sqrt{5}}\langle 4, -2, 5\rangle = \left\langle \frac{4}{3\sqrt{5}}, -\frac{2}{3\sqrt{5}}, \frac{5}{3\sqrt{5}} \right\rangle$$

For $\vec{u} = \langle -1, -2, 2\rangle$:

$$||\vec{u}|| = \sqrt{(-1)^2 + (-2)^2 + 2^2} = \sqrt{1+4+4} = \sqrt{9} = 3$$

$$\hat{u} = \frac{\vec{u}}{||\vec{u}||} = \frac{1}{3}\langle -1, -2, 2\rangle = \left\langle -\frac{1}{3}, -\frac{2}{3}, \frac{2}{3} \right\rangle$$

Problem 17: Normalizing a Vector

Find the unit vector in the direction of $\vec{u} = 5\vec{j} - 12\vec{k}$.

Solution to Problem 17

First, write the vector in component form: $\vec{u} = \langle 0, 5, -12\rangle$. Next, find its magnitude:

$$||\vec{u}|| = \sqrt{0^2 + 5^2 + (-12)^2} = \sqrt{0 + 25 + 144} = \sqrt{169} = 13$$

The unit vector is $\hat{u} = \frac{\vec{u}}{||\vec{u}||} = \frac{1}{13}\langle 0, 5, -12\rangle = \langle 0, \frac{5}{13}, -\frac{12}{13}\rangle$. In standard basis notation, this is $\frac{5}{13}\vec{j} - \frac{12}{13}\vec{k}$.

Problem 18: Creating a Vector of Specific Length

Find a vector that has the same direction as $\vec{a} = \langle -2, 4, 2\rangle$ but has a magnitude of 6.

Solution to Problem 18

First, find the unit vector in the direction of $\vec{a}$.

$$||\vec{a}|| = \sqrt{(-2)^2 + 4^2 + 2^2} = \sqrt{4 + 16 + 4} = \sqrt{24} = 2\sqrt{6}$$

$$\hat{a} = \frac{\vec{a}}{||\vec{a}||} = \frac{1}{2\sqrt{6}}\langle -2, 4, 2\rangle = \left\langle -\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}} \right\rangle$$

Now, multiply this unit vector by the desired magnitude of 6.

$$\vec{v} = 6\hat{a} = 6\left\langle -\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}} \right\rangle = \left\langle -\frac{6}{\sqrt{6}}, \frac{12}{\sqrt{6}}, \frac{6}{\sqrt{6}} \right\rangle$$

Rationalizing the denominator gives:

$$\vec{v} = \langle -\sqrt{6}, 2\sqrt{6}, \sqrt{6}\rangle$$

Problem 19: Application of Vector Operations

Two forces $\vec{F_1} = \langle 10, 5, -2\rangle$ and $\vec{F_2} = \langle -3, -4, 9\rangle$ act on an object. Find the resultant force vector $\vec{F_R}$ and its magnitude.

Solution to Problem 19

The resultant force is the sum of the individual forces:

$$\vec{F_R} = \vec{F_1} + \vec{F_2} = \langle 10 - 3, 5 - 4, -2 + 9\rangle = \langle 7, 1, 7\rangle$$

The magnitude of the resultant force is:

$$||\vec{F_R}|| = \sqrt{7^2 + 1^2 + 7^2} = \sqrt{49 + 1 + 49} = \sqrt{99} = 3\sqrt{11}$$

The resultant force is $\langle 7, 1, 7\rangle$ with a magnitude of $3\sqrt{11}$.

Problem 20: Finding an Endpoint

The vector $\vec{v} = 7\vec{i} - \vec{j} + 3\vec{k}$ starts at the point $P(2, 5, -1)$. Find the coordinates of the terminal point Q.

Solution to Problem 20
Let the coordinates of Q be (x, y, z). The vector from P to Q is given by $\vec{PQ} = Q - P = \langle x-2, y-5, z-(-1)\rangle$. We are given that $\vec{PQ} = \vec{v} = \langle 7, -1, 3\rangle$. Equating the components:

$$x - 2 = 7 \implies x = 9$$

$$y - 5 = -1 \implies y = 4$$

$$z + 1 = 3 \implies z = 2$$

Alternatively, the terminal point Q is the sum of the starting point P and the vector $\vec{v}$:

$$Q = P + \vec{v} = (2, 5, -1) + \langle 7, -1, 3\rangle = (2+7, 5-1, -1+3) = (9, 4, 2)$$

The coordinates of the terminal point Q are $(9, 4, 2)$.

Homework 3 Solutions

Homework 3 - The Dot and Cross Products - Solution Key

Problem 1: Dot Product Calculation

Given the vectors $\vec{u} = \langle 2, -3, 5\rangle$ and $\vec{v} = \langle 1, 4, -2\rangle$, compute the dot product $\vec{u} \cdot \vec{v}$.

Solution:

$$\vec{u} \cdot \vec{v} = (2)(1) + (-3)(4) + (5)(-2) = 2 - 12 - 10 = -20$$

Problem 2: Dot Product with Standard Unit Vectors

Let $\vec{a} = 3\vec{i} + 2\vec{j} - \vec{k}$ and $\vec{b} = 4\vec{i} + 5\vec{k}$. Find $\vec{a} \cdot \vec{b}$.

Solution:
First, write the vectors in component form: $\vec{a} = \langle 3, 2, -1\rangle$ and $\vec{b} = \langle 4, 0, 5\rangle$.

$$\vec{a} \cdot \vec{b} = (3)(4) + (2)(0) + (-1)(5) = 12 + 0 - 5 = 7$$

Problem 3: Finding the Angle Between Vectors

Find the angle θ between the vectors $\vec{p} = \langle 1, 1, 0\rangle$ and $\vec{q} = \langle 2, 0, 2\rangle$. State your answer in both radians and degrees.

Solution:
We use the formula $\cos\theta = \frac{\vec{p}\cdot\vec{q}}{||\vec{p}||||\vec{q}||}$.

- $\vec{p} \cdot \vec{q} = (1)(2) + (1)(0) + (0)(2) = 2$.
- $||\vec{p}|| = \sqrt{1^2 + 1^2 + 0^2} = \sqrt{2}$.

- $||\vec{q}|| = \sqrt{2^2 + 0^2 + 2^2} = \sqrt{8} = 2\sqrt{2}$.

$$\cos\theta = \frac{2}{\sqrt{2} \cdot 2\sqrt{2}} = \frac{2}{4} = \frac{1}{2}$$

$$\theta = \arccos\left(\frac{1}{2}\right) = \frac{\pi}{3} \text{ radians} = 60^\circ$$

Problem 4: Angles and Orthogonality

Consider the vectors $\vec{a} = \langle 2, 2, -1\rangle$, $\vec{b} = \langle 5, -4, 2\rangle$, $\vec{u} = \langle 2, 1, 4\rangle$ and $\vec{v} = \langle 3, -2, -1\rangle$. (a) Find the cosine of the angle between $\vec{a}$ and $\vec{b}$. (b) Determine if the vectors $\vec{u}$ and $\vec{v}$ are orthogonal.

Solution:
(a) $\vec{a} \cdot \vec{b} = (2)(5) + (2)(-4) + (-1)(2) = 10 - 8 - 2 = 0$. Since the dot product is zero, the vectors are orthogonal, and the cosine of the angle between them is $\cos(90^\circ) = 0$. (b) We compute the dot product of $\vec{u}$ and $\vec{v}$:

$$\vec{u} \cdot \vec{v} = (2)(3) + (1)(-2) + (4)(-1) = 6 - 2 - 4 = 0$$

Since their dot product is 0, the vectors $\vec{u}$ and $\vec{v}$ are orthogonal.

Problem 5: Finding an Unknown for Orthogonality

Consider the vectors $\vec{u} = \langle -6, k, 2\rangle$ and $\vec{v} = \langle 2, 5, -1\rangle$. Find the value of the constant k such that $\vec{u}$ and $\vec{v}$ are orthogonal.

Solution:
For the vectors to be orthogonal, their dot product must be zero.

$$\vec{u} \cdot \vec{v} = (-6)(2) + (k)(5) + (2)(-1) = 0$$

$$-12 + 5k - 2 = 0$$

$$5k - 14 = 0 \implies 5k = 14 \implies k = \frac{14}{5}$$

Problem 6: Scalar and Vector Projections

Let $\vec{b} = \langle 2, 3, 1\rangle$ and $\vec{a} = \langle -1, 4, 8\rangle$. (a) Find ${}_{\vec{a}}\vec{b}$. (b) Find ${}_{\vec{a}}\vec{b}$.

Solution:
First, we need $\vec{a} \cdot \vec{b}$ and $||\vec{a}||$.

- $\vec{a} \cdot \vec{b} = (-1)(2) + (4)(3) + (8)(1) = -2 + 12 + 8 = 18$.
- $||\vec{a}|| = \sqrt{(-1)^2 + 4^2 + 8^2} = \sqrt{1 + 16 + 64} = \sqrt{81} = 9$.

(a) The scalar projection is ${}_{\vec{a}}\vec{b} = \frac{\vec{a}\cdot\vec{b}}{||\vec{a}||} = \frac{18}{9} = 2$. (b) The vector projection is ${}_{\vec{a}}\vec{b} = \left(\frac{\vec{a}\cdot\vec{b}}{||\vec{a}||^2}\right)\vec{a} = \left(\frac{18}{81}\right)\langle -1, 4, 8\rangle = \frac{2}{9}\langle -1, 4, 8\rangle = \left\langle -\frac{2}{9}, \frac{8}{9}, \frac{16}{9}\right\rangle$.

Problem 7: Vector Decomposition

Decompose the vector $\vec{u} = \langle 3, 1, -2 \rangle$ into two orthogonal vectors, one parallel to $\vec{v} = \langle 1, 1, 1 \rangle$ and one orthogonal to $\vec{v}$.

Solution:
(a) The parallel component is the vector projection of $\vec{u}$ onto $\vec{v}$.

- $\vec{u} \cdot \vec{v} = (3)(1) + (1)(1) + (-2)(1) = 2$.
- $||\vec{v}||^2 = 1^2 + 1^2 + 1^2 = 3$.

$$\vec{u}_{\parallel} =_{\vec{v}} \vec{u} = \left(\frac{\vec{u} \cdot \vec{v}}{||\vec{v}||^2} \right) \vec{v} = \frac{2}{3} \langle 1, 1, 1 \rangle = \left\langle \frac{2}{3}, \frac{2}{3}, \frac{2}{3} \right\rangle$$

(b) The orthogonal component is found by subtraction: $\vec{u}_{\perp} = \vec{u} - \vec{u}_{\parallel}$.

$$\vec{u}_{\perp} = \langle 3, 1, -2 \rangle - \left\langle \frac{2}{3}, \frac{2}{3}, \frac{2}{3} \right\rangle = \left\langle \frac{9}{3} - \frac{2}{3}, \frac{3}{3} - \frac{2}{3}, -\frac{6}{3} - \frac{2}{3} \right\rangle = \left\langle \frac{7}{3}, \frac{1}{3}, -\frac{8}{3} \right\rangle$$

Verification: $\vec{u}_{\perp} \cdot \vec{v} = (\frac{7}{3})(1) + (\frac{1}{3})(1) + (-\frac{8}{3})(1) = \frac{7+1-8}{3} = 0$. It is orthogonal.

Problem 8: Application - Work

A constant force $\vec{F} = \langle 4, 3, 5 \rangle$ moves an object from $P(1, 0, 2)$ to $Q(3, 1, 5)$. Find the work done.

Solution:
First, find the displacement vector $\vec{D} = \vec{PQ}$.

$$\vec{D} = \langle 3 - 1, 1 - 0, 5 - 2 \rangle = \langle 2, 1, 3 \rangle$$

Work is the dot product of force and displacement:

$$W = \vec{F} \cdot \vec{D} = \langle 4, 3, 5 \rangle \cdot \langle 2, 1, 3 \rangle = (4)(2) + (3)(1) + (5)(3) = 8 + 3 + 15 = 26$$

The work done is 26 Joules.

Problem 9: Cross Product Calculation

Given $\vec{a} = \langle 1, 2, -1 \rangle$ and $\vec{b} = \langle 3, 0, 4 \rangle$, compute $\vec{a} \times \vec{b}$.

Solution:

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 2 & -1 \\ 3 & 0 & 4 \end{vmatrix} = \vec{i} \begin{vmatrix} 2 & -1 \\ 0 & 4 \end{vmatrix} - \vec{j} \begin{vmatrix} 1 & -1 \\ 3 & 4 \end{vmatrix} + \vec{k} \begin{vmatrix} 1 & 2 \\ 3 & 0 \end{vmatrix}$$

$$= \vec{i}(8 - 0) - \vec{j}(4 - (-3)) + \vec{k}(0 - 6) = 8\vec{i} - 7\vec{j} - 6\vec{k} = \langle 8, -7, -6 \rangle$$

Problem 10: Cross Product with Standard Unit Vectors

Let $\vec{u} = \vec{i} + 3\vec{j} - 2\vec{k}$ and $\vec{v} = 4\vec{i} - \vec{j}$. Find $\vec{u} \times \vec{v}$.

Solution:
In component form, $\vec{u} = \langle 1, 3, -2 \rangle$ and $\vec{v} = \langle 4, -1, 0 \rangle$.

$$\vec{u} \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 3 & -2 \\ 4 & -1 & 0 \end{vmatrix} = \vec{i}(0-2) - \vec{j}(0-(-8)) + \vec{k}(-1-12) = -2\vec{i} - 8\vec{j} - 13\vec{k}$$

Problem 11: Orthogonality Property

Let $\vec{p} = \langle 2, -1, 3 \rangle$ and $\vec{q} = \langle 1, 1, 1 \rangle$. Let $\vec{w} = \vec{p} \times \vec{q}$. Show that $\vec{w}$ is orthogonal to both $\vec{p}$ and $\vec{q}$.

Solution:
First, compute $\vec{w} = \vec{p} \times \vec{q}$:

$$\vec{w} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & -1 & 3 \\ 1 & 1 & 1 \end{vmatrix} = \vec{i}(-1-3) - \vec{j}(2-3) + \vec{k}(2-(-1)) = -4\vec{i} + \vec{j} + 3\vec{k} = \langle -4, 1, 3 \rangle$$

Now, check orthogonality using the dot product:

$$\vec{w} \cdot \vec{p} = (-4)(2) + (1)(-1) + (3)(3) = -8 - 1 + 9 = 0$$

$$\vec{w} \cdot \vec{q} = (-4)(1) + (1)(1) + (3)(1) = -4 + 1 + 3 = 0$$

Since both dot products are zero, $\vec{w}$ is orthogonal to both $\vec{p}$ and $\vec{q}$.

Problem 12: Anticommutativity Property

For the vectors $\vec{a}$ and $\vec{b}$ from Problem 9: (a) Compute $\vec{b} \times \vec{a}$. (b) How does it relate to $\vec{a} \times \vec{b}$?

Solution:
(a)

$$\vec{b} \times \vec{a} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 3 & 0 & 4 \\ 1 & 2 & -1 \end{vmatrix} = \vec{i}(0-8) - \vec{j}(-3-4) + \vec{k}(6-0) = -8\vec{i} + 7\vec{j} + 6\vec{k} = \langle -8, 7, 6 \rangle$$

(b) From Problem 9, $\vec{a} \times \vec{b} = \langle 8, -7, -6 \rangle$. We can see that $\vec{b} \times \vec{a} = -\langle 8, -7, -6 \rangle = -(\vec{a} \times \vec{b})$.

Problem 13: Area of a Parallelogram

Find the area of the parallelogram with adjacent sides $\vec{u} = \langle 3, 2, -1 \rangle$ and $\vec{v} = \langle 1, 2, 3 \rangle$.

Solution:
The area is the magnitude of the cross product.

$$\vec{u} \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 3 & 2 & -1 \\ 1 & 2 & 3 \end{vmatrix} = \vec{i}(6-(-2)) - \vec{j}(9-(-1)) + \vec{k}(6-2) = 8\vec{i} - 10\vec{j} + 4\vec{k} = \langle 8, -10, 4 \rangle$$

$$\text{Area} = ||\langle 8, -10, 4 \rangle|| = \sqrt{8^2 + (-10)^2 + 4^2} = \sqrt{64 + 100 + 16} = \sqrt{180} = \sqrt{36 \cdot 5} = 6\sqrt{5}$$

Problem 14: Area of a Triangle

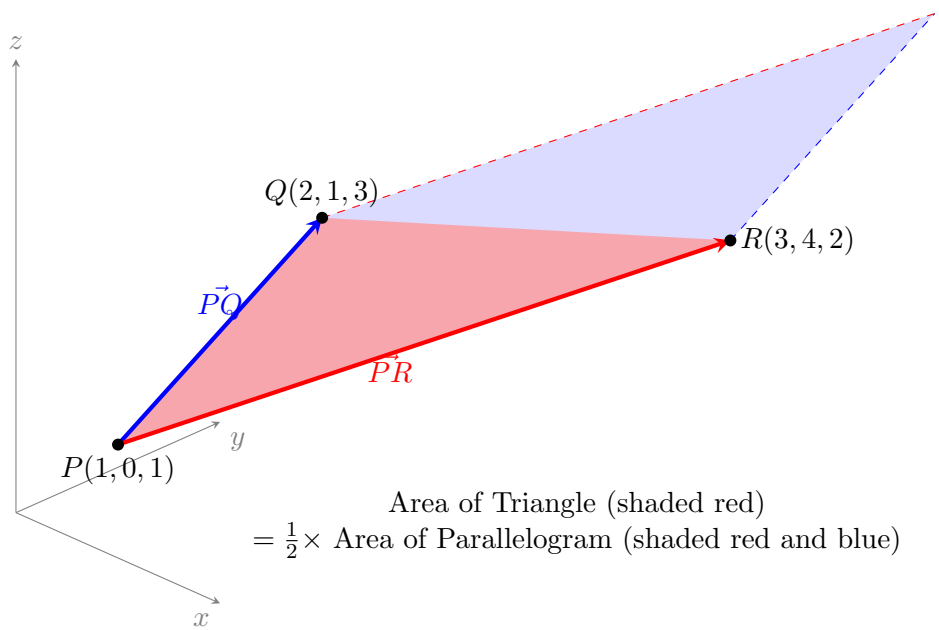
Find the area of the triangle with vertices $P(1,0,1)$, $Q(2,1,3)$, and $R(3,4,2)$.

Solution:
First, find two vectors forming the sides of the triangle, e.g., $\vec{PQ}$ and $\vec{PR}$.

$$\vec{PQ} = \langle 2-1, 1-0, 3-1 \rangle = \langle 1,1,2 \rangle$$

$$\vec{PR} = \langle 3-1, 4-0, 2-1 \rangle = \langle 2,4,1 \rangle$$

The area of the parallelogram formed by these vectors is $||\vec{PQ} \times \vec{PR}||$. The triangle is half of this.



$$\vec{PQ} \times \vec{PR} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 1 & 2 \\ 2 & 4 & 1 \end{vmatrix} = \vec{i}(1-8) - \vec{j}(1-4) + \vec{k}(4-2) = \langle -7, 3, 2 \rangle$$

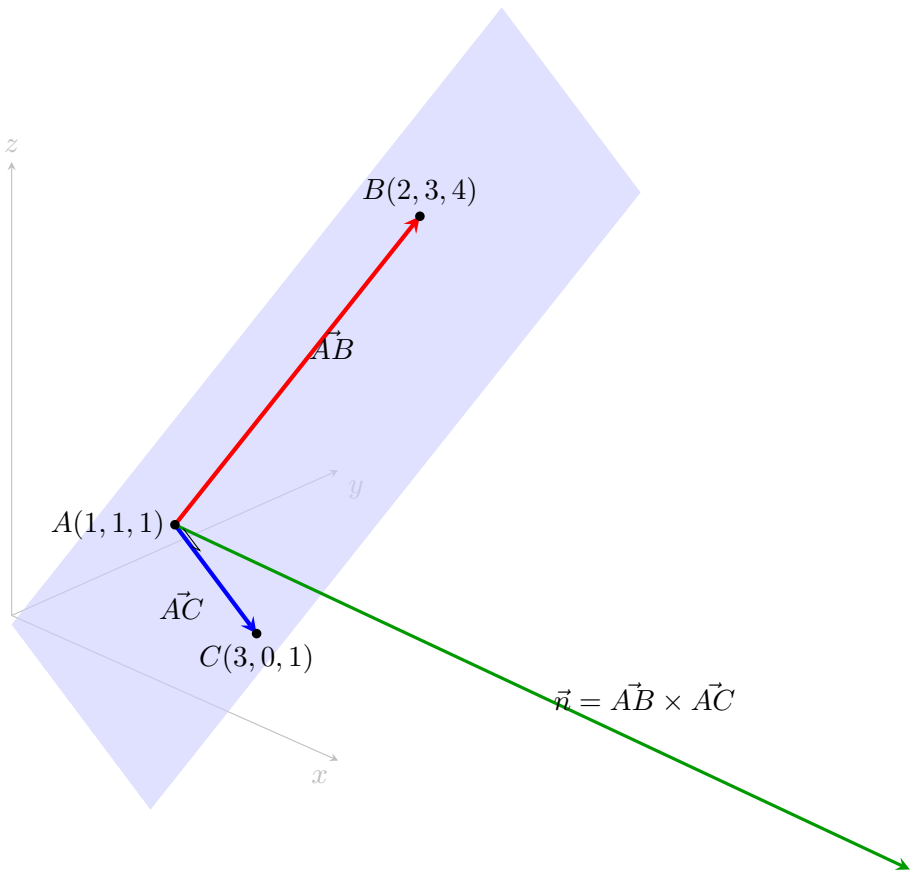
$$\text{Area of Parallelogram} = ||\langle -7,3,2 \rangle|| = \sqrt{(-7)^2 + 3^2 + 2^2} = \sqrt{49+9+4} = \sqrt{62}$$

$$\text{Area of Triangle} = \frac{1}{2}\sqrt{62}$$

Problem 15: Finding a Normal Vector

Find a vector orthogonal to the plane containing the points $A(1,1,1)$, $B(2,3,4)$, and $C(3,0,1)$.

Solution:
The cross product of two vectors in the plane will be orthogonal to the plane.



$$\vec{AB} = \langle 2-1, 3-1, 4-1 \rangle = \langle 1, 2, 3 \rangle$$

$$\vec{AC} = \langle 3-1, 0-1, 1-1 \rangle = \langle 2, -1, 0 \rangle$$

$$\vec{n} = \vec{AB} \times \vec{AC} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 2 & 3 \\ 2 & -1 & 0 \end{vmatrix} = \vec{i}(0-(-3)) - \vec{j}(0-6) + \vec{k}(-1-4) = \langle 3, 6, -5 \rangle$$

The vector $\langle 3, 6, -5 \rangle$ (or any non-zero scalar multiple) is orthogonal to the plane.

Problem 16: Scalar Triple Product Calculation

Let $\vec{a} = \langle 1, 2, 1 \rangle$, $\vec{b} = \langle 0, 3, 4 \rangle$, and $\vec{c} = \langle 5, -1, 0 \rangle$. Compute $\vec{a} \cdot (\vec{b} \times \vec{c})$.

Solution:
The scalar triple product can be computed with a 3x3 determinant.

$$\vec{a} \cdot (\vec{b} \times \vec{c}) = \begin{vmatrix} 1 & 2 & 1 \\ 0 & 3 & 4 \\ 5 & -1 & 0 \end{vmatrix} = 1(0 - (-4)) - 2(0 - 20) + 1(0 - 15) = 4 + 40 - 15 = 29$$

Problem 17: Alternative Scalar Triple Product Calculation

For the vectors in Problem 16, compute $(\vec{a} \times \vec{b}) \cdot \vec{c}$. What do you notice?

Solution:
First, find $\vec{a} \times \vec{b}$:

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 2 & 1 \\ 0 & 3 & 4 \end{vmatrix} = \vec{i}(8 - 3) - \vec{j}(4 - 0) + \vec{k}(3 - 0) = \langle 5, -4, 3 \rangle$$

Now, take the dot product with $\vec{c}$:

$$(\vec{a} \times \vec{b}) \cdot \vec{c} = \langle 5, -4, 3 \rangle \cdot \langle 5, -1, 0 \rangle = (5)(5) + (-4)(-1) + (3)(0) = 25 + 4 + 0 = 29$$

The result is the same as in Problem 16. This demonstrates the identity $\vec{a} \cdot (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) \cdot \vec{c}$.

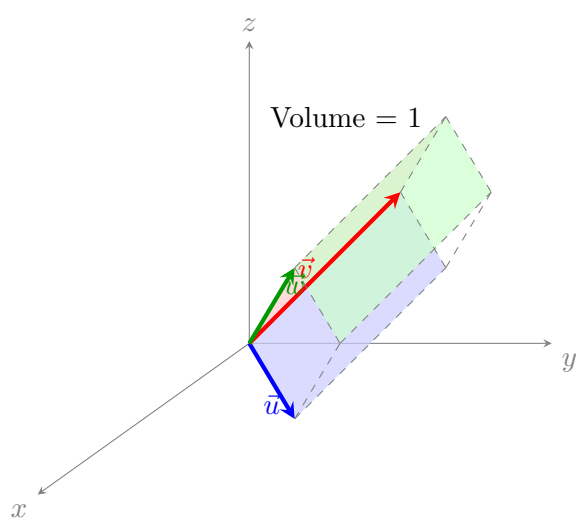
Problem 18: Volume of a Parallelepiped

Find the volume of the parallelepiped with adjacent edges $\vec{u} = \vec{i}+\vec{j}$, $\vec{v} = \vec{j}+\vec{k}$, and $\vec{w} = \vec{i}+\vec{j}+\vec{k}$.

Solution:
The volume is the absolute value of the scalar triple product. $\vec{u} = \langle 1, 1, 0\rangle, \vec{v} = \langle 0, 1, 1\rangle, \vec{w} = \langle 1, 1, 1\rangle$.

$$\text{Volume} = \left|\begin{vmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{vmatrix}\right| = |1(1-1) - 1(0-1) + 0(0-1)| = |0 - (-1) + 0| = |1| = 1$$

The volume is 1 cubic unit.



Problem 19: Checking for Coplanarity

Determine whether the vectors $\vec{a} = \langle 1, 5, -2\rangle$, $\vec{b} = \langle 3, -1, 0\rangle$, and $\vec{c} = \langle 5, 9, -4\rangle$ are coplanar.

Solution:
We check if their scalar triple product is zero.

$$\vec{a} \cdot (\vec{b} \times \vec{c}) = \begin{vmatrix} 1 & 5 & -2 \\ 3 & -1 & 0 \\ 5 & 9 & -4 \end{vmatrix} = 1(4-0) - 5(-12-0) + (-2)(27-(-5))$$

$$= 1(4) - 5(-12) - 2(32) = 4 + 60 - 64 = 0$$

Since the scalar triple product is 0, the volume of the parallelepiped is 0, and the vectors are coplanar.

Problem 20: Application - Torque

A force $\vec{F} = \langle -3, 1, 4\rangle$ is applied at point $P(2, 3, 1)$. Find the torque about the pivot point $Q(1, 1, -1)$.

Solution:
The lever arm vector is $\vec{r} = \vec{QP}$.

$$\vec{r} = \langle 2-1, 3-1, 1-(-1)\rangle = \langle 1, 2, 2\rangle$$

Torque is $\vec{\tau} = \vec{r} \times \vec{F}$.

$$\vec{\tau} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 2 & 2 \\ -3 & 1 & 4 \end{vmatrix} = \vec{i}(8-2) - \vec{j}(4-(-6)) + \vec{k}(1-(-6)) = 6\vec{i} - 10\vec{j} + 7\vec{k}$$

The torque vector is $\langle 6, -10, 7\rangle$.

Homework 4 Solutions

Homework 4 - Lines and Planes in Space - Solution Key

Part 1: Equations of Planes

Problem 1: The Basics of Plane Equations

A plane passes through $P_0(1, -2, 3)$ with normal vector $\vec{n} = \langle 4, 5, -1\rangle$. (a) Find standard form. (b) Find general form. (c) Does $Q(2, 0, 9)$ lie on the plane?

Solution: (a) The standard form is $a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$:

$$4(x-1) + 5(y-(-2)) - 1(z-3) = 0 \implies 4(x-1) + 5(y+2) - (z-3) = 0$$

(b) Distribute the standard form to get the general form $ax + by + cz = d$:

$$4x - 4 + 5y + 10 - z + 3 = 0 \implies 4x + 5y - z + 9 = 0 \implies 4x + 5y - z = -9$$

(c) To check if $Q(2, 0, 9)$ is on the plane, we plug its coordinates into the general form:

$$4(2) + 5(0) - (9) = 8 + 0 - 9 = -1$$

Since $-1 \neq -9$, the point Q does not lie on the plane.

Problem 2: Finding the Equation of a Plane from Three Points

Find the equation of the plane containing the points $A(1, 1, -1)$, $B(2, 0, 2)$, and $C(0, -2, 1)$.

Solution: (a) First, we find two vectors in the plane:

$$\vec{AB} = B - A = \langle 2-1, 0-1, 2-(-1)\rangle = \langle 1, -1, 3\rangle$$

$$\vec{AC} = C - A = \langle 0-1, -2-1, 1-(-1)\rangle = \langle -1, -3, 2\rangle$$

(b) The normal vector $\vec{n}$ is the cross product of these two vectors:

$$\vec{n} = \vec{AB} \times \vec{AC} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & -1 & 3 \\ -1 & -3 & 2 \end{vmatrix} = \vec{i}(-2-(-9)) - \vec{j}(2-(-3)) + \vec{k}(-3-1) = \langle 7, -5, -4\rangle$$

(c) Using point $A(1, 1, -1)$ and $\vec{n} = \langle 7, -5, -4\rangle$:

$$7(x-1) - 5(y-1) - 4(z-(-1)) = 0 \implies 7x - 7 - 5y + 5 - 4z - 4 = 0$$

$$7x - 5y - 4z - 6 = 0 \implies 7x - 5y - 4z = 6$$

Problem 3: Identifying Plane Relationships

Consider P1: $3x - 2y + z = 4$, P2: $x + 2y - z = 3$, P3: $-6x + 4y - 2z = 5$. (a) State normal vectors. (b) Which are parallel? (c) Which are orthogonal?

Solution: (a) The normal vectors are read from the coefficients:

$$\vec{n_1} = \langle 3, -2, 1\rangle, \quad \vec{n_2} = \langle 1, 2, -1\rangle, \quad \vec{n_3} = \langle -6, 4, -2\rangle$$

(b) Planes P1 and P3 are parallel because their normal vectors are scalar multiples: $\vec{n_3} = -2\vec{n_1}$. (c) There is no orthogonal pair. For example, $\vec{n_1} \cdot \vec{n_2} = (3)(1) + (-2)(2) + (1)(-1) = 3 - 4 - 1 = -2 \neq 0$.

Problem 4: Finding the Angle Between Two Planes

Find the angle between the planes $2x + 2y - z = 3$ and $x - 3y + 2z = 1$.

Solution:
The normal vectors are $\vec{n_1} = \langle 2, 2, -1\rangle$ and $\vec{n_2} = \langle 1, -3, 2\rangle$.

$$|\vec{n_1} \cdot \vec{n_2}| = |(2)(1) + (2)(-3) + (-1)(2)| = |2 - 6 - 2| = |-6| = 6$$

$$||\vec{n_1}|| = \sqrt{2^2 + 2^2 + (-1)^2} = \sqrt{4 + 4 + 1} = \sqrt{9} = 3$$

$$||\vec{n_2}|| = \sqrt{1^2 + (-3)^2 + 2^2} = \sqrt{1 + 9 + 4} = \sqrt{14}$$

$$\theta = \arccos\left(\frac{6}{3\sqrt{14}}\right) = \arccos\left(\frac{2}{\sqrt{14}}\right)$$

Problem 5: Finding a Parallel Plane

Find the equation of the plane that passes through $(3, -1, 4)$ and is parallel to $x - 2y + 5z = 10$.

Solution:
Parallel planes share the same normal vector. The normal vector of the given plane is $\vec{n} = \langle 1, -2, 5\rangle$. We use this normal vector and the given point $(3, -1, 4)$:

$$1(x - 3) - 2(y - (-1)) + 5(z - 4) = 0 \implies x - 3 - 2y - 2 + 5z - 20 = 0$$

$$x - 2y + 5z = 25$$

Problem 6: Intercepts of a Plane

Consider $2x + 4y + z = 8$. (a) Find intercepts. (b) Find area of the triangle in the first octant.

Solution:
(a)
x-intercept ($y = 0, z = 0$): $2x = 8 \implies x = 4$. Point is $(4, 0, 0)$.
y-intercept ($x = 0, z = 0$): $4y = 8 \implies y = 2$. Point is $(0, 2, 0)$.
z-intercept ($x = 0, y = 0$): $z = 8$. Point is $(0, 0, 8)$.
(b) The triangle has vertices $A(4, 0, 0)$, $B(0, 2, 0)$, and $C(0, 0, 8)$. We find two vectors from one vertex, e.g., $\vec{AB}$ and $\vec{AC}$.

$$\vec{AB} = \langle -4, 2, 0\rangle, \quad \vec{AC} = \langle -4, 0, 8\rangle$$

The area of the triangle is $\frac{1}{2}||\vec{AB} \times \vec{AC}||$.

$$\vec{AB} \times \vec{AC} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -4 & 2 & 0 \\ -4 & 0 & 8 \end{vmatrix} = \vec{i}(16) - \vec{j}(-32) + \vec{k}(8) = \langle 16, 32, 8 \rangle$$

$$\text{Area} = \frac{1}{2}||\langle 16, 32, 8 \rangle|| = \frac{1}{2}\sqrt{16^2 + 32^2 + 8^2} = \frac{1}{2}\sqrt{256 + 1024 + 64} = \frac{1}{2}\sqrt{1344} = \frac{1}{2}\sqrt{64 \cdot 21} = \frac{8\sqrt{21}}{2} = 4\sqrt{21}$$

Part 2: Equations of Lines

Problem 7

A line passes through $P_0(1, 0, -3)$ and is parallel to $\vec{v} = \langle 2, -4, 5 \rangle$.

Solution: (a) $x = 1 + 2t, \quad y = -4t, \quad z = -3 + 5t$. (b) $\frac{x-1}{2} = \frac{y}{-4} = \frac{z+3}{5}$. (c) Let $t = 1$.

$$x = 1 + 2(1) = 3,$$

$$y = -4(1) = -4,$$

$$z = -3 + 5(1) = 2.$$

Thus another point is $(3, -4, 2)$.

Problem 8

Find parametric equations for the line through $A(3, -1, 4)$ and $B(1, 5, -2)$.

Solution:
The direction vector is $\vec{v} = \vec{AB} = \langle 1 - 3, 5 - (-1), -2 - 4 \rangle = \langle -2, 6, -6 \rangle$. Using point A:

$$x = 3 - 2t, \quad y = -1 + 6t, \quad z = 4 - 6t$$

Problem 9

A line passes through $(2, 1, 5)$ with direction $\vec{v} = \langle 3, 0, -1 \rangle$. Write symmetric equations.

Solution:
Since the y-component of the direction vector is 0, we write that parametric equation separately.

$$y = 1, \quad \frac{x-2}{3} = \frac{z-5}{-1}$$

Problem 10

Are the lines $L_1 : \vec{r_1} = \langle 1 + 2t, 3t, 2 - t \rangle$ and $L_2 : \vec{r_2} = \langle -1 + 4s, 5 + 6s, 3 - 2s \rangle$ parallel?

Solution:
The direction vectors are $\vec{v_1} = \langle 2, 3, -1 \rangle$ and $\vec{v_2} = \langle 4, 6, -2 \rangle$. We check if they are scalar multiples: $\vec{v_2} = 2\langle 2, 3, -1 \rangle = 2\vec{v_1}$. Since the direction vectors are parallel, the lines are parallel.

Problem 11

Consider $L_1 : x = 5 + t, y = 1 + 3t, z = -2 - t$ and $L_2 : x = 2 - s, y = 4 + s, z = 1 + 2s$. Are they parallel? Do they intersect?

Solution:
(a) $\vec{v_1} = \langle 1, 3, -1 \rangle$ and $\vec{v_2} = \langle -1, 1, 2 \rangle$. They are not scalar multiples, so the lines are not parallel.

(b) We set the x, y, z components equal to see if there is a solution:

$$5 + t = 2 - s \implies s = -t - 3$$

$$1 + 3t = 4 + s$$

$$-2 - t = 1 + 2s$$

Substitute the first equation into the second equation:

$$1 + 3t = 4 + (-t - 3)$$

$$1 + 3t = 1 - t$$

$$4t = 0$$

$$t = 0$$

If $t = 0$, then from the first equations,

$$s = -t - 3 = -3.$$

Now we check if these values work in the third equation:

$$-2 - (0) = 1 + 2(-3)$$

$$-2 = 1 - 6$$

$$-2 = -5.$$

This is false. Since there is no solution, the lines do not intersect. Because they are not parallel and do not intersect, they are skew lines.

Problem 12

Find the line through $(0, 1, 2)$ perpendicular to both $\vec{a} = \langle 1, 1, 0 \rangle$ and $\vec{b} = \langle 0, 2, 3 \rangle$.

Solution:
The direction vector of the line must be perpendicular to both $\vec{a}$ and $\vec{b}$. The cross product provides such a vector.

$$\vec{v} = \vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 1 & 0 \\ 0 & 2 & 3 \end{vmatrix} = \vec{i}(3) - \vec{j}(3) + \vec{k}(2) = \langle 3, -3, 2 \rangle$$

Using the point $(0, 1, 2)$ and direction vector $\vec{v}$:

$$x = 3t, \quad y = 1 - 3t, \quad z = 2 + 2t$$

Part 3: Interactions Between Lines and Planes

Problem 13

Find the point where the line $x = 2 - t, y = 1 + 3t, z = 4t$ intersects the plane $2x - y + z = 2$.

Solution:
Substitute the parametric equations into the plane equation:

$$2(2 - t) - (1 + 3t) + (4t) = 2$$

$$4 - 2t - 1 - 3t + 4t = 2$$

$$3 - t = 2$$

$$t = 1$$

Plug $t = 1$ back into the line equations:

$$x = 2 - 1 = 1,$$

$$y = 1 + 3(1) = 4,$$

$$z = 4(1) = 4.$$

The intersection point is $(1, 4, 4)$.

Problem 14

Find the point where the line through $(1, 0, 1)$ and $(4, -2, 2)$ intersects the plane $x + y + z = 6$.

Solution:
First, find the line's equation. Direction vector $\vec{v} = \langle 3, -2, 1\rangle$. Using point $(1, 0, 1)$:

$$x = 1 + 3t, \quad y = -2t, \quad z = 1 + t$$

Substitute into the plane equation:

$$(1 + 3t) + (-2t) + (1 + t) = 6$$

$$2 + 2t = 6$$

$$2t = 4$$

$$t = 2$$

Plug $t = 2$ back into the line equations:

$$x = 1 + 3(2) = 7,$$

$$y = -2(2) = -4,$$

$$z = 1 + 2 = 3.$$

The intersection point is $(7, -4, 3)$.

Problem 15

Find the line of intersection of the planes $x + y + z = 1$ and $x - y + z = 1$.

Solution:
Normal vectors: $\vec{n_1} = \langle 1, 1, 1\rangle$, $\vec{n_2} = \langle 1, -1, 1\rangle$. The direction vector $\vec{v}$ of the line is the cross product of the normal vectors:

$$\vec{v} = \vec{n_1} \times \vec{n_2} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 1 & 1 \\ 1 & -1 & 1 \end{vmatrix} = \langle 1 - (-1), 1 - 1, -1 - 1\rangle = \langle 2, 0, -2\rangle$$

To find a point on the line, let $z = 0$ in both plane equations:

$$x + y = 1 \quad \text{and} \quad x - y = 1$$

Adding the two equations gives $2x = 2 \implies x = 1$, which means $y = 0$. A point on the line is $(1, 0, 0)$. The parametric equations are:

$$x = 1 + 2t, \quad y = 0, \quad z = -2t$$

Problem 16

Find the symmetric equations for the line of intersection of $2x+5z+3 = 0$ and $x-3y+z+2 = 0$.

Solution:
The normal vectors are $\vec{n_1} = \langle 2, 0, 5\rangle$ and $\vec{n_2} = \langle 1, -3, 1\rangle$. The direction vector $\vec{v}$ is their cross product:

$$\vec{v} = \vec{n_1} \times \vec{n_2} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 0 & 5 \\ 1 & -3 & 1 \end{vmatrix} = \langle 0 - (-15), 5 - 2, -6 - 0\rangle = \langle 15, 3, -6\rangle$$

To find a point, let $z = 1$:

$$2x + 5(1) + 3 = 02x = -8$$

$$x = -4$$

$$(-4) - 3y + (1) + 2 = 0$$

$$-1 = 3y$$

$$y = -1/3$$

A point on the line is $(-4, -1/3, 1)$. The symmetric equations are:

$$\frac{x + 4}{15} = \frac{y + 1/3}{3} = \frac{z - 1}{-6}$$

Problem 17

Find the plane containing the lines $L_1 : \frac{x-1}{2} = \frac{y}{3} = \frac{z+1}{-1}$ and $L_2 : \frac{x-1}{-1} = \frac{y}{2} = \frac{z+1}{1}$.

Solution:
The direction vectors are $\vec{v_1} = \langle 2, 3, -1\rangle$ and $\vec{v_2} = \langle -1, 2, 1\rangle$. The normal vector to the plane is their cross product:

$$\vec{n} = \vec{v_1} \times \vec{v_2} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 3 & -1 \\ -1 & 2 & 1 \end{vmatrix} = \langle 3-(-2), 1-2, 4-(-3)\rangle = \langle 5, -1, 7\rangle$$

Both lines pass through the point $(1, 0, -1)$. Using this point and the normal vector:

$$5(x-1) - 1(y-0) + 7(z+1) = 0$$

$$5x - 5 - y + 7z + 7 = 0$$

$$5x - y + 7z = -2$$

Problem 18

Find the plane through $P(1, 2, 3)$ containing the line $x = 3t, y = 1 + t, z = 2 - t$.

Solution:
We need two vectors in the plane. First, the line's direction vector is $\vec{v} = \langle 3, 1, -1\rangle$. Second, we find a vector from a point on the line to the given point P. A point on the line (at t=0) is $P_0(0, 1, 2)$. The vector is $\vec{u} = \vec{P_0P} = \langle 1, 1, 1\rangle$. The normal vector to the plane is the cross product of these two vectors:

$$\vec{n} = \vec{v} \times \vec{u} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 3 & 1 & -1 \\ 1 & 1 & 1 \end{vmatrix} = \langle 1-(-1), -1-3, 3-1\rangle = \langle 2, -4, 2\rangle$$

Using point $P(1, 2, 3)$:

$$2(x-1) - 4(y-2) + 2(z-3) = 0$$

$$2x - 2 - 4y + 8 + 2z - 6 = 0$$

$$2x - 4y + 2z = 0$$

Problem 19

Find the plane through $(6, 0, -2)$ and perpendicular to the line $x = 4 - t, y = 3 + 2t, z = t$.

Solution:
If the plane is perpendicular to the line, the line's direction vector is the plane's normal vector. The direction vector of the line is $\vec{v} = \langle -1, 2, 1\rangle$. This is our normal vector, $\vec{n}$. Using the point $(6, 0, -2)$:

$$-1(x-6) + 2(y-0) + 1(z-(-2)) = 0$$

$$-x + 6 + 2y + z + 2 = 0$$

$$-x + 2y + z = -8$$

Problem 20: Conceptual Question

Line L has direction $\vec{v}$ and plane P has normal $\vec{n}$. (a) What if $\vec{v} \cdot \vec{n} = 0$? (b) What if $\vec{v} \times \vec{n} = \vec{0}$?

Solution:
(a) If $\vec{v} \cdot \vec{n} = 0$, the line's direction vector is orthogonal to the plane's normal vector. This means the **line is parallel to the plane** (or possibly lies within the plane). (b) If $\vec{v} \times \vec{n} = \vec{0}$, the line's direction vector is parallel to the plane's normal vector. This means the **line is orthogonal (perpendicular) to the plane**.

Homework 5 Solutions

Homework 5 - Calculus of Vector-Valued Functions - Solution Key

Problem 1: Derivatives and Tangent Vectors

Consider the vector-valued function $\vec{r}(t) = \langle t^3 - t, \sqrt{2t+1}, \ln(t) \rangle$.

(a) Find the derivative of the vector function, $\vec{r}'(t)$.

(b) Find the tangent vector at the point where $t = 1$.

Solution: (a) We differentiate each component of $\vec{r}(t)$:

$$\vec{r}'(t) = \left\langle \frac{d}{dt}(t^3 - t), \frac{d}{dt}((2t+1)^{1/2}), \frac{d}{dt}(\ln(t)) \right\rangle$$

$$\vec{r}'(t) = \left\langle 3t^2 - 1, \frac{1}{2}(2t+1)^{-1/2} \cdot 2, \frac{1}{t} \right\rangle = \left\langle 3t^2 - 1, \frac{1}{\sqrt{2t+1}}, \frac{1}{t} \right\rangle$$

(b) We evaluate $\vec{r}'(t)$ at $t = 1$:

$$\vec{r}'(1) = \left\langle 3(1)^2 - 1, \frac{1}{\sqrt{2(1)+1}}, \frac{1}{1} \right\rangle = \left\langle 2, \frac{1}{\sqrt{3}}, 1 \right\rangle$$

Problem 2: Finding the Tangent Line

Find the parametric equations for the tangent line to the curve with the given vector equation at the specified point.

$$\vec{r}(t) = \langle e^{-t}, t^2 + 1, \sin(2t) \rangle; \quad t = 0$$

Solution:
(a) The point on the curve is found by evaluating $\vec{r}(t)$ at $t = 0$:

$$\vec{r}(0) = \langle e^{-0}, 0^2 + 1, \sin(2 \cdot 0) \rangle = \langle 1, 1, 0 \rangle$$

(b) The tangent vector (direction vector) is found by evaluating the derivative $\vec{r}'(t)$ at $t = 0$:

$$\vec{r}'(t) = \langle -e^{-t}, 2t, 2\cos(2t) \rangle$$

$$\vec{r}'(0) = \langle -e^{-0}, 2(0), 2\cos(0) \rangle = \langle -1, 0, 2 \rangle$$

(c) The parametric equations for the tangent line are formed using the point $(1, 1, 0)$ and the direction vector $\langle -1, 0, 2 \rangle$:

$$x(t) = 1 - t, \quad y(t) = 1, \quad z(t) = 2t$$

Problem 3: Finding the Unit Tangent Vector

Consider the curve $\vec{r}(t) = \langle 2t, t^2, \frac{1}{3}t^3 \rangle$.

Solution:
(a) The velocity vector $\vec{r}'(t)$ is:

$$\vec{r}'(t) = \langle 2, 2t, t^2 \rangle$$

(b) The speed of the particle is the magnitude of the velocity vector:

$$||\vec{r}'(t)|| = \sqrt{2^2 + (2t)^2 + (t^2)^2} = \sqrt{4 + 4t^2 + t^4}$$

As hinted, this is a perfect square:

$$||\vec{r}'(t)|| = \sqrt{(t^2+2)^2} = t^2 + 2$$

(c) The unit tangent vector $\vec{T}(t)$ is $\vec{r}'(t)$ divided by its magnitude:

$$\vec{T}(t) = \frac{\vec{r}'(t)}{||\vec{r}'(t)||} = \frac{\langle 2, 2t, t^2 \rangle}{t^2+2} = \left\langle \frac{2}{t^2+2}, \frac{2t}{t^2+2}, \frac{t^2}{t^2+2} \right\rangle$$

Problem 4: Finding the Unit Tangent Vector at a Point

Find the unit tangent vector $\vec{T}(t)$ for the curve $\vec{r}(t) = \langle t, 2\cos(t), 2\sin(t) \rangle$ at the point where $t = \pi/3$.

Solution:
First, find the derivative $\vec{r}'(t)$:

$$\vec{r}'(t) = \langle 1, -2\sin(t), 2\cos(t) \rangle$$

Next, find its magnitude:

$$||\vec{r}'(t)|| = \sqrt{1^2 + (-2\sin(t))^2 + (2\cos(t))^2} = \sqrt{1 + 4\sin^2(t) + 4\cos^2(t)} = \sqrt{1 + 4(1)} = \sqrt{5}$$

The unit tangent vector is:

$$\vec{T}(t) = \frac{\vec{r}'(t)}{||\vec{r}'(t)||} = \frac{1}{\sqrt{5}} \langle 1, -2\sin(t), 2\cos(t) \rangle$$

Finally, evaluate at $t = \pi/3$:

$$\vec{T}(\pi/3) = \frac{1}{\sqrt{5}} \langle 1, -2\sin(\pi/3), 2\cos(\pi/3) \rangle = \frac{1}{\sqrt{5}} \left\langle 1, -2\left(\frac{\sqrt{3}}{2}\right), 2\left(\frac{1}{2}\right) \right\rangle = \frac{1}{\sqrt{5}} \langle 1, -\sqrt{3}, 1 \rangle$$

Problem 5: Arc Length

Find the length of the curve given by the vector function $\vec{r}(t) = \langle 2t, \frac{4}{3}t^{3/2}, \frac{1}{2}t^2 \rangle$ from $t = 1$ to $t = 2$.

Solution:
(a) Find the derivative vector $\vec{r}'(t)$:

$$\vec{r}'(t) = \left\langle 2, \frac{4}{3} \cdot \frac{3}{2}t^{1/2}, \frac{1}{2} \cdot 2t \right\rangle = \langle 2, 2\sqrt{t}, t \rangle$$

(b) Find the magnitude (speed), looking for a perfect square:

$$||\vec{r}'(t)|| = \sqrt{2^2 + (2\sqrt{t})^2 + t^2} = \sqrt{4 + 4t + t^2} = \sqrt{(t+2)^2} = t + 2$$

(c) Set up and evaluate the definite integral for arc length L:

$$L = \int_1^2 ||\vec{r}'(t)||dt = \int_1^2 (t+2)dt$$

$$L = \left[\frac{1}{2}t^2 + 2t\right]_1^2 = \left(\frac{1}{2}(2)^2 + 2(2)\right) - \left(\frac{1}{2}(1)^2 + 2(1)\right) = (2+4) - (\frac{1}{2} + 2) = 6 - \frac{5}{2} = \frac{7}{2}$$

Problem 6: Velocity and Acceleration

A particle's position is given by the vector function $\vec{r}(t) = \langle t\sin(t), t\cos(t), t^2 \rangle$.

Solution:
(a) The velocity vector $\vec{v}(t)$ is the first derivative of $\vec{r}(t)$. We use the product rule for the first two components:

$$\vec{v}(t) = \vec{r}'(t) = \langle \sin(t) + t\cos(t), \cos(t) - t\sin(t), 2t \rangle$$

(b) The acceleration vector $\vec{a}(t)$ is the second derivative of $\vec{r}(t)$ (the derivative of $\vec{v}(t)$):

$$\vec{a}(t) = \vec{v}'(t) = \langle \cos(t) + (\cos(t) - t\sin(t)), -\sin(t) - (\sin(t) + t\cos(t)), 2 \rangle$$

$$\vec{a}(t) = \langle 2\cos(t) - t\sin(t), -2\sin(t) - t\cos(t), 2 \rangle$$

Problem 7: Finding the Principal Unit Normal Vector

Consider the circular path $\vec{r}(t) = \langle 5\cos(t), 5\sin(t), 4 \rangle$.

Solution:
(a) Find the unit tangent vector $\vec{T}(t)$:

$$\vec{r}'(t) = \langle -5\sin(t), 5\cos(t), 0 \rangle$$

$$||\vec{r}'(t)|| = \sqrt{(-5\sin t)^2 + (5\cos t)^2 + 0^2} = \sqrt{25(\sin^2 t + \cos^2 t)} = 5$$

$$\vec{T}(t) = \frac{\vec{r}'(t)}{||\vec{r}'(t)||} = \frac{1}{5}\langle -5\sin(t), 5\cos(t), 0 \rangle = \langle -\sin(t), \cos(t), 0 \rangle$$

(b) Find the derivative of the unit tangent vector, $\vec{T}'(t)$:

$$\vec{T}'(t) = \langle -\cos(t), -\sin(t), 0\rangle$$

(c) Find the principal unit normal vector $\vec{N}(t)$:

$$||\vec{T}'(t)|| = \sqrt{(-\cos t)^2 + (-\sin t)^2 + 0^2} = \sqrt{\cos^2 t + \sin^2 t} = 1$$

$$\vec{N}(t) = \frac{\vec{T}'(t)}{||\vec{T}'(t)||} = \frac{1}{1}\langle -\cos(t), -\sin(t), 0\rangle = \langle -\cos(t), -\sin(t), 0\rangle$$

Problem 8: Finding $\vec{N}(t)$ at a Point

Consider the curve $\vec{r}(t) = \langle t^2, \sin(t) - t\cos(t), \cos(t) + t\sin(t)\rangle$. Find the principal unit normal vector $\vec{N}(t)$ at the point where $t = \pi$.

Solution:
First, find $\vec{r}'(t)$:

$$\vec{r}'(t) = \langle 2t, \cos t - (\cos t - t\sin t), -\sin t + (\sin t + t\cos t)\rangle = \langle 2t, t\sin t, t\cos t\rangle$$

Next, find $||\vec{r}'(t)||$:

$$||\vec{r}'(t)|| = \sqrt{(2t)^2 + (t\sin t)^2 + (t\cos t)^2} = \sqrt{4t^2 + t^2(\sin^2 t + \cos^2 t)} = \sqrt{5t^2} = t\sqrt{5}$$

Now find $\vec{T}(t)$:

$$\vec{T}(t) = \frac{\vec{r}'(t)}{||\vec{r}'(t)||} = \frac{\langle 2t, t\sin t, t\cos t\rangle}{t\sqrt{5}} = \frac{1}{\sqrt{5}}\langle 2, \sin t, \cos t\rangle$$

Differentiate $\vec{T}(t)$ to get $\vec{T}'(t)$:

$$\vec{T}'(t) = \frac{1}{\sqrt{5}}\langle 0, \cos t, -\sin t\rangle$$

Find the magnitude $||\vec{T}'(t)||$:

$$||\vec{T}'(t)|| = \frac{1}{\sqrt{5}}\sqrt{0^2 + \cos^2 t + (-\sin t)^2} = \frac{1}{\sqrt{5}}$$

The principal unit normal is $\vec{N}(t) = \vec{T}'(t)/||\vec{T}'(t)||$:

$$\vec{N}(t) = \frac{\frac{1}{\sqrt{5}}\langle 0, \cos t, -\sin t\rangle}{1/\sqrt{5}} = \langle 0, \cos t, -\sin t\rangle$$

Finally, evaluate at $t = \pi$:

$$\vec{N}(\pi) = \langle 0, \cos\pi, -\sin\pi\rangle = \langle 0, -1, 0\rangle$$

Problem 9: Orthogonality of $\vec{T}$ and $\vec{N}$

For the circular path in Problem 7, $\vec{r}(t) = \langle 5\cos(t), 5\sin(t), 4\rangle$, use your computed vectors $\vec{T}(t)$ and $\vec{N}(t)$ to show explicitly that they are orthogonal for all values of t by computing their dot product, $\vec{T}(t) \cdot \vec{N}(t)$.

Solution:
From Problem 7, we have:

$$\vec{T}(t) = \langle -\sin(t), \cos(t), 0\rangle$$
$$\vec{N}(t) = \langle -\cos(t), -\sin(t), 0\rangle$$

Computing the dot product:

$$\vec{T}(t) \cdot \vec{N}(t) = (-\sin t)(-\cos t) + (\cos t)(-\sin t) + (0)(0)$$
$$= \sin t \cos t - \cos t \sin t = 0$$

Since the dot product is 0 for all t, the vectors $\vec{T}(t)$ and $\vec{N}(t)$ are always orthogonal.

Problem 10: Curvature

For the helix $\vec{r}(t) = \langle \cos(t), \sin(t), t\rangle$ calculate the curvature $\kappa(t)$.

Solution:
We use the cross product formula for curvature, $\kappa(t) = \frac{||\vec{r}'(t)\times\vec{r}''(t)||}{||\vec{r}'(t)||^3}$. First, we find the first and second derivatives of $\vec{r}(t)$:

$$\vec{r}'(t) = \langle -\sin t, \cos t, 1\rangle$$
$$\vec{r}''(t) = \langle -\cos t, -\sin t, 0\rangle$$

Next, we compute the cross product $\vec{r}'(t) \times \vec{r}''(t)$:

$$\vec{r}'(t) \times \vec{r}''(t) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -\sin t & \cos t & 1 \\ -\cos t & -\sin t & 0 \end{vmatrix} = \vec{i}(0 - (-\sin t)) - \vec{j}(0 - (-\cos t)) + \vec{k}(\sin^2 t - (-\cos^2 t))$$
$$= \langle \sin t, -\cos t, \sin^2 t + \cos^2 t\rangle = \langle \sin t, -\cos t, 1\rangle$$

Now we find the magnitudes of $\vec{r}'(t)$ and the cross product:

$$||\vec{r}'(t)|| = \sqrt{(-\sin t)^2 + (\cos t)^2 + 1^2} = \sqrt{\sin^2 t + \cos^2 t + 1} = \sqrt{1+1} = \sqrt{2}$$
$$||\vec{r}'(t) \times \vec{r}''(t)|| = \sqrt{(\sin t)^2 + (-\cos t)^2 + 1^2} = \sqrt{\sin^2 t + \cos^2 t + 1} = \sqrt{1+1} = \sqrt{2}$$

Finally, we calculate the curvature:

$$\kappa(t) = \frac{||\vec{r}'(t) \times \vec{r}''(t)||}{||\vec{r}'(t)||^3} = \frac{\sqrt{2}}{(\sqrt{2})^3} = \frac{\sqrt{2}}{2\sqrt{2}} = \frac{1}{2}$$

Problem 11: Arc Length of a Spiral

Find the length of the curve $\vec{r}(t) = \langle e^t\cos(t), e^t\sin(t), e^t\rangle$ from $t = 0$ to $t = 1$.

Solution:
First, find $\vec{r}'(t)$:

$$\vec{r}'(t) = \langle e^t \cos t - e^t \sin t, e^t \sin t + e^t \cos t, e^t \rangle = e^t \langle \cos t - \sin t, \sin t + \cos t, 1 \rangle$$

Next, find its magnitude $||\vec{r}'(t)||$:

$$\begin{aligned} ||\vec{r}'(t)||^2 &= (e^t)^2[(\cos t - \sin t)^2 + (\sin t + \cos t)^2 + 1^2] \\ &= e^{2t}[(\cos^2 t - 2\sin t \cos t + \sin^2 t) + (\sin^2 t + 2\sin t \cos t + \cos^2 t) + 1] \\ &= e^{2t}[(1) + (1) + 1] = 3e^{2t} \end{aligned}$$

$$||\vec{r}'(t)|| = \sqrt{3e^{2t}} = e^t\sqrt{3}$$

Now integrate from $t = 0$ to $t = 1$:

$$L = \int_0^1 e^t\sqrt{3}dt = \sqrt{3}[e^t]_0^1 = \sqrt{3}(e^1 - e^0) = \sqrt{3}(e - 1)$$

Problem 12: Integration and Projectile Motion

A projectile is fired from a height of 100 meters with an initial velocity vector of $\vec{v}(0) = \langle 10, 20, 30 \rangle$ m/s. The only force acting on it is gravity, so its acceleration is given by $\vec{a}(t) = \langle 0, 0, -9.8 \rangle$ m/s^2.

Solution:
(a) Find the velocity vector by integrating the acceleration vector:

$$\vec{v}(t) = \int \vec{a}(t)dt = \int \langle 0, 0, -9.8 \rangle dt = \langle C_1, C_2, -9.8t + C_3 \rangle$$

Use the initial velocity $\vec{v}(0) = \langle 10, 20, 30 \rangle$ to find the constants:

$$\vec{v}(0) = \langle C_1, C_2, C_3 \rangle = \langle 10, 20, 30 \rangle$$

So, the velocity vector is $\vec{v}(t) = \langle 10, 20, 30 - 9.8t \rangle$.

(b) Find the position vector by integrating the velocity vector:

$$\vec{r}(t) = \int \vec{v}(t)dt = \int \langle 10, 20, 30 - 9.8t \rangle dt = \langle 10t + D_1, 20t + D_2, 30t - 4.9t^2 + D_3 \rangle$$

Use the initial position $\vec{r}(0) = \langle 0, 0, 100 \rangle$ to find the constants:

$$\vec{r}(0) = \langle D_1, D_2, D_3 \rangle = \langle 0, 0, 100 \rangle$$

So, the position vector is $\vec{r}(t) = \langle 10t, 20t, 100 + 30t - 4.9t^2 \rangle$.

Problem 13: A Complete Calculation

Find both the unit tangent vector $\vec{T}(t)$ and the principal unit normal vector $\vec{N}(t)$ for the curve $\vec{r}(t) = \langle 3, 3t, t^2 \rangle$.

Solution:
First, we find the unit tangent vector, $\vec{T}(t)$.

$$\vec{r}'(t) = \langle 0, 3, 2t \rangle$$

The magnitude of the derivative is:

$$||\vec{r}'(t)|| = \sqrt{0^2 + 3^2 + (2t)^2} = \sqrt{9 + 4t^2}$$

The unit tangent vector is $\vec{T}(t) = \frac{\vec{r}'(t)}{||\vec{r}'(t)||}$:

$$\vec{T}(t) = \frac{\langle 0, 3, 2t \rangle}{\sqrt{9 + 4t^2}} = \left\langle 0, \frac{3}{\sqrt{9 + 4t^2}}, \frac{2t}{\sqrt{9 + 4t^2}} \right\rangle$$

Next, we find the principal unit normal vector, $\vec{N}(t)$, which requires finding the derivative of $\vec{T}(t)$. We will differentiate each component of $\vec{T}(t)$ separately.

x-component of $\vec{T}'(t)$:

$$\frac{d}{dt}(0) = 0$$

y-component of $\vec{T}'(t)$: We use the quotient rule for $y(t) = \frac{3}{\sqrt{9+4t^2}}$. Let $f = 3$ and $g = \sqrt{9 + 4t^2}$. Then $f' = 0$ and $g' = \frac{1}{2}(9 + 4t^2)^{-1/2}(8t) = \frac{4t}{\sqrt{9+4t^2}}$.

$$\begin{aligned}
\frac{d}{dt}\left(\frac{3}{\sqrt{9 + 4t^2}}\right) &= \frac{f'g - fg'}{g^2} \\
&= \frac{(0)(\sqrt{9 + 4t^2}) - (3)\left(\frac{4t}{\sqrt{9+4t^2}}\right)}{(\sqrt{9 + 4t^2})^2} \\
&= \frac{-12t/\sqrt{9 + 4t^2}}{9 + 4t^2} \\
&= \frac{-12t}{(9 + 4t^2)^{3/2}}
\end{aligned}$$

z-component of $\vec{T}'(t)$: We use the quotient rule for $z(t) = \frac{2t}{\sqrt{9+4t^2}}$. Let $f = 2t$ and $g = \sqrt{9 + 4t^2}$. Then $f' = 2$ and $g' = \frac{4t}{\sqrt{9+4t^2}}$.

$$\begin{aligned}
\frac{d}{dt}\left(\frac{2t}{\sqrt{9 + 4t^2}}\right) &= \frac{f'g - fg'}{g^2} \\
&= \frac{(2)(\sqrt{9 + 4t^2}) - (2t)\left(\frac{4t}{\sqrt{9+4t^2}}\right)}{9 + 4t^2} \\
&= \frac{\frac{2(9+4t^2)}{\sqrt{9+4t^2}} - \frac{8t^2}{\sqrt{9+4t^2}}}{9 + 4t^2} \\
&= \frac{\frac{18+8t^2-8t^2}{\sqrt{9+4t^2}}}{9 + 4t^2} \\
&= \frac{18}{(9 + 4t^2)^{3/2}}
\end{aligned}$$

Combining the components, we get $\vec{T}'(t)$:

$$\vec{T}'(t) = \left\langle 0, \frac{-12t}{(9+4t^2)^{3/2}}, \frac{18}{(9+4t^2)^{3/2}} \right\rangle = \frac{6}{(9+4t^2)^{3/2}} \langle 0, -2t, 3 \rangle$$

Now, we find the magnitude of $\vec{T}'(t)$:

$$||\vec{T}'(t)|| = \frac{6}{(9+4t^2)^{3/2}} \sqrt{0^2 + (-2t)^2 + 3^2} = \frac{6}{(9+4t^2)^{3/2}} \sqrt{4t^2+9} = \frac{6(9+4t^2)^{1/2}}{(9+4t^2)^{3/2}} = \frac{6}{9+4t^2}$$

Finally, the principal unit normal vector is $\vec{N}(t) = \frac{\vec{T}'(t)}{||\vec{T}'(t)||}$:

$$\vec{N}(t) = \frac{\frac{6}{(9+4t^2)^{3/2}} \langle 0, -2t, 3 \rangle}{\frac{6}{9+4t^2}} = \frac{\langle 0, -2t, 3 \rangle}{(9+4t^2)^{3/2}} \cdot (9+4t^2) = \frac{\langle 0, -2t, 3 \rangle}{\sqrt{9+4t^2}}$$

Problem 14: Curvature with the Cross Product Formula

Find $\kappa(t)$ the curvature of the curve $\vec{r}(t) = \langle e^t, e^{-t}, \sqrt{2}t \rangle$.

Solution:
We use the formula $\kappa(t) = \frac{||\vec{r}'(t) \times \vec{r}''(t)||}{||\vec{r}'(t)||^3}$.

$$\vec{r}'(t) = \langle e^t, -e^{-t}, \sqrt{2} \rangle$$

$$\vec{r}''(t) = \langle e^t, e^{-t}, 0 \rangle$$

The cross product is:

$$\vec{r}' \times \vec{r}'' = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ e^t & -e^{-t} & \sqrt{2} \\ e^t & e^{-t} & 0 \end{vmatrix} = \vec{i}(0 - \sqrt{2}e^{-t}) - \vec{j}(0 - \sqrt{2}e^t) + \vec{k}(e^t e^{-t} - (-e^{-t}e^t)) = \langle -\sqrt{2}e^{-t}, \sqrt{2}e^t, 2 \rangle$$

The magnitude of the cross product is:

$$||\vec{r}' \times \vec{r}''|| = \sqrt{2e^{-2t} + 2e^{2t} + 4} = \sqrt{2(e^{2t} + 2 + e^{-2t})} = \sqrt{2(e^t + e^{-t})^2} = \sqrt{2}(e^t + e^{-t})$$

The magnitude of the first derivative is:

$$||\vec{r}'(t)|| = \sqrt{e^{2t} + e^{-2t} + 2} = \sqrt{(e^t + e^{-t})^2} = e^t + e^{-t}$$

Finally, the curvature is:

$$\kappa(t) = \frac{\sqrt{2}(e^t + e^{-t})}{(e^t + e^{-t})^3} = \frac{\sqrt{2}}{(e^t + e^{-t})^2}$$

Problem 15: The TNB Frame

For the helix $\vec{r}(t) = \langle \cos(t), \sin(t), t \rangle$.

Solution:
From Problem 10, we found $\vec{T}(t) = \frac{1}{\sqrt{2}}\langle -\sin t, \cos t, 1\rangle$. The principal unit normal vector is $\vec{N}(t) = \frac{\vec{T}'(t)}{||\vec{T}'(t)||}$. From Problem 10, $\vec{T}'(t) = \frac{1}{\sqrt{2}}\langle -\cos t, -\sin t, 0\rangle$ and $||\vec{T}'(t)|| = 1/\sqrt{2}$.

$$\vec{N}(t) = \frac{\frac{1}{\sqrt{2}}\langle -\cos t, -\sin t, 0\rangle}{1/\sqrt{2}} = \langle -\cos t, -\sin t, 0\rangle$$

(a) The binormal vector $\vec{B}(t) = \vec{T}(t) \times \vec{N}(t)$:

$$\vec{B}(t) = \frac{1}{\sqrt{2}}\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -\sin t & \cos t & 1 \\ -\cos t & -\sin t & 0 \end{vmatrix} = \frac{1}{\sqrt{2}}[\vec{i}(\sin t) - \vec{j}(\cos t) + \vec{k}(\sin^2 t + \cos^2 t)] = \frac{1}{\sqrt{2}}\langle \sin t, -\cos t, 1\rangle$$

(b) Evaluate all three vectors at $t = \pi/2$:

$$\vec{T}(\pi/2) = \frac{1}{\sqrt{2}}\langle -\sin(\pi/2), \cos(\pi/2), 1\rangle = \frac{1}{\sqrt{2}}\langle -1, 0, 1\rangle$$

$$\vec{N}(\pi/2) = \langle -\cos(\pi/2), -\sin(\pi/2), 0\rangle = \langle 0, -1, 0\rangle$$

$$\vec{B}(\pi/2) = \frac{1}{\sqrt{2}}\langle \sin(\pi/2), -\cos(\pi/2), 1\rangle = \frac{1}{\sqrt{2}}\langle 1, 0, 1\rangle$$

Problem 16: Components of Acceleration

Find the tangential and normal components of acceleration, a_T and a_N, for the curve $\vec{r}(t) = \langle 2t, t^2, \frac{1}{3}t^3\rangle$.

Solution:
We find the first and second derivatives:

$$\vec{r}'(t) = \langle 2, 2t, t^2\rangle$$

$$\vec{a}(t) = \vec{r}''(t) = \langle 0, 2, 2t\rangle$$

The tangential component of acceleration is $a_T = \frac{\vec{r}'(t)\cdot\vec{r}''(t)}{||\vec{r}'(t)||}$. The dot product is:

$$\vec{r}'(t) \cdot \vec{r}''(t) = (2)(0) + (2t)(2) + (t^2)(2t) = 4t + 2t^3 = 2t(2 + t^2)$$

The magnitude of the velocity vector is (from Problem 3):

$$||\vec{r}'(t)|| = \sqrt{4 + 4t^2 + t^4} = \sqrt{(t^2 + 2)^2} = t^2 + 2$$

So, the tangential component is:

$$a_T = \frac{2t(t^2 + 2)}{t^2 + 2} = 2t$$

The normal component of acceleration can also be found with the formula $a_N = \frac{||\vec{r}'(t)\times\vec{r}''(t)||}{||\vec{r}'(t)||}$. First, we find the cross product:

$$\vec{r}'(t) \times \vec{r}''(t) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 2t & t^2 \\ 0 & 2 & 2t \end{vmatrix} = \vec{i}(4t^2 - 2t^2) - \vec{j}(4t - 0) + \vec{k}(4 - 0) = \langle 2t^2, -4t, 4\rangle$$

Next, we find its magnitude:

$$||\vec{r}'(t) \times \vec{r}''(t)|| = \sqrt{(2t^2)^2 + (-4t)^2 + 4^2} = \sqrt{4t^4 + 16t^2 + 16}$$

Factoring out a 4 gives:

$$= \sqrt{4(t^4 + 4t^2 + 4)} = 2\sqrt{(t^2 + 2)^2} = 2(t^2 + 2)$$

Now, we can calculate a_N:

$$a_N = \frac{2(t^2 + 2)}{t^2 + 2} = 2$$

The components are $a_T = 2t$ and $a_N = 2$.

Problem 17: Motion with Constant Speed

Consider the motion of a particle along the helix $\vec{r}(t) = \langle 3\cos(t), 3\sin(t), 4t\rangle$. This problem explores the relationship between constant speed and acceleration.

Solution:
(a) Find the velocity vector $\vec{v}(t)$ and show that the speed, $||\vec{v}(t)||$, is a constant value.

$$\vec{v}(t) = \vec{r}'(t) = \langle -3\sin(t), 3\cos(t), 4\rangle$$

$$||\vec{v}(t)|| = \sqrt{(-3\sin t)^2 + (3\cos t)^2 + 4^2} = \sqrt{9\sin^2 t + 9\cos^2 t + 16} = \sqrt{9(1) + 16} = \sqrt{25} = 5$$

The speed is a constant value of 5. (b) Find the acceleration vector $\vec{a}(t)$.

$$\vec{a}(t) = \vec{v}'(t) = \langle -3\cos(t), -3\sin(t), 0\rangle$$

(c) Explicitly calculate the tangential component of acceleration, a_T, using the formula $a_T = \frac{\vec{v}(t)\cdot\vec{a}(t)}{||\vec{v}(t)||}$.

$$\vec{v}(t) \cdot \vec{a}(t) = (-3\sin t)(-3\cos t) + (3\cos t)(-3\sin t) + (4)(0) = 9\sin t\cos t - 9\sin t\cos t = 0$$

$$a_T = \frac{0}{5} = 0$$

This result is expected, as the tangential component of acceleration is the rate of change of speed, and the speed is constant.

Problem 18: Acceleration Components at a Point

Find the tangential and normal components of acceleration, a_T and a_N, for the motion of a particle with position vector $\vec{r}(t) = \langle \cos(t) + t\sin(t), \sin(t) - t\cos(t), t^2\rangle$ at the point $t = \pi$.

Solution:
First, find the velocity and acceleration vectors:

$$\vec{v}(t) = \vec{r}'(t) = \langle -\sin t + (\sin t + t\cos t), \cos t - (\cos t - t\sin t), 2t\rangle = \langle t\cos t, t\sin t, 2t\rangle$$

$$\vec{a}(t) = \vec{v}'(t) = \langle \cos t - t\sin t, \sin t + t\cos t, 2\rangle$$

Now, evaluate these vectors at $t = \pi$:

$$\vec{v}(\pi) = \langle \pi\cos\pi, \pi\sin\pi, 2\pi\rangle = \langle -\pi, 0, 2\pi\rangle$$

$$\vec{a}(\pi) = \langle \cos\pi - \pi\sin\pi, \sin\pi + \pi\cos\pi, 2\rangle = \langle -1, -\pi, 2\rangle$$

Calculate $a_T = \frac{\vec{v}(\pi)\cdot\vec{a}(\pi)}{||\vec{v}(\pi)||}$:

$$\vec{v}(\pi)\cdot\vec{a}(\pi) = (-\pi)(-1) + (0)(-\pi) + (2\pi)(2) = \pi + 4\pi = 5\pi$$

$$||\vec{v}(\pi)|| = \sqrt{(-\pi)^2 + 0^2 + (2\pi)^2} = \sqrt{\pi^2 + 4\pi^2} = \sqrt{5\pi^2} = \pi\sqrt{5}$$

$$a_T = \frac{5\pi}{\pi\sqrt{5}} = \frac{5}{\sqrt{5}} = \sqrt{5}$$

The normal component of acceleration, a_N, can also be found using the cross product formula $a_N = \frac{||\vec{v}(t)\times\vec{a}(t)||}{||\vec{v}(t)||}$. We already have the vectors evaluated at $t = \pi$:

$$\vec{v}(\pi) = \langle -\pi, 0, 2\pi\rangle$$

$$\vec{a}(\pi) = \langle -1, -\pi, 2\rangle$$

First, we find the cross product $\vec{v}(\pi)\times\vec{a}(\pi)$:

$$\vec{v}(\pi)\times\vec{a}(\pi) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -\pi & 0 & 2\pi \\ -1 & -\pi & 2 \end{vmatrix} = \vec{i}(0 - (-2\pi^2)) - \vec{j}(-2\pi - (-2\pi)) + \vec{k}(\pi^2 - 0)$$

$$= \langle 2\pi^2, 0, \pi^2\rangle$$

Next, we find the magnitude of this cross product:

$$||\vec{v}(\pi)\times\vec{a}(\pi)|| = \sqrt{(2\pi^2)^2 + 0^2 + (\pi^2)^2} = \sqrt{4\pi^4 + \pi^4} = \sqrt{5\pi^4} = \pi^2\sqrt{5}$$

We already know from the previous calculation that $||\vec{v}(\pi)|| = \pi\sqrt{5}$. Now we calculate a_N:

$$a_N = \frac{\pi^2\sqrt{5}}{\pi\sqrt{5}} = \pi$$

At $t = \pi$, $a_T = \sqrt{5}$ and $a_N = \pi$.

Problem 19: Finding the Binormal Vector

Find the binormal vector $\vec{B}(t)$ for the curve $\vec{r}(t) = \langle 1, t, t^2\rangle$. You will first need to find $\vec{T}(t)$ and $\vec{N}(t)$ if you have not already.

Solution:
First, find $\vec{T}(t)$:

$$\vec{r}'(t) = \langle 0, 1, 2t\rangle$$

$$||\vec{r}'(t)|| = \sqrt{0^2 + 1^2 + (2t)^2} = \sqrt{1 + 4t^2}$$

$$\vec{T}(t) = \frac{\vec{r}'(t)}{||\vec{r}'(t)||} = \frac{1}{\sqrt{1+4t^2}}\langle 0, 1, 2t\rangle$$

Next, find $\vec{N}(t) = \vec{T}'(t)/||\vec{T}'(t)||$. We have:

$$\vec{T}(t) = \left\langle 0, \frac{1}{\sqrt{1+4t^2}}, \frac{2t}{\sqrt{1+4t^2}}\right\rangle$$

We will differentiate each component of $\vec{T}(t)$ separately using the quotient rule.

x-component of $\vec{T}'(t)$:

$$\frac{d}{dt}(0) = 0$$

y-component of $\vec{T}'(t)$: For $y(t) = \frac{1}{\sqrt{1+4t^2}}$, let $f = 1$ and $g = \sqrt{1+4t^2}$. Then $f' = 0$ and $g' = \frac{4t}{\sqrt{1+4t^2}}$.

$$\begin{aligned}\frac{d}{dt}\left(\frac{1}{\sqrt{1+4t^2}}\right) &= \frac{f'g - fg'}{g^2} \\ &= \frac{(0)(\sqrt{1+4t^2}) - (1)\left(\frac{4t}{\sqrt{1+4t^2}}\right)}{(\sqrt{1+4t^2})^2} \\ &= \frac{-4t/\sqrt{1+4t^2}}{1+4t^2} = \frac{-4t}{(1+4t^2)^{3/2}}\end{aligned}$$

z-component of $\vec{T}'(t)$: For $z(t) = \frac{2t}{\sqrt{1+4t^2}}$, let $f = 2t$ and $g = \sqrt{1+4t^2}$. Then $f' = 2$ and $g' = \frac{4t}{\sqrt{1+4t^2}}$.

$$\begin{aligned}\frac{d}{dt}\left(\frac{2t}{\sqrt{1+4t^2}}\right) &= \frac{f'g - fg'}{g^2} \\ &= \frac{(2)(\sqrt{1+4t^2}) - (2t)\left(\frac{4t}{\sqrt{1+4t^2}}\right)}{1+4t^2} \\ &= \frac{\frac{2(1+4t^2)-8t^2}{\sqrt{1+4t^2}}}{1+4t^2} \\ &= \frac{2+8t^2-8t^2}{(1+4t^2)^{3/2}} = \frac{2}{(1+4t^2)^{3/2}}\end{aligned}$$

Combining the components, we get $\vec{T}'(t)$:

$$\vec{T}'(t) = \left\langle 0, \frac{-4t}{(1+4t^2)^{3/2}}, \frac{2}{(1+4t^2)^{3/2}} \right\rangle = \frac{2}{(1+4t^2)^{3/2}}\langle 0, -2t, 1\rangle$$

$$||\vec{T}'(t)|| = \frac{\sqrt{0^2 + (-4t)^2 + 2^2}}{(1+4t^2)^{3/2}} = \frac{\sqrt{16t^2+4}}{(1+4t^2)^{3/2}} = \frac{2\sqrt{4t^2+1}}{(1+4t^2)^{3/2}} = \frac{2}{1+4t^2}$$

$$\vec{N}(t) = \frac{\vec{T}'(t)}{||\vec{T}'(t)||} = \frac{\langle 0, -4t, 2\rangle}{(1+4t^2)^{3/2}} \cdot \frac{1+4t^2}{2} = \frac{\langle 0, -2t, 1\rangle}{\sqrt{1+4t^2}}$$

Finally, find the binormal vector $\vec{B}(t) = \vec{T}(t) \times \vec{N}(t)$.

$$\vec{B}(t) = \left(\frac{1}{\sqrt{1+4t^2}}\langle 0, 1, 2t\rangle\right) \times \left(\frac{1}{\sqrt{1+4t^2}}\langle 0, -2t, 1\rangle\right)$$

$$\vec{B}(t) = \frac{1}{1+4t^2}\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 1 & 2t \\ 0 & -2t & 1 \end{vmatrix} = \frac{1}{1+4t^2}[\vec{i}(1-(-4t^2)) - \vec{j}(0-0) + \vec{k}(0-0)]$$

$$\vec{B}(t) = \frac{1}{1+4t^2}\langle 1+4t^2, 0, 0\rangle = \langle 1, 0, 0\rangle$$

Problem 20: Comprehensive Analysis

A particle's position is given by $\vec{r}(t) = \langle e^t \cos(t), e^t \sin(t), 2\rangle$.

Solution:

(a) Find the curvature $\kappa(t)$. We use the formula $\kappa(t) = \frac{||\vec{r}'(t)\times\vec{r}''(t)||}{||\vec{r}'(t)||^3}$.

First, find the first and second derivatives of $\vec{r}(t)$:

$$\vec{r}'(t) = \langle e^t \cos t - e^t \sin t, e^t \sin t + e^t \cos t, 0\rangle$$

To find the second derivative, we differentiate each component of $\vec{r}'(t)$:

$$\frac{d}{dt}(e^t \cos t - e^t \sin t) = (e^t \cos t - e^t \sin t) - (e^t \sin t + e^t \cos t) = -2e^t \sin t$$

$$\frac{d}{dt}(e^t \sin t + e^t \cos t) = (e^t \sin t + e^t \cos t) + (e^t \cos t - e^t \sin t) = 2e^t \cos t$$

$$\vec{r}''(t) = \langle -2e^t \sin t, 2e^t \cos t, 0\rangle$$

Next, compute the cross product $\vec{r}'(t) \times \vec{r}''(t)$:

$$\vec{r}' \times \vec{r}'' = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ e^t(\cos t - \sin t) & e^t(\sin t + \cos t) & 0 \\ -2e^t \sin t & 2e^t \cos t & 0 \end{vmatrix}$$

The $\vec{i}$ and $\vec{j}$ components are zero because the z-components of the vectors are zero. We only need to compute the $\vec{k}$ component:

$$\begin{aligned} \vec{k} \text{ comp} &= [e^t(\cos t - \sin t)][2e^t \cos t] - [e^t(\sin t + \cos t)][-2e^t \sin t] \\ &= 2e^{2t}(\cos^2 t - \sin t \cos t) + 2e^{2t}(\sin^2 t + \sin t \cos t) \\ &= 2e^{2t}(\cos^2 t - \sin t \cos t + \sin^2 t + \sin t \cos t) \\ &= 2e^{2t}(\cos^2 t + \sin^2 t) \\ &= 2e^{2t}(1) = 2e^{2t} \end{aligned}$$

So, the cross product is $\vec{r}'(t)\times\vec{r}''(t) = \langle 0, 0, 2e^{2t}\rangle$. The magnitude of the cross product is $||\vec{r}'\times\vec{r}''|| = \sqrt{0^2 + 0^2 + (2e^{2t})^2} = 2e^{2t}$.

Next, find the magnitude of the velocity, $||\vec{r}'(t)||$:

$$\begin{aligned} ||\vec{r}'(t)||^2 &= (e^t(\cos t - \sin t))^2 + (e^t(\sin t + \cos t))^2 + 0^2 \\ &= e^{2t}[(\cos^2 t - 2\sin t \cos t + \sin^2 t) + (\sin^2 t + 2\sin t \cos t + \cos^2 t)] \\ &= e^{2t}[(1) + (1)] = 2e^{2t} \end{aligned}$$

Therefore, $||\vec{r}'(t)|| = \sqrt{2e^{2t}} = \sqrt{2}e^t$.

Finally, calculate the curvature:

$$\kappa(t) = \frac{||\vec{r}' \times \vec{r}''||}{||\vec{r}'(t)||^3} = \frac{2e^{2t}}{(\sqrt{2}e^t)^3} = \frac{2e^{2t}}{2\sqrt{2}e^{3t}} = \frac{1}{\sqrt{2}e^t}$$

(b) Find the tangential component of acceleration, $a_T(t)$, using the dot product formula. The formula is $a_T(t) = \frac{\vec{v}(t)\cdot\vec{a}(t)}{||\vec{v}(t)||} = \frac{\vec{r}'(t)\cdot\vec{r}''(t)}{||\vec{r}'(t)||}$. From part (a), we have:

$$\vec{r}'(t) = \langle e^t(\cos t - \sin t), e^t(\sin t + \cos t), 0\rangle$$

$$\vec{r}''(t) = \langle -2e^t \sin t, 2e^t \cos t, 0 \rangle$$

$$||\vec{r}'(t)|| = \sqrt{2}e^t$$

First, compute the dot product $\vec{r}'(t) \cdot \vec{r}''(t)$:

$$\begin{aligned}
\vec{r}' \cdot \vec{r}'' &= [e^t(\cos t - \sin t)][-2e^t \sin t] + [e^t(\sin t + \cos t)][2e^t \cos t] + (0)(0) \\
&= -2e^{2t} \sin t \cos t + 2e^{2t} \sin^2 t + 2e^{2t} \sin t \cos t + 2e^{2t} \cos^2 t \\
&= 2e^{2t}(\sin^2 t + \cos^2 t) \\
&= 2e^{2t}
\end{aligned}$$

Now, calculate $a_T(t)$:

$$a_T(t) = \frac{2e^{2t}}{\sqrt{2}e^t} = \frac{\sqrt{2}\sqrt{2}e^t e^t}{\sqrt{2}e^t} = \sqrt{2}e^t$$

(c) Find the normal component of acceleration, $a_N(t)$, using the cross product formula. The formula is $a_N(t) = \frac{||\vec{v}(t) \times \vec{a}(t)||}{||\vec{v}(t)||} = \frac{||\vec{r}'(t) \times \vec{r}''(t)||}{||\vec{r}'(t)||}$. From part (a), we have:

$$\vec{r}'(t) \times \vec{r}''(t) = \langle 0, 0, 2e^{2t} \rangle$$

$$||\vec{r}'(t) \times \vec{r}''(t)|| = 2e^{2t}$$

$$||\vec{r}'(t)|| = \sqrt{2}e^t$$

Now, calculate $a_N(t)$:

$$a_N(t) = \frac{2e^{2t}}{\sqrt{2}e^t} = \frac{\sqrt{2}\sqrt{2}e^t e^t}{\sqrt{2}e^t} = \sqrt{2}e^t$$

Homework 6 Solutions

Homework: Partial Derivatives, Gradients, and Tangent Planes - Solution Key

Part 1: Partial Derivatives

Find the requested partial derivatives for each function.

Problem 1: First-Order Partials
Find f_x and f_y for $f(x, y) = 3x^5y^2 - 7x^3y + x - 5y^2 + 9$.

Solution: To find the partial derivative with respect to one variable, we treat all other variables as constants.

To find f_x, we treat y as a constant and differentiate with respect to x:

$$f_x(x, y) = \frac{\partial}{\partial x}(3x^5y^2 - 7x^3y + x - 5y^2 + 9)$$

$$f_x(x, y) = (3y^2)\frac{\partial}{\partial x}(x^5) - (7y)\frac{\partial}{\partial x}(x^3) + \frac{\partial}{\partial x}(x) - \frac{\partial}{\partial x}(5y^2) + \frac{\partial}{\partial x}(9)$$

$$f_x(x, y) = (3y^2)(5x^4) - (7y)(3x^2) + 1 - 0 + 0$$

$$f_x(x, y) = 15x^4y^2 - 21x^2y + 1$$

To find f_y, we treat x as a constant and differentiate with respect to y:

$$f_y(x,y) = \frac{\partial}{\partial y}(3x^5y^2 - 7x^3y + x - 5y^2 + 9)$$

$$f_y(x,y) = (3x^5)\frac{\partial}{\partial y}(y^2) - (7x^3)\frac{\partial}{\partial y}(y) + \frac{\partial}{\partial y}(x) - (5)\frac{\partial}{\partial y}(y^2) + \frac{\partial}{\partial y}(9)$$

$$f_y(x,y) = (3x^5)(2y) - (7x^3)(1) + 0 - 5(2y) + 0$$

$$f_y(x,y) = 6x^5y - 7x^3 - 10y$$

Problem 2: First-Order Partials with Chain Rule

Find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ for $f(x,y) = \cos(x^2y^3)$.

Solution: This problem requires the chain rule. Let the "inside" function be $u(x,y) = x^2y^3$. The "outside" function is $\cos(u)$.

To find $\frac{\partial f}{\partial x}$, we differentiate the outside function with respect to u and multiply by the partial derivative of the inside function with respect to x:

$$\frac{\partial f}{\partial x} = -\sin(x^2y^3) \cdot \frac{\partial}{\partial x}(x^2y^3)$$

$$\frac{\partial f}{\partial x} = -\sin(x^2y^3) \cdot (2xy^3)$$

$$\frac{\partial f}{\partial x} = -2xy^3\sin(x^2y^3)$$

To find $\frac{\partial f}{\partial y}$, we do the same, but differentiate the inside function with respect to y:

$$\frac{\partial f}{\partial y} = -\sin(x^2y^3) \cdot \frac{\partial}{\partial y}(x^2y^3)$$

$$\frac{\partial f}{\partial y} = -\sin(x^2y^3) \cdot (3x^2y^2)$$

$$\frac{\partial f}{\partial y} = -3x^2y^2\sin(x^2y^3)$$

Problem 3: First-Order Partials with Product Rule

Find g_x and g_y for $g(x,y) = y^2e^{xy}$.

Solution: To find g_x, the term y^2 is treated as a constant factor. We only need to differentiate e^{xy} with respect to x using the chain rule.

$$g_x(x,y) = y^2 \cdot \frac{\partial}{\partial x}(e^{xy})$$

$$g_x(x,y) = y^2 \cdot \left(e^{xy} \cdot \frac{\partial}{\partial x}(xy)\right)$$

$$g_x(x,y) = y^2 \cdot (e^{xy} \cdot y)$$

$$g_x(x,y) = y^3 e^{xy}$$

To find g_y, both factors y^2 and e^{xy} contain the variable y. Therefore, we must use the product rule: (first)$'$(second) + (first)(second)$'$.

$$g_y(x,y) = \frac{\partial}{\partial y}(y^2) \cdot e^{xy} + y^2 \cdot \frac{\partial}{\partial y}(e^{xy})$$

$$g_y(x,y) = (2y) \cdot e^{xy} + y^2 \cdot \left(e^{xy} \cdot \frac{\partial}{\partial y}(xy)\right)$$

$$g_y(x,y) = 2ye^{xy} + y^2 \cdot (e^{xy} \cdot x)$$

$$g_y(x,y) = 2ye^{xy} + xy^2 e^{xy}$$

$$g_y(x,y) = ye^{xy}(2 + xy)$$

Problem 4: Second-Order Partials

Find all four second-order partial derivatives ($f_{xx}, f_{yy}, f_{xy}, f_{yx}$) for $f(x,y) = \ln(x^2+y)$. Verify that $f_{xy} = f_{yx}$.

Solution: First, we must find the first-order partial derivatives, f_x and f_y.

$$f_x = \frac{\partial}{\partial x}(\ln(x^2+y)) = \frac{1}{x^2+y} \cdot \frac{\partial}{\partial x}(x^2+y) = \frac{2x}{x^2+y}$$

$$f_y = \frac{\partial}{\partial y}(\ln(x^2+y)) = \frac{1}{x^2+y} \cdot \frac{\partial}{\partial y}(x^2+y) = \frac{1}{x^2+y}$$

Now, we differentiate these first-order partials to find the second-order ones. To find $f_{xx} = \frac{\partial}{\partial x}(f_x)$, we use the quotient rule:

$$f_{xx} = \frac{(2)(x^2+y) - (2x)(2x)}{(x^2+y)^2} = \frac{2x^2+2y-4x^2}{(x^2+y)^2} = \frac{2y-2x^2}{(x^2+y)^2}$$

To find $f_{yy} = \frac{\partial}{\partial y}(f_y)$, we can rewrite f_y as $(x^2+y)^{-1}$ and use the power rule.

$$f_{yy} = \frac{\partial}{\partial y}((x^2+y)^{-1}) = -1(x^2+y)^{-2} \cdot \frac{\partial}{\partial y}(x^2+y) = -\frac{1}{(x^2+y)^2} \cdot 1 = -\frac{1}{(x^2+y)^2}$$

To find $f_{xy} = \frac{\partial}{\partial y}(f_x)$, we differentiate f_x with respect to y.

$$f_{xy} = \frac{\partial}{\partial y}\left(\frac{2x}{x^2+y}\right) = 2x \cdot \frac{\partial}{\partial y}((x^2+y)^{-1}) = 2x \cdot (-1(x^2+y)^{-2} \cdot 1) = -\frac{2x}{(x^2+y)^2}$$

To find $f_{yx} = \frac{\partial}{\partial x}(f_y)$, we differentiate f_y with respect to x.

$$f_{yx} = \frac{\partial}{\partial x}\left(\frac{1}{x^2+y}\right) = \frac{\partial}{\partial x}((x^2+y)^{-1}) = -1(x^2+y)^{-2} \cdot (2x) = -\frac{2x}{(x^2+y)^2}$$

Verification: As expected from Clairaut's Theorem, the mixed partial derivatives are equal.

$$f_{xy} = f_{yx} = -\frac{2x}{(x^2+y)^2}$$

Problem 5: Higher-Order Partials

Find the partial derivative f_{yyy} for $f(x, y) = x^3 e^{2y}$.

Solution: We need to differentiate the function with respect to y three times in succession.

$$f_y = \frac{\partial}{\partial y}(x^3 e^{2y}) = x^3 \cdot \frac{\partial}{\partial y}(e^{2y}) = x^3(2e^{2y}) = 2x^3 e^{2y}$$

$$f_{yy} = \frac{\partial}{\partial y}(2x^3 e^{2y}) = 2x^3 \cdot \frac{\partial}{\partial y}(e^{2y}) = 2x^3(2e^{2y}) = 4x^3 e^{2y}$$

$$f_{yyy} = \frac{\partial}{\partial y}(4x^3 e^{2y}) = 4x^3 \cdot \frac{\partial}{\partial y}(e^{2y}) = 4x^3(2e^{2y}) = 8x^3 e^{2y}$$

Problem 6: Partials of Three Variables

Find f_x, f_y, and f_z for $f(x, y, z) = \frac{xy}{z}$.

Solution: To find f_x, we treat y and z as constants. The term y/z is a constant factor.

$$f_x = \frac{\partial}{\partial x}\left(\frac{y}{z} \cdot x\right) = \frac{y}{z} \cdot 1 = \frac{y}{z}$$

To find f_y, we treat x and z as constants. The term x/z is a constant factor.

$$f_y = \frac{\partial}{\partial y}\left(\frac{x}{z} \cdot y\right) = \frac{x}{z} \cdot 1 = \frac{x}{z}$$

To find f_z, we treat x and y as constants. We can rewrite the function as $f(x, y, z) = xy \cdot z^{-1}$ and use the power rule.

$$f_z = \frac{\partial}{\partial z}(xyz^{-1}) = xy \cdot (-1z^{-2}) = -\frac{xy}{z^2}$$

Part 2: Tangent Planes and Linear Approximations

Problem 7: Finding a Tangent Plane

Find the equation of the plane tangent to the surface $z = x^2 + 3y^2$ at the point $(3, -1)$.

Solution: The equation for a tangent plane at a point (x_0, y_0, z_0) is given by:

$$z = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

Here, $f(x, y) = x^2 + 3y^2$ and the point is $(x_0, y_0) = (3, -1)$.

Step 1: Find z_0.

$$z_0 = f(3, -1) = (3)^2 + 3(-1)^2 = 9 + 3(1) = 12$$

The point of tangency is $(3, -1, 12)$.

Step 2: Find the partial derivatives.

$$f_x(x, y) = \frac{\partial}{\partial x}(x^2 + 3y^2) = 2x$$

$$f_y(x,y) = \frac{\partial}{\partial y}(x^2 + 3y^2) = 6y$$

Step 3: Evaluate the partial derivatives at the point $(3,-1)$**.**

$$f_x(3,-1) = 2(3) = 6$$

$$f_y(3,-1) = 6(-1) = -6$$

Step 4: Assemble the tangent plane equation.

$$z = 12 + 6(x-3) - 6(y-(-1))$$

Problem 8: Finding a Tangent Plane

Find the equation of the plane tangent to the surface $z = \arctan(y/x)$ at the point $(1,1)$.

Solution: The process is the same as in the previous problem. Here $f(x,y) = \arctan(y/x)$ and $(x_0, y_0) = (1,1)$.

Step 1: Find z_0**.**

$$z_0 = f(1,1) = \arctan(1/1) = \arctan(1) = \frac{\pi}{4}$$

The point of tangency is $(1,1,\pi/4)$.

Step 2: Find the partial derivatives. Recall that the derivative of $\arctan(u)$ is $\frac{1}{1+u^2}$.

$$f_x = \frac{1}{1+(y/x)^2} \cdot \frac{\partial}{\partial x}\left(\frac{y}{x}\right) = \frac{1}{1+y^2/x^2} \cdot \left(-\frac{y}{x^2}\right) = \frac{x^2}{x^2+y^2} \cdot \left(-\frac{y}{x^2}\right) = -\frac{y}{x^2+y^2}$$

$$f_y = \frac{1}{1+(y/x)^2} \cdot \frac{\partial}{\partial y}\left(\frac{y}{x}\right) = \frac{1}{1+y^2/x^2} \cdot \left(\frac{1}{x}\right) = \frac{x^2}{x^2+y^2} \cdot \left(\frac{1}{x}\right) = \frac{x}{x^2+y^2}$$

Step 3: Evaluate the partial derivatives at the point $(1,1)$**.**

$$f_x(1,1) = -\frac{1}{1^2+1^2} = -\frac{1}{2}$$

$$f_y(1,1) = \frac{1}{1^2+1^2} = \frac{1}{2}$$

Step 4: Assemble the tangent plane equation.

$$z = \frac{\pi}{4} - \frac{1}{2}(x-1) + \frac{1}{2}(y-1)$$

Problem 9: Finding a Linearization

Find the linearization $L(x,y)$ of the function $f(x,y) = \sqrt{x+e^{4y}}$ at the point $(3,0)$.

Solution: The linearization $L(x,y)$ is just another name for the tangent plane equation, solved for the z-variable.

$$L(x,y) = f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b)$$

Here, $(a,b) = (3,0)$.

Step 1: Find $f(a,b)$.

$$f(3,0) = \sqrt{3+e^{4(0)}} = \sqrt{3+e^0} = \sqrt{3+1} = \sqrt{4} = 2$$

Step 2: Find the partial derivatives.

$$f(x,y) = (x+e^{4y})^{1/2}$$

$$f_x = \frac{1}{2}(x+e^{4y})^{-1/2} \cdot \frac{\partial}{\partial x}(x+e^{4y}) = \frac{1}{2\sqrt{x+e^{4y}}}$$

$$f_y = \frac{1}{2}(x+e^{4y})^{-1/2} \cdot \frac{\partial}{\partial y}(x+e^{4y}) = \frac{4e^{4y}}{2\sqrt{x+e^{4y}}} = \frac{2e^{4y}}{\sqrt{x+e^{4y}}}$$

Step 3: Evaluate the partial derivatives at the point $(3,0)$.

$$f_x(3,0) = \frac{1}{2\sqrt{3+e^0}} = \frac{1}{2\sqrt{4}} = \frac{1}{4}$$

$$f_y(3,0) = \frac{2e^0}{\sqrt{3+e^0}} = \frac{2}{\sqrt{4}} = \frac{2}{2} = 1$$

Step 4: Assemble the linearization formula.

$$L(x,y) = 2 + \frac{1}{4}(x-3) + 1(y-0)$$

Problem 10: Using Linearization to Approximate
Use the linearization you found in Problem 9 to approximate the value of $f(3.05, -0.01)$.

Solution: The core idea of linearization is that for points (x,y) near the center point (a,b), the value of the function is approximately equal to the value of the tangent plane: $f(x,y) \approx L(x,y)$.

We use the linearization $L(x,y) = \frac{1}{4}x + y + \frac{5}{4}$ from Problem 9 and plug in the point $(3.05, -0.01)$.

$$f(3.05, -0.01) \approx L(3.05, -0.01)$$

$$L(3.05, -0.01) = \frac{1}{4}(3.05) + (-0.01) + \frac{5}{4}$$

$$L(3.05, -0.01) = 0.7625 - 0.01 + 1.25$$

$$L(3.05, -0.01) = 2.0025$$

So, we approximate that $f(3.05, -0.01) \approx 2.0025$.

Problem 11: Working Backwards
The equation of the tangent plane to the surface $z = f(x,y)$ at the point $(2,-1)$ is given by $z = 4 - 3(x-2) + 5(y+1)$. Determine the values of $f(2,-1)$, $f_x(2,-1)$, and $f_y(2,-1)$.

Solution: This problem asks us to extract information by comparing the given equation to the general formula for a tangent plane at a point (a, b). The general formula is:

$$z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

The given equation is:

$$z = 4 - 3(x - 2) + 5(y + 1)$$

The point is given as $(a, b) = (2, -1)$. Notice that the terms $(x-2)$ and $(y+1) = (y-(-1))$ match the general form.

By matching the terms between the general formula and the given equation, we can identify the required values.

$$f(a, b) = f(2, -1) \text{ corresponds to the constant term, } 4.$$

$$f(2, -1) = 4$$

$$f_x(a, b) = f_x(2, -1) \text{ corresponds to the coefficient of the } (x - a) \text{ term, } -3.$$

$$f_x(2, -1) = -3$$

$$f_y(a, b) = f_y(2, -1) \text{ corresponds to the coefficient of the } (y - b) \text{ term, } 5.$$

$$f_y(2, -1) = 5$$

Part 3: The Gradient and Steepest Ascent/Descent

Problem 12: Finding a Gradient (2D)
Find the gradient of the function $f(x, y) = x^3 - 2xy + 3y^2$.

Solution: The gradient of a function $f(x, y)$ is a vector containing its first-order partial derivatives: $\nabla f(x, y) = \langle f_x, f_y \rangle$.

Step 1: Find the partial derivatives.

$$f_x = \frac{\partial}{\partial x}(x^3 - 2xy + 3y^2) = 3x^2 - 2y$$

$$f_y = \frac{\partial}{\partial y}(x^3 - 2xy + 3y^2) = -2x + 6y$$

Step 2: Write the gradient vector.

$$\nabla f(x, y) = \langle 3x^2 - 2y, -2x + 6y \rangle$$

Problem 13: Finding a Gradient (3D)
Find the gradient of the function $f(x, y, z) = z^2 \ln(x^2 + y^2)$.

Solution: The gradient of a function $f(x, y, z)$ is the vector $\nabla f(x, y, z) = \langle f_x, f_y, f_z \rangle$.

Step 1: Find the partial derivatives.

$$f_x = z^2 \cdot \frac{1}{x^2 + y^2} \cdot (2x) = \frac{2xz^2}{x^2 + y^2}$$

$$f_y = z^2 \cdot \frac{1}{x^2 + y^2} \cdot (2y) = \frac{2yz^2}{x^2 + y^2}$$

$$f_z = 2z \cdot \ln(x^2 + y^2)$$

Step 2: Write the gradient vector.

$$\nabla f(x, y, z) = \left\langle \frac{2xz^2}{x^2 + y^2}, \frac{2yz^2}{x^2 + y^2}, 2z \ln(x^2 + y^2) \right\rangle$$

Problem 14: Steepest Ascent

Let $f(x, y) = xy^2$. Find the direction of steepest ascent and the maximum rate of change of f at the point $(2, 3)$.

Solution: The direction of steepest ascent at a point is given by the gradient vector at that point. The maximum rate of change is the magnitude of that gradient vector.

Step 1: Find the gradient of $f(x, y)$.

$$f_x = y^2$$

$$f_y = 2xy$$

$$\nabla f(x, y) = \langle y^2, 2xy \rangle$$

Step 2: Evaluate the gradient at the point $(2, 3)$.

$$\nabla f(2, 3) = \langle (3)^2, 2(2)(3) \rangle = \langle 9, 12 \rangle$$

The direction of steepest ascent is the vector $\langle 9, 12 \rangle$. (Any positive scalar multiple, like the unit vector $\langle 9/15, 12/15 \rangle$, is also a correct answer for the direction).

Step 3: Find the magnitude of the gradient at $(2, 3)$.

$$||\nabla f(2, 3)|| = \sqrt{9^2 + 12^2}$$

$$||\nabla f(2, 3)|| = \sqrt{81 + 144} = \sqrt{225} = 15$$

The maximum rate of change is 15.

Problem 15: Steepest Descent

The altitude of a mountain is given by $H(x, y) = 2000 - 0.01(x^2 + y^2)$. If a hiker is standing at the point $(10, 10)$, in which direction should they move to descend the most rapidly? What is the slope of their path in that direction?

Solution: The direction of steepest descent is the opposite of the gradient direction, $-\nabla H$. The slope in that direction is the minimum rate of change, $-||\nabla H||$.

Step 1: Find the gradient of $H(x, y)$.

$$H_x = -0.01(2x) = -0.02x$$

$$H_y = -0.01(2y) = -0.02y$$

$$\nabla H(x, y) = \langle -0.02x, -0.02y \rangle$$

Step 2: Evaluate the gradient at the point $(10, 10)$.

$$\nabla H(10, 10) = \langle -0.02(10), -0.02(10) \rangle = \langle -0.2, -0.2 \rangle$$

Step 3: Find the direction of steepest descent. The direction is $-\nabla H(10, 10)$.

$$-\nabla H(10, 10) = -\langle -0.2, -0.2 \rangle = \langle 0.2, 0.2 \rangle$$

The hiker should move in the direction $\langle 0.2, 0.2 \rangle$ (or simply $\langle 1, 1 \rangle$, as it points in the same direction).

Step 4: Find the slope in that direction. The slope is the minimum rate of change, $-||\nabla H(10, 10)||$.

$$||\nabla H(10, 10)|| = \sqrt{(-0.2)^2 + (-0.2)^2} = \sqrt{0.04 + 0.04} = \sqrt{0.08} = \sqrt{8/100} = \frac{2\sqrt{2}}{10} = \frac{\sqrt{2}}{5}$$

The slope of the path is $-\frac{\sqrt{2}}{5}$.

Problem 16: Steepest Ascent (3D)

The temperature at a point (x, y, z) is given by $T(x, y, z) = 100e^{-x^2-2y^2-3z^2}$. Find the direction of the greatest rate of temperature increase at the point $(1, 1, 1)$.

Solution: The direction of greatest increase is the direction of the gradient vector at the given point.

Step 1: Find the gradient of $T(x, y, z)$. Using the chain rule for each component:

$$T_x = 100e^{-x^2-2y^2-3z^2} \cdot (-2x) = -200xe^{-x^2-2y^2-3z^2}$$

$$T_y = 100e^{-x^2-2y^2-3z^2} \cdot (-4y) = -400ye^{-x^2-2y^2-3z^2}$$

$$T_z = 100e^{-x^2-2y^2-3z^2} \cdot (-6z) = -600ze^{-x^2-2y^2-3z^2}$$

Step 2: Evaluate the gradient at the point $(1, 1, 1)$. First, note the value of the exponential term at $(1, 1, 1)$: $e^{-1^2-2(1)^2-3(1)^2} = e^{-1-2-3} = e^{-6}$.

$$T_x(1, 1, 1) = -200(1)e^{-6} = -200e^{-6}$$

$$T_y(1, 1, 1) = -400(1)e^{-6} = -400e^{-6}$$

$$T_z(1, 1, 1) = -600(1)e^{-6} = -600e^{-6}$$

The gradient vector at $(1, 1, 1)$ is:

$$\nabla T(1, 1, 1) = \langle -200e^{-6}, -400e^{-6}, -600e^{-6} \rangle$$

We can factor out the common term to simplify the direction vector:

$$\nabla T(1, 1, 1) = -200e^{-6}\langle 1, 2, 3 \rangle$$

The direction of greatest increase is $\langle -1, -2, -3 \rangle$.

Problem 17: Rate of Change

For the temperature function in Problem 16, what is the value of the maximum rate of change at the point $(1, 1, 1)$?

Solution: The maximum rate of change is the magnitude of the gradient vector we found in the previous problem.

$$\begin{aligned}
\nabla T(1,1,1) &= \langle -200e^{-6}, -400e^{-6}, -600e^{-6} \rangle \\
||\nabla T(1,1,1)|| &= \sqrt{(-200e^{-6})^2 + (-400e^{-6})^2 + (-600e^{-6})^2} \\
&= \sqrt{(200e^{-6})^2((-1)^2 + (-2)^2 + (-3)^2)} \\
&= \sqrt{(200e^{-6})^2(1+4+9)} \\
&= \sqrt{(200e^{-6})^2(14)} \\
&= 200e^{-6}\sqrt{14}
\end{aligned}$$

The maximum rate of change is $200\sqrt{14}e^{-6}$.

Part 4: Linear Approximation in Three Variables

Problem 18: Finding a Linearization (3D)

Find the linearization $L(x, y, z)$ of the function $f(x, y, z) = x\sqrt{y+z}$ at the point $(2, 1, 3)$.

Solution: The linearization of $f(x, y, z)$ at (a, b, c) is:

$$L(x,y,z) = f(a,b,c) + f_x(a,b,c)(x-a) + f_y(a,b,c)(y-b) + f_z(a,b,c)(z-c)$$

Step 1: Find the value of the function at the point.

$$f(2,1,3) = 2\sqrt{1+3} = 2\sqrt{4} = 4$$

Step 2: Find the partial derivatives.

$$\begin{aligned}
f_x &= \frac{\partial}{\partial x}(x\sqrt{y+z}) = \sqrt{y+z} \\
f_y &= \frac{\partial}{\partial y}(x(y+z)^{1/2}) = x \cdot \frac{1}{2}(y+z)^{-1/2} = \frac{x}{2\sqrt{y+z}} \\
f_z &= \frac{\partial}{\partial z}(x(y+z)^{1/2}) = x \cdot \frac{1}{2}(y+z)^{-1/2} = \frac{x}{2\sqrt{y+z}}
\end{aligned}$$

Step 3: Evaluate the partial derivatives at the point $(2, 1, 3)$.

$$\begin{aligned}
f_x(2,1,3) &= \sqrt{1+3} = 2 \\
f_y(2,1,3) &= \frac{2}{2\sqrt{1+3}} = \frac{2}{4} = \frac{1}{2} \\
f_z(2,1,3) &= \frac{2}{2\sqrt{1+3}} = \frac{2}{4} = \frac{1}{2}
\end{aligned}$$

Step 4: Assemble the linearization formula.

$$\begin{aligned}
L(x,y,z) &= 4 + 2(x-2) + \frac{1}{2}(y-1) + \frac{1}{2}(z-3) \\
L(x,y,z) &= 4 + 2x - 4 + \frac{1}{2}y - \frac{1}{2} + \frac{1}{2}z - \frac{3}{2} \\
L(x,y,z) &= 2x + \frac{1}{2}y + \frac{1}{2}z - 2
\end{aligned}$$

Problem 19: Finding a Linearization (3D)

Find the linearization $L(x, y, z)$ of the function $f(x, y, z) = \sin(x+2y+3z)$ at the point $(0, 0, 0)$.

Solution: Step 1: Find the value of the function at the point.

$$f(0, 0, 0) = \sin(0 + 0 + 0) = \sin(0) = 0$$

Step 2: Find the partial derivatives.

$$f_x = \cos(x + 2y + 3z) \cdot 1 = \cos(x + 2y + 3z)$$

$$f_y = \cos(x + 2y + 3z) \cdot 2 = 2\cos(x + 2y + 3z)$$

$$f_z = \cos(x + 2y + 3z) \cdot 3 = 3\cos(x + 2y + 3z)$$

Step 3: Evaluate the partial derivatives at the point $(0, 0, 0)$.

$$f_x(0, 0, 0) = \cos(0) = 1$$

$$f_y(0, 0, 0) = 2\cos(0) = 2$$

$$f_z(0, 0, 0) = 3\cos(0) = 3$$

Step 4: Assemble the linearization formula.

$$L(x, y, z) = 0 + 1(x - 0) + 2(y - 0) + 3(z - 0)$$

$$L(x, y, z) = x + 2y + 3z$$

Problem 20: Using Linearization to Approximate (3D)

Use the linearization you found in Problem 18 to approximate the value of $f(1.9, 1.1, 3.05)$.

Solution: We use the linearization from Problem 18: $L(x, y, z) = 2x+\frac{1}{2}y+\frac{1}{2}z-2$. We approximate the function value by plugging the point $(1.9, 1.1, 3.05)$ into our linearization.

$$f(1.9, 1.1, 3.05) \approx L(1.9, 1.1, 3.05)$$

$$L(1.9, 1.1, 3.05) = 2(1.9) + \frac{1}{2}(1.1) + \frac{1}{2}(3.05) - 2$$

$$L(1.9, 1.1, 3.05) = 3.8 + 0.55 + 1.525 - 2$$

$$L(1.9, 1.1, 3.05) = 3.875$$

So, we approximate that $f(1.9, 1.1, 3.05) \approx 3.875$.

Homework 7 Solutions

Homework 7: Solution Key

Part 1: Gradients, Directional Derivatives, and Tangent Geometry

Problem 1: Directional Derivative in 2D

Find the directional derivative of the function $f(x, y) = x^3 - 4xy^2$ at the point $P(2, 1)$ in the direction of the vector $\vec{v} = \langle 1, 1\rangle$.

Solution: The formula for the directional derivative is $D_{\vec{u}}f = \nabla f \cdot \vec{u}$, where $\vec{u}$ is a unit vector.

Step 1: Find the gradient of $f(x, y)$.

$$f_x = \frac{\partial}{\partial x}(x^3 - 4xy^2) = 3x^2 - 4y^2$$

$$f_y = \frac{\partial}{\partial y}(x^3 - 4xy^2) = -8xy$$

The gradient is $\nabla f(x, y) = \langle 3x^2 - 4y^2, -8xy\rangle$.

Step 2: Evaluate the gradient at the point $P(2, 1)$.

$$\nabla f(2, 1) = \langle 3(2)^2 - 4(1)^2, -8(2)(1)\rangle = \langle 3(4) - 4, -16\rangle = \langle 12 - 4, -16\rangle = \langle 8, -16\rangle$$

Step 3: Find the unit vector $\vec{u}$ in the direction of $\vec{v}$. First, find the magnitude of $\vec{v} = \langle 1, 1\rangle$.

$$\|\vec{v}\| = \sqrt{1^2 + 1^2} = \sqrt{2}$$

Now, divide $\vec{v}$ by its magnitude to get the unit vector $\vec{u}$.

$$\vec{u} = \frac{\vec{v}}{\|\vec{v}\|} = \frac{\langle 1, 1\rangle}{\sqrt{2}} = \left\langle \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right\rangle$$

Step 4: Compute the dot product $\nabla f(2, 1) \cdot \vec{u}$.

$$D_{\vec{u}}f(2, 1) = \langle 8, -16\rangle \cdot \left\langle \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right\rangle = (8)\left(\frac{1}{\sqrt{2}}\right) + (-16)\left(\frac{1}{\sqrt{2}}\right) = \frac{8}{\sqrt{2}} - \frac{16}{\sqrt{2}} = -\frac{8}{\sqrt{2}}$$

Rationalizing the denominator gives:

$$-\frac{8}{\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} = -\frac{8\sqrt{2}}{2} = -4\sqrt{2}$$

The directional derivative is $-4\sqrt{2}$.

Problem 2: Directional Derivative in 3D

Find the directional derivative of the function $f(x, y, z) = xe^{yz}$ at the point $P(3, 0, 2)$ in the direction of the vector $\vec{v} = \langle 2, -1, 2\rangle$.

Solution: We follow the same four steps as in the previous problem.

Step 1: Find the gradient of $f(x, y, z)$.

$$f_x = \frac{\partial}{\partial x}(xe^{yz}) = e^{yz}$$

$$f_y = \frac{\partial}{\partial y}(xe^{yz}) = xe^{yz} \cdot (z) = xze^{yz}$$

$$f_z = \frac{\partial}{\partial z}(xe^{yz}) = xe^{yz} \cdot (y) = xye^{yz}$$

The gradient is $\nabla f(x, y, z) = \langle e^{yz}, xze^{yz}, xye^{yz} \rangle$.

Step 2: Evaluate the gradient at the point $P(3, 0, 2)$. The exponential term is $e^{(0)(2)} = e^0 = 1$.

$$\nabla f(3, 0, 2) = \langle e^0, (3)(2)e^0, (3)(0)e^0 \rangle = \langle 1, 6, 0 \rangle$$

Step 3: Find the unit vector $\vec{u}$ in the direction of $\vec{v}$.

$$\|\vec{v}\| = \sqrt{2^2 + (-1)^2 + 2^2} = \sqrt{4 + 1 + 4} = \sqrt{9} = 3$$

$$\vec{u} = \frac{\vec{v}}{\|\vec{v}\|} = \frac{\langle 2, -1, 2 \rangle}{3} = \left\langle \frac{2}{3}, -\frac{1}{3}, \frac{2}{3} \right\rangle$$

Step 4: Compute the dot product $\nabla f(3, 0, 2) \cdot \vec{u}$.

$$D_{\vec{u}} f(3, 0, 2) = \langle 1, 6, 0 \rangle \cdot \left\langle \frac{2}{3}, -\frac{1}{3}, \frac{2}{3} \right\rangle = (1)\left(\frac{2}{3}\right) + (6)\left(-\frac{1}{3}\right) + (0)\left(\frac{2}{3}\right) = \frac{2}{3} - \frac{6}{3} + 0 = -\frac{4}{3}$$

The directional derivative is $-4/3$.

Problem 3: Directional Derivative

Find the directional derivative of $f(x, y) = x^2y^3$ at the point $P(2, 1)$ in the direction of the vector $\vec{v} = \langle 4, 3\rangle$.

Solution: Step 1: Find the gradient of $f(x, y)$.

$$\nabla f(x, y) = \langle 2xy^3, 3x^2y^2\rangle$$

Step 2: Evaluate the gradient at $P(2, 1)$.

$$\nabla f(2, 1) = \langle 2(2)(1)^3, 3(2)^2(1)^2\rangle = \langle 4, 12\rangle$$

Step 3: Find the unit vector $\vec{u}$.

$$\|\vec{v}\| = \sqrt{4^2 + 3^2} = \sqrt{16 + 9} = \sqrt{25} = 5$$

$$\vec{u} = \frac{\langle 4, 3\rangle}{5} = \left\langle \frac{4}{5}, \frac{3}{5}\right\rangle$$

Step 4: Compute the dot product.

$$D_{\vec{u}}f(2, 1) = \langle 4, 12\rangle \cdot \left\langle \frac{4}{5}, \frac{3}{5}\right\rangle = (4)\left(\frac{4}{5}\right) + (12)\left(\frac{3}{5}\right) = \frac{16}{5} + \frac{36}{5} = \frac{52}{5}$$

The directional derivative is $52/5$.

Problem 4: Steepest Ascent

For the function $f(x, y) = \ln(x^2 + y^2)$, find the direction of steepest ascent at the point $P(1, 2)$. What is the rate of change in this direction?

Solution: The direction of steepest ascent is the gradient vector, and the rate of change is its magnitude.

Step 1: Find the gradient.

$$f_x = \frac{1}{x^2 + y^2} \cdot (2x) = \frac{2x}{x^2 + y^2}$$

$$f_y = \frac{1}{x^2 + y^2} \cdot (2y) = \frac{2y}{x^2 + y^2}$$

$$\nabla f(x, y) = \left\langle \frac{2x}{x^2 + y^2}, \frac{2y}{x^2 + y^2}\right\rangle$$

Step 2: Evaluate the gradient at $P(1, 2)$. The denominator is $1^2 + 2^2 = 5$.

$$\nabla f(1, 2) = \left\langle \frac{2(1)}{5}, \frac{2(2)}{5}\right\rangle = \left\langle \frac{2}{5}, \frac{4}{5}\right\rangle$$

This is the direction of steepest ascent.

Step 3: Find the magnitude of the gradient.

$$\|\nabla f(1, 2)\| = \sqrt{\left(\frac{2}{5}\right)^2 + \left(\frac{4}{5}\right)^2} = \sqrt{\frac{4}{25} + \frac{16}{25}} = \sqrt{\frac{20}{25}} = \frac{\sqrt{20}}{5} = \frac{2\sqrt{5}}{5}$$

The maximum rate of change is $2\sqrt{5}/5$.

Problem 5: Steepest Descent

For the function $f(x, y, z) = x^2 + y^2 - z^2$, find the direction of steepest descent at the point $P(1, 1, 1)$. What is the rate of change in this direction?

Solution: Steepest descent is in the direction of $-\nabla f$, and the rate is $-\|\nabla f\|$.

Step 1: Find the gradient.

$$\nabla f(x, y, z) = \langle 2x, 2y, -2z \rangle$$

Step 2: Evaluate the gradient at $P(1, 1, 1)$.

$$\nabla f(1, 1, 1) = \langle 2(1), 2(1), -2(1) \rangle = \langle 2, 2, -2 \rangle$$

Step 3: Find the direction of steepest descent. The direction is

$$-\nabla f(1, 1, 1) = -\langle 2, 2, -2 \rangle = \langle -2, -2, 2 \rangle$$

Step 4: Find the rate of steepest descent. First, find the magnitude $\|\nabla f(1, 1, 1)\|$.

$$\|\nabla f(1, 1, 1)\| = \sqrt{2^2 + 2^2 + (-2)^2} = \sqrt{4 + 4 + 4} = \sqrt{12} = 2\sqrt{3}$$

The rate of steepest descent is

$$-\|\nabla f\| = -2\sqrt{3}.$$

Problem 6: Tangent Line to a Level Curve

Find the equation of the tangent line to the level curve $x^2 + 4y^2 = 8$ at the point $(2, 1)$.

Solution: The gradient of the function defining the level curve is normal to the tangent line.

Step 1: Define the function and find its gradient. Let $f(x, y) = x^2 + 4y^2$.

$$\nabla f(x, y) = \langle 2x, 8y \rangle$$

Step 2: Evaluate the gradient at $(2, 1)$ to find the normal vector $\vec{n}$.

$$\vec{n} = \nabla f(2, 1) = \langle 2(2), 8(1) \rangle = \langle 4, 8 \rangle$$

Step 3: Use the point-normal form of a line. The formula is $A(x - x_0) + B(y - y_0) = 0$, where $\vec{n} = \langle A, B \rangle$ and the point is (x_0, y_0).

$$4(x - 2) + 8(y - 1) = 0$$

Problem 7: Another Tangent Line

Find the equation of the tangent line to the curve $y = \cos(x)$ at the point $(\pi/3, 1/2)$. (Hint: Treat the curve as a level curve of $f(x, y) = y - \cos(x)$.)

Solution: Following the hint, we define a level curve.

Step 1: Define the function and find its gradient. Let $f(x, y) = y - \cos(x)$. The curve is the level set $f(x, y) = 0$.

$$\nabla f(x, y) = \langle -(-\sin(x)), 1\rangle = \langle \sin(x), 1\rangle$$

Step 2: Evaluate the gradient at $(\pi/3, 1/2)$ to find the normal vector $\vec{n}$.

$$\vec{n} = \nabla f(\pi/3, 1/2) = \langle \sin(\pi/3), 1\rangle = \left\langle \frac{\sqrt{3}}{2}, 1 \right\rangle$$

Step 3: Use the point-normal form. The formula is $A(x-x_0)+B(y-y_0) = 0$, where $\vec{n} = \langle A, B\rangle$ and the point is (x_0, y_0).

$$\frac{\sqrt{3}}{2}\left(x - \frac{\pi}{3}\right) + 1\left(y - \frac{1}{2}\right) = 0$$

Problem 8: Tangent Plane to a Level Surface

Find the equation of the tangent plane to the level surface $xy + yz + xz = 11$ at the point $(1, 2, 3)$.

Solution: The gradient of the function defining the level surface is normal to the tangent plane.
Step 1: Define the function and find its gradient. Let

$$F(x, y, z) = xy + yz + xz.$$

$$\nabla F(x, y, z) = \langle y + z, x + z, y + x\rangle$$

Step 2: Evaluate the gradient at $(1, 2, 3)$ to find the normal vector $\vec{n}$.

$$\vec{n} = \nabla F(1, 2, 3) = \langle 2 + 3, 1 + 3, 2 + 1\rangle = \langle 5, 4, 3\rangle$$

Step 3: Use the point-normal form of a plane.

$$5(x - 1) + 4(y - 2) + 3(z - 3) = 0$$

Problem 9: Tangent Plane to a Function Graph

Find the equation of the tangent plane to the surface $z = x^2 + 5y^2$ at the point $(2, 1, 9)$.

Solution: We can use the formula

$$z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

Step 1: Identify the function and find its partial derivatives. Let $f(x, y) = x^2 + 5y^2$.

$$f_x = 2x \quad \text{and} \quad f_y = 10y$$

Step 2: Evaluate the partials at the point $(a, b) = (2, 1)$.

$$f_x(2, 1) = 2(2) = 4$$

$$f_y(2, 1) = 10(1) = 10$$

Step 3: Assemble the tangent plane equation.

$$z = 9 + 4(x - 2) + 10(y - 1)$$

Problem 10: Tangent Plane to a Sphere

Find the equation of the tangent plane to the sphere $x^2 + y^2 + z^2 = 14$ at the point $(1, 2, 3)$.

Solution: This is a level surface problem.

Step 1: Define the function and find its gradient. Let

$$F(x, y, z) = x^2 + y^2 + z^2.$$

$$\nabla F(x, y, z) = \langle 2x, 2y, 2z \rangle$$

Step 2: Evaluate the gradient at $(1, 2, 3)$ to find the normal vector $\vec{n}$.

$$\vec{n} = \nabla F(1, 2, 3) = \langle 2(1), 2(2), 2(3) \rangle = \langle 2, 4, 6 \rangle$$

Step 3: Use the point-normal form of a plane.

$$2(x - 1) + 4(y - 2) + 6(z - 3) = 0.$$

Problem 11: Tangent Plane with Exponential Function

Find the equation of the tangent plane to the surface $e^{xyz} = 1$ at the point $(1, 1, 0)$.

Solution: This is a level surface problem.

Step 1: Define the function and find its gradient. Let $F(x, y, z) = e^{xyz}$.

$$\nabla F(x, y, z) = \langle e^{xyz} \cdot yz, e^{xyz} \cdot xz, e^{xyz} \cdot xy \rangle$$

Step 2: Evaluate the gradient at $(1, 1, 0)$ to find the normal vector $\vec{n}$. Recall that $e^0 = 1$.

$$\vec{n} = \nabla F(1, 1, 0) = \langle 1 \cdot (1)(0), 1 \cdot (1)(0), 1 \cdot (1)(1) \rangle = \langle 0, 0, 1 \rangle$$

Step 3: Use the point-normal form of a plane.

$$0(x - 1) + 0(y - 1) + 1(z - 0) = 0$$

$$z = 0$$

Part 2: The Method of Lagrange Multipliers

Problem 12

Find the extreme values of $f(x, y) = x + 2y$ subject to the constraint $x^2 + y^2 = 5$.

Solution: Step 1: Set up the system. Objective: $f(x, y) = x + 2y \implies \nabla f = \langle 1, 2\rangle$. Constraint: $g(x, y) = x^2 + y^2 = 5 \implies \nabla g = \langle 2x, 2y\rangle$. The system is:

$$1 = \lambda(2x)$$

$$2 = \lambda(2y)$$

$$x^2 + y^2 = 5$$

Step 2: Solve the system. From the first two equations (assuming $x, y \neq 0$):

$$\lambda = \frac{1}{2x} \quad \text{and} \quad \lambda = \frac{2}{2y} = \frac{1}{y}$$

Set the expressions for λ equal:

$$\frac{1}{2x} = \frac{1}{y} \implies y = 2x$$

Substitute this into the constraint:

$$x^2 + (2x)^2 = 5$$

$$x^2 + 4x^2 = 5$$

$$5x^2 = 5$$

$$x^2 = 1$$

$$x = \pm 1$$

If $x = 1$, then $y = 2(1) = 2$.

$$(1, 2).$$

If $x = -1$, then $y = 2(-1) = -2$.

$$(-1, -2).$$

Step 3: Evaluate f at the candidate points.

$$f(1, 2) = 1 + 2(2) = 5$$

$$f(-1, -2) = -1 + 2(-2) = -5$$

Maximum is 5, Minimum is -5.

Problem 13

Find the extreme values of $f(x, y) = xy$ subject to the constraint $x^2 + y^2 = 18$.

Solution: Step 1: Set up the system. The objective function is $f(x, y) = xy$, so its gradient is $\nabla f = \langle y, x\rangle$. The constraint function is $g(x, y) = x^2 + y^2 = 18$, so its gradient is $\nabla g = \langle 2x, 2y\rangle$. The Lagrange multiplier system of equations, $\nabla f = \lambda \nabla g$, is:

$$\begin{aligned} y &= \lambda(2x) \\ x &= \lambda(2y) \\ x^2 + y^2 &= 18 \end{aligned}$$

Step 2: Solve the system. Substitute the expression for x from the second equation into the first equation:

$$y = \lambda(2(\lambda(2y)))$$

$$y = 4\lambda^2 y$$

Now, move all terms to one side of the equation:

$$y - 4\lambda^2 y = 0$$

Factor out a y:

$$y(1 - 4\lambda^2) = 0$$

This gives two possibilities: $y = 0$ or $1 - 4\lambda^2 = 0$.

Case 1: $y = 0$. If $y = 0$, we use the second equation, $x = \lambda(2y)$, which gives $x = 0$. The point $(0, 0)$ is not on the constraint curve since $0^2 + 0^2 \neq 18$. So this case yields no solutions.

Case 2: $1 - 4\lambda^2 = 0$. This means $4\lambda^2 = 1$, so $\lambda^2 = 1/4$, which gives $\lambda = \pm 1/2$.

If $\lambda = 1/2$. Substitute this into the first equation $y = \lambda(2x)$:

$$y = \frac{1}{2}(2x)$$

$$y = x$$

Now substitute $y = x$ into the constraint:

$$x^2 + (x)^2 = 18$$

$$2x^2 = 18$$

$$x^2 = 9$$

$$x = \pm 3$$

This gives two points: $(3, 3)$ and $(-3, -3)$.

If $\lambda = -1/2$. Substitute this into the first equation $y = \lambda(2x)$:

$$y = -\frac{1}{2}(2x)$$

$$y = -x$$

Now substitute $y = -x$ into the constraint:

$$x^2 + (-x)^2 = 18$$

$$2x^2 = 18$$

$$x^2 = 9$$

$$x = \pm 3$$

This gives two more points: $(3, -3)$ and $(-3, 3)$.

Step 3: Evaluate f at the candidate points.

$$f(3, 3) = (3)(3) = 9$$

$$f(-3, -3) = (-3)(-3) = 9$$

$$f(3, -3) = (3)(-3) = -9$$

$$f(-3, 3) = (-3)(3) = -9$$

The maximum value is 9 and the minimum value is -9.

Problem 14

Find the extreme values of $f(x, y) = x^2 + y^2$ subject to the constraint $x^2 + 4y^2 = 16$.

Solution: Step 1: Set up the system. Objective: $f(x, y) = x^2 + y^2$, so $\nabla f = \langle 2x, 2y \rangle$. Constraint: $g(x, y) = x^2 + 4y^2 = 16$, so $\nabla g = \langle 2x, 8y \rangle$. The system is:

$$\begin{aligned} 2x &= \lambda(2x) \\ 2y &= \lambda(8y) \\ x^2 + 4y^2 &= 16 \end{aligned}$$

Step 2: Solve the system. Look at the first equation:

$$2x - 2\lambda x = 0$$

$$2x(1 - \lambda) = 0$$

This gives two possibilities: $x = 0$ or $\lambda = 1$.

Case 1: $x = 0$. Substitute $x = 0$ into the constraint equation:

$$(0)^2 + 4y^2 = 16$$

$$4y^2 = 16$$

$$y^2 = 4$$

$$y = \pm 2$$

This gives two candidate points: $(0, 2)$ and $(0, -2)$.

Case 2: $\lambda = 1$. Substitute $\lambda = 1$ into the second equation $2y = \lambda(8y)$:

$$2y = (1)(8y)$$

$$2y - 8y = 0$$

$$-6y = 0$$

$$y = 0$$

Now substitute $y = 0$ into the constraint equation:

$$x^2 + 4(0)^2 = 16$$

$$x^2 = 16$$

$$x = \pm 4$$

This gives two more candidate points: $(4, 0)$ and $(-4, 0)$.

Step 3: Evaluate f at all four candidate points.

$$f(0, 2) = (0)^2 + (2)^2 = 4$$

$$f(0, -2) = (0)^2 + (-2)^2 = 4$$

$$f(4, 0) = (4)^2 + (0)^2 = 16$$

$$f(-4, 0) = (-4)^2 + (0)^2 = 16$$

The maximum value is 16 and the minimum value is 4.

Problem 15

Find the extreme values of $f(x, y) = 4x^2 + y^2$ subject to the constraint $x^2 + y^2 = 1$.

Solution: Step 1: Set up the system. Objective: $f(x, y) = 4x^2 + y^2$, so $\nabla f = \langle 8x, 2y \rangle$. Constraint: $g(x, y) = x^2 + y^2 = 1$, so $\nabla g = \langle 2x, 2y \rangle$. The system is:

$$\begin{aligned} 8x &= \lambda(2x) \\ 2y &= \lambda(2y) \\ x^2 + y^2 &= 1 \end{aligned}$$

Step 2: Solve the system. Look at the second equation:

$$2y - 2\lambda y = 0$$

$$2y(1 - \lambda) = 0$$

This gives two possibilities: $y = 0$ or $\lambda = 1$.

Case 1: $y = 0$. Substitute $y = 0$ into the constraint equation:

$$x^2 + (0)^2 = 1$$

$$x^2 = 1$$

$$x = \pm 1$$

This gives two candidate points: $(1, 0)$ and $(-1, 0)$.

Case 2: $\lambda = 1$. Substitute $\lambda = 1$ into the first equation $8x = \lambda(2x)$:

$$8x = (1)(2x)$$

$$8x - 2x = 0$$

$$6x = 0$$

$$x = 0$$

Now substitute $x = 0$ into the constraint equation:

$$(0)^2 + y^2 = 1$$

$$y^2 = 1$$

$$y = \pm 1$$

This gives two more candidate points: $(0, 1)$ and $(0, -1)$.

Step 3: Evaluate f at all four candidate points.

$$f(1, 0) = 4(1)^2 + (0)^2 = 4$$

$$f(-1, 0) = 4(-1)^2 + (0)^2 = 4$$

$$f(0, 1) = 4(0)^2 + (1)^2 = 1$$

$$f(0, -1) = 4(0)^2 + (-1)^2 = 1$$

The maximum value is 4 and the minimum value is 1.

Problem 16

Find the extreme values of $f(x, y) = 2x + 6y$ subject to the constraint $x^2 + y^2 = 10$.

Solution: Step 1: Set up the system. Objective: $f(x, y) = 2x+6y$, so $\nabla f = \langle 2, 6\rangle$. Constraint: $g(x, y) = x^2 + y^2 = 10$, so $\nabla g = \langle 2x, 2y\rangle$. The system is:

$$\begin{aligned} 2 &= \lambda(2x) \\ 6 &= \lambda(2y) \\ x^2 + y^2 &= 10 \end{aligned}$$

Step 2: Solve the system. Simplify the first two equations:

$$1 = \lambda x$$

$$3 = \lambda y$$

Solve both for λ (assuming $x, y \neq 0$):

$$\lambda = \frac{1}{x} \quad \text{and} \quad \lambda = \frac{3}{y}$$

Set the expressions for λ equal to each other:

$$\frac{1}{x} = \frac{3}{y}$$

Cross-multiply to find the relationship $y = 3x$. Substitute this into the constraint equation:

$$x^2 + (3x)^2 = 10$$

$$x^2 + 9x^2 = 10$$

$$10x^2 = 10$$

$$x^2 = 1$$

$$x = \pm 1$$

We have two cases for x:

Case 1: If $x = 1$, then $y = 3(1) = 3$. This gives the point $(1, 3)$.

Case 2: If $x = -1$, then $y = 3(-1) = -3$. This gives the point $(-1, -3)$.

Step 3: Evaluate f at the candidate points.

$$f(1, 3) = 2(1) + 6(3) = 2 + 18 = 20$$

$$f(-1, -3) = 2(-1) + 6(-3) = -2 - 18 = -20$$

The maximum value is 20 and the minimum value is -20.

Problem 17

Find the extreme values of $f(x, y, z) = x + y + z$ subject to the constraint $x^2 + y^2 + z^2 = 12$.

Solution: Step 1: Set up the system. Objective: $f(x, y, z) = x + y + z$, so $\nabla f = \langle 1, 1, 1 \rangle$. Constraint: $g(x, y, z) = x^2 + y^2 + z^2 = 12$, so $\nabla g = \langle 2x, 2y, 2z \rangle$. The system is:

$$\begin{aligned} 1 &= \lambda(2x) \\ 1 &= \lambda(2y) \\ 1 &= \lambda(2z) \\ x^2 + y^2 + z^2 &= 12 \end{aligned}$$

Step 2: Solve the system. From the first three equations, we can see that:

$$2\lambda x = 1 \quad , \quad 2\lambda y = 1 \quad , \quad 2\lambda z = 1$$

This shows $2\lambda x = 2\lambda y = 2\lambda z$. Since λ cannot be zero (otherwise $1 = 0$), we can divide the entire expression by 2λ, which gives $x = y = z$. Substitute this into the constraint:

$$x^2 + (x)^2 + (x)^2 = 12$$

$$3x^2 = 12$$

$$x^2 = 4$$

$$x = \pm 2$$

This gives two candidate points: $(2, 2, 2)$ and $(-2, -2, -2)$.

Step 3: Evaluate f at the candidate points.

$$f(2, 2, 2) = 2 + 2 + 2 = 6$$

$$f(-2, -2, -2) = -2 - 2 - 2 = -6$$

The maximum value is 6 and the minimum value is -6.

Problem 18

Find the maximum value of $f(x, y, z) = xyz$ subject to the constraint $x + y + z = 6$, assuming $x, y, z > 0$.

Solution: Step 1: Set up the system. Objective: $f = xyz$, so $\nabla f = \langle yz, xz, xy \rangle$. Constraint: $g = x + y + z = 6$, so $\nabla g = \langle 1, 1, 1 \rangle$. The system is:

$$\begin{aligned} yz &= \lambda \\ xz &= \lambda \\ xy &= \lambda \\ x + y + z &= 6 \end{aligned}$$

Step 2: Solve the system. Since $x, y, z > 0$, λ must also be positive. We can set the first two equations equal:

$$yz = xz$$

Since $z > 0$, we can divide by z to get $y = x$. Next, set the second and third equations equal:

$$xz = xy$$

Since $x > 0$, we can divide by x to get $z = y$. This means $x = y = z$. Substitute this into the constraint:

$$x + x + x = 6$$

$$3x = 6$$

$$x = 2$$

This gives the single candidate point $(2, 2, 2)$.

Step 3: Evaluate f at the candidate point. First, we calculate the value of the function at our candidate point $(2, 2, 2)$.

$$f(2, 2, 2) = (2)(2)(2) = 8$$

Since we only found one candidate point, we should verify that it is indeed a maximum. To do this, we can test a nearby point that also satisfies the constraint $x + y + z = 6$. Let's try a point where x is slightly larger, for instance $x = 2.1$, and y is slightly smaller, $y = 1.9$. To satisfy the constraint, z must be $z = 6 - 2.1 - 1.9 = 2$. So our test point is $(2.1, 1.9, 2)$.

Now, we evaluate the function at this new point:

$$f(2.1, 1.9, 2) = (2.1)(1.9)(2)$$

$$f(2.1, 1.9, 2) = (3.99)(2)$$

$$f(2.1, 1.9, 2) = 7.98$$

Since $7.98 < 8$, the function value decreases as we move away from our candidate point. This indicates that $(2, 2, 2)$ is a local maximum. Because our constraint is an open region (since $x, y, z > 0$) and this is the only critical point we found, this local maximum must also be the absolute maximum value.

The maximum value is 8.

Problem 19: Application

You have 100 feet of fencing to build a rectangular enclosure. What is the maximum area you can enclose? (Objective function is $A(x, y) = xy$, constraint is the perimeter).

Solution: Step 1: Set up the system. Let x be the length and y be the width of the rectangle. The objective is to maximize the area, $A(x, y) = xy$. The gradient is $\nabla A = \langle y, x \rangle$. The constraint is the perimeter, $2x + 2y = 100$. We can simplify this constraint to $g(x, y) = x + y = 50$. The gradient of the constraint is $\nabla g = \langle 1, 1 \rangle$. The Lagrange system is:

$$\begin{aligned} y &= \lambda \\ x &= \lambda \\ x + y &= 50 \end{aligned}$$

Step 2: Solve the system. From the first two equations, we see that $x = y$. Substitute this into the constraint:

$$\begin{aligned} x + x &= 50 \\ 2x &= 50 \\ x &= 25 \end{aligned}$$

Since $y = x$, we have $y = 25$. The dimensions are 25 ft by 25 ft (a square).

Step 3: Evaluate the Area.

$$A(25, 25) = (25)(25) = 625$$

The maximum area that can be enclosed is 625 square feet.

Homework 8 Solutions

Homework 8 Solutions: Second Derivative Test Multiple Integrals

Problem 1

Find all critical points of the function $f(x, y) = x^2 + y^2 - 6x + 2y + 7$ and classify each as a local maximum, local minimum, or saddle point.

Solution: Step 1: Find the first partial derivatives.

$$\begin{aligned} f_x &= \frac{\partial}{\partial x}(x^2 + y^2 - 6x + 2y + 7) = 2x - 6 \\ f_y &= \frac{\partial}{\partial y}(x^2 + y^2 - 6x + 2y + 7) = 2y + 2 \end{aligned}$$

Step 2: Find the critical points by setting the partial derivatives to zero.

$$\begin{aligned} f_x = 0 &\implies 2x - 6 = 0 \implies 2x = 6 \implies x = 3 \\ f_y = 0 &\implies 2y + 2 = 0 \implies 2y = -2 \implies y = -1 \end{aligned}$$

The only critical point is $(3, -1)$.

Step 3: Find the second partial derivatives.

$$\begin{aligned} f_{xx} &= \frac{\partial}{\partial x}(2x - 6) = 2 \\ f_{yy} &= \frac{\partial}{\partial y}(2y + 2) = 2 \\ f_{xy} &= \frac{\partial}{\partial y}(2x - 6) = 0 \end{aligned}$$

Step 4: Calculate the discriminant.

$$D(x,y) = f_{xx}f_{yy} - (f_{xy})^2$$
$$D(x,y) = (2)(2) - (0)^2 = 4$$

Step 5: Classify the critical point $(3,-1)$**.** The discriminant is constant for all (x,y), so $D(3,-1) = 4$. Since $D = 4 > 0$, we have a local extremum. We check the sign of f_{xx}.

$$f_{xx}(3,-1) = 2$$

Since $f_{xx} > 0$, the function is concave up at this point. Therefore, the point $\mathbf{(3,-1)}$ is a **local minimum**.

Problem 2

Find all critical points of the function $f(x,y) = xy - x^2 - y^2$ and classify each as a local maximum, local minimum, or saddle point.

Solution: Step 1: Find the first partial derivatives.

$$f_x = y - 2x$$
$$f_y = x - 2y$$

Step 2: Find the critical points. Set $f_x = 0$ and $f_y = 0$:

$$\begin{aligned} &(1) \quad y - 2x = 0 \implies y = 2x \\ &(2) \quad x - 2y = 0 \end{aligned}$$

Substitute the expression for y from (1) into (2):

$$\begin{aligned} x - 2(2x) &= 0 \\ x - 4x &= 0 \\ -3x &= 0 \\ x &= 0 \end{aligned}$$

Now find the corresponding y-value using $y = 2x$:

$$y = 2(0) = 0$$

The only critical point is $(0,0)$.

Step 3: Find the second partial derivatives.

$$f_{xx} = -2$$
$$f_{yy} = -2$$
$$f_{xy} = 1$$

Step 4: Calculate the discriminant.

$$D(x,y) = f_{xx}f_{yy} - (f_{xy})^2$$
$$D(x,y) = (-2)(-2) - (1)^2 = 4 - 1 = 3$$

Step 5: Classify the critical point $(0,0)$**.** The discriminant is $D(0,0) = 3$. Since $D > 0$, we have a local extremum. We check the sign of f_{xx}.

$$f_{xx}(0,0) = -2$$

Since $f_{xx} < 0$, the function is concave down at this point. Therefore, the point $\mathbf{(0,0)}$ is a **local maximum**.

Problem 3

Find all critical points of the function $f(x, y) = x^3 - 12xy + 8y^3$ and classify each as a local maximum, local minimum, or saddle point.

Solution: Step 1: Find the first partial derivatives.

$$f_x = 3x^2 - 12y$$
$$f_y = -12x + 24y^2$$

Step 2: Find the critical points. Set the partials to zero:

$$(1) \quad 3x^2 - 12y = 0 \implies 3x^2 = 12y \implies y = \frac{1}{4}x^2$$
$$(2) \quad -12x + 24y^2 = 0 \implies 24y^2 = 12x \implies y^2 = \frac{1}{2}x$$

Substitute (1) into (2):

$$\left(\frac{1}{4}x^2\right)^2 = \frac{1}{2}x$$
$$\frac{1}{16}x^4 = \frac{1}{2}x$$
$$x^4 = 8x$$
$$x^4 - 8x = 0$$
$$x(x^3 - 8) = 0$$

This gives two solutions for x: $x = 0$ and $x^3 - 8 = 0 \implies x = 2$. Find the corresponding y-values using $y = \frac{1}{4}x^2$:

- If $x = 0$, then $y = \frac{1}{4}(0)^2 = 0$. Critical point: $(0, 0)$.
- If $x = 2$, then $y = \frac{1}{4}(2)^2 = 1$. Critical point: $(2, 1)$.

Step 3: Find the second partial derivatives.

$$f_{xx} = 6x$$
$$f_{yy} = 48y$$
$$f_{xy} = -12$$

Step 4: Calculate the discriminant.

$$D(x, y) = (6x)(48y) - (-12)^2 = 288xy - 144$$

Step 5: Classify the critical points. For $(0, 0)$:

$$D(0, 0) = 288(0)(0) - 144 = -144$$

Since $D < 0$, the point $\mathbf{(0, 0)}$ is a **saddle point**.

For $(2, 1)$:

$$D(2, 1) = 288(2)(1) - 144 = 576 - 144 = 432$$

Since $D > 0$, it's a local extremum. Check f_{xx}:

$$f_{xx}(2, 1) = 6(2) = 12$$

Since $f_{xx} > 0$, the point $\mathbf{(2, 1)}$ is a **local minimum**.

Problem 4

Find all critical points of the function $f(x, y) = e^{2x}\cos(y)$ and classify each as a local maximum, local minimum, or saddle point. (Note: there are infinitely many critical points; describe the general form and test one.)

Solution: Step 1: Find the first partial derivatives.

$$f_x = 2e^{2x}\cos(y)$$
$$f_y = -e^{2x}\sin(y)$$

Step 2: Find the critical points. Set the partials to zero:

$$\begin{aligned} (1) \quad & 2e^{2x}\cos(y) = 0 \\ (2) \quad & -e^{2x}\sin(y) = 0 \end{aligned}$$

Since e^{2x} is never zero, we can divide both equations by it. The system simplifies to:

$$\cos(y) = 0$$
$$\sin(y) = 0$$

There is no value of y for which both $\sin(y)$ and $\cos(y)$ are zero simultaneously. This means there are **no critical points**. Therefore, the function has no local maxima, minima, or saddle points.

Problem 5

Find all critical points of the function $f(x, y) = y\ln(x) - x - 2y$ and classify each as a local maximum, local minimum, or saddle point.

Solution: Step 1: Find the first partial derivatives.

$$f_x = \frac{y}{x} - 1$$
$$f_y = \ln(x) - 2$$

Step 2: Find the critical points. Set the partials to zero:

$$\begin{aligned} (1) \quad & \frac{y}{x} - 1 = 0 \\ (2) \quad & \ln(x) - 2 = 0 \end{aligned}$$

From equation (2):

$$\ln(x) = 2$$
$$x = e^2$$

Substitute this value of x into equation (1):

$$\frac{y}{e^2} - 1 = 0$$
$$\frac{y}{e^2} = 1$$
$$y = e^2$$

The only critical point is (e^2, e^2).

Step 3: Find the second partial derivatives.

$$f_{xx} = \frac{\partial}{\partial x}(yx^{-1} - 1) = -yx^{-2} = -\frac{y}{x^2}$$
$$f_{yy} = \frac{\partial}{\partial y}(\ln(x) - 2) = 0$$
$$f_{xy} = \frac{\partial}{\partial y}(\frac{y}{x} - 1) = \frac{1}{x}$$

Step 4: Calculate the discriminant.

$$D(x, y) = \left(-\frac{y}{x^2}\right)(0) - \left(\frac{1}{x}\right)^2 = -\frac{1}{x^2}$$

Step 5: Classify the critical point (e^2, e^2).

$$D(e^2, e^2) = -\frac{1}{(e^2)^2} = -\frac{1}{e^4}$$

Since $D < 0$, the point $\boldsymbol{(e^2, e^2)}$ is a **saddle point**.

Problem 6

Find all critical points of the function $f(x, y) = x^3 - 3x + y^3 - 12y$ and classify each as a local maximum, local minimum, or saddle point. (This function has four critical points.)

Solution: Step 1: Find the first partial derivatives.

$$f_x = 3x^2 - 3$$
$$f_y = 3y^2 - 12$$

Step 2: Find the critical points.

$$f_x = 0 \implies 3x^2 - 3 = 0 \implies 3x^2 = 3 \implies x^2 = 1 \implies x = \pm 1$$
$$f_y = 0 \implies 3y^2 - 12 = 0 \implies 3y^2 = 12 \implies y^2 = 4 \implies y = \pm 2$$

Since the equations for x and y are independent, we must consider all possible pairings of the x and y solutions. This gives four critical points: $(1, 2)$, $(1, -2)$, $(-1, 2)$, and $(-1, -2)$.

Step 3: Find the second partial derivatives.

$$f_{xx} = 6x$$
$$f_{yy} = 6y$$
$$f_{xy} = 0$$

Step 4: Calculate the discriminant.

$$D(x, y) = (6x)(6y) - (0)^2 = 36xy$$

Step 5: Classify the critical points. For $(1, 2)$: $D(1, 2) = 36(1)(2) = 72 > 0$. And $f_{xx}(1, 2) = 6(1) = 6 > 0$. This is a **local minimum**.

For $(1, -2)$: $D(1, -2) = 36(1)(-2) = -72 < 0$. This is a **saddle point**.

For $(-1, 2)$: $D(-1, 2) = 36(-1)(2) = -72 < 0$. This is a **saddle point**.

For $(-1, -2)$: $D(-1, -2) = 36(-1)(-2) = 72 > 0$. And $f_{xx}(-1, -2) = 6(-1) = -6 < 0$. This is a **local maximum**.

Problem 7

Find all critical points of the function $f(x, y) = x^2y^2 - 4x^2 - y^2$ and classify each as a local maximum, local minimum, or saddle point. (This function has five critical points.)

Solution: Step 1: Find the first partial derivatives.

$$f_x = 2xy^2 - 8x = 2x(y^2 - 4)$$
$$f_y = 2x^2y - 2y = 2y(x^2 - 1)$$

Step 2: Find the critical points. Set the partials to zero:

$$(1) \quad 2x(y^2 - 4) = 0$$
$$(2) \quad 2y(x^2 - 1) = 0$$

From (1), we have two possibilities: $x = 0$ or $y^2 - 4 = 0 \implies y = \pm 2$. Case 1: $x = 0$. Substitute this into equation (2):

$$2y((0)^2 - 1) = 0 \implies 2y(-1) = 0 \implies -2y = 0 \implies y = 0$$

This gives the critical point $(0, 0)$. Case 2: $y = 2$. Substitute this into equation (2):

$$2(2)(x^2 - 1) = 0 \implies 4(x^2 - 1) = 0 \implies x^2 = 1 \implies x = \pm 1$$

This gives the critical points $(1, 2)$ and $(-1, 2)$. Case 3: $y = -2$. Substitute this into equation (2):

$$2(-2)(x^2 - 1) = 0 \implies -4(x^2 - 1) = 0 \implies x^2 = 1 \implies x = \pm 1$$

This gives the critical points $(1, -2)$ and $(-1, -2)$. In total, we have five critical points:

$$(0, 0), (1, 2), (-1, 2), (1, -2), (-1, -2).$$

Step 3: Find the second partial derivatives.

$$f_{xx} = 2y^2 - 8$$
$$f_{yy} = 2x^2 - 2$$
$$f_{xy} = 4xy$$

Step 4: Calculate the discriminant.

$$D(x, y) = (2y^2 - 8)(2x^2 - 2) - (4xy)^2$$

Step 5: Classify the critical points. For $(0, 0)$: $D(0, 0) = (2(0)^2 - 8)(2(0)^2 - 2) - (0)^2 = (-8)(-2) = 16 > 0$. $f_{xx}(0, 0) = 2(0)^2 - 8 = -8 < 0$. This is a **local maximum**.

For $(1, 2)$: $D(1, 2) = (2(2)^2 - 8)(2(1)^2 - 2) - (4(1)(2))^2 = (8 - 8)(2 - 2) - (8)^2 = (0)(0) - 64 = -64 < 0$. This is a **saddle point**.

For $(-1, 2)$: $D(-1, 2) = (2(2)^2 - 8)(2(-1)^2 - 2) - (4(-1)(2))^2 = (0)(0) - (-8)^2 = -64 < 0$. This is a **saddle point**.

For $(1, -2)$: $D(1, -2) = (2(-2)^2 - 8)(2(1)^2 - 2) - (4(1)(-2))^2 = (0)(0) - (-8)^2 = -64 < 0$. This is a **saddle point**.

For $(-1, -2)$: $D(-1, -2) = (2(-2)^2 - 8)(2(-1)^2 - 2) - (4(-1)(-2))^2 = (0)(0) - (8)^2 = -64 < 0$. This is a **saddle point**.

Problem 8

Construct a function $f(x, y)$ that has a single critical point at $(1, 2)$ which is a local minimum. Verify using the Second Derivative Test that your function satisfies these conditions.

Solution: The simplest function with a local minimum is an upward-opening paraboloid. We can construct one with its vertex at $(1, 2)$. Let $f(x, y) = (x-1)^2 + (y-2)^2$. **Step 1 2: Find Critical Points.**

$$f_x = 2(x-1) = 0 \implies x = 1$$
$$f_y = 2(y-2) = 0 \implies y = 2$$

The only critical point is indeed $(1, 2)$. **Step 3, 4, 5: Verify with Second Derivative Test.**

$$f_{xx} = 2$$
$$f_{yy} = 2$$
$$f_{xy} = 0$$

The discriminant is $D(x, y) = (2)(2) - (0)^2 = 4$. At the point $(1, 2)$, we have $D(1, 2) = 4 > 0$ and $f_{xx}(1, 2) = 2 > 0$. The test confirms that $(1, 2)$ is a local minimum.

Part 2: Introduction to Multiple Integration

Problem 9

Evaluate the double integral $\iint_R 7\,dA$ where the region R is the rectangle $R = [-1, 3] \times [2, 5]$.

Solution: This integral represents the volume of a rectangular box with a constant height of 7 over the region R. The length of the x-interval is $3 - (-1) = 4$. The length of the y-interval is $5 - 2 = 3$. The area of the region R is $4 \times 3 = 12$. The volume is (Area of Base) $\times$ (Height) $= 12 \times 7 = 84$.

Alternatively, using iterated integrals:

$$\begin{aligned}
\int_{-1}^{3}\int_{2}^{5} 7\,dy\,dx &= \int_{-1}^{3} [7y]_{y=2}^{y=5}\,dx \\
&= \int_{-1}^{3} (7(5) - 7(2))\,dx \\
&= \int_{-1}^{3} (35 - 14)\,dx \\
&= \int_{-1}^{3} 21\,dx \\
&= [21x]_{x=-1}^{x=3} \\
&= (21(3)) - (21(-1)) \\
&= 63 - (-21) = 84
\end{aligned}$$

Problem 10

Evaluate the iterated integral $\int_0^2 \int_1^3 (x^2 + y)\,dy\,dx$.

Solution: Step 1: Evaluate the inner integral (with respect to y).

$$\begin{aligned}
\int_1^3 (x^2 + y)\,dy &= \left[x^2 y + \frac{1}{2}y^2\right]_{y=1}^{y=3} \\
&= \left(x^2(3) + \frac{1}{2}(3)^2\right) - \left(x^2(1) + \frac{1}{2}(1)^2\right) \\
&= \left(3x^2 + \frac{9}{2}\right) - \left(x^2 + \frac{1}{2}\right) \\
&= 2x^2 + \frac{8}{2} = 2x^2 + 4
\end{aligned}$$

Step 2: Evaluate the outer integral (with respect to x).

$$\begin{aligned}
\int_0^2 (2x^2 + 4)\,dx &= \left[\frac{2}{3}x^3 + 4x\right]_{x=0}^{x=2} \\
&= \left(\frac{2}{3}(2)^3 + 4(2)\right) - (0) \\
&= \frac{2}{3}(8) + 8 = \frac{16}{3} + \frac{24}{3} = \frac{40}{3}
\end{aligned}$$

Problem 11

Evaluate the double integral $\iint_R (2x - 4y^3)\, dA$ where $R = [0,1] \times [0,2]$.

Solution: We set up the iterated integral, for example, in the order $dy\, dx$.

$$\int_0^1 \int_0^2 (2x - 4y^3)\, dy\, dx$$

Step 1: Inner integral (wrt y).

$$\begin{aligned}\int_0^2 (2x - 4y^3)\, dy &= \left[2xy - y^4\right]_{y=0}^{y=2} \\ &= (2x(2) - (2)^4) - (0) \\ &= 4x - 16\end{aligned}$$

Step 2: Outer integral (wrt x).

$$\begin{aligned}\int_0^1 (4x - 16)\, dx &= \left[2x^2 - 16x\right]_{x=0}^{x=1} \\ &= (2(1)^2 - 16(1)) - (0) = 2 - 16 = -14\end{aligned}$$

Problem 12

Evaluate the double integral $\iint_R \frac{xy^2}{x^2+1}\, dA$ where $R = [0,1] \times [-3,3]$.

Solution: Set up the integral. Let's choose the order $dy\, dx$.

$$\int_0^1 \int_{-3}^3 \frac{xy^2}{x^2+1}\, dy\, dx$$

Step 1: Inner integral (wrt y).

$$\begin{aligned}\int_{-3}^3 \frac{xy^2}{x^2+1}\, dy &= \frac{x}{x^2+1} \int_{-3}^3 y^2\, dy \\ &= \frac{x}{x^2+1} \left[\frac{1}{3}y^3\right]_{y=-3}^{y=3} \\ &= \frac{x}{x^2+1} \left(\frac{1}{3}(3)^3 - \frac{1}{3}(-3)^3\right) \\ &= \frac{x}{x^2+1} \left(\frac{27}{3} - \frac{-27}{3}\right) = \frac{x}{x^2+1}(9 - (-9)) = \frac{18x}{x^2+1}\end{aligned}$$

Step 2: Outer integral (wrt x). We need to evaluate $\int_0^1 \frac{18x}{x^2+1}\, dx$. We use a u-substitution. Let $u = x^2 + 1$. Then $du = 2x\, dx$, so $9\, du = 18x\, dx$. Change the bounds: If $x = 0, u = 1$. If $x = 1, u = 2$.

$$\begin{aligned}\int_1^2 \frac{9}{u}\, du &= 9\left[\ln|u|\right]_1^2 \\ &= 9(\ln(2) - \ln(1)) = 9(\ln(2) - 0) = 9\ln(2)\end{aligned}$$

Problem 13

Evaluate $\iint_R 2xy\,dA$ where $R = [0,1]\times[0,2]$ by computing the iterated integral in **both** possible orders ($dy\,dx$ and $dx\,dy$) to explicitly verify Fubini's Theorem.

Solution: Order 1: $dy\,dx$

$$\begin{aligned}\int_0^1\int_0^2 2xy\,dy\,dx &= \int_0^1 \left[xy^2\right]_{y=0}^{y=2}\,dx\\ &= \int_0^1 (x(2)^2 - x(0)^2)\,dx\\ &= \int_0^1 4x\,dx\\ &= \left[2x^2\right]_{x=0}^{x=1}\\ &= 2(1)^2 - 0 = 2\end{aligned}$$

Order 2: $dx\,dy$

$$\begin{aligned}\int_0^2\int_0^1 2xy\,dx\,dy &= \int_0^2 \left[x^2y\right]_{x=0}^{x=1}\,dy\\ &= \int_0^2 ((1)^2y - (0)^2y)\,dy\\ &= \int_0^2 y\,dy\\ &= \left[\frac{1}{2}y^2\right]_{y=0}^{y=2}\\ &= \frac{1}{2}(2)^2 - 0 = 2\end{aligned}$$

The results are the same, verifying Fubini's Theorem for this integral.

Problem 14

Evaluate the double integral $\iint_R y\cos(xy)\,dA$ where $R = [0,1]\times[0,\pi]$. Choose the easier order of integration.

Solution: Let's analyze the two possible inner integrals: 1. $\int y\cos(xy)\,dy$: This requires integration by parts. 2. $\int y\cos(xy)\,dx$: Here, y is a constant, so this is a simple integral. We choose the second order, $dx\,dy$.

$$\int_0^\pi\int_0^1 y\cos(xy)\,dx\,dy$$

Step 1: Inner integral (wrt x).

$$\begin{aligned}\int_0^1 y\cos(xy)\,dx &= y\left[\frac{1}{y}\sin(xy)\right]_{x=0}^{x=1}\\ &= [\sin(xy)]_{x=0}^{x=1}\\ &= \sin(y\cdot 1) - \sin(y\cdot 0) = \sin(y) - \sin(0) = \sin(y)\end{aligned}$$

Step 2: Outer integral (wrt y).

$$\begin{aligned}\int_0^{\pi} \sin(y)\,dy &= [-\cos(y)]_0^{\pi} \\ &= (-\cos(\pi)) - (-\cos(0)) = -(-1) - (-1) = 1 + 1 = 2\end{aligned}$$

Problem 15

Evaluate the double integral $\iint_R xe^{xy}\,dA$ where $R = [0,2] \times [0,1]$. Choose the easier order of integration.

Solution: First, we analyze the two possible orders of integration to determine which is simpler.

Option 1: Order $dx\,dy$ The inner integral would be:

$$\int_0^2 xe^{xy}\,dx$$

To solve this, we would need to use integration by parts with $u = x$ and $dv = e^{xy}dx$. This is complicated.

Option 2: Order $dy\,dx$ The inner integral would be:

$$\int_0^1 xe^{xy}\,dy$$

In this integral, x is treated as a constant. The antiderivative of e^{ay} with respect to y is $\frac{1}{a}e^{ay}$. This is a much simpler, direct integration. We will choose this order.

The setup is:

$$\int_0^2 \int_0^1 xe^{xy}\,dy\,dx$$

Step 1: Evaluate the inner integral (with respect to y).

$$\begin{aligned}\int_0^1 xe^{xy}\,dy &= x\int_0^1 e^{xy}\,dy \\ &= x\left[\frac{1}{x}e^{xy}\right]_{y=0}^{y=1} \\ &= [e^{xy}]_{y=0}^{y=1} \\ &= (e^{x(1)}) - (e^{x(0)}) \\ &= e^x - e^0 \\ &= e^x - 1\end{aligned}$$

Step 2: Evaluate the outer integral (with respect to x).

$$\begin{aligned}\int_0^2 (e^x - 1)\,dx &= [e^x - x]_{x=0}^{x=2} \\ &= (e^2 - 2) - (e^0 - 0) \\ &= e^2 - 2 - 1 \\ &= e^2 - 3\end{aligned}$$

Problem 16

Evaluate the double integral $\iint_R y\sec^2(xy)\,dA$ where $R = [0,1] \times [0,\pi/4]$. Choose the easier order of integration.

Solution: First, we analyze the two possible orders of integration.

Option 1: Order $dy\,dx$ The inner integral is:

$$\int_0^{\pi/4} y\sec^2(xy)\,dy$$

This would require integration by parts, with $u = y$ and $dv = \sec^2(xy)dy$. This is the more difficult path.

Option 2: Order $dx\,dy$ The inner integral is:

$$\int_0^1 y\sec^2(xy)\,dx$$

Here, y is treated as a constant. The antiderivative of $\sec^2(ax)$ with respect to x is $\frac{1}{a}\tan(ax)$. This is a direct integration and is much simpler. We will choose this order.

The setup is:

$$\int_0^{\pi/4}\int_0^1 y\sec^2(xy)\,dx\,dy$$

Step 1: Evaluate the inner integral (with respect to x).

$$\begin{aligned}
\int_0^1 y\sec^2(xy)\,dx &= y\int_0^1 \sec^2(xy)\,dx \\
&= y\left[\frac{1}{y}\tan(xy)\right]_{x=0}^{x=1} \\
&= [\tan(xy)]_{x=0}^{x=1} \\
&= (\tan(y\cdot 1)) - (\tan(y\cdot 0)) \\
&= \tan(y) - \tan(0) \\
&= \tan(y)
\end{aligned}$$

Step 2: Evaluate the outer integral (with respect to y). We now need to evaluate $\int_0^{\pi/4}\tan(y)\,dy$. The antiderivative of $\tan(y)$ is $\ln|\sec(y)|$.

$$\begin{aligned}
\int_0^{\pi/4}\tan(y)\,dy &= [\ln|\sec(y)|]_{y=0}^{y=\pi/4} \\
&= (\ln|\sec(\pi/4)|) - (\ln|\sec(0)|) \\
&= \ln(\sqrt{2}) - \ln(1) \\
&= \ln(\sqrt{2}) - 0 \\
&= \ln(2^{1/2}) \\
&= \frac{1}{2}\ln(2)
\end{aligned}$$

Problem 17

Evaluate the iterated integral $\int_0^1 \int_{-1}^0 \int_1^2 (x+2y-z)\,dz\,dy\,dx$.

Solution: Step 1: Innermost integral (wrt z).

$$\begin{aligned}\int_1^2 (x+2y-z)\,dz &= \left[(x+2y)z - \frac{1}{2}z^2\right]_{z=1}^{z=2} \\ &= \left((x+2y)(2) - \frac{1}{2}(2)^2\right) - \left((x+2y)(1) - \frac{1}{2}(1)^2\right) \\ &= (2x+4y-2) - (x+2y-\frac{1}{2}) = x+2y-\frac{3}{2}\end{aligned}$$

Step 2: Middle integral (wrt y).

$$\begin{aligned}\int_{-1}^0 \left(x+2y-\frac{3}{2}\right)\,dy &= \left[xy+y^2-\frac{3}{2}y\right]_{y=-1}^{y=0} \\ &= (0) - \left(x(-1)+(-1)^2-\frac{3}{2}(-1)\right) \\ &= -(-x+1+\frac{3}{2}) = x-\frac{5}{2}\end{aligned}$$

Step 3: Outermost integral (wrt x).

$$\begin{aligned}\int_0^1 \left(x-\frac{5}{2}\right)\,dx &= \left[\frac{1}{2}x^2-\frac{5}{2}x\right]_{x=0}^{x=1} \\ &= \left(\frac{1}{2}(1)^2-\frac{5}{2}(1)\right) - (0) = \frac{1}{2}-\frac{5}{2} = -\frac{4}{2} = -2\end{aligned}$$

Problem 18

Evaluate the triple integral $\iiint_B (y+z)^2\,dV$ where the box B is given by $B = [0,2]\times[0,1]\times[0,1]$.

Solution: Expand the integrand first: $(y+z)^2 = y^2+2yz+z^2$. Set up the integral, e.g., in the order $dx\,dy\,dz$.

$$\int_0^1 \int_0^1 \int_0^2 (y^2+2yz+z^2)\,dx\,dy\,dz$$

Step 1: Innermost integral (wrt x).

$$\begin{aligned}\int_0^2 (y^2+2yz+z^2)\,dx &= \left[(y^2+2yz+z^2)x\right]_{x=0}^{x=2} \\ &= (y^2+2yz+z^2)(2) - 0 = 2y^2+4yz+2z^2\end{aligned}$$

Step 2: Middle integral (wrt y).

$$\begin{aligned}\int_0^1 (2y^2+4yz+2z^2)\,dy &= \left[\frac{2}{3}y^3+2y^2z+2z^2y\right]_{y=0}^{y=1} \\ &= (\frac{2}{3}(1)^3+2(1)^2z+2z^2(1)) - (0) = \frac{2}{3}+2z+2z^2\end{aligned}$$

Step 3: Outermost integral (wrt z).

$$\int_0^1 \left(\frac{2}{3} + 2z + 2z^2\right) dz = \left[\frac{2}{3}z + z^2 + \frac{2}{3}z^3\right]_{z=0}^{z=1}$$
$$= (\frac{2}{3}(1) + (1)^2 + \frac{2}{3}(1)^3) - (0) = \frac{2}{3} + 1 + \frac{2}{3} = \frac{4}{3} + \frac{3}{3} = \frac{7}{3}$$

Problem 19

Evaluate the triple integral $\iiint_B ze^{x+y}\, dV$ where $B = [0,1] \times [0,1] \times [0,1]$.

Solution: The integrand can be separated: $f(x,y,z) = e^x \cdot e^y \cdot z$. When the variables are separable and the domain is a box, we can separate the integrals.

$$\iiint_B ze^{x+y}\, dV = \left(\int_0^1 e^x\, dx\right)\left(\int_0^1 e^y\, dy\right)\left(\int_0^1 z\, dz\right)$$

Let's evaluate each part:

$$\int_0^1 e^x\, dx = [e^x]_0^1 = e^1 - e^0 = e - 1$$
$$\int_0^1 e^y\, dy = [e^y]_0^1 = e^1 - e^0 = e - 1$$
$$\int_0^1 z\, dz = \left[\frac{1}{2}z^2\right]_0^1 = \frac{1}{2} - 0 = \frac{1}{2}$$

Now, multiply the results:

$$(e-1) \cdot (e-1) \cdot \frac{1}{2} = \frac{1}{2}(e-1)^2$$

Problem 20

A solid box is bounded by the planes $x = 0, x = 2, y = 0, y = 1, z = 0, z = 1$. The density of the box at any point (x,y,z) is given by the function $\rho(x,y,z) = 1 + 2y$. Find the total mass of the box by setting up and evaluating a triple integral.

Solution: The mass M is the triple integral of the density function over the box B. The box is $B = [0,2] \times [0,1] \times [0,1]$.

$$M = \iiint_B (1+2y)\, dV = \int_0^2 \int_0^1 \int_0^1 (1+2y)\, dz\, dy\, dx$$

Step 1: Innermost integral (wrt z).

$$\int_0^1 (1+2y)\, dz = [(1+2y)z]_{z=0}^{z=1}$$
$$= (1+2y)(1) - 0 = 1 + 2y$$

Step 2: Middle integral (wrt y).

$$\int_0^1 (1+2y)\, dy = \left[y + y^2\right]_{y=0}^{y=1}$$
$$= (1 + 1^2) - 0 = 2$$

Step 3: Outermost integral (wrt x).

$$\begin{aligned}\int_0^2 2\,dx &= [2x]_{x=0}^{x=2} \\ &= 2(2) - 0 = 4\end{aligned}$$

The total mass of the box is 4.

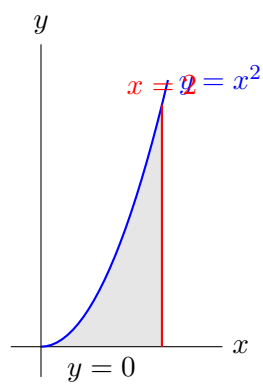
Homework 9 Solutions

Homework 9 Key: Double Integrals over General & Polar Regions

Problem 1: Type I Integral

Evaluate the iterated integral $\int_0^2 \int_0^{x^2} xy\,dy\,dx$.

The region of integration is bounded by $y = 0$, $y = x^2$, and $x = 2$.



We first evaluate the inner integral with respect to y.

$$\begin{aligned}\int_0^{x^2} xy\,dy &= x\left[\frac{1}{2}y^2\right]_{y=0}^{y=x^2} \\ &= x\left(\frac{1}{2}(x^2)^2 - \frac{1}{2}(0)^2\right) \\ &= x\left(\frac{1}{2}x^4\right) \\ &= \frac{1}{2}x^5\end{aligned}$$

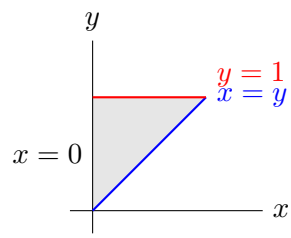
Now, we evaluate the outer integral with respect to x.

$$\begin{aligned}\int_0^2 \frac{1}{2}x^5\,dx &= \frac{1}{2}\left[\frac{1}{6}x^6\right]_0^2 \\ &= \frac{1}{12}\left[x^6\right]_0^2 \\ &= \frac{1}{12}(2^6 - 0^6) \\ &= \frac{64}{12} \\ &= \frac{16}{3}\end{aligned}$$

Problem 2: Type II Integral

Evaluate the iterated integral $\int_0^1 \int_0^y (x+y)\,dx\,dy$.

The region of integration is bounded by $x = 0$, $x = y$, and $y = 1$.



First, we evaluate the inner integral with respect to x.

$$\begin{aligned}
\int_0^y (x+y)\,dx &= \left[\frac{1}{2}x^2 + yx\right]_{x=0}^{x=y} \\
&= \left(\frac{1}{2}(y)^2 + y(y)\right) - \left(\frac{1}{2}(0)^2 + y(0)\right) \\
&= \frac{1}{2}y^2 + y^2 \\
&= \frac{3}{2}y^2
\end{aligned}$$

Now, we evaluate the outer integral with respect to y.

$$\begin{aligned}
\int_0^1 \frac{3}{2}y^2\,dy &= \frac{3}{2}\left[\frac{1}{3}y^3\right]_0^1 \\
&= \frac{1}{2}[y^3]_0^1 \\
&= \frac{1}{2}(1^3 - 0^3) \\
&= \frac{1}{2}
\end{aligned}$$

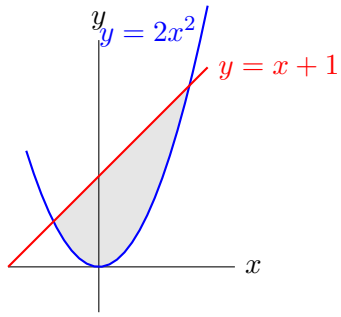
Problem 3: Volume over a General Region

Find the volume of the solid that lies under the plane $z = x$ and above the region D bounded by the parabola $y = 2x^2$ and the line $y = x + 1$.

First, we find the intersection points of the boundary curves.

$$\begin{aligned}
2x^2 &= x + 1 \\
2x^2 - x - 1 &= 0 \\
(2x+1)(x-1) &= 0
\end{aligned}$$

The intersection points are $x = -1/2$ and $x = 1$. In this interval, the line $y = x + 1$ is above the parabola $y = 2x^2$.



The volume is the integral of the height function $z = x$.

$$V = \int_{-1/2}^{1} \int_{2x^2}^{x+1} x \, dy \, dx$$

Inner integral:

$$\begin{aligned} \int_{2x^2}^{x+1} x \, dy &= x \, [y]_{y=2x^2}^{y=x+1} \\ &= x((x+1) - 2x^2) \\ &= -2x^3 + x^2 + x \end{aligned}$$

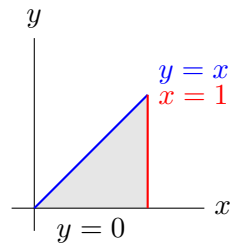
Outer integral:

$$\begin{aligned} V &= \int_{-1/2}^{1} (-2x^3 + x^2 + x) \, dx \\ &= \left[-\frac{2x^4}{4} + \frac{x^3}{3} + \frac{x^2}{2} \right]_{-1/2}^{1} \\ &= \left(-\frac{1}{2}(1)^4 + \frac{1}{3}(1)^3 + \frac{1}{2}(1)^2 \right) - \left(-\frac{1}{2}(-\frac{1}{2})^4 + \frac{1}{3}(-\frac{1}{2})^3 + \frac{1}{2}(-\frac{1}{2})^2 \right) \\ &= \left(-\frac{1}{2} + \frac{1}{3} + \frac{1}{2} \right) - \left(-\frac{1}{32} - \frac{1}{24} + \frac{1}{8} \right) \\ &= \left(\frac{1}{3} \right) - \left(\frac{-3 - 4 + 12}{96} \right) \\ &= \frac{1}{3} - \frac{5}{96} \\ &= \frac{32}{96} - \frac{5}{96} = \frac{27}{96} = \frac{9}{32} \end{aligned}$$

Problem 4: Volume under a Surface

Find the volume of the solid that lies under the plane $z = 2x + 1$ and above the triangular region D in the xy-plane with vertices $(0, 0)$, $(1, 0)$, and $(1, 1)$.

The region D is a triangle bounded by $y = 0$, $x = 1$, and $y = x$.



The volume is the integral of the height function $z = 2x + 1$.

$$V = \int_0^1 \int_0^x (2x + 1)\, dy\, dx$$

Inner integral:

$$\begin{aligned}\int_0^x (2x + 1)\, dy &= (2x + 1)[y]_0^x \\ &= (2x + 1)(x - 0) = 2x^2 + x\end{aligned}$$

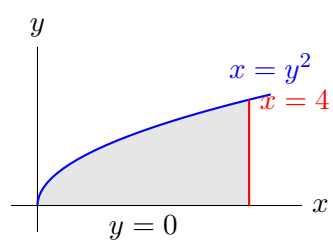
Outer integral:

$$\begin{aligned}V = \int_0^1 (2x^2 + x)\, dx &= \left[\frac{2x^3}{3} + \frac{x^2}{2}\right]_0^1 \\ &= \left(\frac{2}{3} + \frac{1}{2}\right) - (0) = \frac{7}{6}\end{aligned}$$

Problem 5: Integral over a Type II Region

Evaluate the double integral $\iint_D x^2 y\, dA$ where D is the region bounded by $x = y^2$, $x = 4$, and the x-axis ($y = 0$).

The intersection of $x = y^2$ and $x = 4$ is at $y = 2$ (in the first quadrant).



This is a Type II region with bounds $0 \le y \le 2$ and $y^2 \le x \le 4$.

$$\iint_D x^2 y\, dA = \int_0^2 \int_{y^2}^4 x^2 y\, dx\, dy$$

Inner integral:

$$\begin{aligned}\int_{y^2}^4 x^2 y\, dx &= y \left[\frac{x^3}{3}\right]_{x=y^2}^{x=4} \\ &= \frac{y}{3}(4^3 - (y^2)^3) = \frac{y}{3}(64 - y^6) = \frac{64y - y^7}{3}\end{aligned}$$

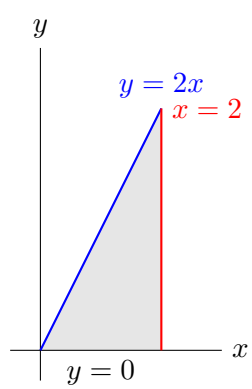
Outer integral:

$$\begin{aligned}\int_0^2 \frac{64y - y^7}{3}\,dy &= \frac{1}{3}\left[32y^2 - \frac{y^8}{8}\right]_0^2 \\ &= \frac{1}{3}\left((32(2^2) - \frac{2^8}{8}) - 0\right) \\ &= \frac{1}{3}\left(128 - \frac{256}{8}\right) \\ &= \frac{1}{3}(128 - 32) = \frac{96}{3} = 32\end{aligned}$$

Problem 6: Volume over a Triangle

Find the volume of the solid under the surface $z = xy$ and above the triangular region with vertices (0,0), (2,0), and (2,4).

The region is a right triangle bounded by $y = 0$, $x = 2$, and the line connecting (0,0) to (2,4), which is $y = 2x$.



The bounds are $0 \le x \le 2$ and $0 \le y \le 2x$.

$$V = \int_0^2 \int_0^{2x} xy\,dy\,dx$$

Inner integral:

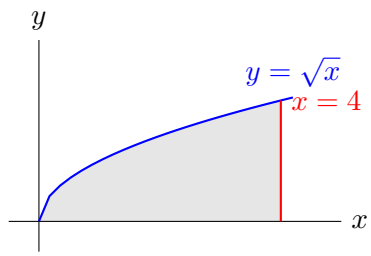
$$\begin{aligned}\int_0^{2x} xy\,dy &= x\left[\frac{y^2}{2}\right]_0^{2x} \\ &= \frac{x}{2}((2x)^2 - 0) = \frac{x}{2}(4x^2) = 2x^3\end{aligned}$$

Outer integral:

$$\begin{aligned}V = \int_0^2 2x^3\,dx &= 2\left[\frac{x^4}{4}\right]_0^2 \\ &= \frac{1}{2}[x^4]_0^2 = \frac{1}{2}(16) = 8\end{aligned}$$

Problem 7: Mass of a Lamina

Find the mass of the lamina that occupies the region D bounded by $y = \sqrt{x}$, $y = 0$, and $x = 4$, with density function $\rho(x, y) = y$.



$$m = \int_0^4 \int_0^{\sqrt{x}} y \, dy \, dx$$

Inner integral:

$$\int_0^{\sqrt{x}} y \, dy = \left[\frac{1}{2}y^2\right]_0^{\sqrt{x}} = \frac{1}{2}(\sqrt{x})^2 = \frac{1}{2}x$$

Outer integral:

$$m = \int_0^4 \frac{1}{2}x \, dx = \frac{1}{2}\left[\frac{x^2}{2}\right]_0^4 = \frac{1}{4}(16) = 4$$

Problem 8: Center of Mass

Find the center of mass $(\bar{x}, \bar{y})$ of the lamina from Problem 7.

We have $m = 4$ and $\rho = y$.

$$\begin{aligned} M_x &= \int_0^4 \int_0^{\sqrt{x}} y \cdot y \, dy \, dx = \int_0^4 \int_0^{\sqrt{x}} y^2 \, dy \, dx \\ &= \int_0^4 \left[\frac{y^3}{3}\right]_0^{\sqrt{x}} dx = \int_0^4 \frac{x^{3/2}}{3} dx \\ &= \frac{1}{3}\left[\frac{2}{5}x^{5/2}\right]_0^4 = \frac{2}{15}(4^{5/2}) = \frac{2}{15}(32) = \frac{64}{15} \end{aligned}$$

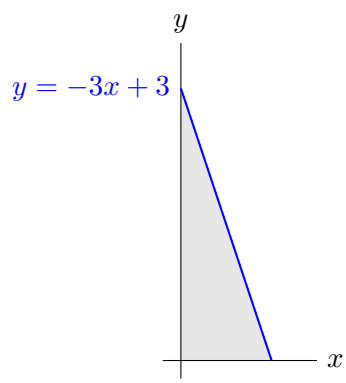
$$\begin{aligned} M_y &= \int_0^4 \int_0^{\sqrt{x}} x \cdot y \, dy \, dx \\ &= \int_0^4 x \left[\frac{y^2}{2}\right]_0^{\sqrt{x}} dx = \int_0^4 x \frac{(\sqrt{x})^2}{2} dx \\ &= \int_0^4 \frac{x^2}{2} dx = \frac{1}{2}\left[\frac{x^3}{3}\right]_0^4 = \frac{1}{6}(64) = \frac{32}{3} \end{aligned}$$

$$\bar{x} = \frac{M_y}{m} = \frac{32/3}{4} = \frac{8}{3}, \quad \bar{y} = \frac{M_x}{m} = \frac{64/15}{4} = \frac{16}{15}$$

Problem 9: Mass of a Lamina

Find the mass of the lamina that occupies the triangular region D with vertices $(0,0)$, $(0,3)$, and $(1,0)$, with density function $\rho(x,y) = x$.

The line connecting $(0,3)$ and $(1,0)$ is $y = -3x + 3$.



$$m = \int_0^1 \int_0^{-3x+3} x\,dy\,dx$$

Inner integral:

$$\begin{aligned}\int_0^{-3x+3} x\,dy &= x[y]_0^{-3x+3} \\ &= x(-3x+3) = 3x - 3x^2\end{aligned}$$

Outer integral:

$$\begin{aligned}m = \int_0^1 (3x - 3x^2)\,dx &= \left[\frac{3x^2}{2} - x^3\right]_0^1 \\ &= \frac{3}{2} - 1 = \frac{1}{2}\end{aligned}$$

Problem 10: Center of Mass

Find the center of mass $(\bar{x}, \bar{y})$ of the lamina from Problem 9.

We have $m = 1/2$ and $\rho = x$.

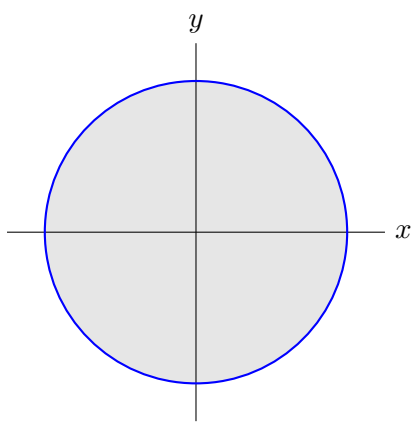
$$\begin{aligned}M_x &= \int_0^1 \int_0^{-3x+3} yx\,dy\,dx \\ &= \int_0^1 x\left[\frac{y^2}{2}\right]_0^{-3x+3} dx = \frac{1}{2}\int_0^1 x(-3x+3)^2 dx \\ &= \frac{9}{2}\int_0^1 x(1-x)^2 dx = \frac{9}{2}\int_0^1 (x - 2x^2 + x^3) dx \\ &= \frac{9}{2}\left[\frac{x^2}{2} - \frac{2x^3}{3} + \frac{x^4}{4}\right]_0^1 = \frac{9}{2}\left(\frac{1}{2} - \frac{2}{3} + \frac{1}{4}\right) = \frac{9}{2}\left(\frac{1}{12}\right) = \frac{3}{8}\end{aligned}$$

$$
\begin{aligned}
M_y &= \int_0^1 \int_0^{-3x+3} x^2 \, dy \, dx \\
&= \int_0^1 x^2 [y]_0^{-3x+3} dx = \int_0^1 x^2(3-3x) dx \\
&= \int_0^1 (3x^2 - 3x^3) dx = \left[x^3 - \frac{3x^4}{4} \right]_0^1 = 1 - \frac{3}{4} = \frac{1}{4}
\end{aligned}
$$

$$
\bar{x} = \frac{M_y}{m} = \frac{1/4}{1/2} = \frac{1}{2}, \quad \bar{y} = \frac{M_x}{m} = \frac{3/8}{1/2} = \frac{3}{4}
$$

Problem 11: Converting to Polar Coordinates

Evaluate the integral $\iint_D (x^2 + y^2 + 1) \, dA$ where D is the disk $x^2 + y^2 \leq 4$.
Hint: *The region is a full disk of radius 2. The polar bounds are* $0 \leq r \leq 2$ *and* $0 \leq \theta \leq 2\pi$.



The integrand is $r^2 + 1$.

$$
\int_0^{2\pi} \int_0^2 (r^2 + 1) r \, dr \, d\theta
$$

Inner:

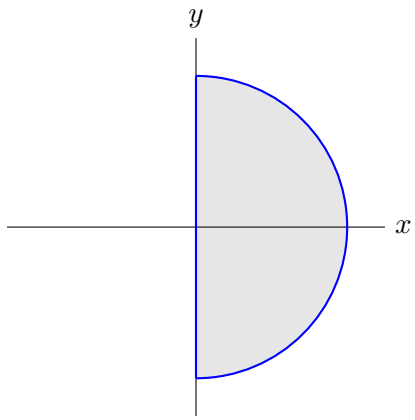
$$
\int_0^2 (r^3 + r) \, dr = \left[\frac{r^4}{4} + \frac{r^2}{2} \right]_0^2 = \left(\frac{16}{4} + \frac{4}{2} \right) - 0 = 6
$$

Outer:

$$
\int_0^{2\pi} 6 \, d\theta = [6\theta]_0^{2\pi} = 12\pi
$$

Problem 12: Polar Integral

Evaluate the integral $\iint_D e^{-x^2-y^2} \, dA$ where D is the region bounded by the semicircle $x = \sqrt{4 - y^2}$ and the y-axis.
Hint: *The region is the right half of a disk of radius 2. The polar bounds are* $0 \leq r \leq 2$ *and* $-\pi/2 \leq \theta \leq \pi/2$.



The integrand is e^{-r^2}.

$$\int_{-\pi/2}^{\pi/2} \int_0^2 e^{-r^2} r \, dr \, d\theta$$

Inner (u-sub $u = -r^2$):

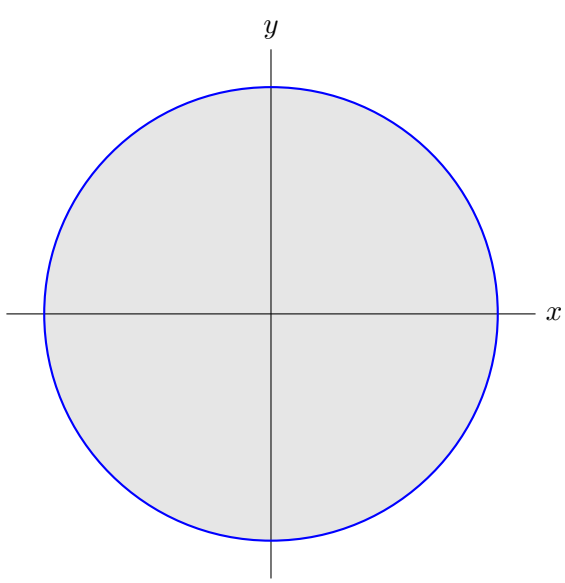
$$\int_0^2 e^{-r^2} r \, dr = \left[-\frac{1}{2} e^{-r^2} \right]_0^2 = -\frac{1}{2}(e^{-4} - e^0) = \frac{1}{2}(1 - e^{-4})$$

Outer:

$$\int_{-\pi/2}^{\pi/2} \frac{1}{2}(1 - e^{-4}) \, d\theta = \frac{1}{2}(1 - e^{-4})[\theta]_{-\pi/2}^{\pi/2} = \frac{\pi}{2}(1 - e^{-4})$$

Problem 13: Volume in Polar Coordinates

Use a polar double integral to find the volume of the solid that lies under the paraboloid $z = x^2 + y^2$ and above the disk $x^2 + y^2 \leq 9$.
Hint: *The region of integration is a full disk of radius 3. The polar bounds are* $0 \leq r \leq 3$ *and* $0 \leq \theta \leq 2\pi$.



The height is $z = r^2$.

$$V = \int_0^{2\pi} \int_0^3 (r^2) r \, dr \, d\theta$$

Inner:

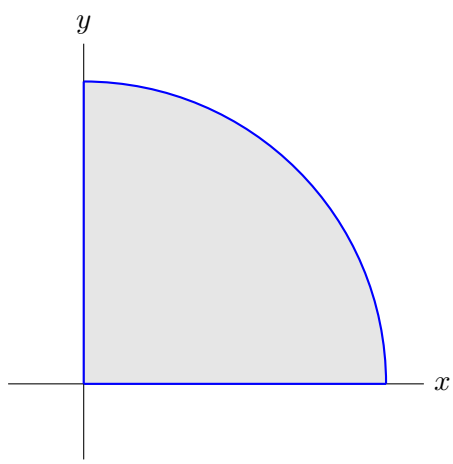
$$\int_0^3 r^3 \, dr = \left[\frac{r^4}{4} \right]_0^3 = \frac{81}{4}$$

Outer:

$$\int_0^{2\pi} \frac{81}{4} \, d\theta = \frac{81}{4} [\theta]_0^{2\pi} = \frac{81\pi}{2}$$

Problem 14: Volume in Polar Coordinates

Find the volume of the solid bounded by the paraboloid $z = 16 - x^2 - y^2$ and the xy-plane, for the portion that lies in the first octant.
Hint: *The base of the solid is a quarter-disk of radius 4 in the first quadrant. The polar bounds are* $0 \leq r \leq 4$ *and* $0 \leq \theta \leq \pi/2$.



The height is $z = 16 - r^2$.

$$V = \int_0^{\pi/2} \int_0^4 (16 - r^2) r \, dr \, d\theta$$

Inner:

$$\int_0^4 (16r - r^3) \, dr = \left[8r^2 - \frac{r^4}{4} \right]_0^4 = 8(16) - \frac{256}{4} = 128 - 64 = 64$$

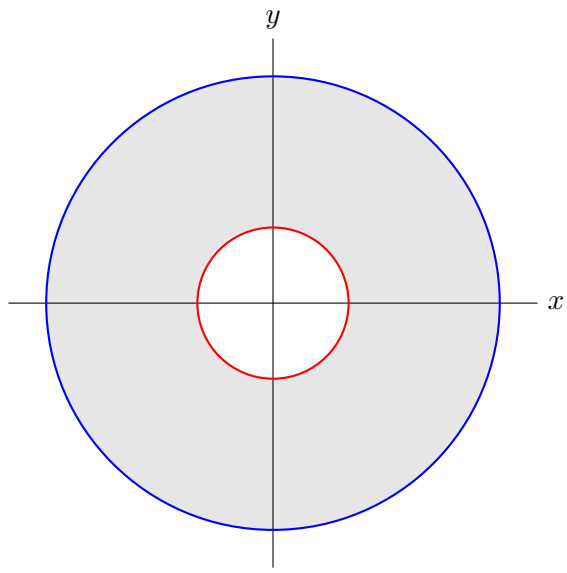
Outer:

$$V = \int_0^{\pi/2} 64 \, d\theta = 64[\theta]_0^{\pi/2} = 32\pi$$

Problem 15: Mass of a Polar Lamina

Find the mass of a lamina that covers the annular region D given by $1 \leq x^2 + y^2 \leq 9$. The density is $\rho(x, y) = \frac{k}{\sqrt{x^2+y^2}}$.

Hint: *The region is an annulus (a ring) between circles of radius 1 and 3. The polar bounds are* $1 \leq r \leq 3$ *and* $0 \leq \theta \leq 2\pi$.



The density is $\rho = k/r$.

$$m = \int_0^{2\pi} \int_1^3 \frac{k}{r} \cdot r \, dr \, d\theta$$

Inner:

$$\int_1^3 k \, dr = k[r]_1^3 = k(3 - 1) = 2k$$

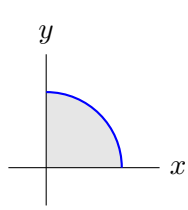
Outer:

$$m = \int_0^{2\pi} 2k \, d\theta = 2k[\theta]_0^{2\pi} = 4k\pi$$

Problem 16: Center of Mass of a Polar Lamina

Find the center of mass of a lamina that occupies the region D in the first quadrant bounded by the circle $x^2 + y^2 = 1$. The density is $\rho(x, y) = k\sqrt{x^2 + y^2}$.

Hint: *The region is a quarter-disk of radius 1 in the first quadrant. The polar bounds are* $0 \leq r \leq 1$ *and* $0 \leq \theta \leq \pi/2$.



The density is $\rho = kr$.

$$m = \int_0^{\pi/2} \int_0^1 (kr)r\,dr\,d\theta = k\left(\int_0^{\pi/2} d\theta\right)\left(\int_0^1 r^2 dr\right) = k(\pi/2)(1/3) = k\pi/6$$

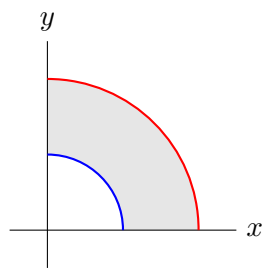
$$M_y = \int_0^{\pi/2} \int_0^1 (r\cos\theta)(kr)r\,dr\,d\theta = k\left(\int_0^{\pi/2} \cos\theta d\theta\right)\left(\int_0^1 r^3 dr\right) = k(1)(1/4) = k/4$$

$$M_x = \int_0^{\pi/2} \int_0^1 (r\sin\theta)(kr)r\,dr\,d\theta = k\left(\int_0^{\pi/2} \sin\theta d\theta\right)\left(\int_0^1 r^3 dr\right) = k(1)(1/4) = k/4$$

$$\bar{x} = \frac{M_y}{m} = \frac{k/4}{k\pi/6} = \frac{3}{2\pi}, \quad \bar{y} = \frac{M_x}{m} = \frac{k/4}{k\pi/6} = \frac{3}{2\pi}$$

Problem 17: Integral over a Polar Sector

Evaluate $\iint_D y\,dA$ where D is the region in the first quadrant between the circles $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$.
Hint: *This region is a sector of an annulus. The polar bounds are* $1 \le r \le 2$ *and* $0 \le \theta \le \pi/2$.

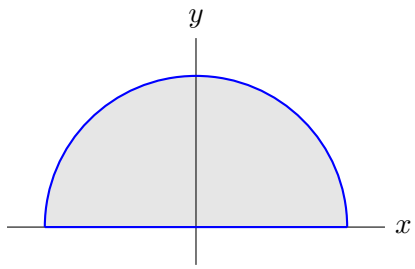


The integrand is $y = r\sin\theta$.

$$\begin{aligned}\iint_D y\,dA &= \int_0^{\pi/2} \int_1^2 (r\sin\theta)r\,dr\,d\theta \\ &= \left(\int_0^{\pi/2} \sin\theta d\theta\right)\left(\int_1^2 r^2 dr\right) \\ &= \left([-\cos\theta]_0^{\pi/2}\right)\left([\frac{r^3}{3}]_1^2\right) \\ &= (0-(-1))(\frac{8}{3} - \frac{1}{3}) = (1)\left(\frac{7}{3}\right) = \frac{7}{3}\end{aligned}$$

Problem 18: Integral over a Semicircle

Evaluate the double integral $\iint_D x^2\,dA$ where D is the top half of the disk centered at the origin with radius 2.
Hint: *The region is a semicircle of radius 2 above the x-axis. The polar bounds are* $0 \le r \le 2$ *and* $0 \le \theta \le \pi$.



The integrand is $x^2 = r^2 \cos^2 \theta$.

$$\iint_D x^2 \, dA = \int_0^{\pi} \int_0^2 (r^2 \cos^2 \theta) r \, dr \, d\theta$$
$$= \left(\int_0^{\pi} \cos^2 \theta d\theta \right) \left(\int_0^2 r^3 dr \right)$$

Theta integral:

$$\int_0^{\pi} \frac{1}{2}(1 + \cos(2\theta)) d\theta = \frac{1}{2} \left[\theta + \frac{1}{2} \sin(2\theta) \right]_0^{\pi} = \frac{\pi}{2}$$

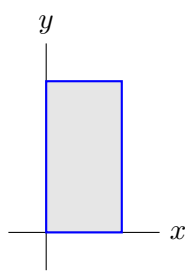
R integral:

$$\int_0^2 r^3 dr = \left[\frac{r^4}{4} \right]_0^2 = 4$$

The result is $(\pi/2)(4) = 2\pi$.

Problem 19: Mass of a Solid Box

Find the total mass of the solid box $B = [0, 1] \times [0, 2] \times [0, 3]$ if the density at any point (x, y, z) is given by the function $\rho(x, y, z) = x^2 y$.



$$m = \int_0^3 \int_0^2 \int_0^1 x^2 y \, dx \, dy \, dz$$

Innermost (x):

$$\int_0^1 x^2 y \, dx = y \left[\frac{x^3}{3} \right]_0^1 = \frac{y}{3}$$

Middle (y):

$$\int_0^2 \frac{y}{3} \, dy = \frac{1}{3} \left[\frac{y^2}{2} \right]_0^2 = \frac{4}{6} = \frac{2}{3}$$

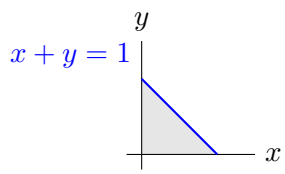
Outermost (z):

$$m = \int_0^3 \frac{2}{3} \, dz = \frac{2}{3} [z]_0^3 = 2$$

Problem 20: Mass of a Solid Wedge

Find the total mass of the solid tetrahedron (a wedge in the first octant) bounded by the coordinate planes and the plane $x+y+z=1$. The density of the solid is given by $\rho(x,y,z)=y$.

The base of the solid is a triangle in the xy-plane bounded by $x=0, y=0, x+y=1$.



$$m=\int_0^1\int_0^{1-x}\int_0^{1-x-y} y\,dz\,dy\,dx$$

Innermost (z):

$$\int_0^{1-x-y} y\,dz = y[z]_0^{1-x-y} = y(1-x-y) = y-xy-y^2$$

Middle (y):

$$\int_0^{1-x}(y-xy-y^2)\,dy = \left[\frac{y^2}{2}-x\frac{y^2}{2}-\frac{y^3}{3}\right]_0^{1-x} = \frac{(1-x)^3}{6}$$

Outermost (x):

$$m=\int_0^1\frac{(1-x)^3}{6}\,dx = \frac{1}{6}\left[-\frac{(1-x)^4}{4}\right]_0^1 = -\frac{1}{24}(0-1) = \frac{1}{24}$$

Homework 10 Solutions

Homework 10: Line Integrals, Conservative Fields, and FTC
Due: Sunday, November 16

Problem 1: Work (Line Segment 1)

Evaluate the line integral $\int_C \vec{F}\cdot d\vec{r}$ where $\vec{F}=\langle 3xy, x^2+y\rangle$ and C is the line segment from $(0,0)$ to $(1,2)$.

Step 1: Parameterize the curve C.
A line segment from point A to point B can be parameterized as $\vec{r}(t)=A(1-t)+Bt$ for $0\le t\le 1$. Here, $A=(0,0)$ and $B=(1,2)$, so $\vec{r}(t)=\langle 0,0\rangle(1-t)+\langle 1,2\rangle t=\langle t,2t\rangle$. So, we have $x(t)=t$ and $y(t)=2t$ for $0\le t\le 1$.

Step 2: Find $d\vec{r}$.
$d\vec{r}=\vec{r}'(t)\,dt=\langle 1,2\rangle\,dt$.

Step 3: Express $\vec{F}$ in terms of t.
Substitute $x(t)$ and $y(t)$ into $\vec{F}$:

$$\vec{F}(\vec{r}(t))=\langle 3(t)(2t),(t)^2+(2t)\rangle=\langle 6t^2,t^2+2t\rangle$$

Step 4: Compute the dot product $\vec{F}\cdot d\vec{r}$.

$$\vec{F}\cdot\vec{r}'(t)=\langle 6t^2,t^2+2t\rangle\cdot\langle 1,2\rangle=(6t^2)(1)+(t^2+2t)(2)=6t^2+2t^2+4t=8t^2+4t$$

Step 5: Evaluate the integral.

$$\int_C \vec{F} \cdot d\vec{r} = \int_0^1 (8t^2 + 4t)\, dt = \left[\frac{8t^3}{3} + \frac{4t^2}{2}\right]_0^1 = \left[\frac{8t^3}{3} + 2t^2\right]_0^1$$
$$= \left(\frac{8(1)^3}{3} + 2(1)^2\right) - (0) = \frac{8}{3} + 2 = \frac{8}{3} + \frac{6}{3} = \frac{14}{3}$$

Problem 2: Work (Line Segment 2)

Evaluate the line integral $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = \langle z, x, y\rangle$ and C is the line segment from $(0, 0, 0)$ to $(1, 3, -1)$.

Step 1: Parameterize the curve C.
Using the line segment formula for $A = (0, 0, 0)$ and $B = (1, 3, -1)$: $\vec{r}(t) = \langle t, 3t, -t\rangle$ for $0 \le t \le 1$. This gives $x(t) = t$, $y(t) = 3t$, and $z(t) = -t$.

Step 2: Find $d\vec{r}$.
$d\vec{r} = \vec{r}'(t)\, dt = \langle 1, 3, -1\rangle\, dt$.

Step 3: Express $\vec{F}$ in terms of t.

$$\vec{F}(\vec{r}(t)) = \langle z(t), x(t), y(t)\rangle = \langle -t, t, 3t\rangle$$

Step 4: Compute the dot product $\vec{F} \cdot d\vec{r}$.

$$\vec{F} \cdot \vec{r}'(t) = \langle -t, t, 3t\rangle \cdot \langle 1, 3, -1\rangle = (-t)(1) + (t)(3) + (3t)(-1) = -t + 3t - 3t = -t$$

Step 5: Evaluate the integral.

$$\int_C \vec{F} \cdot d\vec{r} = \int_0^1 (-t)\, dt = \left[-\frac{t^2}{2}\right]_0^1 = -\frac{1^2}{2} - 0 = -\frac{1}{2}$$

Problem 3: Work (Parabola)

Evaluate the line integral $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = \langle y^2, -x\rangle$ and C is the parabola $y = x^2$ from $(-1, 1)$ to $(1, 1)$.

Step 1: Parameterize the curve C.
Let $x = t$. Then $y = t^2$. Since x goes from -1 to 1, the parameter t goes from -1 to 1. $\vec{r}(t) = \langle t, t^2\rangle$ for $-1 \le t \le 1$.

Step 2: Find $d\vec{r}$.
$d\vec{r} = \vec{r}'(t)\, dt = \langle 1, 2t\rangle\, dt$.

Step 3: Express $\vec{F}$ in terms of t.

$$\vec{F}(\vec{r}(t)) = \langle (t^2)^2, -t\rangle = \langle t^4, -t\rangle$$

Step 4: Compute the dot product $\vec{F} \cdot d\vec{r}$.

$$\vec{F} \cdot \vec{r}'(t) = \langle t^4, -t\rangle \cdot \langle 1, 2t\rangle = (t^4)(1) + (-t)(2t) = t^4 - 2t^2$$

Step 5: Evaluate the integral.

$$\int_C \vec{F} \cdot d\vec{r} = \int_{-1}^{1} (t^4 - 2t^2)\, dt = \left[\frac{t^5}{5} - \frac{2t^3}{3}\right]_{-1}^{1}$$
$$= \left(\frac{1^5}{5} - \frac{2(1)^3}{3}\right) - \left(\frac{(-1)^5}{5} - \frac{2(-1)^3}{3}\right) = \left(\frac{1}{5} - \frac{2}{3}\right) - \left(-\frac{1}{5} + \frac{2}{3}\right)$$
$$= \frac{2}{5} - \frac{4}{3} = \frac{6-20}{15} = -\frac{14}{15}$$

Problem 4: Work (Exponential Curve)

Evaluate the line integral $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = \langle ye^x, x^2 \rangle$ and C is the curve $\vec{r}(t) = \langle t, e^t \rangle$ for $0 \le t \le 1$.

Step 1: The curve C is already parameterized.
$x(t) = t$, $y(t) = e^t$ for $0 \le t \le 1$.

Step 2: Find $d\vec{r}$.
$d\vec{r} = \vec{r}'(t)\, dt = \langle 1, e^t \rangle\, dt$.

Step 3: Express $\vec{F}$ in terms of t.

$$\vec{F}(\vec{r}(t)) = \langle (e^t)e^t, t^2 \rangle = \langle e^{2t}, t^2 \rangle$$

Step 4: Compute the dot product $\vec{F} \cdot d\vec{r}$.

$$\vec{F} \cdot \vec{r}'(t) = \langle e^{2t}, t^2 \rangle \cdot \langle 1, e^t \rangle = e^{2t} + t^2 e^t$$

Step 5: Evaluate the integral.

$$\int_C \vec{F} \cdot d\vec{r} = \int_0^1 (e^{2t} + t^2 e^t)\, dt = \int_0^1 e^{2t}\, dt + \int_0^1 t^2 e^t\, dt$$

The first integral is straightforward: $\int_0^1 e^{2t}\, dt = \left[\frac{1}{2}e^{2t}\right]_0^1 = \frac{1}{2}e^2 - \frac{1}{2}e^0 = \frac{e^2-1}{2}$. The second integral requires integration by parts twice. For $\int t^2 e^t\, dt$: Let $u = t^2, dv = e^t dt$. Then $du = 2tdt, v = e^t$.

$$\int t^2 e^t\, dt = t^2 e^t - \int 2te^t\, dt$$

For $\int 2te^t\, dt$: Let $u = 2t, dv = e^t dt$. Then $du = 2dt, v = e^t$.

$$\int 2te^t\, dt = 2te^t - \int 2e^t\, dt = 2te^t - 2e^t$$

Putting it together: $\int t^2 e^t\, dt = t^2 e^t - (2te^t - 2e^t) = (t^2 - 2t + 2)e^t$. Now evaluate from 0 to 1:

$$\int_0^1 t^2 e^t\, dt = \left[(t^2 - 2t + 2)e^t\right]_0^1 = ((1 - 2 + 2)e^1) - ((0 - 0 + 2)e^0) = e - 2$$

Total integral $= (\frac{e^2-1}{2}) + (e - 2) = \frac{e^2}{2} - \frac{1}{2} + e - 2 = \frac{e^2}{2} + e - \frac{5}{2}$.

Problem 5: Work (Closed Triangle)

Evaluate the line integral $\oint_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = \langle x+y, x-y \rangle$ and C is the triangle with vertices $(0,0)$, $(1,0)$, and $(0,1)$, oriented counterclockwise.

We can solve this by parameterizing the three line segments or by using Green's Theorem. Green's Theorem is more efficient.

Method 1: Green's Theorem

Green's Theorem states $\oint_C P\,dx + Q\,dy = \iint_D (\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y})\,dA$. Here, $P = x+y$ and $Q = x-y$.

$$\frac{\partial Q}{\partial x} = 1 \quad \text{and} \quad \frac{\partial P}{\partial y} = 1$$

So, $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 1 - 1 = 0$. The integral becomes:

$$\iint_D (0)\,dA = 0$$

The value of the line integral is 0.

Method 2: Direct Calculation

C_1: From $(0,0)$ to $(1,0)$.

$$\vec{r}(t) = \langle t, 0 \rangle, \quad 0 \le t \le 1$$
$$d\vec{r} = \langle 1, 0 \rangle dt$$
$$\vec{F} = \langle t, t \rangle$$
$$\int_{C_1} \vec{F} \cdot d\vec{r} = \int_0^1 \langle t, t \rangle \cdot \langle 1, 0 \rangle dt = \int_0^1 t\,dt = \frac{1}{2}.$$

C_2: From $(1,0)$ to $(0,1)$.

$$\vec{r}(t) = \langle 1-t, t \rangle, 0 \le t \le 1$$
$$d\vec{r} = \langle -1, 1 \rangle dt$$
$$\vec{F} = \langle (1-t)+t, (1-t)-t \rangle = \langle 1, 1-2t \rangle$$
$$\int_{C_2} \vec{F} \cdot d\vec{r} = \int_0^1 \langle 1, 1-2t \rangle \cdot \langle -1, 1 \rangle dt = \int_0^1 (-1+1-2t)\,dt = \int_0^1 -2t\,dt = -1.$$

C_3: From $(0,1)$ to $(0,0)$.

$$\vec{r}(t) = \langle 0, 1-t \rangle, 0 \le t \le 1$$
$$d\vec{r} = \langle 0, -1 \rangle dt$$
$$\vec{F} = \langle 1-t, -(1-t) \rangle$$
$$\int_{C_3} \vec{F} \cdot d\vec{r} = \int_0^1 \langle 1-t, t-1 \rangle \cdot \langle 0, -1 \rangle dt = \int_0^1 (1-t)\,dt = \frac{1}{2}.$$

Thus: $\oint_C \vec{F} \cdot d\vec{r} = \frac{1}{2} - 1 + \frac{1}{2} = 0$.

Problem 6: Work (Closed Box Boundary)

Evaluate the line integral $\oint_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = \langle y, -x, z^2 \rangle$ and C is the boundary of the square in the plane $z = 1$ with vertices $(0,0,1)$, $(1,0,1)$, $(1,1,1)$, and $(0,1,1)$, oriented counterclockwise

when viewed from above.

C_1 (bottom): $(0,0,1) \to (1,0,1)$.

$$\vec{r}(t) = \langle t, 0, 1\rangle, 0 \le t \le 1$$

$$d\vec{r} = \langle 1, 0, 0\rangle dt$$

$$\vec{F} = \langle 0, -t, 1\rangle$$

$$\int_{C_1} \vec{F} \cdot d\vec{r} = \int_0^1 0\, dt = 0$$

C_2 (right): $(1,0,1) \to (1,1,1)$.

$$\vec{r}(t) = \langle 1, t, 1\rangle, 0 \le t \le 1$$

$$d\vec{r} = \langle 0, 1, 0\rangle dt$$

$$\vec{F} = \langle t, -1, 1\rangle$$

$$\int_{C_2} \vec{F} \cdot d\vec{r} = \int_0^1 -1\, dt = -1$$

C_3 (top): $(1,1,1) \to (0,1,1)$.

$$\vec{r}(t) = \langle 1-t, 1, 1\rangle, 0 \le t \le 1$$

$$d\vec{r} = \langle -1, 0, 0\rangle dt$$

$$\vec{F} = \langle 1, -(1-t), 1\rangle$$

$$\int_{C_3} \vec{F} \cdot d\vec{r} = \int_0^1 -1\, dt = -1$$

C_4 (left): $(0,1,1) \to (0,0,1)$.

$$\vec{r}(t) = \langle 0, 1-t, 1\rangle, 0 \le t \le 1$$

$$d\vec{r} = \langle 0, -1, 0\rangle dt$$

$$\vec{F} = \langle 1-t, 0, 1\rangle$$

$$\int_{C_4} \vec{F} \cdot d\vec{r} = \int_0^1 0\, dt = 0$$

Thus:

$$\int_C \vec{F} \cdot d\vec{r} = 0 + (-1) + (-1) + 0 = -2$$

Problem 7: Work (Helix)

Evaluate the line integral $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = \langle z, y, x \rangle$ and C is the helix $\vec{r}(t) = \langle \cos t, \sin t, t \rangle$ for $0 \leq t \leq \pi$.

Step 1: The curve is parameterized.
$x(t) = \cos t$, $y(t) = \sin t$, $z(t) = t$.

Step 2: Find $d\vec{r}$.
$d\vec{r} = \vec{r}'(t)\, dt = \langle -\sin t, \cos t, 1 \rangle\, dt$.

Step 3: Express $\vec{F}$ in terms of t.

$$\vec{F}(\vec{r}(t)) = \langle z(t), y(t), x(t) \rangle = \langle t, \sin t, \cos t \rangle$$

Step 4: Compute the dot product $\vec{F} \cdot d\vec{r}$.

$$\begin{aligned} \vec{F} \cdot \vec{r}'(t) &= \langle t, \sin t, \cos t \rangle \cdot \langle -\sin t, \cos t, 1 \rangle \\ &= -t \sin t + \sin t \cos t + \cos t \end{aligned}$$

Step 5: Evaluate the integral.

$$\int_0^\pi (-t \sin t + \sin t \cos t + \cos t)\, dt$$

We integrate term by term:

1. $\int -t \sin t\, dt$: Use integration by parts. $u = -t, dv = \sin t dt \implies du = -dt, v = -\cos t$.

$$\int -t \sin t\, dt = (-t)(-\cos t) - \int (-\cos t)(-dt) = t \cos t - \int \cos t\, dt = t \cos t - \sin t$$

2. $\int \sin t \cos t\, dt$: Use u-substitution. $u = \sin t, du = \cos t dt$.

$$\int u\, du = \frac{u^2}{2} = \frac{\sin^2 t}{2}$$

3. $\int \cos t\, dt = \sin t$.

The full antiderivative is $(t \cos t - \sin t) + (\frac{\sin^2 t}{2}) + (\sin t) = t \cos t + \frac{\sin^2 t}{2}$. Now evaluate from 0 to π:

$$\begin{aligned} \left[t \cos t + \frac{\sin^2 t}{2} \right]_0^\pi &= \left(\pi \cos \pi + \frac{\sin^2 \pi}{2} \right) - \left(0 \cos 0 + \frac{\sin^2 0}{2} \right) \\ &= (\pi(-1) + 0) - (0 + 0) = -\pi \end{aligned}$$

Problem 8: Line Integral (Scalar Function)

Evaluate the line integral with respect to arc length $\int_C (x + y^2)\, ds$, where C is the upper half of the circle $x^2 + y^2 = 4$.

Step 1: Parameterize the curve C.
The circle has radius 2. The upper half corresponds to angles from 0 to π. $\vec{r}(t) = \langle 2 \cos t, 2 \sin t \rangle$ for $0 \leq t \leq \pi$.

Step 2: Find the arc length element ds.
$ds = ||\vec{r}'(t)||\,dt$. First find $\vec{r}'(t)$:

$$\vec{r}'(t) = \langle -2\sin t, 2\cos t\rangle$$

Now find its magnitude:

$$||\vec{r}'(t)|| = \sqrt{(-2\sin t)^2 + (2\cos t)^2} = \sqrt{4\sin^2 t + 4\cos^2 t} = \sqrt{4(1)} = 2$$

So, $ds = 2\,dt$.

Step 3: Express the function $(x + y^2)$ in terms of t.

$$x(t) + y(t)^2 = (2\cos t) + (2\sin t)^2 = 2\cos t + 4\sin^2 t$$

Step 4: Set up and evaluate the integral.

$$\int_C (x + y^2)\,ds = \int_0^\pi (2\cos t + 4\sin^2 t)(2\,dt) = \int_0^\pi (4\cos t + 8\sin^2 t)\,dt$$

Use the identity $\sin^2 t = \frac{1-\cos(2t)}{2}$.

$$\int_0^\pi \left(4\cos t + 8\left(\frac{1-\cos(2t)}{2}\right)\right)\,dt = \int_0^\pi (4\cos t + 4 - 4\cos(2t))\,dt$$

Now integrate:

$$\left[4\sin t + 4t - 4\frac{\sin(2t)}{2}\right]_0^\pi = [4\sin t + 4t - 2\sin(2t)]_0^\pi$$
$$= (4\sin\pi + 4\pi - 2\sin(2\pi)) - (4\sin 0 + 4(0) - 2\sin 0)$$
$$= (0 + 4\pi - 0) - (0) = 4\pi$$

Problem 9: Flux (Line Segment)

Compute the flux of the vector field $\vec{F} = \langle x, y\rangle$ across the line segment C from $(0, 0)$ to $(2, 1)$.

The flux of $\vec{F} = \langle P, Q\rangle$ across a curve C is given by $\int_C P\,dy - Q\,dx$.

Step 1: Parameterize the curve C.
$\vec{r}(t) = \langle 2t, t\rangle$ for $0 \le t \le 1$. This gives $x(t) = 2t$ and $y(t) = t$.

Step 2: Find dx and dy.
$dx = x'(t)\,dt = 2\,dt$. $dy = y'(t)\,dt = 1\,dt$.

Step 3: Express $\vec{F} = \langle P, Q\rangle$ in terms of t.
$P(t) = x(t) = 2t$. $Q(t) = y(t) = t$.

Step 4: Set up and evaluate the flux integral.

$$\int_C P\,dy - Q\,dx = \int_0^1 ((2t)(1\,dt) - (t)(2\,dt)) = \int_0^1 (2t - 2t)\,dt = \int_0^1 0\,dt = 0$$

Problem 10: Flux (Semicircle)

Compute the flux of the vector field $\vec{F} = \langle y, -x\rangle$ across the upper semicircle of $x^2 + y^2 = 1$, oriented clockwise.

Flux is $\int_C P\,dy - Q\,dx$, with $P = y, Q = -x$.

Step 1: Parameterize the curve C.
The standard counter-clockwise parameterization is $\vec{r}(t) = \langle \cos t, \sin t\rangle$ for $0 \le t \le \pi$. For a clockwise orientation, we can reverse the bounds of integration, from π to 0. So, $x(t) = \cos t$ and $y(t) = \sin t$ for t from π to 0.

Step 2: Find dx and dy.
$dx = -\sin t\,dt$. $dy = \cos t\,dt$.

Step 3: Express P and Q in terms of t.
$P(t) = y(t) = \sin t$. $Q(t) = -x(t) = -\cos t$.

Step 4: Set up and evaluate the flux integral.

$$\int_C P\,dy - Q\,dx = \int_\pi^0 ((\sin t)(\cos t\,dt) - (-\cos t)(-\sin t\,dt))$$
$$= \int_\pi^0 (\sin t\cos t - \sin t\cos t)\,dt = \int_\pi^0 0\,dt = 0$$

This result makes sense because the vector field $\vec{F} = \langle y, -x\rangle$ represents pure rotation and is always tangent to the circle, so there is no flow *across* the circular boundary.

Problem 11: Flux (Rectangle Boundary)

Compute the flux of $\vec{F} = \langle 2x + y, x - 3y\rangle$ across the boundary of the rectangle C defined by $0 \le x \le 1$, $0 \le y \le 2$, oriented counterclockwise.

For a closed curve, it is easiest to use the flux form of Green's Theorem: $\oint_C \vec{F}\cdot\vec{n}\,ds = \iint_D (\nabla\cdot\vec{F})\,dA$.

Step 1: Compute the divergence of $\vec{F}$.
$\vec{F} = \langle P, Q\rangle = \langle 2x + y, x - 3y\rangle$.

$$\nabla\cdot\vec{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} = \frac{\partial}{\partial x}(2x + y) + \frac{\partial}{\partial y}(x - 3y) = 2 - 3 = -1$$

Step 2: Set up the double integral over the region D.
The region D is the rectangle $0 \le x \le 1$, $0 \le y \le 2$.

$$\iint_D (-1)\,dA$$

Step 3: Evaluate the integral.
The integral of a constant over a region is the constant times the area of the region. Area of the rectangle = (width) $\times$ (height) $= 1 \times 2 = 2$.

$$\iint_D (-1)\,dA = -1 \times (\text{Area of D}) = -1 \times 2 = -2$$

The total outward flux is -2.

Problem 12: Conservative Test (2D)

Determine if the vector field $\vec{F} = \langle 2xy, x^2\rangle$ is conservative.

For a 2D vector field $\vec{F} = \langle P(x,y), Q(x,y)\rangle$ to be conservative, we must have $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$. Here, $P(x,y) = 2xy$ and $Q(x,y) = x^2$. Compute the partial derivatives:

$$\frac{\partial P}{\partial y} = \frac{\partial}{\partial y}(2xy) = 2x$$

$$\frac{\partial Q}{\partial x} = \frac{\partial}{\partial x}(x^2) = 2x$$

Since $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$, the vector field is conservative.

Problem 13: Find Potential Function (2D)

If the vector field from Problem 12 is conservative, find its scalar potential function $f(x,y)$.

We need to find a function $f(x,y)$ such that $\nabla f = \vec{F}$, which means $\frac{\partial f}{\partial x} = 2xy$ and $\frac{\partial f}{\partial y} = x^2$.

Step 1: Integrate the first component with respect to x.

$$f(x,y) = \int \frac{\partial f}{\partial x}\, dx = \int 2xy\, dx = x^2y + g(y)$$

The "constant" of integration, $g(y)$, can be any function of y.

Step 2: Differentiate the result with respect to y and set it equal to the second component.

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}(x^2y + g(y)) = x^2 + g'(y)$$

We know $\frac{\partial f}{\partial y}$ must equal $Q(x,y) = x^2$.

$$x^2 + g'(y) = x^2$$

$$g'(y) = 0$$

Step 3: Find $g(y)$ and the potential function $f(x,y)$.

If $g'(y) = 0$, then $g(y)$ must be a constant, K. Substituting this back into our expression for f:

$$f(x,y) = x^2y + K$$

Problem 14: Conservative Test (2D)

Determine if the vector field $\vec{F} = \langle y^2, x^2\rangle$ is conservative.

For $\vec{F} = \langle P, Q\rangle = \langle y^2, x^2\rangle$, we check if $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$.

$$\frac{\partial P}{\partial y} = \frac{\partial}{\partial y}(y^2) = 2y$$

$$\frac{\partial Q}{\partial x} = \frac{\partial}{\partial x}(x^2) = 2x$$

Since $2y \neq 2x$, the vector field is not conservative.

Problem 15: Conservative Test (3D)

Determine if the vector field $\vec{F} = \langle yz, xz, xy \rangle$ is conservative.

A 3D vector field $\vec{F}$ is conservative if its curl is the zero vector, i.e., $\nabla \times \vec{F} = \vec{0}$.

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ yz & xz & xy \end{vmatrix}$$

$$= \vec{i}(\frac{\partial}{\partial y}(xy) - \frac{\partial}{\partial z}(xz)) - \vec{j}(\frac{\partial}{\partial x}(xy) - \frac{\partial}{\partial z}(yz)) + \vec{k}(\frac{\partial}{\partial x}(xz) - \frac{\partial}{\partial y}(yz))$$

$$= \vec{i}(x - x) - \vec{j}(y - y) + \vec{k}(z - z) = \langle 0, 0, 0 \rangle$$

Since the curl is the zero vector, the field is conservative.

Problem 16: Find Potential Function (3D)

If the vector field from Problem 15 is conservative, find its scalar potential function $f(x, y, z)$.

We seek $f(x, y, z)$ such that $\frac{\partial f}{\partial x} = yz$, $\frac{\partial f}{\partial y} = xz$, and $\frac{\partial f}{\partial z} = xy$.

Step 1: Integrate the first component with respect to x.

$$f(x, y, z) = \int yz \, dx = xyz + g(y, z)$$

Step 2: Differentiate with respect to y and compare to the second component.

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}(xyz + g(y, z)) = xz + \frac{\partial g}{\partial y}$$

Set this equal to the second component, xz:

$$xz + \frac{\partial g}{\partial y} = xz \implies \frac{\partial g}{\partial y} = 0$$

This means g does not depend on y, so it must be a function of z alone, $g(z)$. Our potential function is now $f(x, y, z) = xyz + g(z)$.

Step 3: Differentiate with respect to z and compare to the third component.

$$\frac{\partial f}{\partial z} = \frac{\partial}{\partial z}(xyz + g(z)) = xy + g'(z)$$

Set this equal to the third component, xy:

$$xy + g'(z) = xy \implies g'(z) = 0$$

This means $g(z)$ is a constant, K.

Step 4: Write the final potential function.

$$f(x, y, z) = xyz + K$$

Problem 17: Conservative Test (3D)

Determine if the vector field $\vec{F} = \langle e^z, xe^z, xe^y \rangle$ is conservative.

We compute the curl of $\vec{F}$.

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ e^z & xe^z & xe^y \end{vmatrix}$$

$$= \vec{i}(\frac{\partial}{\partial y}(xe^y) - \frac{\partial}{\partial z}(xe^z)) - \vec{j}(\frac{\partial}{\partial x}(xe^y) - \frac{\partial}{\partial z}(e^z)) + \vec{k}(\frac{\partial}{\partial x}(xe^z) - \frac{\partial}{\partial y}(e^z))$$

$$= \vec{i}(xe^y - xe^z) - \vec{j}(e^y - e^z) + \vec{k}(e^z - 0)$$

Since the resulting vector is not $\langle 0, 0, 0 \rangle$ (for example, the first component is not always zero), the vector field is not conservative.

Problem 18: FTC for Line Integrals

First, confirm the vector field $\vec{F} = \langle 2x+3y, 3x-2y \rangle$ is conservative. Then, use the Fundamental Theorem of Line Integrals to evaluate $\int_C \vec{F} \cdot d\vec{r}$ where C is any path from $(-1, 1)$ to $(2, -1)$.

Step 1: Confirm the field is conservative.
$P = 2x + 3y$ and $Q = 3x - 2y$.

$$\frac{\partial P}{\partial y} = 3 \quad \text{and} \quad \frac{\partial Q}{\partial x} = 3$$

Since the partials are equal, the field is conservative.

Step 2: Find the potential function $f(x, y)$.

$$f(x, y) = \int (2x + 3y)\, dx = x^2 + 3xy + g(y)$$

Differentiate with respect to y:

$$\frac{\partial f}{\partial y} = 3x + g'(y)$$

Set equal to Q:

$$3x + g'(y) = 3x - 2y \implies g'(y) = -2y$$

Integrate to find $g(y)$:

$$g(y) = \int -2y\, dy = -y^2$$

So, the potential function is $f(x, y) = x^2 + 3xy - y^2$ (we can ignore the constant K).

Step 3: Apply the Fundamental Theorem of Line Integrals.
The theorem states $\int_C \nabla f \cdot d\vec{r} = f(\text{end point}) - f(\text{start point})$. Start point $A = (-1, 1)$. End point $B = (2, -1)$.

$$f(B) = f(2, -1) = (2)^2 + 3(2)(-1) - (-1)^2 = 4 - 6 - 1 = -3$$

$$f(A) = f(-1, 1) = (-1)^2 + 3(-1)(1) - (1)^2 = 1 - 3 - 1 = -3$$

The value of the integral is:

$$f(B) - f(A) = -3 - (-3) = 0$$

Problem 19: FTC with Given Potential

Evaluate the integral $\int_C \nabla f \cdot d\vec{r}$, where $f(x, y, z) = x\ln(y) + z^2$ and C is any path from $(1, e, 0)$ to $(3, 1, 2)$.

The integral is of the form $\int_C \nabla f \cdot d\vec{r}$, which is the line integral of a conservative vector field. By the Fundamental Theorem of Line Integrals, its value is the difference in the potential function f evaluated at the endpoints of the path.

Start point $A = (1, e, 0)$. End point $B = (3, 1, 2)$.

Step 1: Evaluate f at the end point.

$$f(B) = f(3, 1, 2) = 3\ln(1) + (2)^2 = 3(0) + 4 = 4$$

Step 2: Evaluate f at the start point.

$$f(A) = f(1, e, 0) = 1\ln(e) + (0)^2 = 1(1) + 0 = 1$$

Step 3: Compute the difference.

$$\int_C \nabla f \cdot d\vec{r} = f(B) - f(A) = 4 - 1 = 3$$

Problem 20: FTC Over a Closed Loop

Evaluate $\oint_C \vec{F} \cdot d\vec{r}$ for the vector field $\vec{F} = \langle \frac{x}{x^2+y^2}, \frac{y}{x^2+y^2} \rangle$, where C is the boundary of the square with vertices $(1, 1)$, $(-1, 1)$, $(-1, -1)$, and $(1, -1)$ oriented counterclockwise. (Assume the field is conservative on a suitable domain).

The Fundamental Theorem of Line Integrals states that the line integral of a conservative vector field over a closed path is zero.

Step 1: Verify the field is conservative.

Let $f(x, y) = \frac{1}{2}\ln(x^2 + y^2)$. Then

$$\frac{\partial f}{\partial x} = \frac{1}{2} \cdot \frac{2x}{x^2 + y^2} = \frac{x}{x^2 + y^2}$$

$$\frac{\partial f}{\partial y} = \frac{1}{2} \cdot \frac{2y}{x^2 + y^2} = \frac{y}{x^2 + y^2}$$

So, $\vec{F} = \nabla f$, which confirms the field is conservative.

Step 2: Check the domain.

The vector field is defined everywhere except at the origin $(0, 0)$. The given path is a square that goes from $(1, 1)$ to $(-1, 1)$ to $(-1, -1)$ to $(1, -1)$ and back to $(1, 1)$. This path does not contain or pass through the origin. Therefore, the field is conservative on the entire path and the region it encloses.

Step 3: Apply the theorem for a closed loop.

Since $\vec{F}$ is conservative and C is a closed loop in a region where $\vec{F}$ is defined, the integral must be zero.

$$\oint_C \vec{F} \cdot d\vec{r} = 0$$

Problem 21: The Jacobian Determinant

For a transformation from the uv-plane to the xy-plane, given by $x = g(u, v)$ and $y = h(u, v)$, Compute the Jacobian $\frac{\partial(x,y)}{\partial(u,v)}$ determinant for each of the following transformations.

(a) $x = 5u - 2v, \quad y = 3u + v$

(b) $x = u^2 - v^2, \quad y = 2uv$

(c) $x = u^2 \cos(v), \quad y = u^2 \sin(v)$

(d) $x = e^u \cos(v), \quad y = e^u \sin(v)$

The Jacobian determinant is given by the formula $J(u, v) = \det \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix} = \frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u}$.

(a) $x = 5u - 2v, \quad y = 3u + v$

$$\frac{\partial x}{\partial u} = 5, \quad \frac{\partial x}{\partial v} = -2$$

$$\frac{\partial y}{\partial u} = 3, \quad \frac{\partial y}{\partial v} = 1$$

$$J = (5)(1) - (-2)(3) = 5 + 6 = 11$$

(b) $x = u^2 - v^2, \quad y = 2uv$

$$\frac{\partial x}{\partial u} = 2u, \quad \frac{\partial x}{\partial v} = -2v$$

$$\frac{\partial y}{\partial u} = 2v, \quad \frac{\partial y}{\partial v} = 2u$$

$$J = (2u)(2u) - (-2v)(2v) = 4u^2 + 4v^2 = 4(u^2 + v^2)$$

(c) $x = u^2 \cos(v), \quad y = u^2 \sin(v)$

$$\frac{\partial x}{\partial u} = 2u\cos(v), \quad \frac{\partial x}{\partial v} = -u^2 \sin(v)$$

$$\frac{\partial y}{\partial u} = 2u\sin(v), \quad \frac{\partial y}{\partial v} = u^2 \cos(v)$$

$$J = (2u\cos v)(u^2 \cos v) - (-u^2 \sin v)(2u \sin v)$$

$$= 2u^3 \cos^2 v + 2u^3 \sin^2 v = 2u^3(\cos^2 v + \sin^2 v) = 2u^3$$

(d) $x = e^u \cos(v), \quad y = e^u \sin(v)$

$$\frac{\partial x}{\partial u} = e^u \cos(v), \quad \frac{\partial x}{\partial v} = -e^u \sin(v)$$

$$\frac{\partial y}{\partial u} = e^u \sin(v), \quad \frac{\partial y}{\partial v} = e^u \cos(v)$$

$$J = (e^u \cos v)(e^u \cos v) - (-e^u \sin v)(e^u \sin v)$$

$$= e^{2u} \cos^2 v + e^{2u} \sin^2 v = e^{2u}(\cos^2 v + \sin^2 v) = e^{2u}$$

Homework 11 Solutions

Homework 11: Green's Theorem, Curl/Divergence, and Surfaces
Due: Sunday, November 30

Problem 1: Divergence and Curl

Compute the divergence ($\nabla \cdot \vec{F}$) and curl ($\nabla \times \vec{F}$) for the vector field $\vec{F} = \langle x^2y, yz^2, zx^2 \rangle$.

Let $P = x^2y$, $Q = yz^2$, and $R = zx^2$.

Divergence: The formula for divergence is $\nabla \cdot \vec{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$.

$$\frac{\partial P}{\partial x} = \frac{\partial}{\partial x}(x^2y) = 2xy$$

$$\frac{\partial Q}{\partial y} = \frac{\partial}{\partial y}(yz^2) = z^2$$

$$\frac{\partial R}{\partial z} = \frac{\partial}{\partial z}(zx^2) = x^2$$

$$\nabla \cdot \vec{F} = 2xy + z^2 + x^2$$

Curl: The formula for curl is $\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}$.

$$\nabla \times \vec{F} = \vec{i}\left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}\right) - \vec{j}\left(\frac{\partial R}{\partial x} - \frac{\partial P}{\partial z}\right) + \vec{k}\left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)$$

$$\frac{\partial R}{\partial y} = \frac{\partial}{\partial y}(zx^2) = 0$$

$$\frac{\partial Q}{\partial z} = \frac{\partial}{\partial z}(yz^2) = 2yz$$

$$\frac{\partial R}{\partial x} = \frac{\partial}{\partial x}(zx^2) = 2zx$$

$$\frac{\partial P}{\partial z} = \frac{\partial}{\partial z}(x^2y) = 0$$

$$\frac{\partial Q}{\partial x} = \frac{\partial}{\partial x}(yz^2) = 0$$

$$\frac{\partial P}{\partial y} = \frac{\partial}{\partial y}(x^2y) = x^2$$

$$\nabla \times \vec{F} = \vec{i}(0 - 2yz) - \vec{j}(2zx - 0) + \vec{k}(0 - x^2) = \langle -2yz, -2zx, -x^2 \rangle$$

Problem 2: Divergence and Curl

Compute the divergence and curl for the vector field $\vec{F} = \langle e^{xy}, \cos(yz), z^3 \rangle$.

Let $P = e^{xy}$, $Q = \cos(yz)$, and $R = z^3$.

Divergence:

$$\frac{\partial P}{\partial x} = \frac{\partial}{\partial x}(e^{xy}) = ye^{xy}$$
$$\frac{\partial Q}{\partial y} = \frac{\partial}{\partial y}(\cos(yz)) = -z\sin(yz)$$
$$\frac{\partial R}{\partial z} = \frac{\partial}{\partial z}(z^3) = 3z^2$$
$$\nabla \cdot \vec{F} = ye^{xy} - z\sin(yz) + 3z^2$$

Curl:

$$\frac{\partial R}{\partial y} = 0, \quad \frac{\partial Q}{\partial z} = -y\sin(yz)$$
$$\frac{\partial R}{\partial x} = 0, \quad \frac{\partial P}{\partial z} = 0$$
$$\frac{\partial Q}{\partial x} = 0, \quad \frac{\partial P}{\partial y} = xe^{xy}$$
$$\nabla \times \vec{F} = \vec{i}(0 - (-y\sin(yz))) - \vec{j}(0 - 0) + \vec{k}(0 - xe^{xy}) = \langle y\sin(yz), 0, -xe^{xy} \rangle$$

Problem 3: Divergence and Curl

Compute the divergence and curl for the vector field $\vec{F} = \langle y\ln(z), x\ln(z), xy/z \rangle$.

Let $P = y\ln(z)$, $Q = x\ln(z)$, and $R = xy/z$.

Divergence:

$$\frac{\partial P}{\partial x} = 0$$
$$\frac{\partial Q}{\partial y} = 0$$
$$\frac{\partial R}{\partial z} = \frac{\partial}{\partial z}(xyz^{-1}) = -xyz^{-2} = -\frac{xy}{z^2}$$
$$\nabla \cdot \vec{F} = 0 + 0 - \frac{xy}{z^2} = -\frac{xy}{z^2}$$

Curl:

$$\frac{\partial R}{\partial y} = \frac{x}{z}, \quad \frac{\partial Q}{\partial z} = \frac{x}{z}$$
$$\frac{\partial R}{\partial x} = \frac{y}{z}, \quad \frac{\partial P}{\partial z} = \frac{y}{z}$$
$$\frac{\partial Q}{\partial x} = \ln(z), \quad \frac{\partial P}{\partial y} = \ln(z)$$
$$\nabla \times \vec{F} = \vec{i}\left(\frac{x}{z} - \frac{x}{z}\right) - \vec{j}\left(\frac{y}{z} - \frac{y}{z}\right) + \vec{k}(\ln(z) - \ln(z)) = \langle 0, 0, 0 \rangle$$

Since the curl is the zero vector, this field is conservative.

Problem 4: Divergence and Curl

Compute the divergence and curl for the vector field $\vec{F} = \langle \cos(xy), -\sin(xy), z \rangle$.

Let $P = \cos(xy)$, $Q = -\sin(xy)$, and $R = z$.

Divergence:

$$\frac{\partial P}{\partial x} = -y\sin(xy)$$
$$\frac{\partial Q}{\partial y} = -x\cos(xy)$$
$$\frac{\partial R}{\partial z} = 1$$
$$\nabla \cdot \vec{F} = -y\sin(xy) - x\cos(xy) + 1$$

Curl:

$$\frac{\partial R}{\partial y} = 0, \quad \frac{\partial Q}{\partial z} = 0$$
$$\frac{\partial R}{\partial x} = 0, \quad \frac{\partial P}{\partial z} = 0$$
$$\frac{\partial Q}{\partial x} = -y\cos(xy), \quad \frac{\partial P}{\partial y} = -x\sin(xy)$$
$$\nabla \times \vec{F} = \vec{i}(0-0) - \vec{j}(0-0) + \vec{k}(-y\cos(xy) - (-x\sin(xy)))$$
$$= \langle 0, 0, x\sin(xy) - y\cos(xy) \rangle$$

Problem 5: Divergence and Curl

Compute the divergence and curl for the vector field $\vec{F} = \langle 3x, 5y^2, -2z \rangle$.

Let $P = 3x$, $Q = 5y^2$, and $R = -2z$.

Divergence:

$$\frac{\partial P}{\partial x} = 3$$
$$\frac{\partial Q}{\partial y} = 10y$$
$$\frac{\partial R}{\partial z} = -2$$
$$\nabla \cdot \vec{F} = 3 + 10y - 2 = 1 + 10y$$

Curl: The components P, Q, R are functions of only x, y, z respectively. All mixed partial derivatives will be zero.

$$\frac{\partial R}{\partial y} = 0, \quad \frac{\partial Q}{\partial z} = 0$$
$$\frac{\partial R}{\partial x} = 0, \quad \frac{\partial P}{\partial z} = 0$$
$$\frac{\partial Q}{\partial x} = 0, \quad \frac{\partial P}{\partial y} = 0$$
$$\nabla \times \vec{F} = \vec{i}(0-0) - \vec{j}(0-0) + \vec{k}(0-0) = \langle 0, 0, 0 \rangle$$

This field is conservative.

Problem 6: Divergence and Curl

Compute the divergence and curl for the vector field $\vec{F} = \langle \frac{x}{x^2+y^2+z^2}, \frac{y}{x^2+y^2+z^2}, \frac{z}{x^2+y^2+z^2} \rangle$.

Let $\rho^2 = x^2 + y^2 + z^2$. Then $\vec{F} = \langle x/\rho^2, y/\rho^2, z/\rho^2 \rangle$. This is a radial field. This field is the gradient of $f(x, y, z) = \frac{1}{2}\ln(x^2 + y^2 + z^2)$, so its curl must be the zero vector. Let's verify.

Curl: We only need to check one component. For the $\vec{k}$ component, we need $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$.

$$\frac{\partial Q}{\partial x} = \frac{\partial}{\partial x}\left(\frac{y}{x^2+y^2+z^2}\right) = \frac{0 - y(2x)}{(x^2+y^2+z^2)^2} = \frac{-2xy}{(\rho^2)^2} = \frac{-2xy}{\rho^4}$$

$$\frac{\partial P}{\partial y} = \frac{\partial}{\partial y}\left(\frac{x}{x^2+y^2+z^2}\right) = \frac{0 - x(2y)}{(x^2+y^2+z^2)^2} = \frac{-2xy}{(\rho^2)^2} = \frac{-2xy}{\rho^4}$$

The $\vec{k}$ component is $\frac{-2xy}{\rho^4} - \frac{-2xy}{\rho^4} = 0$. By symmetry, the other components are also zero.

$$\nabla \times \vec{F} = \vec{0}$$

Divergence: We compute $\frac{\partial P}{\partial x}$ using the quotient rule.

$$\frac{\partial P}{\partial x} = \frac{\partial}{\partial x}\left(\frac{x}{x^2+y^2+z^2}\right) = \frac{1(x^2+y^2+z^2) - x(2x)}{(x^2+y^2+z^2)^2} = \frac{x^2+y^2+z^2-2x^2}{\rho^4} = \frac{-x^2+y^2+z^2}{\rho^4}$$

By symmetry, the other partials are:

$$\frac{\partial Q}{\partial y} = \frac{x^2-y^2+z^2}{\rho^4} \quad \text{and} \quad \frac{\partial R}{\partial z} = \frac{x^2+y^2-z^2}{\rho^4}$$

Summing them:

$$\nabla \cdot \vec{F} = \frac{(-x^2+y^2+z^2) + (x^2-y^2+z^2) + (x^2+y^2-z^2)}{\rho^4}$$

$$= \frac{x^2+y^2+z^2}{\rho^4} = \frac{\rho^2}{\rho^4} = \frac{1}{\rho^2} = \frac{1}{x^2+y^2+z^2}$$

(Note: The divergence is zero only if $x^2 + y^2 + z^2 = 0$, which is only at the origin where the field is undefined).

Problem 7: Green's Theorem (Circulation)

Use Green's theorem to evaluate the line integral $\oint_C (3y\,dx - 5x\,dy)$, where C is the boundary of the rectangle with vertices $(0, 0), (4, 0), (4, 2), (0, 2)$, oriented counterclockwise.

Step 1: Identify P, Q and apply Green's Theorem. Green's Theorem (Circulation Form): $\oint_C P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA$. Here $P = 3y$ and $Q = -5x$.

Step 2: Compute the partial derivatives.

$$\frac{\partial Q}{\partial x} = \frac{\partial}{\partial x}(-5x) = -5$$

$$\frac{\partial P}{\partial y} = \frac{\partial}{\partial y}(3y) = 3$$

The integrand is $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = -5 - 3 = -8$.

Step 3: Set up and evaluate the double integral. The region D is the rectangle $0 \le x \le 4$, $0 \le y \le 2$.

$$\iint_D (-8)\, dA = -8 \iint_D dA = -8 \times (\text{Area of D})$$

The area of the rectangle is $4 \times 2 = 8$.

$$\oint_C (3y\, dx - 5x\, dy) = -8 \times 8 = -64$$

Problem 8: Green's Theorem (Circulation)

Use Green's theorem to evaluate $\oint_C (x^2 y\, dx + x^3\, dy)$, where C is the ellipse $x^2 + 4y^2 = 4$, oriented counterclockwise.

Step 1: Identify P, Q and apply Green's Theorem. $P = x^2 y$ and $Q = x^3$.

$$\frac{\partial Q}{\partial x} = \frac{\partial}{\partial x}(x^3) = 3x^2$$

$$\frac{\partial P}{\partial y} = \frac{\partial}{\partial y}(x^2 y) = x^2$$

The integral becomes $\iint_D (3x^2 - x^2)\, dA = \iint_D 2x^2\, dA$.

Step 2: Set up the integral over the elliptical region D. The ellipse is $\frac{x^2}{4} + \frac{y^2}{1} = 1$. We use a change of variables: $x = 2r\cos\theta$, $y = r\sin\theta$. The domain for these new coordinates is $0 \le r \le 1$, $0 \le \theta \le 2\pi$. The Jacobian of this transformation is $\frac{\partial(x,y)}{\partial(r,\theta)} = \det \begin{pmatrix} 2\cos\theta & -2r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix} = 2r\cos^2\theta + 2r\sin^2\theta = 2r$. So, $dA = 2r\, dr\, d\theta$.

Step 3: Evaluate the integral.

$$\iint_D 2x^2\, dA = \int_0^{2\pi} \int_0^1 2(2r\cos\theta)^2 (2r\, dr\, d\theta)$$

$$= \int_0^{2\pi} \int_0^1 2(4r^2\cos^2\theta)(2r)\, dr\, d\theta = \int_0^{2\pi} \int_0^1 16r^3\cos^2\theta\, dr\, d\theta$$

$$= 16 \int_0^{2\pi} \cos^2\theta \left[\frac{r^4}{4}\right]_0^1 d\theta = 16 \int_0^{2\pi} \cos^2\theta \left(\frac{1}{4}\right) d\theta = 4 \int_0^{2\pi} \cos^2\theta\, d\theta$$

Using the identity $\cos^2\theta = \frac{1+\cos(2\theta)}{2}$:

$$4 \int_0^{2\pi} \frac{1+\cos(2\theta)}{2}\, d\theta = 2\left[\theta + \frac{\sin(2\theta)}{2}\right]_0^{2\pi} = 2((2\pi + 0) - (0 + 0)) = 4\pi$$

Problem 9: Area with Green's Theorem

Use a line integral to find the area of the region bounded by the curves $y = x^2$ and $y = x$.
Hint: *Use the formula Area* $= \frac{1}{2}\oint_C x\, dy - y\, dx$.

The curves intersect when $x^2 = x \implies x(x-1) = 0$, so at $x = 0$ and $x = 1$. The points are $(0,0)$ and $(1,1)$. The boundary C is traversed counter-clockwise. This means going from $(0,0)$ to $(1,1)$ along $C_1 : y = x^2$ and then from $(1,1)$ back to $(0,0)$ along $C_2 : y = x$.

Step 1: Integrate along C_1. Parameterize C_1: $x = t, y = t^2$ for $0 \le t \le 1$. $dx = dt$, $dy = 2t\,dt$.

$$\int_{C_1} x\,dy - y\,dx = \int_0^1 t(2t\,dt) - t^2(dt) = \int_0^1 (2t^2 - t^2)\,dt = \int_0^1 t^2\,dt = \left[\frac{t^3}{3}\right]_0^1 = \frac{1}{3}$$

Step 2: Integrate along C_2. We must go from $(1,1)$ to $(0,0)$. Parameterize C_2: $x = 1-t, y = 1-t$ for $0 \le t \le 1$. $dx = -dt$, $dy = -dt$.

$$\int_{C_2} x\,dy - y\,dx = \int_0^1 (1-t)(-dt) - (1-t)(-dt) = \int_0^1 0\,dt = 0$$

Step 3: Calculate the total area. The total line integral is $\oint_C = \int_{C_1} + \int_{C_2} = \frac{1}{3} + 0 = \frac{1}{3}$.

$$\text{Area} = \frac{1}{2}\oint_C x\,dy - y\,dx = \frac{1}{2}\left(\frac{1}{3}\right) = \frac{1}{6}$$

Problem 10: Green's Theorem (Flux)

Use the flux form of Green's theorem to find the flux of $\vec{F} = \langle x^2, y^2\rangle$ out of the triangular region with vertices $(0,0), (2,0)$, and $(0,3)$.

The flux form of Green's Theorem is $\oint_C \vec{F}\cdot\vec{n}\,ds = \iint_D (\nabla\cdot\vec{F})\,dA$. Here $\vec{F} = \langle P, Q\rangle = \langle x^2, y^2\rangle$.

Step 1: Compute the divergence of $\vec{F}$.

$$\nabla\cdot\vec{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} = \frac{\partial}{\partial x}(x^2) + \frac{\partial}{\partial y}(y^2) = 2x + 2y$$

Step 2: Set up the double integral over the triangle D. The vertices are $(0,0), (2,0), (0,3)$. The line connecting $(2,0)$ and $(0,3)$ is $y = -\frac{3}{2}x + 3$. The bounds for the integral are $0 \le x \le 2$ and $0 \le y \le -\frac{3}{2}x + 3$.

$$\iint_D (2x+2y)\,dA = \int_0^2 \int_0^{-3/2x+3} (2x+2y)\,dy\,dx$$

Step 3: Evaluate the integral.

$$\int_0^2 \left[2xy + y^2\right]_0^{-3/2x+3} dx = \int_0^2 \left(2x(-\frac{3}{2}x+3) + (-\frac{3}{2}x+3)^2\right) dx$$

$$= \int_0^2 \left(-3x^2 + 6x + \frac{9}{4}x^2 - 9x + 9\right) dx = \int_0^2 \left(-\frac{3}{4}x^2 - 3x + 9\right) dx$$

$$= \left[-\frac{3}{4}\frac{x^3}{3} - \frac{3x^2}{2} + 9x\right]_0^2 = \left[-\frac{x^3}{4} - \frac{3x^2}{2} + 9x\right]_0^2$$

$$= \left(-\frac{8}{4} - \frac{3(4)}{2} + 9(2)\right) - (0) = -2 - 6 + 18 = 10$$

Problem 11: Green's Theorem (Flux)

Use the flux form of Green's theorem to find the flux of $\vec{F} = \langle \sin(y) + 3x, e^x - 5y \rangle$ across the unit circle $x^2 + y^2 = 1$.

Flux is given by $\iint_D (\nabla \cdot \vec{F})\, dA$. Here $P = \sin(y) + 3x$ and $Q = e^x - 5y$.

Step 1: Compute the divergence of $\vec{F}$.

$$\frac{\partial P}{\partial x} = \frac{\partial}{\partial x}(\sin(y) + 3x) = 3$$

$$\frac{\partial Q}{\partial y} = \frac{\partial}{\partial y}(e^x - 5y) = -5$$

$$\nabla \cdot \vec{F} = 3 + (-5) = -2$$

Step 2: Set up and evaluate the integral. The region D is the unit disk.

$$\iint_D (-2)\, dA = -2 \iint_D dA = -2 \times (\text{Area of D})$$

The area of the unit disk is $\pi r^2 = \pi(1)^2 = \pi$.

$$\text{Flux} = -2 \times \pi = -2\pi$$

Problem 12: Parameterize a Plane

Parameterize the surface S which is the part of the plane $z = 4 - x - 2y$ that lies over the rectangle $D = [0, 2] \times [0, 1]$. Then, compute the normal vector $\vec{r}_x \times \vec{r}_y$.

Step 1: Parameterize the surface. For a surface given as a function $z = g(x, y)$, we can use x and y as the parameters.

$$\vec{r}(x, y) = \langle x, y, z \rangle = \langle x, y, 4 - x - 2y \rangle$$

The domain is the given rectangle $0 \leq x \leq 2$, $0 \leq y \leq 1$.

Step 2: Compute the partial derivative vectors.

$$\vec{r}_x = \frac{\partial \vec{r}}{\partial x} = \langle 1, 0, -1 \rangle$$

$$\vec{r}_y = \frac{\partial \vec{r}}{\partial y} = \langle 0, 1, -2 \rangle$$

Step 3: Compute the cross product (the normal vector).

$$\vec{r}_x \times \vec{r}_y = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & -1 \\ 0 & 1 & -2 \end{vmatrix} = \vec{i}(0 - (-1)) - \vec{j}(-2 - 0) + \vec{k}(1 - 0)$$

$$= \langle 1, 2, 1 \rangle$$

Since the $\vec{k}$ component is positive (1 ¿ 0), this is an upward-pointing normal vector.

Problem 13: Parameterize a Cylinder

Parameterize the surface S which is the part of the cylinder $x^2 + y^2 = 9$ for $0 \leq z \leq 5$. Then, compute the outward-pointing normal vector $\vec{r}_\theta \times \vec{r}_z$.

Step 1: Parameterize the surface. We use cylindrical coordinates. The radius is fixed at $r = \sqrt{9} = 3$. The parameters are angle θ and height z.

$$x = 3\cos\theta, \quad y = 3\sin\theta, \quad z = z$$

$$\vec{r}(\theta, z) = \langle 3\cos\theta, 3\sin\theta, z\rangle$$

The domains are $0 \leq \theta \leq 2\pi$ and $0 \leq z \leq 5$.

Step 2: Compute the partial derivative vectors.

$$\vec{r}_\theta = \frac{\partial \vec{r}}{\partial \theta} = \langle -3\sin\theta, 3\cos\theta, 0\rangle$$

$$\vec{r}_z = \frac{\partial \vec{r}}{\partial z} = \langle 0, 0, 1\rangle$$

Step 3: Compute the cross product.

$$\vec{r}_\theta \times \vec{r}_z = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -3\sin\theta & 3\cos\theta & 0 \\ 0 & 0 & 1 \end{vmatrix} = \vec{i}(3\cos\theta - 0) - \vec{j}(-3\sin\theta - 0) + \vec{k}(0 - 0)$$

$$= \langle 3\cos\theta, 3\sin\theta, 0\rangle$$

This vector points radially outward from the z-axis, so it is the correct outward-pointing normal vector.

Problem 14: Parameterize a Paraboloid

Parameterize the surface S which is the part of the paraboloid $z = x^2 + y^2$ that lies below the plane $z = 4$. Then, compute the upward-pointing normal vector.

Step 1: Parameterize the surface. Cylindrical coordinates are natural here. Let $x = r\cos\theta$ and $y = r\sin\theta$. Then $z = x^2 + y^2 = r^2$. The condition $z < 4$ means $r^2 < 4$, so $0 \leq r \leq 2$. The angle θ goes from 0 to 2π. The parameterization is:

$$\vec{r}(r, \theta) = \langle r\cos\theta, r\sin\theta, r^2\rangle$$

Step 2: Compute the partial derivative vectors.

$$\vec{r}_r = \frac{\partial \vec{r}}{\partial r} = \langle \cos\theta, \sin\theta, 2r\rangle$$

$$\vec{r}_\theta = \frac{\partial \vec{r}}{\partial \theta} = \langle -r\sin\theta, r\cos\theta, 0\rangle$$

Step 3: Compute the cross product.

$$\vec{r}_r \times \vec{r}_\theta = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \cos\theta & \sin\theta & 2r \\ -r\sin\theta & r\cos\theta & 0 \end{vmatrix}$$

$$= \vec{i}(0 - 2r^2\cos\theta) - \vec{j}(0 - (-2r^2\sin\theta)) + \vec{k}(r\cos^2\theta - (-r\sin^2\theta))$$
$$= \langle -2r^2\cos\theta, -2r^2\sin\theta, r(\cos^2\theta + \sin^2\theta)\rangle$$
$$= \langle -2r^2\cos\theta, -2r^2\sin\theta, r\rangle$$

The problem asks for an upward-pointing normal vector. The $\vec{k}$ component is r, which is positive for $r > 0$.

Problem 15: Surface Flux (Plane)

Evaluate $\iint_S \vec{F}\cdot d\vec{S}$ where $\vec{F} = \langle 1,1,1\rangle$ and S is the surface from **Problem 12**, oriented upward.

The surface flux integral is $\iint_D \vec{F}(\vec{r}(x,y))\cdot(\vec{r}_x \times \vec{r}_y)\,dA$.

Step 1: Recall the parameterization and normal vector. From Problem 12, the surface is $\vec{r}(x,y) = \langle x, y, 4 - x - 2y\rangle$. The upward normal vector is $\vec{r}_x \times \vec{r}_y = \langle 1,2,1\rangle$. The domain of integration D is the rectangle $0 \le x \le 2$, $0 \le y \le 1$.

Step 2: Compute the dot product. The vector field $\vec{F}$ is constant, $\vec{F} = \langle 1,1,1\rangle$.

$$\vec{F}\cdot(\vec{r}_x \times \vec{r}_y) = \langle 1,1,1\rangle \cdot \langle 1,2,1\rangle = 1(1) + 1(2) + 1(1) = 4$$

Step 3: Evaluate the integral.

$$\iint_D 4\,dA = 4 \times (\text{Area of D}) = 4 \times (2 \times 1) = 8$$

Problem 16: Surface Flux (Cylinder)

Evaluate $\iint_S \vec{F}\cdot d\vec{S}$ where $\vec{F} = \langle x,y,z\rangle$ and S is the lateral surface of the cylinder from **Problem 13**, oriented outward.

The integral is $\iint_D \vec{F}(\vec{r}(\theta,z))\cdot(\vec{r}_\theta \times \vec{r}_z)\,dA$.

Step 1: Recall the parameterization and normal vector. From Problem 13, $\vec{r}(\theta,z) = \langle 3\cos\theta, 3\sin\theta, z\rangle$. The outward normal vector is $\vec{r}_\theta \times \vec{r}_z = \langle 3\cos\theta, 3\sin\theta, 0\rangle$. The domain is $0 \le \theta \le 2\pi$ and $0 \le z \le 5$.

Step 2: Express $\vec{F}$ in terms of the parameters.

$$\vec{F}(\vec{r}(\theta,z)) = \langle x,y,z\rangle = \langle 3\cos\theta, 3\sin\theta, z\rangle$$

Step 3: Compute the dot product.

$$\vec{F}\cdot(\vec{r}_\theta \times \vec{r}_z) = \langle 3\cos\theta, 3\sin\theta, z\rangle \cdot \langle 3\cos\theta, 3\sin\theta, 0\rangle$$
$$= 9\cos^2\theta + 9\sin^2\theta + 0 = 9(\cos^2\theta + \sin^2\theta) = 9$$

Step 4: Evaluate the integral.

$$\iint_D 9\,dA = \int_0^5\int_0^{2\pi} 9\,d\theta\,dz = 9\int_0^5 [\theta]_0^{2\pi}\,dz = 9\int_0^5 2\pi\,dz$$
$$= 18\pi\int_0^5 dz = 18\pi[z]_0^5 = 18\pi(5) = 90\pi$$

Problem 17: Surface Flux (Paraboloid)

Evaluate $\iint_S \vec{F} \cdot d\vec{S}$ where $\vec{F} = \langle 0, 0, z \rangle$ and S is the paraboloid surface from **Problem 14**, oriented upward.

The upward normal is

$$d\vec{S} = \langle -2r^2 \cos(\theta), -2r^2 \sin(\theta), r \rangle \, d\theta dr$$

The vector field is $\vec{F} = \langle 0, 0, z \rangle$. On the surface, $z = r^2$, so $\vec{F} = \langle 0, 0, r^2 \rangle$.

$$\vec{F} \cdot \langle -2r^2 \cos(\theta), -2r^2 \sin(\theta), r \rangle = \langle 0, 0, r^2 \rangle \cdot \langle -2r^2 \cos(\theta), -2r^2 \sin(\theta), r \rangle = r^3$$

$$\iint_S \vec{F} \cdot d\vec{S} = \int_0^{2\pi} \int_0^2 (r^2)(r \, dr \, d\theta) = \int_0^{2\pi} \int_0^2 r^3 \, dr \, d\theta$$

$$= \int_0^{2\pi} \left[\frac{r^4}{4} \right]_0^2 d\theta = \int_0^{2\pi} \left(\frac{16}{4} - 0 \right) d\theta = \int_0^{2\pi} 4 \, d\theta = 4[\theta]_0^{2\pi} = 8\pi$$

Problem 18: Surface Flux (Hemisphere)

Evaluate $\iint_S \vec{F} \cdot d\vec{S}$ where $\vec{F} = \langle x, y, 0 \rangle$ and S is the upper hemisphere $x^2 + y^2 + z^2 = 1$, $z \geq 0$, oriented outward (upward).

Step 1: Parameterize the surface. Use spherical coordinates with radius $\rho = 1$. Parameters are ϕ and θ. $\vec{r}(\phi, \theta) = \langle \sin\phi \cos\theta, \sin\phi \sin\theta, \cos\phi \rangle$. For the upper hemisphere, $0 \leq \phi \leq \pi/2$ and $0 \leq \theta \leq 2\pi$. The outward normal vector is $\vec{r}_\phi \times \vec{r}_\theta = \sin\phi \vec{r} = \langle \sin^2\phi \cos\theta, \sin^2\phi \sin\theta, \sin\phi \cos\phi \rangle$.

Step 2: Express $\vec{F}$ in terms of parameters.

$$\vec{F}(\vec{r}(\phi, \theta)) = \langle x, y, 0 \rangle = \langle \sin\phi \cos\theta, \sin\phi \sin\theta, 0 \rangle$$

Step 3: Compute the dot product.

$$\vec{F} \cdot (\vec{r}_\phi \times \vec{r}_\theta) = (\sin\phi \cos\theta)(\sin^2\phi \cos\theta) + (\sin\phi \sin\theta)(\sin^2\phi \sin\theta) + 0$$

$$= \sin^3\phi \cos^2\theta + \sin^3\phi \sin^2\theta = \sin^3\phi(\cos^2\theta + \sin^2\theta) = \sin^3\phi$$

Step 4: Evaluate the integral.

$$\int_0^{2\pi} \int_0^{\pi/2} \sin^3\phi \, d\phi \, d\theta = 2\pi \int_0^{\pi/2} \sin^3\phi \, d\phi$$

To integrate $\sin^3\phi$, we write it as $\sin\phi(1 - \cos^2\phi) = \sin\phi - \sin\phi\cos^2\phi$.

$$\int (\sin\phi - \sin\phi\cos^2\phi) \, d\phi = -\cos\phi + \frac{\cos^3\phi}{3}$$

$$2\pi \left[-\cos\phi + \frac{\cos^3\phi}{3} \right]_0^{\pi/2} = 2\pi \left[(0 + 0) - (-1 + \frac{1}{3}) \right] = 2\pi \left[-(-\frac{2}{3}) \right] = \frac{4\pi}{3}$$

Problem 19: Surface Flux (Simple Surface)

Evaluate $\iint_S \vec{F} \cdot d\vec{S}$ where $\vec{F} = \langle z, x, y \rangle$ and S is the part of the plane $z = x$ that lies over the square $D = [0,1] \times [0,1]$, oriented upward.

Step 1: Identify components for the flux integral. The surface is $z = g(x,y) = x$. The upward normal vector is $d\vec{S} = \langle -g_x, -g_y, 1 \rangle \, dA = \langle -1, 0, 1 \rangle \, dA$. The vector field on the surface is $\vec{F} = \langle z, x, y \rangle = \langle x, x, y \rangle$. The domain is the square $0 \le x \le 1$, $0 \le y \le 1$.

Step 2: Compute the dot product.

$$\vec{F} \cdot \langle -1, 0, 1 \rangle = \langle x, x, y \rangle \cdot \langle -1, 0, 1 \rangle = -x + 0 + y = y - x$$

Step 3: Evaluate the integral.

$$\iint_D (y - x) \, dA = \int_0^1 \int_0^1 (y - x) \, dx \, dy$$

$$= \int_0^1 \left[yx - \frac{x^2}{2} \right]_0^1 dy = \int_0^1 \left(y - \frac{1}{2} \right) dy$$

$$= \left[\frac{y^2}{2} - \frac{y}{2} \right]_0^1 = \left(\frac{1}{2} - \frac{1}{2} \right) - (0) = 0$$

Problem 20: Surface Flux (Cone)

Evaluate $\iint_S \vec{F} \cdot d\vec{S}$ where $\vec{F} = \langle x, y, -2z \rangle$ and S is the part of the cone $z = \sqrt{x^2 + y^2}$ between the planes $z = 1$ and $z = 2$, oriented upward.

Step 1: Parameterize the surface. In cylindrical coordinates, $z = r$. The condition $1 \le z \le 2$ implies $1 \le r \le 2$. The parameterization is $\vec{r}(r, \theta) = \langle r\cos\theta, r\sin\theta, r \rangle$. The domain is $1 \le r \le 2$, $0 \le \theta \le 2\pi$.

Step 2: Compute the normal vector.

$$\vec{r}_r = \langle \cos\theta, \sin\theta, 1 \rangle$$

$$\vec{r}_\theta = \langle -r\sin\theta, r\cos\theta, 0 \rangle$$

$$\vec{r}_r \times \vec{r}_\theta = \langle -r\cos\theta, -r\sin\theta, r\cos^2\theta + r\sin^2\theta \rangle = \langle -r\cos\theta, -r\sin\theta, r \rangle$$

The $\vec{k}$ component is r, which is positive. This is the upward normal vector.

Step 3: Express $\vec{F}$ and compute the dot product. On the surface, $\vec{F} = \langle r\cos\theta, r\sin\theta, -2r \rangle$.

$$\vec{F} \cdot (\vec{r}_r \times \vec{r}_\theta) = (r\cos\theta)(-r\cos\theta) + (r\sin\theta)(-r\sin\theta) + (-2r)(r)$$

$$= -r^2\cos^2\theta - r^2\sin^2\theta - 2r^2 = -r^2(\cos^2\theta + \sin^2\theta) - 2r^2 = -r^2 - 2r^2 = -3r^2$$

Step 4: Evaluate the integral.

$$\int_0^{2\pi} \int_1^2 (-3r^2) \, dr \, d\theta = -3 \int_0^{2\pi} \left[\frac{r^3}{3} \right]_1^2 d\theta$$

$$= -3 \int_0^{2\pi} \left(\frac{8}{3} - \frac{1}{3} \right) d\theta = -3 \int_0^{2\pi} \frac{7}{3} \, d\theta = -7 \int_0^{2\pi} d\theta = -7(2\pi) = -14\pi$$